

glutamine-synthetase adenylyltransferase

glutamine-synthetase adenylyltransferase *see* [glutamate-ammonia-ligase] adenylyltransferase.

glutamine transaminase L *see* glutamine-pyruvate aminotransferase.

glutamino 1 N^2 -glutamino the alkylamino group, $H_2N-CO-[CH_2]_2-CH(COOH)-NH-$, derived from glutamine by loss of a hydrogen atom from its α -amino group. 2 N^5 -glutamino the acylamino group, $H_2N-CH(COOH)-[CH_2]_2-CO-NH-$, derived from glutamine by loss of a hydrogen atom from its amide group.

glutaminy the acyl group, $H_2N-CO-[CH_2]_2-CH(NH_2)-CO-$, derived from glutamine. *Compare* isoglutaminy.

glutamo the alkylamino group, $HOOC-[CH_2]_2-CH(COOH)-NH-$, derived from glutamic acid.

glutamoyl the diacyl group, $-CO-[CH_2]_2-CH(NH_2)-CO-$, derived from glutamic acid.

α -glutamyl glutam-1-yl; the α -monoacyl group, $HOOC-[CH_2]_2-CH(NH_2)-CO-$, derived from glutamic acid.

γ -glutamyl or (formerly) isoglutamyl, glutam-5-yl; the γ -monoacyl group, $-CO-[CH_2]_2-CH(NH_2)-COOH$, derived from glutamic acid.

glutamyl aminopeptidase EC 3.4.11.7; *other names*: aminopeptidase A; angiotensinase A; differentiation antigen gp160; an enzyme that removes an N-terminal glutamyl residue from a peptide. It can also hydrolyse N-terminal aspartase, and can thus inactivate angiotensin II, being expressed in kidney, as well as on cells of the capillaries and intestine, and on lymphocytes Example from human: database code AMPE_HUMAN, 957 amino acids (109.24 kDa).

γ -glutamyl cycle an integrated cycle of chemical events that uses the γ -carboxyl group of glutamate to transport amino acids actively across (mammalian) cell membranes. Glutathione supplies the activated γ -glutamyl group, which reacts by transpeptidation with the amino acid to be transported. The resulting γ -glutamylamino acid enters the cell, where it is split into the free amino acid and 5-oxoproline. The latter is then converted to glutamate in an ATP-dependent reaction. The cysteinylglycine formed from glutathione in the initial transpeptidation step is hydrolysed by a peptidase. Glutathione is then regenerated in two ATP-dependent steps. *See also* glutamyltransferase.

γ -glutamylcysteine synthetase *see* glutamate-cysteine ligase.

glutamyl endopeptidase EC 3.4.21.19; *other names*: staphylococcal serine proteinase; V8 proteinase; protease V8; endoproteinase Glu-C. An enzyme that catalyses the preferential cleavage: Asp-[Xaa, Glu]-Xaa. Example (precursor) from *Staphylococcus aureus*: database code STSP_STAAU, 336 amino acids (36.29 kDa).

glutamyltransferase *abbr.* (in clinical chemistry): GMT; EC 2.3.2.2; *recommended name*: γ -glutamyltransferase; *systematic name*: (S-L-glutamyl)-peptide:amino acid 5-glutamyltransferase; *other name*: γ -glutamyl transpeptidase. An enzyme that reversibly transfers a γ -L-glutamyl residue (linked to the substrate by an isopeptide bond) from the N terminus of a peptide, normally glutathione, to the amino group of an amino acid. It is present in high concentrations in liver, kidney, prostate, and pancreas, and is useful diagnostically in liver or biliary disease, levels being five times normal or more in cholestasis, hepatitis, and cirrhosis, and also in pancreatitis; it is also elevated in the plasma of alcoholics. The level in normal human (male) plasma is $<60 \text{ IU L}^{-1}$. *See also* γ -glutamyl cycle.

glutaraldehyde pentanedial; a compound that is useful as a tissue fixative in electron microscopy, as a crosslinking agent for compounds containing amino groups, and as a sporicide.

glutaramic acid the trivial name for 4-carbamoylbutanoic acid; $H_2N-CO-[CH_2]_3-COOH$; the monoamide of glutaric acid (*see* glutarate).

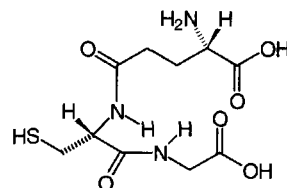
glutarate 1 trivial name for propane-1,3-dicarboxylate; pentanedioate; $-OOC-[CH_2]_3-COO-$; the dianion of glutaric acid (pentanedioic acid). 2 any mixture of free glutaric acid and its

mono- and dianions. 3 any (partial or full) salt or ester of glutaric acid.

glutaredoxin a monomeric polypeptide with a similar function to thioredoxin. It can replace thioredoxin, or may be the main reactant, in the pathway for nucleotide reduction. Example from *Escherichia coli*: database code GLRX_ECOLI, 85 amino acids, (9.67 kDa).

glutaryl-CoA dehydrogenase EC 1.3.99.7; a flavoprotein decarboxylase that catalyses the reaction between glutaryl-CoA and acceptor to form crotonoyl-CoA, CO_2 , and reduced acceptor. Example from *Pseudomonas* sp. (fragment): database code A48367, 20 amino acids (2.10 kDa).

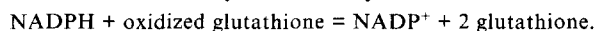
glutathione *abbr.*: GSH; *symbol*: Glu(-Cys-Gly); γ -glutamylcysteinylglycine, a tripeptide that is widely distributed in most if not all cells. It acts as a coenzyme for some enzymes and as an antioxidant in the protection of sulfhydryl groups in enzymes and other proteins; it has a specific role in the reduction of hydrogen peroxide and oxidized ascorbate, and it participates in the γ -glutamyl cycle. Glutathione may be oxidized with the formation of a disulfide bridge linking two molecules and generation of the compound known as oxidized glutathione (or the oxidized form of glutathione) (*abbr.*: GSSG). For clarity, glutathione is sometimes termed reduced glutathione (or the reduced form of glutathione).



glutathione dehydrogenase (ascorbate) EC 1.8.5.1; an enzyme that restores ascorbate to its reduced state after exercise of its function of reducing agent. It catalyses the reduction by two molecules of glutathione of dehydroascorbate to form oxidized glutathione and ascorbate.

glutathione peroxidase EC 1.11.1.9; *systematic name*: glutathione:hydrogen-peroxide oxidoreductase. An enzyme that catalyses the oxidation by hydrogen peroxide of two molecules of glutathione to form oxidized glutathione and two molecules of H_2O . It has a UGA-encoded selenocysteine residue at its active site. The main role of the enzyme is to protect hemoglobin from oxidative breakdown: the oxidized glutathione is reduced by glutathione reductase, the system thus contributing to a reduction of peroxide levels in the cell. Example from human: database code GSHC_HUMAN, 201 amino acids (21.87 kDa).

glutathione reductase *abbr.* (in clinical chemistry): GTD; EC 1.6.4.2; *recommended name*: glutathione reductase (NADPH); *systematic name*: NADPH:oxidized-glutathione oxidoreductase. An enzyme that catalyses the reaction:



It is a dimeric protein, has FAD as cofactor, and is important in maintaining glutathione in the reduced state. Example from *Escherichia coli*: database code GSHR_ECOLI, 450 amino acids (48.72 kDa).

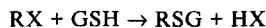
glutathione synthase EC 6.3.2.3; *other name*: glutathione synthetase; an enzyme that catalyses the formation of glutathione from ATP, γ -L-glutamyl-L-cysteine, and glycine with release of ADP and orthophosphate. Example from *Escherichia coli*: database code GSH2_ECOLI, 316 amino acids (35.52 kDa); 3-D structure of parts is known.

glutathione transferase EC 2.5.1.18; *systematic name*: glutathione R-transferase; *other names*: glutathione S-alkyltransferase; glutathione S-aryltransferase; S-(hydroxyalkyl)glutathione lyase; glutathione S-aralkyltransferase;

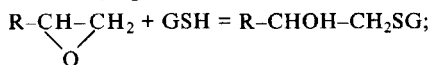


glutathione *S*-transferase. Glutathione transferases catalyse various glutathione-dependent reactions:

(1) nucleophilic substitution, e.g.



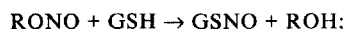
where RX is an aryl- or alkyl-halide (EC 2.5.1.18); (2) addition of GSH across activated unsaturated carbon-carbon or carbon-nitrogen bonds (often α,β -unsaturated ketones/thiones) or oxirane, e.g.



(3) organic hydroperoxide reduction (selenium-independent GSH peroxidase):



(4) organic nitrite reduction



(5) isomerizations in which GSH acts as cofactor, e.g. conversion of prostaglandin H to prostaglandin D. The proteins also bind a variety of hydrophobic anions, e.g. hematin, fatty acid and steroid metabolites, and various xenobiotics/metabolites including the GSH conjugates which may be their catalytic products. Such binding is responsible for their retention by S-linked GSH-Sepharose, hence the widely used simple purification of **GST fusion proteins**. GSH transferases were originally identified as binding proteins (*see ligandin*) and are widely distributed, abundant dimeric soluble enzymes or trimeric membrane-bound forms. They are generally considered an important part of phase II metabolism of xenobiotics. Five classes of soluble GSH transferase subunits have been distinguished (α , μ , π , θ , and σ) and heterodimers occur at least within classes yielding, for example, about 20 different isoenzymes in the rat. Distinct soluble mitochondrial and bacterial forms also occur. Example, GST 5.7 from mouse: database code GTSA_MOUSE, 222 amino acids (25.55 kDa). Example, GST 1-1 from *Drosophila melanogaster*: database code GT_DROME, 209 amino acids (23.84 kDa).

glutelin any member of a group of simple proteins found especially in the seeds of cereal plants. They are insoluble in water, dilute salt solutions, and ethanol but are soluble in dilute acids or alkalis. The group includes the **glutenins** of wheat, the hordenins of barley, and the oryzemins of rice. They serve as storage proteins, and have Gln-rich domains. Example from rice, homohexamer of subunits (acidic and basic chain derived from one precursor) with Gln-rich domains; glutelin I precursor: database code GLU1_ORYSA, 499 amino acids (56.15 kDa). *Compare prolamins*.

gluten a proteinaceous fraction obtained from certain cereal seeds, especially wheat, consisting of a considerable number of proteins that are all either **glutenins** or **prolamins**. It forms a sticky mass when wet, and gives dough its characteristic consistency. *See also celiac disease*.

glutenin any **glutelin** of the endosperm of wheat seeds. Glutenins form aggregates of high molecular mass in which they are linked together by disulfide bridges; they are thought to be responsible for the viscoelastic properties of wheat dough. The sequences contain repeats of two motifs,



and



Example, high molecular mass DY10 precursor: database code GLT0_WHEAT, 648 amino acids (69.55 kDa). *Compare gliadin*.

Glx general symbol for a residue of any α -amino (or imino) acid that yields L-glutamic acid on acid hydrolysis. It was formerly used only for either of the α -amino acids L-glutamic acid (*specific symbol*: Glu) or L-glutamine (*specific symbol*: Gln), but may now be used to represent any such residues, including

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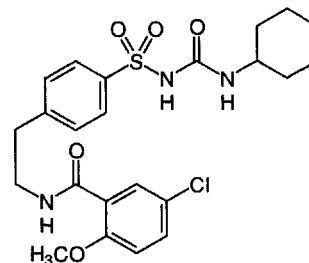
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those of L-4-carboxyglutamic acid (*specific symbol*: Gla) and 5-oxoproline (= pyrroglutamic acid) (*specific symbol*: Glp); (alternative to Z).

Gly symbol for a residue of the achiral (α -)amino acid glycine (alternative to G).

glyburide 1-[[p-[2-(chloro-*o*-anisamido)ethyl]phenyl]sulfonyl]-3-cyclohexylurea; glibenclamide; a sulfonylurea with oral hypoglycemic actions similar to **tolbutamide**.



glyc+ a variant form of **glyco+** (before a vowel).

glycaemia a variant spelling (*esp. UK*) of **glycemia**.

glycamine any ω -amino- ω -deoxyalditol, i.e. any compound formally derived from an **alditol** by replacement of a terminal hydroxyl group by an amino group.

glycan or **polysaccharide** any polymer consisting of more than about 10 monosaccharide (i.e. glucose) residues joined to each other by glycosidic linkages. A **homoglycan** (or **homopolysaccharide**) is composed of only one type of monosaccharide residue, whereas a **heteroglycan** (or **heteropolysaccharide**) is composed of two or more different kinds of monosaccharide residue. A homoglycan is named, generally, after the constituent monosaccharide, with the +ose ending replaced by +an; hence, cellulose, dextran, dextrin, and starch are all glucans, after glucose. Distinguish from **glycosan**.

glycaric acid an obsolescent name for **aldaric acid**.

glycated hemoglobin or **glycohemoglobin** any derivative of hemoglobin that is formed nonenzymically by reaction at the N terminus with glucose. In the normal adult human such derivatives constitute a few percent of the total erythrocyte hemoglobin, the most abundant being **hemoglobin A_{1c}**, which increases severalfold in concentration in diabetes mellitus. Glycation of other proteins such as lens crystallins, collagen and low-density lipoproteins has been implicated in the pathogenesis of diabetic complications. The extent of glycation of hemoglobin is a useful measure of the degree to which the diabetes has been controlled by the administration of insulin. It is sometimes improperly termed **glucosylated hemoglobin** or **glycosylated hemoglobin**. *See also glycation*.

glycated proteins glycation is the post-translational modification of a protein by the covalent attachment of a sugar residue resulting from the spontaneous amino-carbonyl reaction (the **Maillard reaction**) between these two moieties. The reaction is initiated by the amino-carbonyl reaction of $-\text{NH}_2$ and $-\text{C}=\text{O}$ groups and occurs principally between the free amino groups of proteins and the carbonyl group of reducing sugars as first described by Louis-Camille Maillard in 1912. The initial condensation to form a Schiff's base is a spontaneous but reversible reaction that is rapidly followed by a complex series of irreversible reactions ultimately leading to the production of Amadori products of brown pigments and the cross-linking of the polypeptide chains involved. The importance of the Maillard reaction has long been recognized in the food industry where it causes spoilage and the production of colour and off-flavour in cooked foods. In 1977 Cerami identified an Amadori product in variants of human hemoglobin. Glycation of various proteins has now been implicated in the etiology of various diseases such as the development of



glycation

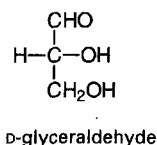
Alzheimer-type pathology. *See* **Alzheimer's disease**. *See also* **glycation**.

glycation any chemical reaction, whether enzyme-catalysed or not, that links a glucose (i.e. a sugar) to a peptide or protein and thereby forms a glycopeptide or glycoprotein. The term thus includes (enzymic or nonenzymic) **glycosylation** (and hence **glucosylation**) of a peptide or protein. However, it is usefully applied in addition to any (nonenzymic) reaction of glucose (or another sugar) with a peptide or protein where the product is not a glycoside but results from condensation of the sugar with free amino groups to form a **Schiff base** which then undergoes an **Amadori rearrangement**. —glycated *adj.*

glycemia or **glucemia** or (*esp. UK*) **glycaemia** or **glucaeemia** 1 the presence of glucose (strictly sugar) in the blood. 2 the concentration of (strictly) sugar but, in fact, glucose in the blood. When this concentration is within normal limits the condition of the blood (or of the corresponding individual) is termed **normoglycemia**. An abnormally elevated concentration, **hyperglycemia**, may result in thirst, whereas an abnormally depressed concentration, **hypoglycemia**, may produce sweating, trembling, hunger, irritability, and in extreme cases confusion or coma. *See also* **blood sugar**, **glucose-tolerance test**, **insulin-tolerance test**. —glycemic or glycaemic or glucemic or glucaeemic *adj.*

glycerol an alternative name for **glyceraldehyde**.

glyceraldehyde *symbol*: Gra; *glycerol*; 2,3-dihydroxypropanal; *formerly sometimes known as glyceric aldehyde; sometimes (not recommended) called glycerose*. It is the simplest chiral aldose and a constitutional isomer of **glycerone**. The naturally occurring form is D-glyceraldehyde, the absolute configuration of which is known and is taken as the configurational reference standard for the D-, and hence also for the L-, forms of monosaccharides.



D-glyceraldehyde 3-phosphate *abbr.*: D-glyceraldehyde-3-P; an important intermediate in glycolysis. It is isomeric with **glycerone phosphate**.

glyceraldehyde-3-phosphate dehydrogenase (NADP⁺) (phosphorylating) EC 1.2.1.13; *other name*: triosephosphate dehydrogenase (NADP⁺); an enzyme that catalyses the conversion of D-glyceraldehyde 3-phosphate, orthophosphate, and NADP⁺ to 3-phospho-D-glyceroyl phosphate and NADPH. Example (fragment) from chloroplast (spinach): database code S33872, 39 amino acids (4.46 kDa).

glyceraldehyde-3-phosphate dehydrogenase (phosphorylating) *abbr.*: GAPDH; EC 1.2.1.12; *systematic name*: D-glyceraldehyde-3-phosphate:NAD⁺ oxidoreductase (phosphorylating); *other name*: triosephosphate dehydrogenase. An enzyme that catalyses a reaction between D-glyceraldehyde 3-phosphate, orthophosphate, and NAD⁺ to form 3-phospho-D-glyceroyl phosphate and NADH. It is an important enzyme of the pathways of glycolysis and gluconeogenesis. It has five motifs and is a homotetramer. Example from yeast (GAPDH 1): database code G3P1_YEAST, 331 amino acids (35.56 kDa).

glycerate *symbol*: Gri; 1 2,3-dihydroxypropanoate; HOCH₂—CH(OH)—COO[−]; the anion derived from glyceric acid. The naturally occurring form is D-glycerate. 2 any salt or ester of glyceric acid.

glyceric aldehyde *former name for glyceraldehyde*.

glyceride the former name for **acylglycerol**; i.e. any ester of glycerol, especially the mono-, di-, or triesters of glycerol with

fatty acids, formerly termed mono-, di-, or triglycerides respectively. These are now known as mono-, di-, or tri-*O*-acylglycerols.

glycerin(e) *see* **glycerol**.

glycero+ *prefix or infix (in chemical nomenclature)* 1 denoting a residue of glycerol. 2 denoting the acyl group derived from glyceric acid.

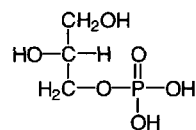
glycero- *prefix (in chemical nomenclature)* indicating the presence of a single CHO group, in the D or L configuration as specified.

glycerokinase *see* **glycerol kinase**.

glycerol *symbol*: Gro; 1,2,3-propanetriol; a sweet, hygroscopic, viscous liquid, widely distributed in nature as a constituent of many lipids (*see* **acylglycerol**). The older name **glycerin(e)** is now obsolete except in non-technical usage and sometimes in pharmacy.

glycerol kinase EC 2.7.1.30; *systematic name*: ATP:glycerol 3-phosphotransferase; *other name*: glycerokinase. An enzyme that catalyses the phosphorylation by ATP of glycerol to form *sn*-glycerol 3-phosphate with release of ADP. In tissues and organisms ranging from human liver to microorganisms, this is the rate-limiting step in glycerol utilization. The enzyme is induced, e.g. in *Escherichia coli*, by *sn*-glycerol 3-phosphate; it is a homotetramer; example (*E. coli*): database code GLPK_ECOLI, 501 amino acids (56.04 kDa).

glycerol phosphate or **glycerophosphate** any of three isomeric phosphoric monoesters of glycerol; of these, two are chiral and form an enantiomeric pair, whereas the third is achiral. The earlier nomenclature of the chiral esters has been confusing; ambiguities are avoided by the adoption of **stereospecific numbering** (*symbol*: *sn*-). (1) *sn*-Glycerol 3-phosphate or *sn*-glycero-3-phosphate or 3-phospho-*sn*-glycerol or 3-*O*-phosphono-*sn*-glycerol (*abbr.* *sn*-glycerol-3-*P*) are alternative recommended names for the chiral ester (R)-[glycerol 1-(dihydrogen phosphate)]; (R)-[2,3-dihydroxypropyl dihydrogen phosphate]. It was previously known as L-glycerol 3-phosphate or D-glycerol 1-phosphate, and originally named D-glycerol 1-phosphate or L-*a*-glycerophosphate or L-*a*-glycerophosphoric acid. It is a component of many phospholipids. The absolute configuration is illustrated. (2) *sn*-Glycerol 1-phosphate or *sn*-glycero-1-phosphate or 1-phospho-*sn*-glycerol or 1-*O*-phosphono-*sn*-glycerol (*abbr.* *sn*-glycerol-1-*P*) are alternative recommended names for the chiral ester (S)-[glycerol 1-(dihydrogen phosphate)]; (S)-[2,3-dihydroxypropyl dihydrogen phosphate]. It was previously known as L-glycerol 1-phosphate or D-glycerol 3-phosphate, and originally named L-glycerol 1-phosphate or D-*a*-glycerophosphate or D-*a*-glycerophosphoric acid. This enantiomer is rarely encountered. (3) Glycerol 2-phosphate or glycero-2-phosphate or 2-phosphoglycerol or 2-*O*-phosphonoglycerol are alternative names for the achiral ester glycerol 2-(dihydrogen phosphate); di(hydroxymethyl)methyl dihydrogen phosphate. It was originally named β-glycerophosphate or β-glycerophosphoric acid. It is an artefact occurring in hydrolysates of glycerophospholipids that arises by migration of the phosphate group from C-1 to C-2 of the glycerol moiety. *Distinguish from* **phosphoglycerate**.



sn-glycerol 3-phosphate

glycerol-3-phosphate cycle or **glycerol-3-phosphate shuttle** or **α-glycerophosphate cycle** an intracellular metabolic cycle that can transfer reducing equivalents from the cytoplasm to the mitochondria. In the cytoplasm NAD-linked **glycerol-3-**

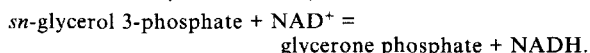
glycerol-3-phosphate cytidylyltransferase

phosphate dehydrogenase (NAD⁺) catalyses the reduction of glycerone phosphate (formed together with glyceraldehyde 3-phosphate in glycolysis) to *sn*-glycerol 3-phosphate. The glycerol 3-phosphate thus formed enters the mitochondria where it is oxidized to glycerone phosphate by the FAD-linked mitochondrial glycerol-3-phosphate dehydrogenase. The glycerone phosphate then returns to the cytoplasm to complete the cycle. The overall result is the oxidation of cytoplasmic NADH + H⁺ to NAD⁺ and the reduction of mitochondrial enzyme-FAD to enzyme-FADH₂.

glycerol-3-phosphate cytidylyltransferase EC 2.7.7.39; *other name*: CDPglycerol pyrophosphorylase; an enzyme that catalyses the formation of CDPglycerol from CTP and *sn*-glycerol 3-phosphate with release of pyrophosphate. Example, probable CDPglycerol pyrophosphorylase (teichoic acid biosynthesis protein D) from *Bacillus subtilis*: database code TAGD_BACSU, 129 amino acids (15.25 kDa).

glycerol-3-phosphate dehydrogenase *abbr.*: G3PDH; EC 1.1.99.5; a flavoprotein enzyme that catalyses the oxidation by an electron acceptor of *sn*-glycerol 3-phosphate to glycerone phosphate. The enzyme is part of the **glycerol 3-phosphate cycle**. Examples from *Escherichia coli* in which there are two unrelated enzymes: (1) anaerobic G3PDH, which has two subunits: database code GLPA_ECOLI, 542 amino acids (58.90 kDa); and database code GLPB_ECOLI, 419 amino acids (45.31 kDa); (2) aerobic G3PDH: database code GLPD_ECOLI, 501 amino acids (56.69 kDa).

glycerol-3-phosphate dehydrogenase (NAD⁺) EC 1.1.1.8; *systematic name*: *sn*-glycerol-3-phosphate:NAD⁺ 2-oxido-reductase. An enzyme that catalyses the reaction:



Its role is especially important in the provision of glycerol 3-phosphate for lipid synthesis, using glycerone phosphate derived from glucose by the **glycolytic** pathway, and as a component of the **glycerol-3-phosphate cycle**. Example, **flightin**, homodimer from *Drosophila melanogaster*: database code GPDA_DROME, 351 amino acids (38.32 kDa).

glycerol-3-phosphate O-acyltransferase EC 2.3.1.15; *systematic name*: acyl-CoA:*sn*-glycerol-3-phosphate 1-O-acyltransferase; it catalyses the acylation by acyl-CoA of glycerol 3-phosphate to form 1-acyl-*sn*-glycerol 3-phosphate (lysophosphatidic acid) with release of CoA. This is the first step in phospholipid biosynthesis and may be involved in regulation of membrane biogenesis. Plants exhibit oleate-selective acyltransferases; the enzyme from chilling-resistant plants discriminates against palmitic acid (a fatty acid with high transition temperature), and selects oleic acid (leading to more fluid membranes), whereas the acyltransferase from sensitive plants accepts both fatty acids. Example (precursor) from pea: database code PLSB_PEA, 457 amino acids (50.70 kDa).

glycerol-3-phosphate oxidase EC 1.1.3.21; *systematic name*: *sn*-glycerol-3-phosphate:oxygen 2-oxidoreductase. An enzyme that catalyses the oxidation by dioxygen of *sn*-glycerol 3-phosphate to glycerone phosphate and hydrogen peroxide; FAD is a cofactor. It is a mitochondrial protein. In insects it plays an important role in the transfer of energy for flight, along with **glycerol-3-phosphate dehydrogenase (NAD⁺)** (flightin).

glycerol trinitrate *see* trinitroglycerine.

glycerone *symbol*: Grn; the currently recommended trivial name for dihydroxyacetone; 1,3-dihydroxy-2-propanone. It is the simplest ketose, is achiral, and is a constitutional isomer of glyceraldehyde.

glycerone phosphate the currently recommended trivial name for dihydroxyacetone phosphate; 1,3-dihydroxy-2-propanone 1-phosphate; the monoester formed by phosphorylation of dihydroxyacetone, glycerone. Glycerone phosphate and D-glyceraldehyde 3-phosphate are together described as triose phosphates and are interconvertible by triose-phosphate isomerase, EC 5.3.1.1. They are important components of the

glycolytic pathway. Glycerone phosphate is a precursor of the glycerol moiety of glycerolipids.

glycerone transferase *see* transaldolase.

glycerophosphate an alternative name for glycerol phosphate.

α-glycerophosphate cycle an alternative name for glycerol-3-phosphate cycle.

glycerophosphate phosphatidyltransferase EC 2.7.8.5; *recommended name*: CDPdiacylglycerol-glycerol-3-phosphate 3-phosphatidyltransferase. An enzyme that catalyses a reaction between CDPdiacylglycerol and *sn*-glycerol-3-phosphate to form 3-(3-*sn*-phosphatidyl)-*sn*-glycerol 1-phosphate, with release of CMP. It is a membrane protein of endoplasmic reticulum and component of the pathway for the biosynthesis of acidic phospholipids; its product loses phosphate to form phosphatidylglycerol from which **cardiolipin** is formed. Example from *Escherichia coli*: database code PGSA_ECOLI, 215 amino acids (24.64 kDa).

glycerophospholipid *general name* for any derivative of glycerophosphate that contains at least one O-acyl, O-alkyl, or O-alkenyl group attached to the glycerol residue. Formerly known as phosphoglyceride.

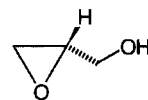
glycerophosphoric acid *see* glycerol phosphate.

glycerose alternative name (not recommended) for glyceraldehyde.

glyceroyl 2,3-dihydroxypropanoyl; the acyl group, HOCH₂-CH(OH)-CO-, derived from glyceric acid.

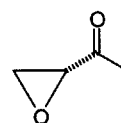
glyceryl any group derived from glycerol by replacing or removing one or more of its hydroxyl groups.

glycidol the trivial name for 3-hydroxypropylene oxide; 2,3-epoxy-1-propanol.



S(-)-glycidol

glycidoyl 2,3-epoxypropanoyl; the acyl group derived from glycidic acid.



(R)-glycidoyl group

glycidyl 2,3-epoxypropyl; the alkyl group: derived from glycidol by removal of its hydroxyl group.

glycinamide the trivial name for the amide of glycine, aminoacetamide; aminoethanamide; H₂N-CH₂-CO-NH₂. It is predominantly cationic at neutrality and is useful as a buffer substance; pK_a (20 °C) = 8.20.

glycinamide ribonucleotide synthetase (*abbr.* GARS) *see* phosphoribosylamine-glycine ligase.

glycinamide ribotide *N*-glycyl-5-phosphoribosylamine; an early intermediate in the biosynthesis of purines.

glycinate 1 glycine anion; the anion, NH₂-CH₂-COO⁻, derived from **glycine**. 2 any salt containing glycine anion. 3 any ester of glycine.

glycine *symbol*: G or Gly; the trivial name for aminoacetic acid; aminoethanoic acid; H₂N-CH₂-COOH; an achiral (α)-amino acid and the simplest naturally occurring amino acid. It is a coded amino acid found in peptide linkage in proteins; codon: GGA, GGC, GGG, or GGU. In mammals it is a nonessential dietary amino acid and is **glucogenic**. It is

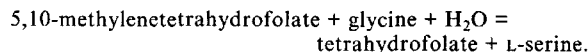


glycine amidinotransferase

sweet-tasting, hence its name. [From **glyc(o)**+ (def. 1) + **+ine** (def. 1)]

glycine amidinotransferase EC 2.1.4.1; an enzyme that catalyses the formation of guanidinoacetate, a precursor of **creatine**, from L-arginine and glycine, also forming L-ornithine. Example (precursor) from human: database code S41734, 423 amino acids (48.40 kDa).

glycine hydroxymethyltransferase EC 2.1.2.1; *systematic name*: 5,10-methylenetetrahydrofolate:glycine hydroxymethyltransferase; *other names*: serine hydroxymethylase; threonine aldolase; serine aldolase. An enzyme that catalyses the reaction:



It is a pyridoxal phosphate-requiring enzyme that interconverts glycine and serine; it transfers a hydroxymethyl group from N^5,N^{10} -methylenetetrahydrofolate to a glycine-pyridoxal phosphate Schiff base or from a serine-pyridoxal phosphate Schiff base to tetrahydrofolate. Example, cytosolic enzyme from *Neurospora crassa*: database code GLY1_NEUCR, 479 amino acids (52.76 kDa).

glycine receptor a complex of integral membrane proteins that binds glycine and mediates its effects as an inhibitory neurotransmitter in lower brain areas and spinal cord. The receptor is a pentamer of α (four isoforms, $\alpha 1$ – $\alpha 4$) and β subunits. The receptor forms an intrinsic Cl^- channel, blocked by strychnine (distinguishing it from the GABA_A Cl^- channel). All subunits have four transmembrane segments, M1–M4, M3 and M4 being separated by a cytoplasmic hydrophilic loop; this is comparable to nicotinic acetylcholine receptors (*see cholinergic receptor*) and GABA_A receptors (*see GABA receptor*), with which some sequence similarities also exist. Glycine also binds to a site on the NMDA (glutamate) receptor, which must be occupied by glycine to enable activation of the receptor by NMDA, and is also blocked by strychnine. Examples (all precursors) from *Rattus norvegicus*: $\alpha 1$, database code GRA1_RAT, 449 amino acids (51.68 kDa); $\alpha 2$, database code GRA2_RAT, 452 amino acids (52.05 kDa); $\alpha 3$, database code GRA3_RAT, 464 amino acids (53.67 kDa); β , database code GRB_RAT, 496 amino acids (55.93 kDa).

glycinin a 350 kDa **globulin** that is the chief protein constituent of the soya bean, *Glycine max*. It resembles **arachin**, **edestin**, and **excelsin** structurally.

glycinium glycine cation; the cation, $\text{H}_3\text{N}^+\text{CH}_2\text{COOH}$, derived from **glycine**.

glycino the alkylamino group, $\text{HOOC-CH}_2\text{-NH-}$, derived from **glycine**.

glycitol an obsolescent name for **alditol**.

glyco+ or (before a vowel) **glyc+** *comb. form* 1 indicating sweet-tasting. 2 *a* indicating sugar. 3 *(in medicine)* indicating glucose. *See also gluco+*. 4 *(in chemical nomenclature)* indicating covalently linked to carbohydrate. 5 indicating combination with glycine; e.g. **glycocholate**, **glycocyamine**.

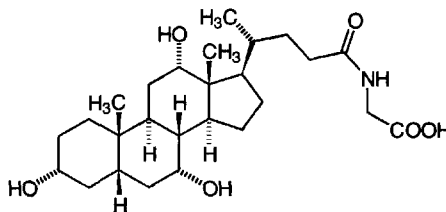
glycoamino acid any compound in which a saccharide is combined with an amino acid through any kind of covalent linkage. The term includes **glycosylamino acids**.

glycobiology the branch of biochemistry dealing with **glycoconjugates**, including their structure, analysis, function, and metabolism.

glycocalicin a carbohydrate-rich hydrophilic fragment, M_r 130 000, cleaved from the external portion of the α chain of **glycoprotein lb** by protease action.

glycocalyx or **cell coat** a zone surrounding many plant and animal cells that is rich in glycoproteins and proteoglycans.

glycocholate 1 the anion of glycocholic acid; *N*-cholyglycine; a bile acid. The anions of similar glycine conjugates of other bile acids are known, e.g. glycochenodeoxycholate, **glycodeoxycholate**, glycolithocholate. 2 any salt of glycocholic acid. Sodium glycocholate is the chief **bile salt** in humans and herbivorous animals. It is useful as a detergent; aggregation number 2.1, CMC 7.1 mM.



glycocholic acid

glycocoll an obsolete name for **glycine**.

glycoconjugate or (formerly) **glucoconjugate** any large and complex biological molecule that contains a carbohydrate moiety. The term comprehends glycolipid, glycopeptide, glycoprotein, peptide glycan, proteoglycan, and heteroglycan. *See also glyco+* (def. 3), **conjugate** (def. 5).

glycocyamine an older name for *N*-amidinoglycine; guanidinoacetic acid. In polychaete worms, the guanidine group can undergo phosphorylation to phosphoglycocyamine, which acts as a **phosphagen**.

glycodeoxycholate 1 glycodeoxycholic acid; *N*-deoxycholyglycine; a bile acid. 2 any salt of glycodeoxycholic acid; sodium glycodeoxycholate is useful as a detergent; aggregation number 2, CMC 2.1 mM.

glycoform any of several differently glycosylated variants of a specified glycoprotein. Glycoforms may vary in the type, number, and/or position of the sugar residues, differ in their physiological properties (e.g. rate of action), and occur together in a given tissue.

glycogen a polydisperse, highly branched glucan or polyglucose, composed of chains of D-glucose residues in $\alpha(1\rightarrow4)$ glycosidic linkage, joined together by $\alpha(1\rightarrow6)$ glycosidic linkages; a small number of $\alpha(1\rightarrow3)$ glycosidic linkages and some cumulative $\alpha(1\rightarrow6)$ links also may occur. Its structure is similar to that of **amylopectin** but with more, though rather shorter, branches; those in glycogen variously contain 8 to 12 glucose residues. The size of the molecule varies greatly according to source and method of measurement, with average M_r values of 2.5×10^5 to upwards of 1.5×10^7 reported. Glycogen serves as a cellular store of glucose; it occurs, frequently as granules, in all animal tissues, especially liver and skeletal muscle, and also in some bacteria and yeasts. The glycogen molecule is linked to the protein **glycogenin** which is required for the *ab initio* synthesis of glycogen by **glycogen synthase**.

glycogenase an alternative name for either α - or β -amylase.

glycogen branching enzyme *see branching enzyme*.

glycogen disease or **glycogen storage disease** or **glycogenosis** any human disease in which there is an inherited abnormality in glycogen metabolism. Such diseases have been classified into eight types. Type I glycogen disease (or **von Gierke's disease**) involves a deficiency of hepatic **glucose-6-phosphatase**. It is an inherited autosomal recessive disease of infants, characterized by massive hepatomegaly, failure to thrive, severe hypoglycemia, and increased hepatic content of normal glycogen. Type II glycogen disease (or **Pompe's disease**) involves a generalized deficiency of lysosomal α -**glucosidase**. It is an inherited autosomal recessive disease of infants, characterized by massive cardiomegaly, hypotonia, and an increased content of normal glycogen in virtually all tissues. Type III glycogen disease (or **limit dextrinosis** or **Cori's disease**) involves a generalized deficiency of amylo-1,6-**glucosidase** (debrancher enzyme) (EC 3.2.1.33). It is an inherited autosomal recessive disease of infants that is similar to, but milder than, type I glycogen disease. It is characterized by an elevated content of glycogen in muscle, liver, and erythrocytes, the glycogen having abnormally short outer branches. A number of subtypes

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glycogenesis

glycolytic patl

have been recognized. Type IV glycogen disease (*or amylopectinosis or Andersen's disease*) is a rare condition of infants in which there is a generalized deficiency of 1,4- α -glucan branching enzyme (EC 2.4.1.18). It is characterized by hepatic cirrhosis, hepatosplenomegaly, ascites, and glycogen with abnormally long outer branches; the glycogen may not accumulate excessively. Type V glycogen disease (*or McArdle's disease*) involves a deficiency of glycogen phosphorylase (EC 2.4.1.1) in voluntary muscle. The disease is probably inherited as an autosomal recessive. It is characterized by myoglobinuria and painful cramps occurring during strenuous exercise, symptoms that do not usually appear until 20 years of age. There is no hypoglycemia but muscular exercise fails to elevate the blood lactate concentration. Type VI glycogen disease (*or Hers's disease*) involves an increased hepatic glycogen content and a lowered hepatic glycogen phosphorylase (EC 2.4.1.1) activity but a normal phosphorylase-activating system. The clinical course is mild. Type VII glycogen disease involves a deficiency of muscle 6-phosphofructokinase. The disease is probably inherited as an autosomal recessive and is clinically identical to Type V glycogen disease. There are raised contents in muscles of normal glycogen, glucose 6-phosphate, and fructose 6-phosphate, together with a lower than normal concentration of fructose 1,6-bisphosphate. Type VIII glycogen disease involves a reduced **phosphorylase kinase** activity in the liver and in leukocytes. Thought to be inherited as an X-linked character, the disease is characterized by mild hepatomegaly, increased liver glycogen content, and mild hypoglycemia.

glycogenesis a former term for 1 **glyconeogenesis**. 2 **gluconeogenesis**. 3 **glucogenesis**. 4 the metabolic formation of glycogen from sugar (i.e. glucose).

glycogenetic of or relating to the production of glycogen. (*Note:* The term is often used confusingly in the senses of **glycogenic** (def. 1, 2).)

glycogenic 1 of, relating to, or involving glycogen. 2 of or relating to the formation of sugar (i.e. glucose), especially in animals; glucogenic. (*Note:* the term is often used confusingly in the sense of **glycogenetic**.)

glycogenin EC 2.4.1.186; *recommended name:* glycogenin glucosyltransferase; a protein that is required for the initiation of glycogen synthesis by glycogen synthase. Glycogenin catalyses formation of a glycosidic link between a glucose and one of its own tyrosines (Tyr¹⁹⁴ in human) and then catalyses addition of a further five glucose molecules. UDPglucose is the donor of the glucose units. This forms a primer which is then extended with further glucose molecules by **glycogen synthase**. Glycogen synthase is only catalytically efficient when it is bound to glycogenin. This has two consequences: first, the number of glycogen granules is determined by the number of molecules of glycogenin, and second, elongation stops when the synthase is sterically prevented from contact with glycogenin, which forms the core of the particle. Hence, the glycogenin-synthase interaction limits the size of glycogen granules. Glycogenin is a monomeric glycoprotein (via its reactivity), tightly bound to the 86 kDa catalytic subunit of glycogen synthase. Example from *Oryctolagus cuniculus*: database code GLYG_RABIT, 332 amino acids (37.27 kDa).

glycogenolysis the breakdown of glycogen into glucose, either by hydrolysis or via glucose 1-phosphate and glucose 6-phosphate. The term is sometimes restricted to the phosphorylation of glycogen to glucose 1-phosphate. —**glycogenolytic adj.**

glycogenesis an alternative name for **glycogen disease**.

glycogen phosphorylase see **phosphorylase**.

glycogen storage disease an alternative term for **glycogen disease**.

glycogen synthase EC 2.4.1.11; *recommended name* (example): glycogen (starch) synthase (the nature of the synthetic product should be included in the name); *systematic name:* UDPglucose:glycogen 4- α -D-glucosyltransferase; *other name:* UDPglucose-glycogen glucosyltransferase. An enzyme that

catalyses the reaction between UDPglucose and (1,4- α -D-glucosyl)_n to form (1,4- α -D-glucosyl)_{n+1} with release of UDP. The mammalian enzyme requires glucosylated **glycogenin** as a primer (product of glycogenin glucosyltransferase, EC 2.4.1.186), and is susceptible to allosteric activation by glucose 6-phosphate, and phosphorylation by a cyclic AMP-dependent kinase. Example from human muscle: database code UGST_HUMAN, 737 amino acids (83.77 kDa). *See also dependent form of glycogen synthase, glycogenin, starch synthase.*

[glycogen-synthase-D] phosphatase EC 3.1.3.42; an enzyme that hydrolyses phosphate from glycogen synthase D, thereby activating it.

glycoglycerolipid any **glycolipid** containing one or more glycerol residues.

glycohemoglobin or (*esp. UK*) **glycohaemoglobin** an alternative term for **glycated hemoglobin**.

glycoinositolphospholipid see **GPI anchor**.

glycol 1 ethylene glycol; ethane-1,2-diol; HOCH₂-CH₂OH. 2 or **glychol** any dihydric alcohol, i.e. any compound in which two alcoholic hydroxyl groups are on different carbon atoms (which are usually but not necessarily adjacent).

glycolaldehyde hydroxyacetaldehyde; hydroxyethanal; HOCH₂-CHO.

glycolate or (*formerly*) **glycollate** 1 the trivial name for hydroxyacetate; hydroxyethanoate; the anion, HOCH₂-COO⁻, derived from **glycolic acid**. It is an important metabolite in C₃ plants. 2 any mixture of glycolic acid and its anion. 3 any salt or ester of glycolic acid. It is formed from phosphoglycolate, a product of **ribulose biphosphate carboxylase** when this is acting as an oxidase. It is converted to glyoxylate by glycolate oxidase (*see* 2-hydroxy-acid oxidase).

glycolate oxidase an FMN-containing enzyme that oxidizes glycolate to glyoxylate in plant peroxisomes. It is now included with 2-hydroxy-acid oxidase.

glycolic acid or (*formerly*) **glycollic acid** hydroxyacetic acid; hydroxyethanoic acid; HOCH₂-COOH; a constituent of sugar-cane juice.

glycolipid any compound containing (usually) one to four linked glycoside (monosaccharide) residues joined by a glycosyl linkage to a lipid moiety (e.g. to a mono- or diacylglycerol, to a long-chain base such as sphingosine, or to a ceramide) or (in a bacterial glycolipid) joined in ester linkage to one or more fatty-acid molecules. The term embraces **galactolipid**. *See also cerebroside, ganglioside, lipopolysaccharide.*

glycollate an obsolete variant spelling of **glycolate**.

glycoloyl *symbol:* Gc; the acyl group, HOCH₂-CO-, derived from **glycolic acid**.

glycolytic pathway or **Embden-Meyerhof(-Parnas) pathway** the widely distributed, extra-mitochondrial, cellular metabolic pathway by which sugars, sugar phosphates, or their precursors are broken down with an overall negative free energy change favourable to formation of ATP; the pathway yields usually pyruvate aerobically or lactate anaerobically in animal tissues. By extension, the term is also used of similar processes in other organisms yielding, sometimes, other products, e.g. ethanol, glycerol, etc. Overall there is a net yield of 2 ATP for each glucose metabolized, since phosphorylation of the hexose substrates requires 2 ATP, but conversion to pyruvate of the two 3-carbon substrates formed yields 2 ATP for each. The pathway from glucose involves its conversion to glucose 6-phosphate, thence successively to fructose 6-phosphate, fructose 1,6-bisphosphate, glyceraldehyde 3-phosphate plus glyceraldehyde 3-phosphate (which is converted to glyceraldehyde 3-phosphate in situations in which it is further degraded), 1,3-bisphosphoglycerate, 3-phosphoglycerate, 2-phosphoglycerate, phosphoenolpyruvate, and pyruvate. The enzymes involved are hexokinase, glucose-6-phosphate isomerase, phosphofructokinase, fructose-bisphosphate aldolase, triose-phosphate isomerase, glyceraldehyde-3-phosphate dehydrogenase, phosphoglycerate kinase, phosphoglycerate mutase, enolase, and pyruvate kinase (*see* individual entry for each enzyme). Glyco-



glycolytic oscillator

gen breakdown (**glycogenolysis**) by phosphorylase yields glucose 1-phosphate, which is converted by phosphoglucomutase to glucose 6-phosphate, which in muscle is then degraded by the pathway outlined above. The pathway is subject to hormonal regulation, especially by insulin, glucagon, and epinephrine. Flux through the pathway is under powerful regulation by phosphofructokinase, the main modulator of which is **fructose 2,6-bisphosphate**.

glycolytic oscillator any metabolic activity that produces periodic regular oscillations in the concentrations of intermediates of the **glycolytic pathway**. Such oscillations have been observed in glycolysing yeast cells and extracts of skeletal muscle and heart. These oscillations reflect periodic changes in the activity of **phosphofructokinase** in response to changes in the levels of its activators and inhibitors in the system being studied.

glycon or (formerly) **glycone** the carbohydrate moiety of any **glycoside** that also contains a noncarbohydrate component.

glyconeogenesis 1 an alternative name for **gluconeogenesis**. 2 (sometimes) the formation of glycogen from noncarbohydrate precursors.

glyconic acid an obsolescent name for **aldonic acid**.

glycopeptide any compound in which carbohydrate is covalently attached to an oligopeptide composed of residues of L- and/or D-amino acids. The term usually denotes a product of proteolytic degradation of a glycoprotein but includes glycosylated peptide. It should be distinguished from **peptidoglycan**, for which it is sometimes considered to be a synonym.

glycophorin any of several type I membrane proteins of the erythrocyte membrane. They are carbohydrate-rich sialoglycoproteins (SGPs – hence the nomenclature for glycophorins: α SGP, etc.), and stain with periodic acid–Schiff (PAS) reagent (hence the nomenclature PAS 1, etc. for bands on electrophoresis). The glycosylated moiety serves as an antigenic determinant and as a receptor for viruses and plant agglutinins. A glycophorin was the first membrane protein to have its full sequence elucidated. There is a single membrane-spanning domain. **Glycophorin A** (or α SGP) is the major sialoglycoprotein of the human erythrocyte membrane. On electrophoresis it occurs in PAS 1 (dimer). It has a molecular mass of ≈ 31 kDa, contains 60% carbohydrate by weight, and consists of a 131-residue polypeptide chain that spans the cell membrane; all the carbohydrate is attached to the N-terminal half and is exposed to the exterior of the cell. Glycophorin A displays either M or N blood group activity (see **MNSs system**), depending on which of two alternative pairs of amino-acid residues occurs at positions 1 and 5 in the sequence. It binds influenza virus. Database code (precursor) GLPA_HUMAN, 150 amino acids (16.25 kDa). **Glycophorin B** (or δ SGP), also from the human erythrocyte membrane, is a minor sialoglycoprotein similar to glycophorins A and E. It occurs in PAS 3 and has an amino-acid sequence identical in its N-terminal segment with that of glycophorin A_N. It carries the immunodeterminants for the N and Ss antigens (see **MNSs system**). Database code (precursor) GLPB_HUMAN, 91 amino acids (9.78 kDa). **Glycophorin C** (or β SGP) occurs in PAS 2, glycoconnectin; it is a minor sialoglycoprotein that probably contains blood group Gerbich antigens and receptors for *Plasmodium falciparum* (malaria) merozoites. Database code (precursor) GLPC_HUMAN, 128 amino acids (13.80 kDa). **Glycophorin E** is similar to glycophorins A and B. Database code GLPE_HUMAN, 78 amino acids (8.52 kDa).

glycoprotein any protein that contains covalently bound glucose (i.e. monosaccharide) residues other than as a moiety of nucleic acid; the glycoside occurs most commonly as oligosaccharide or fairly small polysaccharide but occasionally as monosaccharide. Glycoproteins are widespread in animals and in the primary cell walls of plants. The term includes glycosylated protein and is now considered also to include **proteoglycan**. Compare **mucoprotein**.

glycoprotein hormone general name for any member of a family of closely related mammalian hormones of the pituitary

and placenta, believed to be derived by biochemical evolution from a common primitive molecule. The group comprises the three gonadotropins, namely pituitary **follicle-stimulating hormone**, pituitary **LH**, and placental **chorionic gonadotropin**, plus pituitary **thyrotropin**; all have two peptide chains, α and β , each with attached, variable carbohydrate substituent groups that account for 15–31% of the molecular mass. In any given species, the amino-acid sequences of the α chains are identical, or nearly so, in all these hormones, whereas the β chains have greater differences in amino-acid sequence and confer hormonal specificity on the molecules.

glycoprotein Ib abbr.: GP-Ib; a surface membrane protein of platelets that participates in the formation of platelet plugs by binding to von Willebrand factor. It is a disulfide-linked heterodimer of α and β subunits. Examples (precursors) from human: α subunit, database code GPBA_HUMAN, 626 amino acids (68.88 kDa); β subunit, database code GPBB_HUMAN, 206 amino acids (21.69 kDa).

glycoprotein N-palmitoyltransferase EC 2.3.1.96; an enzyme that catalyses a reaction between palmitoyl-CoA and a glycoprotein to give N-palmitoylglycoprotein with release of CoA.

glycosaemia a variant spelling (esp. Brit) of **glycosemia** (see **glycemia**).

glycosamine an older name for **aminodeoxysugar**, now restricted to some trivial names. Distinguish from **glycosylamine**.

glycosaminoglycan any glycan (i.e. polysaccharide) containing a substantial proportion of aminomonosaccharide residues.

glycosaminopeptide (sometimes) an alternative name for **peptidoglycan**.

glycosan an old term (not recommended) for any anhydrosugar whose formation involves its reducing group. Distinguish from **glycan**.

glycose 1 a less frequently used term for any monosaccharide.

2 an obsolete name for **glucose** (see also **glyco+** (def. 2)).

3 (sometimes) an alternative name for **corn syrup**.

glycosemia or (esp. Brit) **glycosaemia** an alternative name for **glycemia**.

glycosidase any hydrolase of subclass EC 3.2, enzymes that hydrolyse glycosidic linkages. Glycosidases are subdivided into those that hydrolyse O-glycosyl compounds (EC 3.2.1), N-glycosyl compounds (EC 3.2.2), or S-glycosyl compounds (EC 3.2.3). Some glycosidases can also transfer glycosyl residues to oligosaccharides, polysaccharides, or other acceptor alcohols. See also (DNA) **glycosylase**, **glucosidase**.

glycoside or **oside** 1 any compound in which a **glycosyl** group is substituted into a hydroxyl group, a thiol group, or a selenol group in another compound; such a substance may be named, respectively, O-glycoside, S-glycoside (or thioglycoside), or Se-glycoside (or selenoglycoside). A glycoside is either a mixed acetal (if derived from a cyclic form of an aldose) or a mixed ketal (if derived from a cyclic form of a ketose). By extension, any compound formed by attachment of a glycosyl group to a hydrocarbyl group has been termed a C-glycoside and that by attachment of a glycosyl group to an amino or substituted amino group has been termed also N-glycoside (or nitrogen glycoside), but these terms are misnomers and should not be used; the preferred terms are **C-glycosyl compound** and **glycosylamine**, respectively. See also **cardiac glycoside**, **nucleoside**. 2 an obsolete term for **glucoside**.

glycosidic 1 of, pertaining to, or being a **glycoside**. 2 an alternative term for **anomeric** (def. 2).

glycosidic bond the covalent bond between the anomeric carbon atom of a saccharide and some other group or molecule with which it forms a glycoside.

glycosiduronate 1 the anion of a **glycosiduronic acid**. 2 any salt or ester of a glycuronic acid.

glycosiduronic acid or (formerly) **glycuronide** a glycoside of any uronic acid.

glycosome a microbody-like organelle present in *Trypano-*

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glycosphingolipid

soma brucei and containing nine enzymes involved in glycolysis, and glycerol kinase.

glycosphingolipid any compound containing residues of a sphingoid and at least one monosaccharide. Glycosphingolipids are subdivided into: (1) neutral glycosphingolipids, comprising monoglycosyl- and oligoglycosylsphingoids and monoglycosyl- and oligoglycosylceramides; and (2) acidic glycosphingolipids, comprising sialosylglycosylsphingolipids (**gangliosides**) and sulfoglycosylsphingolipids (formerly known as **sulfatides**).

glycostatic maintaining a constant level of sugar, e.g. of glucose or glycogen. *Compare* **glycotropic**.

glycosuria the presence of abnormally high concentrations of glucose and/or other sugars in the urine. *See also* **glucosuria**, **fructosuria**.

glycosyl any chemical group formed by detaching the anomeric, i.e. glycosidic, hydroxyl group from the cyclic form of a monosaccharide or monosaccharide derivative, or (by extension) of a lower oligosaccharide or oligosaccharide derivative.

glycosylamine any compound formed by the replacement of the anomeric, i.e. glycosidic, hydroxyl group of a cyclic form of a monosaccharide or monosaccharide derivative by an amino group or a substituted amino group. (The term **nitrogen glycoside** (or **N-glycoside**) is a misnomer for glycosylamine and should not be used.) *Distinguish from* **glycosamine**.

glycosylamino acid any **glycoamino acid** in which the saccharide moiety is in glycosyl (*O*-, *N*-, or *S*-) linkage to the amino-acid moiety. These are obtained by enzymic or chemical cleavage of glycoproteins or by chemical or enzymic synthesis.

glycosylase *see* (DNA) **glycosylase**.

glycosylasparaginase *see* **N⁶-(β-N-acetylglucosaminyl)-L-asparaginase**.

glycosylated hemoglobin a misnomer for **glycated hemoglobin**.

glycosylation substitution of one or more **glycosyl** groups into a chemical compound or group. —**glycosylated** *adj.* *See also* **glycation**.

C-glycosyl compound any compound formed by the replacement of the anomeric, i.e. glycosidic, hydroxyl group of a cyclic form of a monosaccharide or monosaccharide derivative by an organyl group. The term **C-glycoside** is a misnomer for C-glycosyl compound and should not be used.

glycosyloxy any chemical group formed by detaching the hydrogen atom from the anomeric, i.e. glycosidic, hydroxyl group of the cyclic form of a monosaccharide or monosaccharide derivative.

glycosylphosphatidylinositol *see* **GPI anchor**.

glycosyltransferase any enzyme transferring a glycosyl group. Glycosyltransferases belong to subclass EC 2.4, which includes hexosyltransferases (EC 2.4.1.1 to 2.4.1.206); pentosyltransferases (EC 2.4.2.1 to 2.4.2.37); and enzymes transferring other glycosyl groups (EC 2.4.99.1 to 2.4.99.11).

glycotropic increasing the concentration of glucose in the blood. *Compare* **glycostatic**.

glycuronan or (formerly) **polyuronide** any glycan composed entirely of uronic-acid residues.

glycuronic acid an obsolete term for **uronic acid**.

glycuronide the former term for **glycosiduronic acid**.

glycyl the acyl group, $\text{NH}_2\text{-CH}_2\text{-CO-}$, derived from **glycine**.

glycyl chain a name sometimes applied to the **A chain** of any of those species of insulin whose amino-acid sequences were the first to be determined. It is so-called by virtue of the N-terminal glycine residue occurring in those instances.

glycylglycine a dipeptide useful for the preparation of buffers in the pH range around its pK_a (20 °C) of 8.40.

glycyrrhiza also called **licorice** or (esp. US) **licorice**; a preparation obtained from the dried roots of the leguminous shrub, *Glycyrrhiza glabra*, with flavouring and medical uses.

glycyrrhizic acid 20β-carboxy-11-oxo-30-norolean-12-en-3β-yl-2-O-β-D-glucopyranuronosyl-α-D-glucopyranosiduronic acid; a glycosiduronic acid extracted from *Glycyrrhiza* (*see* **glycyrrhiza**); it has an intensely sweet taste.

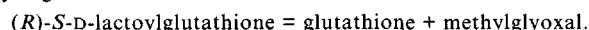
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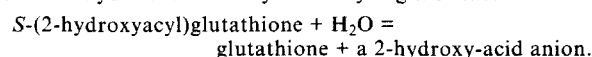
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glyoxal ethanedial; OHC-CHO ; the dialdehyde analogue of glycol and oxalic acid.

glyoxalase I EC 4.4.1.5; *recommended name*: lactoylglutathione lyase, *other name*: methylglyoxalase. An enzyme catalysing the reaction:



glyoxalase II EC 3.1.2.6; *recommended name*: hydroxyacylglutathione hydrolase. An enzyme catalysing the reaction:



glyoxalate a former name for **glyoxylate**.

glyoxaline an alternative name for **imidazole**.

glyoxylate or (formerly) **glyoxalate** 1 the anion, OHC-COO^- , derived from glyoxylic acid. 2 any salt or ester of glyoxylic acid.

glyoxylate cycle or **glyoxylate pathway** a modification of the **tricarboxylic-acid cycle** occurring in some plants and microorganisms, in which isocitrate, instead of being dehydrogenated, is cleaved as described under **isocitrate lyase** to glyoxylate and succinate. This serves to generate glyoxylate from the acetyl moiety of acetyl-CoA. The glyoxylate can then react with another molecule of acetyl-CoA to form malate, a reaction catalysed by **malate synthase**. These two reactions taken together allow a four-carbon molecule, malate, and hence carbohydrate, to be formed from acetyl-CoA, which, in turn, may be formed from fatty acids. In some plants the enzymes of the glyoxylate pathway are present in **glyoxysomes**.

glyoxysome a cellular organelle, of density 1.25 g cm^{-3} , found in the organs of fatty seedlings where fat is being converted to carbohydrate. It contains the enzymes of the **glyoxylate cycle** and most of the particulate beta-oxidation activity.

glyphosate *N*-(phosphonomethyl)glycine; a potent, systemically acting, nonselective herbicide, present as its isopropylamine salt in some commercial weedkillers. It functions by inhibiting 5-enolpyruvylshikimate-3-phosphate (EPSP) synthase. *See also* **shikimate**. *Proprietary names* include Roundup and Tumbleweed.

glypiation the attachment or process of attachment of glycosylphosphatidylinositol to proteins. *See* **GPI anchor**.

GM or **G-M** *abbr.* for Geiger-Müller (e.g. in **GM counter** for Geiger-Müller counter).

GM-CSF *abbr.* for granulocyte/macrophage colony stimulating factor. *See* **colony stimulating factor**.

GMD (in clinical chemistry) *abbr.* for **glutamate dehydrogenase (NAD(P)⁺)**.

GMP *abbr.* for guanosine monophosphate i.e. guanosine phosphate.

GMP synthase EC 6.3.5.2; *recommended name*: GMP synthase (glutamine-hydrolysing); *systematic name*: xanthosine-5-phosphate:L-glutamine amido-ligase (AMP-forming); *other names*: GMP synthetase (glutamine-hydrolysing); glutamine amido-transferase. An enzyme that catalyses a reaction between ATP, xanthosine 5'-phosphate, L-glutamine, and H_2O to form AMP, pyrophosphate, guanosine monophosphate, and L-glutamate. It is inhibited by azaserine and 6-diazo-5-oxonorleucine, in common with other glutamine amidotransferases. It has four motifs and a region conserved in amidotransferases (one motif). Example from *Escherichia coli*: database code GUAA_ECOLI, 525 amino acids (58.61 kDa).

GMT (in clinical chemistry) *abbr.* for **glutamyltransferase**.

gnotobiotic describing an organism, especially a higher animal, or its environment, that has been deliberately rendered free of microorganisms and other organisms, including parasites, commensals, and symbionts, that it would normally harbour, or that harbours only a few, known, such organisms.

GnRH *abbr.* for gonadotropin-releasing hormone.

GNTR a transcriptional repressor for the **gluconate (gnt)** operon in *Bacillus subtilis*. The **gnt** operon is induced by gluconic acid β-lactone; this was the first of a family of **gnt** re-



goblin

pressors with **helix-turn-helix** motifs. Database code GNTR_BACSU, 243 amino acids (28.25 kDa).

goblin a protein of avian erythrocyte plasma membranes and M_r 260 000. It undergoes phosphorylation, and is highly homologous with **ankyrin**, of which it is the avian equivalent.

GOD (*sometimes*) *abbr.* for glucose oxidase.

godnose a name proposed for ascorbic acid by A. Szent-Györgyi, as an alternative to **ignose**, before he knew its structure.

GOGAT *abbr.* for glutamine (amide):2-oxoglutarate aminotransferase (oxidoreductase NADP⁺). It is now known to be two enzymes, renamed **glutamate synthase (ferredoxin)** and **glutamate-ammonia ligase**.

goitre or (*US*) **goiter** any swelling of the thyroid gland. Although common in areas where dietary iodine is insufficient, the condition is not related to thyroid state and patients with goitre may be euthyroid, hyperthyroid, or hypothyroid.

goitrogen any agent or factor causing goitre.

Goldstein, Joseph Leonard (1940–), US physician and molecular geneticist notable for his discovery (in collaboration with M. S. Brown) of cellular receptors for low-density lipoproteins and their role in the removal of cholesterol from the bloodstream; Nobel Laureate in Physiology or Medicine (1985) jointly with M. S. Brown 'for their discoveries concerning the regulation of cholesterol metabolism'.

Golgi, Camillo (1843–1926), Italian anatomist, cytologist, neurologist, and pathologist remembered for his cytological studies of the central nervous system, for the introduction of techniques for staining cells with metallic salts, and in particular for his discovery thereby of the cellular organelle named after him (*see* **Golgi apparatus**); Nobel Laureate in Physiology or Medicine (1906) jointly with S. Ramón y Cajal 'in recognition of their work on the structure of the nervous system'.

Golgi apparatus or **Golgi body** or **Golgi complex** a compound membranous cytoplasmic organelle of eukaryotic cells, consisting of flattened, ribosome-free vesicles arranged in a more or less regular stack. The Golgi apparatus differs from the endoplasmic reticulum in often having slightly thicker membranes, appearing in sections as a characteristic shallow semicircle so that the convex side (*cis* or entry face) abuts the endoplasmic reticulum, secretory granules emerging from the concave side (*trans* or exit face). In vertebrate cells there is usually one such organelle, while in invertebrates and plants, where they are known usually as **dictyosomes**, there may be several scattered in the cytoplasm. The Golgi apparatus processes proteins produced on the ribosomes of the rough endoplasmic reticulum; such processing includes modification of the core oligosaccharides of glycoproteins, and the sorting and packaging of proteins for transport to a variety of cellular locations, e.g. to storage granules, lysosomes, or secretory vesicles. The Golgi is a major site of synthesis of polysaccharides, including the **pectin** and hemicellulose of the plant cell wall and most of the glycosaminoglycans of the extracellular matrix in animals. Three different regions of the Golgi are now recognized both in terms of structure and function: *cis*, in the vicinity of the *cis* face, *trans*, in the vicinity of the *trans* face, and *medial*, lying between the *cis* and *trans* regions. While the **luminal proteins** of the endoplasmic reticulum leave it together with the proteins to be secreted, they are retrieved from the *cis* network and returned to the endoplasmic reticulum. Thus transport occurs in both directions. The drug **brefeldin A** blocks protein secretion through the Golgi but does not appear to block the return pathway. **Galactosyltransferase** is a marker enzyme for the *trans* membranes while **N-acetylglucosaminyltransferase III** serves this role for the medial membranes. [After Camillo Golgi, who, in 1898, was the first to describe it.]

Golgi mannosidase EC 3.2.1.114; *recommended name*: mannosyl-oligosaccharide 1,3-1,6- α -mannosidase; *systematic name*: 1,3-(1,6)-mannosyl-oligosaccharide α -D-mannohydrolase; *other name*: mannosidase II. An enzyme that catalyses the hydrolysis of the terminal 1,3- and 1,6-linked α -D-mannose residues

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in the mannosyl-oligosaccharide Man₅GlcNAc₃. It is a type II membrane protein, and a glycosyl hydrolase enzyme in all tissues, mostly in adrenal and thymus. It belongs to a family of glycosyl hydrolases. Example from mouse: database code MAN2_MOUSE, 1150 amino acids (131.44 kDa).

golgiosome an alternative name for **dictyosome**.

gonad any animal organ that produces gametes. In some species the gonads also produce hormones. —**gonadal** *adj.*

gonadoliberin *recommended name for gonadotropin-releasing hormone*.

gonadotropic hormone or **gonadotrophic hormone** a former name for **gonadotropin**.

gonadotropin or (*formerly*) **gonadotrophin** or **gonadotropic hormone** or **gonadotrophic hormone** 1 any mammalian glycoprotein hormone that stimulates gonadal function. Strictly, gonadotropins comprise the adenohipophyseal hormones **follicle-stimulating hormone** (FSH) and **luteinizing hormone** (LH) together with the placental hormone **choriogonadotropin**, but the term is often extended to include also the adenohipophyseal hormone **prolactin** (which is not a glycoprotein but has gonadotropic activity). 2 any single substance having the combined activities of FSH and LH, such as **choriogonadotropin** or the gonadotropic hormones of cold-blooded vertebrates. 3 any impure hormonal preparation displaying the separate activities of FSH and LH, such as (human) **menopausal gonadotropin** (i.e. **urogonadotropin**) and **pregnant mare's serum gonadotropin**. —**gonadotropic** *adj.*

gonadotropin-releasing factor *abbr.*: GnRF; *another name for gonadotropin-releasing hormone*.

gonadotropin-releasing hormone (*abbr.*: GnRH) or **gonadoliberin** (*recommended name*) or **luteinizing hormone/follicle-stimulating hormone releasing factor** (*abbr.*: LH/FSH-RF) a family of decapeptide amides, of which there are at least seven members. They have been found in many vertebrate species from jawless fish (lamprey) to higher mammals. The structure is highly conserved; the sequence Glp-His-Trp-Ser-Tyr-Gly-Leu-Arg-Pro-Gly-NH₂ is common to all mammals; among other species, position 8 is most variable, positions 1, 2, 4, 9, and 10 being invariant. It is named after its function in mammals, in which it is released by the hypothalamus into the hypophyseal portal circulation in response to neural and/or chemical stimuli, and is considered to be the sole neuropeptide causing the release of FSH and LH by the anterior pituitary. In mammals it is also expressed in the placenta, and its functions across phylogenetic classes may include actions as a neurotransmitter, neuromodulator or local hormone. It is synthesized as a much larger precursor, example from rat: database code GONL_RAT, 93 amino acids (10.5 kDa); residues 1–23 are the signal peptide, 24–33 GnRH (35 and 36 are a dibasic cleavage site, Gly³⁴ provides the amide group) and 37–92 are a peptide (GnRH-associated peptide, *abbr.*: GAP) initially thought to inhibit secretion of prolactin-releasing hormone, but its role is not well understood and it is poorly conserved.

gonadotropin-releasing hormone receptor *abbr.*: GnRHR; any of a family of integral membrane proteins that bind GnRH and mediate its intracellular effects. They are found not only in pituitary, but also in ovary, testis, breast, and prostate. These are seven-transmembrane-helix G-protein-coupled receptors, that stimulate the phosphatidylinositol-Ca²⁺ second messenger system. Example from *Rattus norvegicus*: database code GRHR_RAT, 327 amino acids (37.75 kDa).

gonane the parent 17-carbon tetracyclic hydrocarbon of steroids and bile acids.

gondoic acid *see* **eicosenoic acid**.

gonosome any motile animal germ cell, e.g. a spermatozoon.

Good buffer substance or **Good's buffer** (*substance*) any member of a collection of **zwitterionic buffer** substances selected or devised for suitability in experimental biological systems according to a number of predetermined criteria. These in-



clude: ease of preparation and purification; high solubility in water; low temperature coefficient of pK_a ; and high transparency to light or ultraviolet light of wavelengths greater than 230 nm. The original lists included substances having the following trivial names (pK_a values at 20 °C are in parentheses): **Mes** (6.15), **Ada** (6.6), **Pipes** (6.8), **Aces** (6.9), **Bes** (7.15), **Mops** (7.2), **Tes** (7.5), **Hepes** (7.55), **Hepps** (8.0), **Tricine** (8.15), **Bicine** (8.35), and **Taps** (8.4); **Ches** (9.3 at 25 °C) and **Caps** (10.4) have similar properties. [After Norman Everett Good (1917–), US plant pathologist.]

gossypol a pigment found in cottonseed and containing two naphthalene rings with multiple –OH and other groups. It has antispermatic activity, and has been used (in China) as a male contraceptive. It is a potent inhibitor of contractions of strips of guinea-pig lung induced by platelet-activating factor or leukotriene, and it inhibits 5- and 12-lipoxygenase.

GOT (*in clinical chemistry*) *abbr.* for glutamate-oxaloacetate transaminase (AST preferred). *See* **transaminase**.

gout a condition characterized by inflammation of the joints and cartilage due to hyperuricemia and consequent deposition of monosodium urate crystals in the joints, often within polymorphonuclear leukocytes. The hyperuricemia results from overproduction of uric acid, which may be due to elevated activity of ATP phosphoribosyltransferase (PRPP synthetase; EC 2.4.2.17), an abnormal **amidophosphoribosyltransferase**, or deficiency of hypoxanthine phosphoribosyltransferase (EC 2.4.2.8). **Allopurinol** is useful in treatment.

gp *abbr.* for glycoprotein used in conjunction with the relative molecular mass (e.g. gp115).

gp32 an envelope glycoprotein from HIV-related viruses including **SIV** types. *See* **HIV protein**.

G1P or **G-1-P** *abbr.* for glucose 1-phosphate.

G6P or **G-6-P** *abbr.* for glucose 6-phosphate.

GPD (*in clinical chemistry*) *abbr.* for glucose-6-phosphate dehydrogenase.

G₀ phase G-'zero' phase; a state of withdrawal from the eukaryotic **cell-division cycle**. It occurs if no stimulus for cell division occurs during G₁ phase, leaving the cell in a quiescent G₁ phase. It is often seen in differentiated cells.

G₁ phase Gap 1 phase; the phase of the eukaryotic **cell-division cycle** between the end of **cytokinesis** and the start of DNA synthesis, i.e. the gap between M phase and S phase.

G₂ phase Gap 2 phase; the phase of the eukaryotic **cell-division cycle**, between the end of DNA synthesis and the beginning of mitosis, i.e. the gap between S phase and M phase.

GPI (*in clinical chemistry*) *abbr.* for glucose-6-phosphate isomerase.

GPI anchor *abbr.* for glycosylphosphatidylinositol anchor; a molecular mechanism for attaching a membrane protein to the lipid bilayer of a cell membrane. Structurally it consists of a molecule of phosphatidylinositol (*abbr.*: PI) to which is linked, via the C-6 hydroxyl of the inositol, a carbohydrate chain. This chain is in turn linked to the protein through an ethanolamine phosphate moiety, the amino group of which is in amide linkage with the C-terminal carboxyl of the protein chain, the phosphate group being esterified to the C-6 hydroxyl of the terminal mannose of the core carbohydrate chain. The carbohydrate chain varies in composition between species, but invariably contains a core tetrasaccharide, the first component of which is a glucosamine residue in glycosidic linkage with the inositol C-6 hydroxyl. The tetrasaccharide has the structure Man(α 1-2)Man(α 1-6)Man(α 1-4)GlcNH₂. Variation frequently occurs in the presence of additional α -mannose residues on the anchors of *Trypanosoma cruzi*, yeast, slime mould, and mammalian proteins. Anchors of *Trypanosoma brucei* have more elaborate carbohydrate side-chains containing galactose, glucosamine, and sialic-acid residues. In the case of metazoan eukaryotes, additional ethanolamine phosphate substituents may occur on mannose residues, and the inositol is often palmitoylated. Biosynthesis begins with attachment of the glucosamine moiety to phosphatidylinosi-

tol, followed by sequential addition of other sugars; the GPI moiety is attached to protein during processing in the endoplasmic reticulum, the signal being a hydrophobic sequence at the C terminus, which is removed during the attachment process. GPI-anchored proteins are released from membranes by PI-specific phospholipases.

While the general function of the GPI anchor is to anchor the protein to the membrane, wider functions may include: location of the protein at specific sites, particularly the apical membrane of polarized epithelial cells; promotion of sequestration into membrane microdomains, these proteins being excluded from clathrin-associated regions but found in **caveolae**; anchoring of proteins of the glycocalyx, and cell signalling and cell recognition proteins.

In addition to protein-linked GPIs, *Leishmania* spp. also synthesize two classes of free GPIs, the polydisperse **lipophosphoglycans** (LPGs) and the low molecular mass **glycoinositolphospholipids** (GIPLs). LPGs are restricted in occurrence (e.g. promastigote stage) but GIPLs are relatively abundant on the cell surface of promastigote and amastigote stages. GIPLs are also found on the surface of many other trypanosomatid parasites, e.g. *T. cruzi*, *Leptomonas*, *Herpetomonas*, and *Phytomonas*.

gp 43 processing protease *see* **oviductin**.

G-protein guanine nucleotide binding protein; any of a family of heterotrimeric GTP-binding and hydrolysing proteins; they belong to a superfamily of GTPases that includes monomeric proteins such as EF-Tu (*see* **EF-T**) and p21^{ras} (*see* **RAS**). The heterotrimeric G-proteins are a component of receptor mediated activation or inhibition of **adenylate cyclase** and other **second messenger** systems, e.g. **phosphatidylinositol cycle**. They mediate the intracellular actions of many hormones, growth factors, and cytokines. The system is widely distributed but unique to eukaryotes. G-proteins consist of three subunits; the α subunit (39–46 kDa) contains the guanine nucleotide binding site and possesses GTPase activity; the β (37 kDa) and γ (8 kDa) subunits are tightly associated and function as a $\beta\gamma$ heterodimer. α , β , and γ polypeptides are encoded by at least 16, 4, and 7 genes respectively. The classes of G-protein and subunits are subscripted: thus the α subunit of G-protein S (a stimulatory G-protein that results in activation of **adenylate cyclase**) is G_{S α} ; other G-proteins include G_i, an inhibitory **G-protein** which differs from G_S structurally (different type of α subunit) and results in inhibition of **adenylate cyclase**; the binding of **somatostatin** to its receptor results in G_i activation; G_o, a G-protein detected by cDNA probes, of wide tissue distribution in mammals but of unclear role; G_q, a G-protein that leads to activation of phospholipase C (*see* **phospholipase**), release of Ca²⁺ and activation via **calmodulin** and **protein kinase C** (*see also* **phosphatidylinositol cycle**). Binding of G_q to activated receptor occurs when GDP is bound to the G-protein ($\alpha\beta\gamma$ -GDP). Association of the heterotrimer with activated receptor causes exchange of GDP for GTP. Binding of GTP causes the heterotrimer to dissociate into α -GTP and $\beta\gamma$ units, either of which may interact with an effector. The GTPase activity of α yields α -GDP which inactivates it. α -GDP has a high affinity for $\beta\gamma$ thus reforming $\alpha\beta\gamma$ -GDP which is once again available for association with an activated receptor. G_T, *see* **transducin**. Examples: G_{S α} from human and dog; database code GBAS_HUMAN, 394 amino acids (45.61 kDa); β subunit from G_S and transducin (human); database code GBB2_HUMAN, 340 amino acids (37.29 kDa); γ brain and adrenal form for G_S, G_i, G_o (bovine); database code GBG2_BOVIN, 71 amino acids (7.84 kDa). *See also* **Appendix E**.

GPT (*in clinical chemistry*) *abbr.* for glutamate-pyruvate transaminase (EC 2.6.1.2) (ALT preferred).

G-quartet or **G-tetrad** a cyclic H-bonded array of guanines that can occur in both RNA and DNA to form a quadruple helix structure. Sequences that form such structures are found in telomeric sites, the switch regions of immunoglobulin heavy-chain genes, and elsewhere. *See* **KEM1**.

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Gra *symbol for a residue of glyceraldehyde.*

Graafian follicle a spherical vesicle occurring in the mature mammalian ovary and in which the ovum develops. When the ovum matures it is discharged after rupture of the follicle, which is then transformed into a **corpus luteum**. [After Reijnier (Regnier) de Graaf (1641–73), Dutch physician and anatomist.]

grade **1** a degree or position in a scale, as of quality, size, etc., e.g. analytical grade, commercial grade. **2** a unit of angle equal to one-hundredth of a right angle, i.e. 0.9°.

gradient the change in the value of a specified (chemical or physical) property or value per unit distance in a specified direction.

gradient elution a chromatographic elution procedure in which the concentration of a component of the eluent is increased continuously. The gradient may be linear, concave – i.e. increasing slowly at first and more rapidly as elution proceeds – or convex – i.e. increasing rapidly at first and more slowly as elution proceeds. More complex or discontinuous (stepped) elution gradients are sometimes used.

gradient former or **gradient mixer** any device for producing solutions of varying, but defined, composition containing one or more components.

gradient gel electrophoresis or **gradient acrylamide gel electrophoresis** or **polyacrylamide gradient gel electrophoresis** (*abbr.*: PAGE) a type of **zone electrophoresis** in which the supporting medium is a pore-gradient gel, i.e. one where an increasing concentration of polyacrylamide in the direction of electrophoretic mobility decreases the sizes of the pores in the gel in that direction to the point where molecules of more than a certain size are prevented from moving further in the electric field. The concentration gradient may be either discontinuous, i.e. stepped, or continuous. Advantages of the technique over electrophoresis in uniform gels (*see* **gel electrophoresis**) are sharper zones and an ability to resolve, on a single gel, mixtures of macromolecules of a wider range of sizes.

gradient layer or **gradient packing** a chromatographic layer, or column packing, incorporating a continuous change of some property that affects separations, e.g. a pH gradient or a pore gradient.

graft **1** any part of an organism, animal or plant, that is inserted into and unites with a larger part of another or the same organism. **2** to transplant (tissue) to another organism or to another site on the same organism.

graft versus host disease a condition where immunocompetent donor cells recognize and react against the recipient's tissues, because the recipient is immunosuppressed or cannot recognize the different genetic constitution of the donor cells.

Graham, Thomas (1805–69), Scottish chemist, one of the founders of physical chemistry; he conducted research on absorption of gases, osmosis, diffusion, colloids, and dialysis.

Graham's law of diffusion a law stating that at equal temperatures and pressures the rates of diffusion of gases are inversely proportional to the square roots of their densities and hence to the square roots of their relative molecular masses (M_r s). Thus for two gases A and B ,

$$J_A/J_B = \bar{v}_A/\bar{v}_B = M_{rB}^{1/2}/M_{rA}^{1/2},$$

where J_A and J_B , $\bar{v}_A$ and $\bar{v}_B$, and M_{rA} and M_{rB} are the fluxes, mean velocities, and relative molecular masses of A and B , respectively.

grain **1** any small, hard particle, e.g. a grain of sand. **2** any simple single-seeded indehiscent fruit in which the seed wall fuses with the carpel wall during development. Technically called a **caryopsis**, it is typical of the cereals and other grasses. **3** (*in photography*) any of the many particles of silver halide in a photographic emulsion. The resolution possible in the image is inversely proportional to the grain size in the emulsion. **4** any of the dried insect bodies present in dyes such as cochineal.

gram *symbol*: g; a unit of mass equal to a thousandth part of

the kilogram in both the *Système International* (SI) and the metric system (cgs or mks) of units.

+gram *comb. form* indicating a drawing, something written or recorded; the record produced by an instrument, e.g. spectrogram; the visible result of a procedure, e.g. autoradiogram, chromatogram.

gram-atom or **gram-atomic weight** *abbr.*: g-atom; the atomic weight of an element expressed in grams; it has now been replaced by amount of substance, measured in **moles**.

gram-equivalent or (*formerly*) **gram-equivalent weight** *abbr.*: g-equiv; an **equivalent** (*def.* 3) expressed in grams.

gramicidin any of a group of linear polypeptide antibiotics that are active against Gram-positive microorganisms and are isolated from cultures of *Bacillus brevis*. The commercial material is a mixture of gramicidins A, B, C, and D, all of which consist of chains of 15 amino-acid residues in a pattern of alternating D- and L-forms; all have a formyl group at the N terminus and an ethanolamine residue at the C terminus. Gramicidins increase the permeability of biological membranes to protons and alkali-metal cations by forming transient dimeric ionophoric channels through the membrane. *See also* **gramicidin S**.

gramicidin S a homodetic cyclic decapeptide antibiotic, *cyclo*(-Val-Orn-Leu-DPhe-Pro-Val-Orn-Leu-DPhe-Pro-), produced by a strain of *Bacillus brevis* and active against Gram-positive bacteria. Gramicidin S is more closely related to **tyrocidins** in its biological and chemical properties than to true **gramicidins**. Its biosynthesis occurs by a mechanism that is not dependent on the presence of ribosomes or mRNA. It was the first peptide antibiotic to be chemically synthesized, and is so named because it was first isolated in Soviet Russia, where it was widely used.

gram-molecule or **gram-molecular weight** *symbol*: g-mol; the molecular weight of a compound (or element) expressed in grams; it has now been replaced by the **mole**.

Gram-negative describing a bacterial cell that, during **Gram staining**, does not retain the basic dye in its cell wall. The walls of Gram-negative organisms are thin, about 10 nm across, but complex, having a lipopolysaccharide layer outside the peptidoglycan layer; they are also resistant to digestion by lysozyme. *Compare* **Gram-positive**.

Gram-positive describing a bacterial cell that, during **Gram staining**, retains the basic dye in its cell wall. The walls of Gram-positive organisms are relatively thick, 15–80 nm across, and consist of a network of peptidoglycan; they are readily digested by lysozyme. *Compare* **Gram-negative**.

Gram staining or **Gram stain** a staining procedure used for bacteria in which the cells are first heat-fixed and stained with a basic dye, e.g. Crystal Violet, then treated with I₂-KI solution to fix the stain, washed with ethanol or acetone, and finally counterstained with another dye, e.g. carbolfuchsin or safranin. Cells that retain the basic dye after this procedure are called **Gram-positive**; those that do not are called **Gram-negative**. [After Hans Christian Joachim Gram (1853–1938), Danish physician and bacteriologist, who described the method in 1884.]

grancalcin a Ca²⁺-binding protein containing four **EF hands** and abundant in neutrophils and monocytes. It exhibits strong homology to **sorcin**. It is a homodimer. Example from human: database code A42578, 217 amino acids (23.98 kDa).

β granin *see* **chromogranin**.

Granit, Ragnar Arthur (1900–91), Finnish-born Swedish neurophysiologist; Nobel Laureate in Physiology or Medicine (1967) jointly with H. K. Hartline and G. Wald 'for their discoveries concerning the primary physiological and chemical visual processes in the eye'.

granular or **granulose** or **granulate** being composed of, or having the character of small particles or **grains** (*def.* 1).

granular endoplasmic reticulum or **rough endoplasmic reticulum** *see* **endoplasmic reticulum**.

granulate **1** to make (something) into **grains** (*def.* 1). **2** (of a wound, ulcer, etc.) to form **granulation tissue**. **3** *another word for* **granular**.

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granulation

granulation 1 the act or process of subdividing a solid into small, usually rounded **grains** (def. 1). 2 the formation of one or more small, rounded lumps (grains (def. 1)) in a tissue. 3 the development of **granulation tissue**.

granulation tissue the mass of young connective tissue and blood vessels formed on the surface of a wound or ulcer in the early stages of healing.

granule any small **grain** (def. 1), especially any small intracellular particle containing stored material that can be seen by (electron) microscopy, e.g. insulin-containing β granules in the B cells of the islets of Langerhans, starch granules in potato tubers.

granulocyte a category of white blood cells distinguished by the presence of conspicuous cytoplasmic granules. Granulocytes include **neutrophils**, **basophils**, and **eosinophils**.

granulocyte-colony stimulating factor *see* **colony stimulating factor**.

granulose 1 another word for **granular**. 2 an obsolete name for starch or amylose.

granum (*pl.* **grana**) a stack of disclike **thylakoids** resembling a pile of coins. Grana occur in **chloroplasts**, especially in higher plants.

granzyme A enzyme of sub-subclass EC 3.4.21; *other names*: cytotoxic T-lymphocyte proteinase 1; Hanukkah factor (HF); granzyme 1; CTL tryptase; fragmentin 1; trypsin-like serine proteinase. An enzyme necessary for target-cell lysis in cell-mediated immune responses; it cleaves on the carbonyl side of Lys or Arg. It may be involved in **apoptosis**. It occurs in cytoplasmic granules. The molecule is a homodimer, disulfide linked. Example (precursor) from human: database code GRAA_HUMAN, 262 amino acids (28.93 kDa).

grape sugar a former name for D-glucose. *See* **glucose**.

graph 1 a diagram showing the relation between two variable quantities, or the relation between a mathematical function and different values of one of the variables in the function, usually plotted between axes at right angles to each other. 2 to plot or draw a graph.

+graph *comb. form* 1 denoting (something) drawn, recorded, or written (in the way specified), e.g. autoradiograph, photograph. 2 denoting an instrument or device that draws or records (observations), e.g. spectrograph. 3 denoting the act or process of producing such a drawing or record. *Compare* **+gram**. — **+graphic** or **+graphical** *adj. comb. form*.

graphics (*functioning as sing.*) 1 the use of diagrams in calculation. 2 the production of diagrams, representations of molecular structures, etc., often by computer.

+graphy *comb. form* denoting a method, process, or technique that produces a drawing or record, e.g. chromatography. *See* **+graph**.

grating *see* **diffraction grating**.

Graves' disease or **Basedow's disease** a disease characterized by diffuse goitre, thyrotoxicosis, and exophthalmia. It is an autoimmune condition associated with the presence of thyroid-stimulating antibodies in the blood. [After Robert James Graves (1796–1853), Irish physician.]

Graves' disease carrier protein *see* **GDC**.

gravimetric analysis quantitative analysis of the chemical composition of substances or materials based on the separation of each component of interest and weighing either the purified compound or a derivative of it.

gravimetry measurement by weighing. — **gravimetric** or **gravimetric** *adj.*

gravitation the attractive force that bodies exert on one another as a result of their mass; any process or result caused by such an attractive force. — **gravitational** *adj.*

gravitational constant *symbol*: G ; the proportionality constant relating the attractive force, F , between two bodies of mass m_1 and m_2 , and the distance between them, s , in Newton's law of gravitation:

$$G = F s / m_1 m_2 = 6.672\,59\,(85) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

gravitational field any **field** (def. 1) associated with a mass by which a force is exerted on a second mass placed in this field.

gravitational interaction any interaction between bodies or particles that results from their masses.

gravitational mass *see* **mass**.

gravity 1 the attractive force that moves or tends to move bodies towards the centre of any celestial body such as the Earth or the Moon. 2 the attribute of having weight. 3 **specific gravity**.

gray *symbol*: Gy; the SI derived unit of **absorbed dose** of radiation, equal to one joule per kilogram of living tissue;

$$1 \text{ Gy} = 1 \text{ J kg}^{-1} = 1 \text{ m}^2 \text{ s}^{-2}.$$

Compare **sievert**. *See also* **rad**. [After Louis Harold Gray (1905–65), British physicist and radiobiologist.]

Grb2 growth factor receptor-bound protein 2; *also called* **Ash protein**; a cytoplasmic protein, related to **sem-5** (*see* **sem-5**), that participates in downstream signalling after activation of a variety of cell receptors. It binds directly to activated epidermal growth factor and platelet-derived growth factor receptors; indirectly it can be recruited by G-protein-coupled receptors through tyrosine phosphorylation of **Shc** (*see* **SHC**). Grb2 binds to activated **Shc** and the complex then recruits **Sos** (*see* **sos**). The **Shc-Grb2-Sos** complex binds to the plasma membrane where **Sos** catalyses **Ras** guanine nucleotide exchange, leading to MAP kinase activation. Example from human (identical in rat): database code GRB2_HUMAN, 217 amino acids (25.18 kDa).

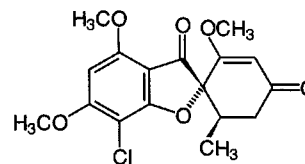
GRE *abbr. for* glucocorticoid response element.

GRF *abbr. (sometimes) for* growth hormone releasing factor (i.e. **GHRH**).

Gri *symbol for* glyceric acid or glycerate.

grid 1 a framework made up of fine wires arranged in two sets at right angles to each other, especially a device for supporting the specimen in an electron microscope. 2 an electrode controlling the current flowing through a thermionic valve.

griseofulvin [2*S*]-*trans*-7-chloro-2',4,6-trimethoxy-6'-methylspiro[benzofuran-2(3*H*),1'-(2)cyclohexene]-3,4'-dione; a chlorine-containing **polyketide** metabolite of *Penicillium griseofulvum* and *P. janczewskii*. It has antifungal properties and is used therapeutically, e.g. for treatment of ringworm. It inhibits spindle formation in mitosis, microtubule function, and possibly nucleic-acid synthesis.



Grn *symbol for* glycerone.

gRNA *abbr. for* guide RNA; small RNA molecules, 60–80 nucleotides long, that may be involved in editing of RNA by supplying information concerning nucleotide deletion or addition. *See also* **editosome**.

gro the symbol for any of a group of related genes found in bacteria (especially *Escherichia coli*), mutations in which affect the growth of phages.

Gro *symbol for* glycerol.

groEL the symbol for any of a family of bacterial genes encoding **chaperonins**. They have eukaryotic counterparts. Example of a **groEL** protein product from *Escherichia coli*: oligomer of 14 subunits composed of two stacked rings of seven subunits; it has ATPase activity; database code CH60_ECOLI, 547 (57.02 kDa).

groES the symbol for any of a family of bacterial genes encoding **chaperonins**. They have eukaryotic counterparts. Example of a **groES** protein product from *Escherichia coli*: six to eight subunits arranged in a ring; it binds to the **groEL** protein prod-



Grotthus and Draper law

uct to suppress the latter's ATPase activity; database code CH10_ECOLI, 97 amino acids (10.38 kDa)

Grotthus and Draper law a law stating that only those radiations absorbed by a system are effective in producing chemical change in the system.

groundplasm or **cytoplasmic matrix** the sub-electron-microscopic material of the **cytoplasm**, excluding the cell organelles, cell particles, and other microscopically resolvable structures. It contains the cytoplasmic enzymes and metabolites together with components of the **cytoskeleton**.

ground state the state of lowest energy of any system.

ground substance an alternative name for the **matrix** (def. 2) of connective tissue.

ground tissue a collective term for plant tissue other than the epidermis or peridermis and vascular tissue.

group (in chemistry) or (formerly) **radical 1** any defined continuously linked collection of atoms within a molecular entity. In organic chemical nomenclature the meaning within this general sense is restricted to any covalently bonded collection of atoms that behaves as a unit in chemical reactions; **functional group**, **substituent group**. **2** any of the columns of chemical elements in the **periodic table**. **3** a **blood group**.

group pK the **pK** of a specific group in a molecule with two or more acid groups. In general the term applies only to a particular ionic form of the molecule, because the **pK** of a group changes when any other group in the molecule changes its ionization state, and thus changes as the substance is titrated. Compare **molecular pK**, **titration pK**.

group potential the tendency of a **group** (def. 1) to escape from a molecular entity.

group-transfer potential the Gibbs energy change (free energy change, ΔG) occurring when a **group** (def. 1) is transferred from a donor compound to the nucleophile water, i.e. when the group is removed by hydrolysis of the donor. It reflects the thermodynamic tendency for the group to be transferred.

group-transfer reaction any chemical reaction in which a **group** (def. 1) is transferred from a donor compound to an acceptor compound. Such reactions may be catalysed by a **transferase** enzyme of EC class 2.

group translocation any process in which a substance to be transported across a biological membrane undergoes covalent modification prior to crossing the membrane.

growth curve the change, as a function of time, in the number of cells of a growing culture or in some measure of the size of an organism.

growth factor 1 any specific substance that must be present in a culture medium for multiplication of the cultured cells to take place. **2** a specific substance that is a necessary constituent of the diet for the normal growth of a particular animal, or is required in the soil for the normal growth of a particular plant.

growth factor receptor-bound protein 2 see **Grb2**.

growth hormone another name for **somatotropin**.

growth hormone inhibitory factor *abbr.*: GHIF or GIF; a former name for **somatostatin**.

growth hormone release-inhibiting hormone *abbr.*: GH-RIH; a former name for **somatostatin**.

growth hormone releasing factor *abbr.*: GHRF or GRF; another name for **GHRH**.

growth hormone releasing hormone see **GHRH**.

GRP see **Grb2**.

GRP 75 see **mortalin**.

GRS (in clinical chemistry) *abbr.* for β -glucuronidase.

Grunstein-Hogness procedure an alternative name for **colony hybridization**.

GSH *abbr.* for **glutathione**.

GSSG *abbr.* for oxidized glutathione (see **glutathione**).

GST *abbr.* for **glutathione S-transferase** (see **glutathione transferase**).

GST fusion protein a **fusion protein** in which a synthetic oligonucleotide of a defined sequence of the protein of interest

is grafted onto the glutathione transferase (GST) gene. The fused protein is then expressed in a recombinant system, purified by **affinity chromatography** (making use of the properties of GST), and the GST removed to release the pure protein.

GT-AG rule a rule stating that in RNA there is a requirement for a GT at the 5' side of a splice junction and an AG at the 3' side.

GTD (in clinical chemistry) *abbr.* for **glutathione reductase**.

G-tetrad an alternative name for **G-quartet**.

GTF *abbr.* for **glucose-tolerance factor**.

GTP *abbr.* for **guanosine triphosphate**.

GTPase any enzymic activity that hydrolyses GTP to GDP and orthophosphate. It is associated with **G-proteins**, **EF-Ts**, and **HIV protein**.

GTP-binding protein any of various proteins that bind GTP as a regulatory molecule. This binding is frequently associated with a **GTPase** activity. GTP-binding proteins include elongation and initiation factors (e.g. **EF-Tu**), **G-proteins**, **G25K GTP-binding protein**, **Ras proteins**, and yeast **CDC42** (see **CDC markers**). Binding of GTP normally activates the protein, an effect that is reversed by hydrolysis of the GTP to GDP. Example from human: database code G25B_HUMAN, 191 amino acids (21.29 kDa).

GTP[S] or **GTP γ S** *abbr.* for **guanosine 5'- γ -thiotriphosphate**.

GTR1 a representative 12-transmembrane domain transport protein that effects the facilitated diffusion of glucose in certain mammalian tissues. Example from human erythrocytes: database code GTR1_HUMAN, 492 amino acids (54.04 kDa).

GTT *abbr.* for **glucose tolerance test**.

Gu symbol for **guanidine**.

Gua symbol for a residue of the purine base **guanine** (alternative to **G**).

GUA a codon in mRNA for **L-valine**.

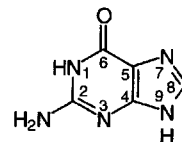
guanidine symbol: **Gu**; the strongly basic substance, imino-urea; aminomethanamide; $(\text{H}_2\text{N})_2\text{C}=\text{NH}$.

guanidinium the cation, $[\text{C}(\text{NH}_2)_3]^+$, derived from **guanidine** by uptake of a hydron.

guanidino the amidinoimino group, $-\text{HN}-\text{C}(=\text{NH})-\text{NH}_2$, derived from **guanidine**.

guanidinoacetate N-methyltransferase EC 2.1.1.2; an enzyme of the pathway for *de novo* synthesis of **creatine**, by reaction of *S*-adenosyl-L-methionine with **guanidinoacetate** releasing *S*-adenosyl-L-homocysteine. Example from rat: database code **GAMT_RAT**, 235 amino acids (26.25 kDa).

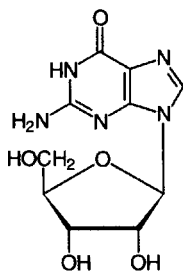
guanine symbol: **G** or **Gua**; 2-amino-6-hydroxypurine; 2-amino-1,7-dihydro-6H-purin-6-one; a purine that is one of the five main bases found in nucleic acids and a component of a number of phosphorylated guanosine derivatives whose metabolic or regulatory functions are important.



guanine nucleotide binding protein see **GTP-binding protein**.

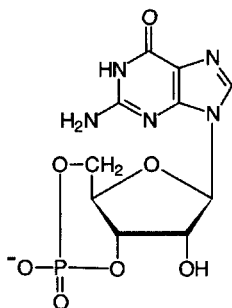
guanine nucleotide releasing protein *abbr.*: **GNRP**; or **guanine nucleotide releasing factor**, *abbr.*: **GRF**; a protein that binds to a **Ras** protein (see **Ras family of proteins**) or **Ras-related GTP-binding protein** and activates it by stimulating it to release its bound **GDP** and bind **GTP** in its place. The C-terminal 310 amino acids show homologies to **Cdc25** (see **cdc gene**) and **Sos**, the product of *src*. Example from *Rattus norvegicus*: database code **GNRP_RAT**, 1244 amino acids (142.67 kDa).

guanosine symbol: **G** or **Guo**; **guanine riboside**; 9- β -D-ribofuranosylguanine; a **nucleoside** with a wide species distribution.



guanosine

guanosine 2',3'-(cyclic)phosphate *see* **guanosine phosphate**.
guanosine 3',5'-(cyclic)phosphate *or* **guanosine 3',5'-cyclophosphate** *see* **guanosine 3',5'-phosphate**.



guanosine 5'-diphosphate *symbol:* Guo5'PP or ppG; *the recommended name for* guanosine diphosphate (*abbr.:* GDP), 5'-diphosphoguanosine, 5'-guanylyl phosphate, guanosine 5'-(trihydrogen diphosphate), a universally distributed nucleotide that occurs in both the free state and as a component of **guanosinediphosphosugars**. It is a metabolic precursor of guanosine 5'-triphosphate, GTP, and is a substrate, a product, or a regulatory agent in many metabolic systems.

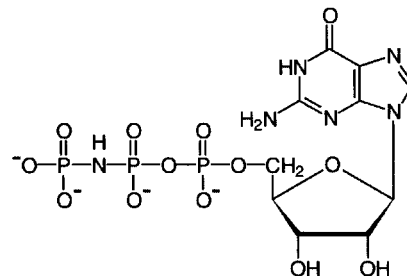
guanosine 3'-diphosphate 5'-diphosphate *symbol:* ppGpp; *see* **relA**, **spoT**.

guanosinediphosphomannose *or* **guanosine(5')diphospho(1)-α-D-mannose** *symbol:* GuoPPMan; *the alternative recommended names for* guanosine diphosphate mannose (*abbr.:* GDPM *or* GDPmannose); guanosine 5'-(α-D-mannopyranosyl diphosphate); α-D-mannopyranosyl 5'-guanylyl phosphate; the most important **guanosinediphosphosugar**. It is synthesized from α-D-mannose 1-phosphate and guanosine 5'-triphosphate by the action of **mannose-1-phosphate guanylyltransferase** (EC 2.7.7.13). It is an intermediate in the synthesis of mannose-containing homo- and heteroglycans in plants. In animal cells GDPmannose is an intermediate in the incorporation of residues of both mannose and fucose into N-linked oligosaccharides of glycoproteins, its conversion to GDP-L-fucose having first taken place in the case of fucose.

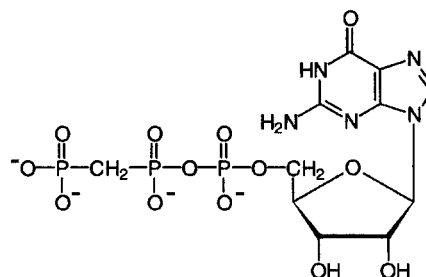
guanosinediphosphosugar *or* **guanosine diphosphate sugar** *abbr.* GDPsugar; any of several **nucleosidediphosphosugars** in which the distal phosphoric residue of guanosine 5'-diphosphate is in glycosidic linkage with a monosaccharide (or a derivative of a monosaccharide). Included are GDP-L-fucose, GDP-D-glucose, GDP-L-guluronic acid, GDP-D-mannose (*see* **guanosinediphosphomannose**), and GDP-D-mannuronic acid.

guanosine 5'-[β,γ-imido]triphosphate *symbol:* Guo-PP[NH]P *or* p[NH]ppG; *the recommended name for* β,γ-imido-

doguanosine 5'-triphosphate (*abbr.:* GTP[β,γ-NH]), 5'-guanylyl imidodiphosphate (*abbr.:* GMP-PNP), 5'-guanylyl iminodiphosphonate, guanosine (5'→O³)-1,2-μ-imidotriphosphate, a synthetic nonhydrolysable analogue of guanosine triphosphate, GTP, in which the oxy group between the latter's intermediate and terminal phosphorus atoms is replaced by an imido group. It stimulates G-proteins and systems activated thereby.



guanosine 5'-[β,γ-methylene]triphosphate *symbol:* GuoPP[CH₂]P *or* p[CH₂]ppG; *the recommended name for* β,γ-methyleneguanosine 5'-triphosphate, (*abbr.:* GTP[β,γ-CH₂]), 5'-guanylyl methylenediphosphonate (*abbr.:* GMP-PCP), guanosine (5'→O³)-1,2-μ-methylenetriphosphate, a synthetic nonhydrolysable analogue of guanosine 5'-triphosphate, GTP, in which the oxy group between the latter's intermediate and terminal phosphorus atoms is replaced by a methylene group. It is an effective activator of ADP-ribosylation factor but not of G_s (*see* **G-protein**).



guanosine monophosphate *abbr.:* GMP; *an alternative name for* any guanosine phosphate, but in particular for **guanosine 5'-phosphate**, especially when its distinction from guanosine (5'-)diphosphate and guanosine (5'-)triphosphate requires emphasis.

guanosine phosphate *symbol:* GuoP; guanosine monophosphate (*abbr.:* GMP); any phosphoric monoester or diester of guanosine. There are three monoesters – guanosine 2'-phosphate, guanosine 3'-phosphate, and **guanosine 5'-phosphate** – and two diesters – guanosine 2',3'-phosphate and **guanosine 3',5'-phosphate** – although guanosine 5'-phosphate, guanylic acid is the ester commonly denoted (the locant being omitted if no ambiguity may arise). Guanosine 2',3'-phosphate (*symbol:* Guo-2',3'-P), also named guanosine 2',3'-(cyclic)phosphate *or* cyclic monophosphate, guanylic acid (*abbr.:* 2',3'-cyclic GMP), is formed as an intermediate during the alkaline hydrolysis of ribonucleic acid. This diester then readily hydrolyses to a mixture of the two monoesters guanosine 2'-phosphate (*symbol:* Guo2'P), also named guanosine 2'-monophosphate (*abbr.:* 2'GMP) *or* 2'-guanylic acid *or* guanylic acid a, and guanosine 3'-phosphate (*symbol:* Guo3'P), also named guanosine 3'-monophosphate (*abbr.:* 3'GMP) *or* 3'-guanylic acid *or* guanylic acid b.

guanosine 2'-phosphate *see* **guanosine phosphate**.

guanosine 2',3'-phosphate

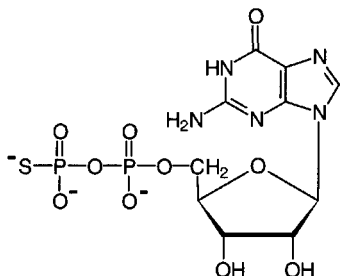
guanosine 2',3'-phosphate *see* **guanosine phosphate**.

guanosine 3'-phosphate *see* **guanosine phosphate**.

guanosine 3',5'-phosphate *symbol:* Guo-3',5'-P; *the recommended name for cyclic guanosine 3',5'-monophosphate (abbr.: 3',5'-cyclic GMP or cyclic GMP or cGMP), guanosine 3',5'-cyclophosphate, guanosine 3',5'-(cyclic) phosphate, a monophosphoric diester of guanosine found in practically all mammalian tissues. It is a cyclic nucleotide similar in structure and the regulation of its metabolism to adenosine 3',5'-phosphate, cyclic AMP, but the levels in tissues are commonly less than 5 percent of those of cyclic AMP. Also like cyclic AMP, it acts as a second messenger in many systems, and it activates a number of distinct cyclic-GMP-dependent protein kinases that share some of their properties with their cyclic AMP counterparts; in some instances it antagonizes the action of cyclic AMP. Cyclic GMP is produced by the action of guanylate cyclase on guanosine 5'-triphosphate, GTP, and it mediates signal transduction for systems that activate this enzyme.*

guanosine 5'-phosphate or **5'-guanylic acid** or **5'-phosphoguanosine** or **5'-O-phosphoguanosine** *symbol:* Guo5'P; *alternative recommended names for guanosine 5'-monophosphate (abbr.: 5'GMP), guanosine 5'-(dihydrogen phosphate), guanine (mono)ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) It is biosynthesized from inosine 5'-phosphate, 5'-inosinic acid, via xanthosine 5'-phosphate and also from free guanine by the salvage pathway.*

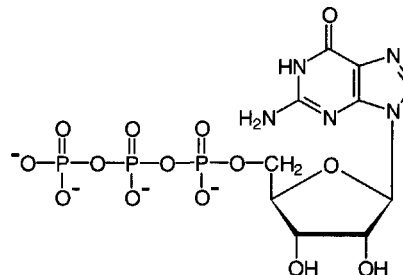
guanosine 5'-β-thiodiphosphate *symbol:* Guo5'PP[S] or [S]ppG; *the recommended name for guanosine 5'-[β-thio]diphosphate (abbr.: GDPβS), guanosine 5'-(2-thiodiphosphate) (abbr.: GDP[S]), guanosine (5'→O²)-1-thiodiphosphate, a synthetic nonhydrolysable analogue of guanosine 5'-diphosphate, GDP, in which an oxygen atom of its terminal phosphoric residue is replaced by a sulfur atom. It is a competitive inhibitor of the activation of G-proteins by guanosine 5'-triphosphate and of the stimulation of adenylate cyclase by guanosine 5'-triphosphate and fluoride.*



guanosine 5'-γ-thiotriphosphate *symbol:* GuoPPP[S] or [S]pppG; *the recommended name for guanosine 5'-[γ-thio] triphosphate (abbr.: GTP-γ-S), guanosine 5'-(3-thiophosphate) (abbr.: GTP[S]), guanosine (5'→O³)-1-thiotriphosphate, a synthetic nonhydrolysable analogue of guanosine 5'-triphosphate, GTP, in which an oxygen atom of its terminal phosphoric residue is replaced by a sulfur atom. It is used in membrane studies to stimulate G-proteins and in studies of DNA synthesis.*

guanosine 5'-triphosphate *symbol:* Guo5'PPP or pppG; *the recommended name for guanosine triphosphate (abbr.: GTP), 5'-triphosphoguanosine, 5'-guanylyl diphosphate, guanosine 5'-(tetrahydrogen triphosphate), an important coenzyme and enzyme regulator present in all cells. It is formed from guanosine 5'-diphosphate by substrate-level phosphorylation or by transfer of the terminal phosphoric residue from adenosine 5'-triphosphate through the action of nucleoside-diphosphate kinase (EC 2.7.4.6). Reactions in which GTP participates are*

often driven in the direction leading to hydrolysis, the free energy change then being utilized for energy-requiring purposes, e.g. microtubule assembly or protein biosynthesis. GTP also participates in various synthetic reactions by the transfer of a phosphoric residue to other metabolites, e.g. to oxaloacetate to form phosphoenolpyruvate in gluconeogenesis.



guanosyl any chemical group formed by the loss of a 2', a 3', or a 5'-hydroxyl group from the ribose moiety of guanosine. *Compare* **guanylyl**.

guanylyl *misnomer for* 1 **guanylyl**. 2 **guanylate**.

guanylate 1 the mono- or the di-anion of **guanylic acid**. 2 any mixture of guanylic acid and its anions. 3 any salt or ester of guanylic acid.

guanylate cyclase EC 4.6.1.2; *systematic name:* GTP pyrophosphate-lyase (cyclizing); *other names:* guanylyl cyclase (not recommended but often used); guanyl cyclase (incorrect). A phosphorus-oxygen lyase enzyme that catalyses the elimination of pyrophosphate from guanosine 5'-triphosphate, GTP, by formation of guanosine 3',5'-phosphate, cyclic GMP. There are two types: (1) a soluble form, which exists as a heterodimer of α and β chains; human examples: α-3, database code CYG5_HUMAN, 717 amino acids (81.28 kDa); β-1, database code CYG1_HUMAN, 619 amino acids (70.43 kDa); and (2) a membrane-bound form, typical of some receptors; e.g. human atrial **natriuretic peptide receptor A** (precursor): database code ANPA_HUMAN, 1061 amino acids (118.78 kDa).

guanyl cyclase *incorrect name for* **guanylate cyclase**.

guanylic acid *the trivial name for* any phosphoric monoester of guanosine. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant. *See* **guanosine phosphate**. However, 5'-guanylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Guanylic acid is also an alternative recommended name for **guanosine 5'-phosphate**.

guanylin a pentadecapeptide hormone from rat jejunum that activates the **guanylate cyclase** component of the intestinal heat-stable **enterotoxin** receptor.

guanylyl the guanosine[mono]phospho group; the acyl group derived from **guanylic acid**.

guanylylation the process of introducing guanylyl groups into a substance.

guanylyl cyclase *see* **guanylate cyclase**.

guaran a mucilaginous **galactomannan** of ≈220 kDa that serves as a storage polysaccharide in the seeds of guar and is the prime constituent of **guar gum**. It consists of long chains of (1→4)-linked β-D-mannopyranose residues, to every alternate one of which a single α-D-galactopyranose residue is attached in (1→6)-linkage.

guard cell either of a pair of crescent-shaped cells that surround a **stoma** in plants.

guard column a small column fitted in front of the main column as a device to prolong the life of columns in high-performance liquid chromatography by removing from the sample any particulate matter, aggressive agents, or components that bind irreversibly to the column packing.

guar gum a greyish-white powder consisting predominantly of



the polysaccharide **guaran** and obtained by grinding the endosperm of the seeds of guar, an Indian leguminous plant, *Cyamopsis tetragonoloba* (*psoralioides*), also called cluster bean. It hydrates and disperses readily in cold water to give colloidal solutions of very high viscosity that are stable over a very wide pH range and have very good suspending and lubricating properties. It is used for sizing paper and textiles, and as a stabilizing, suspending, and thickening agent in the food and pharmaceutical industries, etc. It has also been suggested as a nonabsorbable dietary additive to improve glycemic control in diabetes mellitus, possibly by slowing the uptake of ingested absorbable carbohydrate.

GUC a codon in mRNA for L-valine.

GUG a codon in mRNA for L-valine.

Guggenheim plot a procedure that makes it possible to obtain rate constants for first-order reactions when neither the initial nor the final reactant concentrations are known. Measurements of the concentration of the reactant (or of some physical quantity that measures it), $c_1, c_2, c_3, \dots, c_n$, are made at times $t_1, t_2, t_3, \dots, t_n$ and also at times $(t_1 + \delta t), (t_2 + \delta t), (t_3 + \delta t), \dots, (t_n + \delta t)$, when the concentrations of the reactant are $c_1', c_2', c_3', \dots, c_n'$, where δt is a constant time difference, then

$$\ln(c_i - c_i') = \ln(c_0 - c_\infty)(1 - e^{-k\delta t}) - kt,$$

or

$$\ln(c_i - c_i') = \text{a constant} - kt_i,$$

hence a plot of $\ln(c_i - c_i')$ versus t_i has a slope of $-k$. [After Edward Armand Guggenheim (1901–70), British physical chemist.]

guide RNA *see* gRNA.

Guillemin, Roger Charles Louis (1924–), French-born US physician and endocrinologist; Nobel Laureate in Physiology or Medicine (1977) jointly with A. V. Schally 'for their discoveries concerning the peptide hormone production of the brain' [prize shared with R. S. Yalow].

guinea pig any of several species of wild or domesticated hystriocomorph rodents, or caviae, in the genus *Cavia*, especially *Cavia porcellus*, which is widely used in biomedical research and testing.

Guinier plot a method of plotting the results of small-angle X-ray scattering to obtain the radius of gyration of a molecule, r_G , in the absence of information about the relative molecular mass of the component in question. It can be shown that

$$\ln(I_\theta/I_0) = -(4\pi r_G \sin\theta/\lambda)^2/3,$$

where I_θ and I_0 are the intensities of the scattered radiation at angles θ and 0 respectively, and λ is the wavelength of the radiation; hence a plot of $\ln(I_\theta/I_0)$ versus $-(4\pi \sin\theta/\lambda)^2/3$ will yield r_G .

Gul *symbol for a residue (or sometimes a molecule) of the aldohexose gulose.*

GulA *symbol for a residue (or sometimes a molecule) of the uronic acid guluronic acid.*

gulo- *prefix (in chemical nomenclature) indicating a particular configuration of a set of (usually) four contiguous >CHOH groups, as in the acyclic form of D- or L-gulose. See monosaccharide.*

L-gulonolactone oxidase EC 1.1.3.8; *other name:* L-gulonolactone oxidase; a flavoprotein enzyme that catalyses the oxidation by dioxygen of L-gulonolactone to L-xylohexulonolactone (I) and H_2O_2 . I isomerizes spontaneously to L-ascorbate. The flavin is FAD. The enzyme is essential for the formation of ascorbate, and a deficiency of the enzyme, as in humans, results in dependency on a dietary supply of ascorbate (*see ascorbic acid*). Example from rat: GGLO_RAT, 439 amino acids (50.42 kDa).

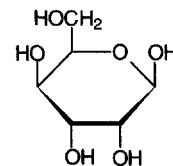
gulose *symbol:* Gul; *the trivial name for the aldohexose gulohexose; a stereoisomer of glucose in which the configurations at both C-3 and C-4 are inverted. There are two enantiomers. (Illustrated opposite.)*

guluronic acid *symbol:* GulA; *the uronic acid resulting from the oxidation of the hydroxymethylene group at C-6 of gulose to a carboxyl group. L-Guluronic acid is a component of alginic acid.*

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gulose (β -D-pyranose form)

gum any of numerous water-soluble or water-dispersible heteroglycans, frequently containing uronic acids, that are produced as exudates from certain plants, especially trees, following physical injury. Gums are tacky, slimy substances that harden when dry. *See also gum arabic, gum ghatti.*

gum arabic a gum exuded from trees of the genus *Acacia*. Structurally it is a heteroglycan containing D-galactose, L-arabinose, L-rhamnose, and D-glucuronic acid. It is used in industry, e.g. in the manufacture of thickening agents, emulsifiers, inks, and pills. It is also used in glues and pastes.

gum ghatti a gum from the tree *Anogeissus latifolia*. It has a complex structure and contains as major components D-mannose, D-galactose, and L-arabinose, the latter in both furanose and pyranose arrangement. Small amounts of D-xylose and D-glucuronic acid are also present.

Gunn rat any member of a mutant strain of Wistar rats with hereditary hyperbilirubinemia. The defect affects the rate of glucuronidation of a number of substrates, but not equally. [After C. H. Gunn.]

Güntelberg-Müller charging process a method of calculating the electrical energy of an ion. According to the process, only the ion under consideration is supposed to receive a charge; the surrounding ions are assumed to be already charged, and the only change that they undergo during the process is a gradual redistribution as the central ion receives its charge.

Guo *symbol for a residue of the ribonucleoside guanosine (alternative to G).*

GuoP *symbol for guanosine phosphate.*

Guo2'P *symbol for guanosine 2'-phosphate; see guanosine phosphate.*

Guo-2',3'-P *symbol for guanosine 2',3'-phosphate; see guanosine phosphate.*

Guo3'P *symbol for guanosine 3'-phosphate; see guanosine phosphate.*

Guo-3',5'-P *symbol for guanosine 3',5'-phosphate.*

Guo5'P *symbol for guanosine 5'-phosphate.*

Guo5'PP *symbol for guanosine diphosphate (alternative to ppG).*

GuoPP[CH₂]P *symbol for guanosine 5'-[β , γ -methylene] triphosphate (alternative to p[CH₂]ppG).*

GuoPPMan *symbol for guanosinediphosphomannose.*

GuoPP[NH]P *symbol for guanosine 5'-[β , γ -imido]triphosphate (alternative to p[NH]ppG).*

Guo5'PPP *symbol for guanosine 5'-triphosphate (alternative to pppG).*

Guo5'PPP[S] *symbol for guanosine 5'- γ -thiotriphosphate.*

Guo5'PP[S] *symbol for guanosine 5'- β -thiodiphosphate.*

gustducin a taste-cell-specific G-protein α subunit closely related to the transducins and comprising 354 amino acids.

gut a term variously used to describe the whole of the alimentary canal or only the intestine, i.e. the part from the stomach to the anus.

gut glucagon (*sometimes*) an alternative term for enteroglucagon.

gut hormone a general, inaccurate, but widely used term for all the various regulatory peptides of the gut. They are neither confined to the gut, nor are they exclusively hormones.

Guthrie test a microbiological test in which a strain of *Bacillus*

**gutta-percha**

subtilis that requires phenylalanine for growth is used to test for excessive levels of plasma phenylalanine in conditions such as **phenylketonuria**. [After R. Guthrie (1916–).]

gutta-percha a *trans*-1,4-polyisoprene derived by coagulation of the latex of various Malayan trees of the genus *Palaquium*. It has a lower molecular mass than rubber and is thermoplastic.

GUU a codon in mRNA for L-valine.

g-value a factor expressing the size of the magnetic moment of a paramagnetic chemical species. It specifies the position at

which the absorption in electron spin resonance of the paramagnetic species occurs. See **electron spin resonance spectroscopy**.

Gy symbol for gray.

gynaminic acid formerly, a name for a constituent of human milk, now known to be identical with **sialic acid**.

gynogamone see **gamone**.

gyrase see type II **DNA topoisomerase**.

gyromagnetic ratio or **magnetogyric ratio** symbol: γ ; the ratio of the magnetic moment, μ , of a given atomic particle to its angular momentum, L ; i.e. $\gamma = \mu/L$. See also **proton magnetogyric ratio**.

Hh

h symbol for 1 hecto+ (SI prefix denoting 10^2 times). 2 hour (supersedes hr.).

h symbol for 1 Planck constant. 2 Hill coefficient (alternative to n_H). 3 height.

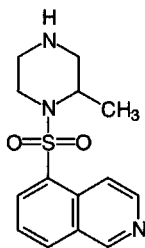
h+ or (not recommended) **H+** prefix to abbreviations for peptide hormone (names) denoting human, i.e. of (or as if of) human origin; e.g. hGH (for human growth hormone).

H symbol for 1 hydrogen; see also **protium**, **deuterium**, **tritium**. 2 a residue of the α -amino acid **histidine** (alternative to His). 3 a residue of an incompletely specified base in a nucleic-acid sequence that may be adenine, cytosine, or either thymine (in DNA) or uracil (in RNA). 4 henry. 5 histone. 6 histamine (see **histamine receptor**).

H+ see **h+**.

H1 a DNA-binding protein from *Escherichia coli* belonging to the **H-NS** family of proteins.

H7 1-(5-isoquinolinesulfonyl)-2-methylpiperazine; an inhibitor of protein kinase C ($K_i = 6.0 \mu\text{M}$), protein kinase A ($K_i = 3.0 \mu\text{M}$), and cyclic GMP-dependent protein kinase ($K_i = 5.8 \mu\text{M}$).



H₂ symbol for 1 dihydrogen. 2 the prefix **dihydro+** in a trivial chemical name, e.g. **H₂folate** for dihydrofolate.

H₄ symbol for the prefix **tetrahydro+** in a trivial chemical name, e.g. **H₄folate** for tetrahydrofolate.

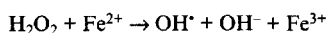
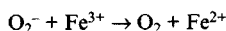
H symbol for 1 fluence (alternative to *F*). 2 enthalpy. 3 magnetic field strength (light italic in nonvector equations; bold italic in vector equations).

-H conformational descriptor designating the half-chair conformation of a six-membered ring form of a monosaccharide or monosaccharide derivative. Locants of ring atoms that lie on the side of the structure's reference plane from which the numbering appears clockwise are indicated by left superscripts and those that lie on the other side of the reference plane by right subscripts; e.g. methyl 2,3-anhydro-5-thio- β -L-lyxopyranoside-5-*H_s*. See also **conformation**.

H- or (sometimes) **H** (in chemical nomenclature) symbol for indicated hydrogen (at the position specified by a prefixed locant).

Haber, Fritz (1868–1934), German chemist; Nobel Laureate in Chemistry (1919) 'for the synthesis of ammonia from its elements'.

Haber-Weiss reaction the iron-catalysed reaction of a superoxide anion with hydrogen peroxide to form oxygen, a hydroxyl radical, and a hydroxyl ion. It is thought to proceed as follows:



[After F. Haber and Joseph Weiss.]

HAc abbr. for acetic acid.

HACBP see **calreticulin**.

haem a variant spellings (esp. Brit.) of **heme**.

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haema+ or (before a vowel) **haem+** variant spellings (esp. Brit.) of **hema+**.

haemadin an anticoagulant peptide from the Indian leech, *Haemadipsa sylvestrus*. It is a slow tight-binding inhibitor of thrombin, but does not inhibit trypsin, chymotrypsin, factor Xa, or plasmin. It comprises 57 amino acids.

haemato+ or (before a vowel) **haemat+** variant spellings of (esp. Brit.) of **hemato+**.

haemo+ a variant spelling (esp. Brit.) of **hemo+**.

haemoglobin a variant spelling (esp. Brit.) of **hemoglobin**.

Hageman factor an alternative name for factor XII; see **blood coagulation**.

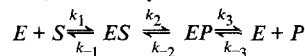
hairpin any part of a linear molecular structure, e.g. of a polynucleotide strand or a prostaglandin, in which two adjacent segments of the molecule are folded back one on the other and held in that conformation by secondary molecular forces such as hydrogen bonds or van der Waals interactions. **Hairpin DNA** (or foldback DNA or loopback DNA) contains **inverted repeats**; when denatured it renatures extremely rapidly by concentration-independent intrachain base-pairing between complementary sequences of the inverted repeat. Similar structures occur in certain large RNA molecules. The hairpin conformation of a prostaglandin molecule is thought to be necessary for expression of its hormonal activity; in this conformation the α and κ chains of each molecule are extended, parallel to one another, at van der Waals contact distance for their full lengths. See also **ribozyme**, **beta bend**.

Haldane coefficient the difference, per bound dioxygen molecule, in the number of hydrons bound by oxy- and deoxy-hemoglobin, at constant pH. [After John Scott Haldane (1860–1936), British physiologist.]

Haldane effect the observation that oxygenated blood absorbs less carbon dioxide than deoxygenated blood. It is the reciprocal of the **Bohr effect**.

Haldane gas-analysis apparatus an apparatus once used for the estimation of the carbon dioxide and dioxygen contents of a gas sample.

Haldane relation a relation for reversible enzyme reactions. In the single-substrate, essentially reversible reaction:



the overall equilibrium constant,

$$K = C_P/C_S = k_1 k_2 k_3 / k_{-1} k_{-2} k_{-3},$$

where C_S and C_P are the concentrations of substrate, S, and product, P, respectively. It can further be shown that

$$K = V_S K_{MP} / V_P K_{MS},$$

where V_S and V_P are the maximum velocities of the reactions $S \rightarrow P$ in the absence of P and of $P \rightarrow S$ in the absence of S, and K_{MS} and K_{MP} are the respective Michaelis constants for these reactions. [After John Burdon Sanderson Haldane (1892–1964), British biochemist, physiologist, and geneticist]

half-boat conformation either of the **conformations** of a nonplanar monounsaturated six-membered ring compound when the two ring atoms not directly bound to the doubly bonded atoms lie on the same side of the plane containing the other four (adjacent) atoms.

half-cell one half of an **electric cell**, consisting of one reversible electrode inserted into a solution. Electrical connection between this solution and that in another half-cell is required to complete the electric circuit.

half-chair conformation 1 either of the **conformations** of a nonplanar monounsaturated six-membered ring compound when the two ring atoms not directly bound to the doubly bonded atoms lie on opposite sides of the plane containing the



half-cystine

other four (adjacent) atoms. **2** any conformation of the six-membered ring form of a monosaccharide or monosaccharide derivative when four adjacent ring atoms are coplanar and the other two lie on opposite sides of the plane. The conformational descriptor *-H* may be added to the name of such compounds.

half-cystine or **hemicystine** *symbol:* Cys or Cys^s ; the group formally derived by removal of a hydrogen atom from the thiol group on the side chain in cysteine (or a cysteine residue), or by homolysis of the dithio linkage in cystine (or a cystine residue). It is used in describing or depicting the content or the linear sequence of amino-acid residues in a polypeptide when it contains more than two apparent cysteine residues and the positions of any dithio linkages between them are not known, or when only a coded sequence is being considered.

half-cystyl or **hemicystyl** the acyl group, $\cdot S-CH_2-CH(NH_2)-CO-$, derived from the **half-cystine** group. *Compare* **cysteinyl**.

half-ester any monoester of a dibasic acid.

half-life **1** *symbol:* $t_{1/2}$ or $T_{1/2}$; the time for one half of the atoms of an amount of radionuclide to undergo radioactive decay. **2** a similar measure of the stability (i.e. rate of decay) of an excited atom or molecule, a radical, an unstable elementary particle, etc. **3** the time for one half of the amount of an administered substance to be metabolized or excreted. If the substance is radioactive, the time required for one half of the dose to be eliminated biochemically or physiologically is termed the **biological half-life** and that required for one half to disappear by radioactive decay as well as by the elimination is the **effective half-life**. **4** or (sometimes) **half-time** *symbol:* $t_{1/2}$; the time required for the concentration of a reactant in a chemical reaction to reach a value that is the arithmetic mean of its initial and final (equilibrium) values. **5** the time for one half of the number of cells in a tissue or organ to be replaced by new cells.

half-of-the-sites reactivity or **half-site reactivity** a phenomenon shown by many enzymes that exhibit **negative cooperativity**, in which the maximal stoichiometric yield of either an enzyme-substrate intermediate or a product in a single turnover amounts to only half the number of apparently equivalent active sites. This may be caused by the binding of substrate to one site of a dimeric enzyme, e.g., inducing a conformational change in the enzyme so as to abolish completely the binding affinity of the other site.

half-reaction either of the two coupled chemical changes that together constitute an oxidation-reduction reaction. In one half-reaction there is a gain of electrons and in the other half-reaction a corresponding loss of electrons.

half-site reactivity an alternative name for **half-of-the-sites reactivity**.

half-thickness the thickness of a piece of a specified material that will reduce the intensity of a beam of transmitted electromagnetic radiation to one half that of the incident beam.

half-time an alternative name for **half-life** (def. 4) for a first-order process.

half-time of exchange the time taken for one half of the exchangeable atoms (or molecules) to be exchanged in a reaction involving such an exchange.

half-wave potential the electrical potential at the mid-point of the current-voltage curve in **polarography**.

halide or **halogenide** **1** an anion of any halogen. **2** a salt of any halogen acid (halogen hydride). **3** any compound containing halogen atoms in organic linkage.

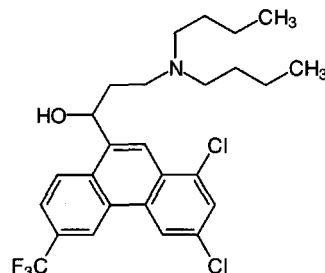
hallucinogen any drug or other substance that causes hallucinations. —**hallucinogenic** *adj.*

halo- or (sometimes before a vowel) **hal+** *comb. form* **1** relating to or containing a halogen. **2** relating to the sea or salt.

halochromism the phenomenon or property of the formation of strongly coloured compounds on the addition of strong acids or certain metallic salts to colourless or faintly coloured substances.

halofantrine 1,3-dichloro- α -[2-(dibutylamino)-ethyl]-6-

(trifluoromethyl)-9-phenanthrenemethanol; a derivative of 9-phenanthrenemethanol that is effective against chloroquine-resistant malarial infections.

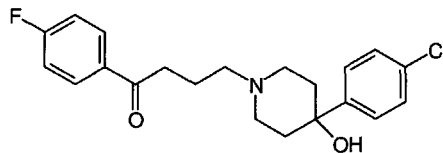


halogen any of the monovalent chemical elements of group 17 of the IUPAC **periodic table**: fluorine, chlorine, bromine, iodine, and astatine.

halogenate to treat or react (a substance) with a halogen; to introduce one or more halo groups into an organic compound. —**halogenation** *n.*

halogenide an alternative name for **halide**.

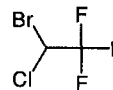
haloperidol 4-[4-(4-chlorophenyl)-4-hydroxy-1-piperidinyl]-1-(4-fluorophenyl)-1-butanone; one of the butyrophenone group of **neuroleptic** drugs.



halophile **1** an organism that grows in or tolerates saline conditions, especially concentrations of sodium chloride equal to or greater than that in sea water. **2** describing such an organism. —**halophilic** *adj.*

halorhodopsin an energy-transducing pigment identified in membranes of *Halobacterium halobium*. It is similar to, but distinct from **bacteriorhodopsin** and is associated with a light-driven pump for sodium ions.

halothane the trivial name for 2-bromo-2-chloro-1,1,1-trifluoroethane, a widely used, volatile, lipid-soluble general inhalational anesthetic agent.



hamamelose 2-C-(hydroxymethyl)-D-ribose; a substance found principally in **tannin** of witch hazel (*Hamamelis virginiana*).

Hamburger shift an alternative name for **chloride shift**.

Hamiltonian operator *symbol:* H ; a function used to express the total energy of a system in joules: $H = T + V$, where T is the kinetic energy operator and V the potential energy operator. For a particle of mass m and momentum p ,

$$H = p^2/2m + V.$$

[After William Rowan Hamilton (1805–65), Irish astronomer and mathematician.]

hammerhead *see* **ribozyme**.

handedness a form of terminology (based on the human left and right hands) for describing objects and compounds that

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handle

are mirror images of each other and hence nonsuperimposable. A pair of **enantiomers** display molecular handedness. *See* **chirality**.

handle a colloquial term for any chemical group attached to a specified substance with the purpose of making it more easily identified, more reactive, or more easily operated upon, or one that may serve in targeting or specific binding. The attachment of a handle may be brought about chemically or metabolically.

handle technique a technique used to separate excess alkylating agents from products in peptide synthesis. For instance, the C terminus of a growing peptide chain can be protected with a basic function, e.g. by forming a 4-picolyl ester of the carboxyl group, thereby allowing separation of the products on cation-exchange resin columns or by extraction into aqueous acidic solutions.

Hanes plot a graphical method for treating enzyme kinetic data using the following form of the Michaelis equation (*see* **Michaelis kinetics**):

$$s/v = K_m/V + s/V$$

where s is the substrate concentration, v is the velocity of the reaction, and V is the limiting rate. If s/v is plotted against s , the slope of the resulting straight line is $1/V$, the intercept on the s/v axis is K_m/V , and the intercept on the s axis is $-K_m$. [After Charles Samuel Hanes (1903–90), British biochemist.]

Hanks' balanced salt solution *abbr.*: HBSS; a balanced salt solution used in tissue culture to provide a suitable ionic and osmotic environment for cell growth and development. It contains, per 100 mL of distilled water, 0.8 g NaCl, 0.04 g KCl, 0.01855 g $\text{CaCl}_2 \cdot \text{H}_2\text{O}$, 0.02 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.006 g KH_2PO_4 , 0.00475 g Na_2HPO_4 , 0.0350 g NaHCO_3 , 0.1 g glucose, and 0.0017 g sodium Phenol Red. The pH is 7.0–7.2.

Hansch equation an equation relating the differences in the partitions of variously substituted compounds between an organic and an aqueous phase to the **hydrophobicity constant**, π , of the substituent. If P_0 is the ratio of the solubility of the parent compound in an organic phase to that in the aqueous phase, and P is the corresponding ratio for the substituted compound, then: $\pi = \log(P/P_0)$.

Hansenula *see* **methylotrophic yeasts**.

H antigen *see* **blood-group substance**.

haploid describing a cell, an organism or a nucleus of a cell having a single genome or a single set of homologous chromosomes; i.e. containing half the **diploid** number; having a **ploidy** of one. —**haploidy** *n*.

haplotype a set of genes located on a single chromosome; the term is used also to denote the characteristics dependent on those genes. In outbred populations the maternal and paternal chromosomes usually differ, so an individual has two haplotypes, one derived from each parent.

'happiness hormone' a jocular name for any **endorphin** or **enkephalin**, so-named because of the supposed ability of such substances to promote a sense of well-being.

hapt+ a variant form of **hapto+** (before a vowel).

haptén any, usually small, substance that can combine specifically with antibody but that is not immunogenic unless bound to a **carrier** (def. 8). Most haptens carry only one or two antigenic determinants. —**hapténic** *adj*.

haptén-inhibition test a serological test for characterizing an antigenic determinant. It involves the use of haptens of known structure to block the combining site of a particular antibody.

haptén-sandwich labelling a variant of the **sandwich technique** for amplification of an antigen-antibody reaction. In this modification, the primary antibody, specific for the antigen of interest, is combined with a number of residues of a hapten, e.g. *p*-azobenzene arsonate. The secondary (labelled or marked) antibody is a specific anti-hapten antibody; on reaction of secondary antibody molecules with a molecule of primary antibody, the hapten residues become sandwiched between the large molecules of antibody. By the use of different

haptens and two or more non-cross-reactive anti-hapten antibodies bearing distinguishable markers, two or more antigens on the same or different cells may simultaneously be labelled.

hapticity *symbol*: η ; the attribute of a ligand entity or group that combines with a central (usually metal) atom to form a coordination entity in which the central atom is bonded to two or more contiguous atoms of the ligand rather than to a specific atom. Such a coordination entity is frequently (but not exclusively) a **pi adduct**. *See also* **hapto**.

hapto or (less commonly) **eta** (in chemical nomenclature) the name by which to read the symbol η - (Greek eta), when prefixed to the name of the ligand group in a coordination entity (with added locants if appropriate) to indicate hapticity.

hapto+ or (before a vowel) **hapt+** *comb. form* indicating ability to bind or combine.

haptocorrin a blood cobalamin-binding glycoprotein that may prevent uptake of bacterial cobalamin analogues. Example from *Sus scrofa* (fragment): database code HAPC_PIG, 416 amino acids (46.91 kDa).

haptoglobin *abbr.*: Hp; any α_2 globulin of blood plasma that can combine with free oxyhemoglobin to form a stable, ≈ 300 kDa, complex. Such combinations occur when hemoglobin is liberated in the blood, and the complexes so formed, which have weak peroxidase activity and cannot be filtered by the kidney, are rapidly removed and degraded in the tissues. Three human haptoglobin phenotypes are known, designated Hp 1-1, Hp 2-2, and Hp 2-1, each of which contains two pairs of non-identical polypeptide chains, α and β , joined by disulfide bonds. There are two kinds of α chain, α_1 of 84 amino-acid residues (9 kDa) and α_2 of 143 residues (17 kDa), and a single kind of β chain, of 244 residues (in human). The α_2 chain, found solely in human haptoglobin, is identical in sequence to that of residues 1–71 or 1–72 of an α_1 chain linked to residues 11 or 12 to 84. The basic subunit structure of Hp 1-1 is $(\alpha_1\beta)_2$, that of Hp 2-2, $(\alpha_2\beta)_2$, and that of Hp 2-1, $(\alpha_2\beta)(\alpha_1\beta)$. However, the α_2 chain has an odd number of half-cystine residues, as a consequence of which larger oligomers of Hp 2-2 and Hp 2-1 may arise by formation of disulfide bonds with α_2 chains of other molecules or of $(\alpha_2\beta)$ half-molecules. Example (precursor) from human: database code HPT1_HUMAN, 347 amino acids (38.45 kDa).

haptomer any substance (such as a lectin or the B fragment of diphtheria toxin) that can interact with the cell membrane and that binds or can be bound to an **effectomer** (such as gelonin or the A fragment of diphtheria toxin), thereby allowing the effectomer to exert its toxic action on intact cells. —**haptomeric** *adj*.

haptophore a term introduced by Ehrlich to denote the specific chemical group in a toxin (i.e. antigen) molecule that attaches it to an antitoxin (i.e. antibody) molecule. Unlike the **toxophore**, it is not destroyed on conversion of the toxin to a toxoid.

H⁺(aq) *symbol* for the hydrated hydrogen ion, the normal state of the hydrogen ion in aqueous solutions, consisting of a hydron surrounded by a shell of water molecules held by ion-dipole interaction. *Compare* **oxonium** (def. 1).

hard (in radiation physics) describing corpuscular or electromagnetic ionizing radiation (especially beta particles or X- or gamma-radiation) having relatively high energy, hence high ionizing power and long-range or high penetrating power. Hard X- or gamma-radiation has short wavelengths. *Compare* **soft** (def. 1).

Harden, (Sir) Arthur (1865–1940), British chemist and biochemist distinguished for his discoveries (with W. J. Young) that fermentation of sugar by yeast juice involved a heat-stable and separable fraction (later named cozymase by H. von Euler), required the presence of phosphate, and (also with R. Robison) resulted in the intermediate formation of two hexose phosphates (*see* **Harden-Young ester**, **Robison ester**); Nobel Laureate in Chemistry (1929), the prize being shared with H. von Euler 'for their investigations on the fermentation of sugar and fermentative enzymes'.

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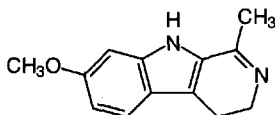
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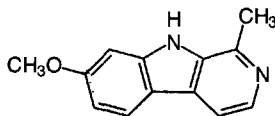
Harden-Young ester a former name for D-fructose 1,6-bisphosphate (before its chemical structure had been elucidated). [After A. Harden and William John Young (1878–1942), British biochemists.]

Hardy-Weinberg law a mathematical statement about the relation between gene frequencies and genotype frequencies within a population. If the genotypes are AA, Aa, and aa, and p is the frequency of allele A in the population, and q the frequency of allele a, so that $p + q = 1$, then the relative proportions of the three genotypes in the population, AA:Aa:aa, is given by $p^2:2pq:q^2$. [After Godfrey Harold Hardy (1877–1947), British mathematician, and Wilhelm Weinberg (1862–1937), German physician.]

harmaline 4,9-dihydro-7-methoxy-1-methyl-3H-pyrido[3,4-b]indole; dihydroharmine; a cardioactive alkaloid from Syrian rue, *Peganum harmala* that also has hallucinogenic properties. See also **harmine**.



harmine 7-methoxy-1-methyl-9H-pyrido[3,4-b]indole; a β -carboline alkaloid found in, e.g., *Peganum harmala* (Syrian rue). It inhibits monoamine oxidase and is a central nervous system stimulant. See also **harmaline**.



HART abbr. for hybrid-arrested translation.

Hartley filter funnel a development of the Büchner funnel comprising three pieces, the reservoir, a perforated plate to support the filter disk, and a conical piece to collect the filtrate. [After (Sir) Percival Hartley (1881–1957), British bacteriologist.]

Hartline, Haldan Keffer (1903–83), US neurophysiologist; Nobel Laureate in Physiology or Medicine (1967) jointly with R. Granit and G. Wald 'for their discoveries concerning the primary physiological and chemical visual processes in the eye'.

Hartnup disease a rare, autosomal recessively inherited condition of humans in which there is a specific hyperaminoaciduria and indoluria, due to a diminished capacity for renal reabsorption of a group of monoamino-monocarboxylic acids, especially tryptophan. There may also be an impaired transport of the same group of amino acids across the gut wall. The aminoaciduria is often associated with a pellagra-like skin rash, due to inefficient tryptophan utilization and metabolism. The pellagra may often be resolved by oral nicotinic acid. The primary defect remains unknown. [After E. Hartnup, the first patient with the condition to be described.]

hashish a resinous extract obtained from the dried flower tops of the hemp plant (*Cannabis sativa*). The term may sometimes be used for various other extracts. See **cannabis**.

Hassel, Odd (1897–1981), Norwegian physical chemist; Nobel Laureate in Chemistry (1969), the prize being shared with D. H. R. Barton 'for their contributions to the development of the concept of conformation and its application in chemistry'.

HAT abbr. for hypoxanthine-aminopterin-thymidine (medium); a selective medium used in tissue culture. It is of particular use in the production of **hybridomas**. The medium was originally devised to select against cells deficient in hypoxanthine-guanine phosphoribosyltransferase (HGPRT⁺ cells).

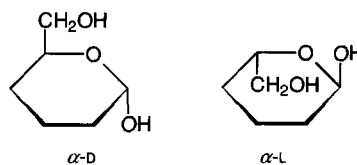
Amethopterin blocks *de novo* synthesis of purines and HGPRT⁺ cells are unable to utilize the salvage pathway, and hence are killed.

hatching enzyme see **envelysin**.

Hatch-Slack pathway or **C₄ cycle** or **C₄ pathway** a metabolic pathway for carbon-dioxide translocation occurring in **C₄ plants**. In the mesophyll cells **phosphoenolpyruvate carboxylase** catalyses the condensation of CO₂ with phosphoenolpyruvate to form oxaloacetate; this is then reduced to malate by light-generated NADPH. The malate passes into the bundle-sheath cells where it is oxidatively decarboxylated by the **malic enzyme** to pyruvate and CO₂; the pyruvate returns to the mesophyll cells and is reconverted to phosphoenolpyruvate by ATP and pyruvate dikinase; the CO₂ remains in the bundle-sheath cells where it enters the **reductive pentose phosphate cycle**. The overall effect of the pathway is to carry CO₂ from the mesophyll cells, which are in contact with air, to the bundle-sheath cells, which are the major sites of photosynthesis.

Haworth, (Sir) Walter Norman (1883–1950), British organic chemist noted for his structural studies on monosaccharides (see also **Haworth representation**) and polysaccharides and for producing the first chemical synthesis of ascorbic acid, vitamin C; Nobel Laureate in Chemistry (1937) 'for his investigations on carbohydrates and vitamin C' [prize shared with P. Karrer].

Haworth representation a diagrammatic way of unambiguously representing the structural formulae of the cyclic forms of monosaccharides to show on a plane surface the relative configurational arrangements of the atoms and groups in space. Pyranoses are represented by a hexagon (five carbons and one oxygen) and furanoses by a pentagon (four carbons and one oxygen). A Haworth representation is derived from a **Fischer projection** as follows. The monosaccharide in question is depicted with the carbon-chain horizontal and in the plane of the paper, the potential carbonyl group being to the right. The oxygen bridge is then depicted as being formed behind the plane of the paper. The heterocyclic ring is therefore located in a plane approximately perpendicular to the plane of the paper and the groups attached to the carbon atoms of that ring are above and below the ring. The edge of the ring nearer the viewer is indicated by thickened lines. For clarity, the carbon atoms of the ring and their attached hydrogen atoms are not shown. Groups that, in the Fischer projection, appear to the right of the vertical chain then appear in the Haworth representation below the plane of the ring. However, at the asymmetric carbon atom involved via oxygen in ring formation



(e.g. C-5 in glucopyranose, C-4 in glucofuranose), a rotation about the bond to the preceding carbon atom must be envisaged in order to bring that oxygen atom into the correct orientation to link to the carbon atom of the carbonyl group. Thus, if the hydroxyl group that is engaged in ring formation lies to the right in the Fischer projection, the group that lies below it must point upwards in the Haworth representation, whereas if it lies to the left, it must point down in the Haworth representation. In representing the α and β anomers in the Haworth convention, it should be noted that for the α anomer, the anomeric hydroxyl points in the opposite direction to the R group (see structure); for D-sugars, the R group points upwards, so the α -hydroxyl must point down; conversely, if the R group points down (as in α -L-glucose), the α anomeric hydroxyl must point upwards. The above rules only obtain if the



ring is numbered clockwise from the anomeric carbon, as is frequently the case, and it is important to appreciate that Haworth representations, unlike Fischer projections, cannot be rotated in the plane of the paper but must be reoriented in space.

Hb *abbr. and symbol for hemoglobin*; hence, e.g., HbO_2 symbolizes oxyhemoglobin, and HbCO symbolizes carbonmonoxy-hemoglobin.

Hb A *abbr. for adult hemoglobin.*

H band *an alternative name for H zone (see sarcomere).*

HBD (*in clinical chemistry*) *abbr. for α -hydroxybutyrate dehydrogenase*, the LD_1 isoenzyme of **lactate dehydrogenase**, or 'heart-specific' lactate dehydrogenase, which is assayed using 2-hydroxybutanoate as substrate, a substance that does not occur naturally.

Hb F *abbr. for fetal hemoglobin.*

Hb S *abbr. for sickle-cell hemoglobin; see sickle cell.*

HBSS *abbr. for Hanks' balanced salt solution.*

hCG or **HCG** *abbr. for human chorionadotropin (see chorionadotropin).*

H chain *see heavy chain.*

HCO *symbol for the formyl group (alternative to For).*

HCR *abbr. for host-cell reactivation.*

Hcy *symbol for a residue of the α -amino acid L-homocysteine.*

HDEL a four-residue motif, His-Asp-Glu-Leu, of similar function to the **KDEL** motif.

HDH *abbr. for histidinol dehydrogenase.*

HDL *abbr. for high-density lipoprotein.*

HDP *abbr. for helix-destabilizing protein.*

He *symbol for helium.*

head 1 the foremost or uppermost part of the body of an animal; in vertebrates it comprises the skull containing the mouth, brain, and the organs of hearing, sight, smell, and taste. 2 the part of a molecular structure that bears a specific functional group or that is polar, reactive, bulky, or in some other characteristic way differs from the remainder of the molecule. 3 a common name for a centrifuge rotor.

headgroup of a polar lipid, that part of the molecule that expresses its polar character. For a phospholipid, this comprises the phosphate group together with any polar entity attached to it. For a glycolipid, it comprises the carbohydrate moiety.

headspace analysis a technique used in the identification of microorganisms involving gas-chromatographic analysis of the vapour in the headspace above the specimen or culture.

heat *symbol: q or Q* ; the energy possessed by a system in the form of kinetic energy of atomic or molecular translation, rotation, or vibration. It is measured in joules or calories. Heat energy can be transferred from points of higher temperature to points of lower temperature by conduction, convection, or radiation. When the temperature of a system changes, its enthalpy (heat content) changes by an amount equal to the product of its mass, specific heat capacity, and its change in temperature.

heat capacity *symbol: C* ; the amount of heat required to raise the temperature of a body or a system by 1 kelvin; it is usually measured in joules per kelvin. The **heat capacity at constant pressure**, C_p , is given by: $C_p = (\partial H / \partial T)_p$ J K⁻¹ and the **heat capacity at constant volume**, C_v , is given by: $C_v = (\partial U / \partial T)$, J K⁻¹ where H is the enthalpy, U the internal energy, and T the thermodynamic temperature.

heat content *another name for enthalpy.*

heat-labile describing molecules that decompose at elevated temperatures or that lose biological activity when heated to a moderate temperature, e.g. 50 °C.

heat of activation the difference in heat content between a transition state and the state of its reactants.

heat of combustion the amount of heat evolved when a given amount, usually one mole, of a substance is completely burnt in dioxygen.

heat of evaporation *an alternative name for heat of vaporization.*

heat of formation the amount of heat absorbed or evolved when one mole of a substance is formed from its elements.

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Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

heat of fusion or **heat of melting** the amount of heat absorbed or evolved when a specified quantity of substance, usually one mole, is converted from the solid to the liquid state at a specified pressure and a specified temperature. If the pressure is 1 atm the temperature is the melting point of the substance.

heat of neutralization the amount of heat absorbed or evolved when one mole of an acid or base is completely neutralized.

heat of reaction the amount of heat absorbed or evolved in the course of a (bio)chemical reaction, usually expressed per mole of reacting substances.

heat of solution the amount of heat absorbed or evolved when one mole of a substance is dissolved in a large volume of specified solvent.

heat of vaporization or **heat of evaporation** the amount of heat absorbed or evolved when a specified amount of substance, usually 1 mole, is converted from the liquid to the vapour state at a specified pressure and a specified temperature. If the pressure is 1 atm the temperature is the boiling point.

heat-shock protein *abbr.: hsp*; any of a group of specific proteins that are synthesized by both prokaryotic and eukaryotic cells, e.g. bacteria, yeasts, and *Drosophila*, after they have been exposed to a temperature that is higher than normal. Other stresses, such as free-radical damage, have a similar effect. Many members of the hsp family are not induced but are normally present in all cells. Human cells in culture also produce heat-shock proteins but the induction of heat-shock proteins has not been clearly demonstrated in the human body. The heat-shock proteins are classified, according to their size, in three classes, Hsp60, Hsp70, and Hsp90. They are characterized by their role as molecular **chaperones**. A major chaperone in the lumen of the endoplasmic reticulum is **binding protein**, a member of the Hsp70 class. These proteins are highly conserved in structure so that the proteins from *Escherichia coli* and human are 50% identical in primary structure. The Hsp70 proteins possess ATPase and peptide binding domains. The other members of the Hsp70 class are Grp75 in mitochondria and **DnaK** in bacterial cytosol. In the Hsp60 class is GroEL in bacterial cytosol (*see groEL*). Example, hsp71: database code HS71_HUMAN, 641 amino acids (69.97 kDa).

heat-shock regulatory element or **heat-shock response element** *abbr.: HSE*; a base sequence in prokaryotic and eukaryotic genomes responsible for regulating gene expression in response to heat-shock and related stress situations.

heat-shock response element *see HSE.*

heat-stable describing something that retains its (bio)chemical activity when heated to a moderate temperature, e.g. 50 °C.

heavy (*in physics*) of more than the usual mass. *Compare light* (def. 2).

heavy atom or **heavy isotope** or **heavy nuclide** an isotopic form of an atom that contains more than the common number of neutrons and is thus of greater relative atomic mass than the most abundant or most commonly observed isotope.

heavy carbon *see carbon-13.*

heavy chain or **H chain** the heavier of the two types of polypeptide chain found in **immunoglobulin** molecules. Each heavy chain is linked, usually by disulfide bonds, to a **light chain** and to another, identical heavy chain. Heavy chains differ in relative molecular mass according to the type of immunoglobulin: in humans the M_r is $\approx 50\,000$ in IgG and $\approx 70\,000$ in IgM. Each heavy chain consists of an **Fc fragment** and an **Fd fragment**. Heavy chains carry the antigenic determinants that differentiate the various immunoglobulin classes.

heavy-chain disease a rare disease of humans in which there are tumours of lymphoid tissue associated with the presence of free immunoglobulin fragments in plasma and urine.

heavy-chain switch or **H-chain switch** the change in the expression of the immunoglobulin **heavy chain** (H chain) gene system as manifested by the predominance of IgM (μ chain class) in the primary immune response and by the predominance of IgG (γ chain class) in the **secondary immune response**.



heavy hydrogen

heavy hydrogen *an alternative name for deuterium.*

heavy meromyosin *abbr.: H-meromyosin or HMM*; a 350 kDa fragment produced when **myosin** is subjected to tryptic digestion. It contains part of the helical, rod-shaped tail and both the attached globular heads of the original myosin molecule. Heavy meromyosin retains both the ATPase and the actin-binding activity of myosin.

heavy nitrogen the stable nuclide ^{15}N (nitrogen-15); relative abundance in natural nitrogen 0.37 atom percent. It is used as a tracer in studies of nitrogen metabolism.

heavy oxygen the stable nuclide ^{18}O (oxygen-18), usually accompanied by the stable nuclide ^{17}O (oxygen-17); relative abundances in natural oxygen: ^{18}O , 0.20 atom percent; ^{17}O , 0.037 atom percent. ^{18}O is used as a tracer in studies of reaction mechanisms and of oxygen metabolism.

heavy strand or H strand 1 any polynucleotide chain labelled with a heavy isotope, e.g. nitrogen-15. 2 any naturally occurring chain in a polynucleotide duplex that is heavier or has a greater density than the complementary strand.

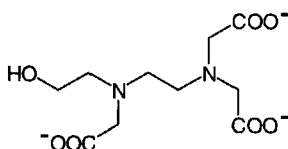
heavy water deuterium oxide; *symbol:* $^2\text{H}_2\text{O}$ or D_2O ; water in which the hydrogen atoms in all of the molecules have been replaced by deuterium; or water in which the hydrogen is appreciably enriched in deuterium. The M_r of pure $^2\text{H}_2\text{O}$ is 20.028. Heavy water is prepared from natural water by exchange techniques or by fractional distillation or electrolysis. It is used as a tracer in metabolic studies, in investigation of reaction mechanisms, and as a solvent in resonance spectroscopy.

hect+ *see* **hecto+** (def. 2).

hecto+ 1 *symbol:* h; an SI prefix denoting 10^2 times. 2 *or (before a vowel)* **hect+** denoting 100 or 100 times.

hedgehog protein a transmembrane protein involved in segment polarity and cell-to-cell signalling during embryogenesis and metamorphosis in *Drosophila melanogaster*. Database code (precursor): HH_DROME, 471 amino acids (52.03 kDa). *See also* **sonic hedgehog protein**.

HEDTA *abbr. for* *N*-(2-hydroxyethyl)ethylenediamine triacetate, or its trisodium salt, or for its partially or fully protonated forms. It is a chelating agent, especially for ferric ions in the pH range 7.0–10.0.



HEETP *abbr. for* height equivalent to an effective theoretical plate.

Heidelberger, Michael (1888–1991), US pathologist described as the father of modern immunology. The recipient of numerous awards, he continued to carry out research almost to the end of his life at age 103.

height equivalent to an effective theoretical plate *abbr.: HEETP; symbol:* *H*; (in chromatography) the column length divided by the **effective theoretical plate number**.

height equivalent to a theoretical plate or plate height *abbr.: HETP; symbol:* *h*; (in chromatography) the column length divided by the **theoretical plate number**.

Heisenberg, Werner Karl (1901–76), German theoretical physicist renowned for his work on quantum theory and atomic structure and for advancing (in 1927) the theory known later as the **Heisenberg uncertainty principle**; Nobel Laureate in Physics (1932) 'for the creation of quantum mechanics', the application of which has, *inter alia*, led to the discovery of the allotropic forms of hydrogen.

Heisenberg uncertainty principle the principle that the simultaneous precise determination of the velocity (or any

related property, e.g. momentum or energy) of a material particle and its position is impossible; the smaller the particle the greater the degree of uncertainty. [After Werner Karl Heisenberg (1901–76), German physicist; Nobel Laureate in Physics (1932).]

HeLa cell an established tissue culture strain of human epidermoid carcinoma cells containing 70–80 chromosomes per cell (compared with 46 in normal cells). It is much used for biochemical work. HeLa cells have been in continuous culture since 8 February 1951, and were derived from tissue removed from a patient named Henrietta Lacks.

helical cross a name sometimes given to the X-shaped pattern of X-ray scattering from a continuous helical molecule.

helicase a name sometimes given to the protein (**repA**) that promotes the ATP-dependent unwinding of the parental DNA duplex during DNA replication. *See also* **DNA helicase**, **RNA helicase**.

helicorubin a b-type cytochrome that occurs in the hepatopancreas of the snail *Helix pomatia* and related species.

helix (*pl. helices*) 1 a coiled or spiral structure, e.g. the thread of a bolt or a coil (tubular) spring. *See also* **alpha helix**, **beta helix**, **gamma helix**. 2 the curve traced on the surface of a cylinder or cone by a point crossing its right sections at a constant angle. 3 a space curve with turns of constant angle to the base and constant distance from the axis. —**helical** *adj.*; **helicity** *n*.

helix breaker any amino-acid residue that, whenever it occurs in a polypeptide chain, interrupts the α -helical structure; e.g. proline, hydroxyproline, and, sometimes, other residues.

helix-destabilizing protein *abbr.: HDP*; any of a group of proteins that bind tightly and preferentially to single-stranded DNA. They have, theoretically, the following properties: (1) they lower the melting temperature, T_m , for thermally induced **helix-to-coil transition**; (2) they lower the optimum temperature for duplex renaturation; (3) they sterically hinder base pairing and the degradation of single strands by nucleases; (4) they induce a conformation for the single-stranded DNA that is optimal for its function in some processes, e.g. as a template in replication; and (5) they interact with other proteins (enzymes) to potentiate some process involving single-stranded DNA and these proteins.

helix former any amino-acid residue that, whenever it is present in a polypeptide chain, tends to promote an α -helical structure. *Compare* **helix breaker**.

helix-loop-helix *abbr.: HLH*; an alternative term for **helix-turn-helix**.

helix-to-coil transition the transition of a macromolecule (polypeptide or polynucleotide) from an ordered to a disordered structure; it is equivalent to **denaturation**. Helix-to-coil transitions are usually detected by monitoring a change in some physical property of the macromolecule such as intrinsic viscosity, optical absorbance, or sedimentation coefficient. *See also* **melting temperature**.

helix-turn-helix (*abbr.: HTH*) or **helix-loop-helix** (*abbr.: HLH*) a protein structural feature consisting of two helices separated by a short **beta turn**. It is found in several DNA-binding regulatory proteins; there are two sequence motifs.

helix winding number *see* **supercoil**.

helodermin *see* **exendin**.

helospectin *see* **exendin**.

helper cell *abbr.: T_H cell*; a **T lymphocyte**, carrying the surface markers CD4 and CD5 in the human (L3T4 and Lyl in the mouse), that acts as an inducer of the effector cells for both **humoral** and **cell-mediated immunity**. Helper cells recognize and bind to antigen in combination with class II **major histocompatibility complex** molecules on the surface of antigen-presenting cells, and generate lymphokines that stimulate effector cytotoxic cells and B cells. Other cells of the helper phenotype, e.g. CD4⁺ cells, are known as **T-suppressor** cells and are responsible for suppression, exerting negative feedback control on the helper cells.

helper peak-1 *abbr.: HP-1*; a former name for **interleukin 1**.

helper virus

helper virus any virus that supplies one or more of the functions that a **defective virus** is unable to perform. When a cell is infected with both a defective virus and a helper virus, the former is able to replicate, a property that can be exploited in various molecular biological techniques.

hem+ or (*esp. Brit.*) **haem+** a variant form of **hemo+** (before a vowel).

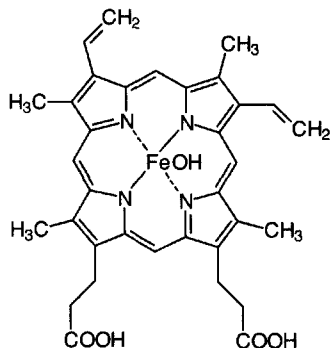
hema+ or (*esp. Brit.*) **haema+** a variant form of **hemo+**.

hemagglutination or (*esp. Brit.*) **haemagglutination** the agglutination of red blood cells.

hemagglutinin or (*esp. Brit.*) **haemagglutinin** **1** any **agglutinin** of red blood cells. **2** any nonantibody substance, e.g. a lectin or a surface component of some virus particles, that can agglutinate red blood cells. **3** a viral envelope glycoprotein responsible for the attachment of virus to cell receptors and the initiation of infection. Example from influenza A virus (strain A/Aichi/2/68) (precursor): database code HEMA_IAAIC, 566 amino acids (63.34 kDa); 3-D structure known (much β structure).

hemat+ or (*esp. Brit.*) **haemat+** a variant form of **hemato+** (before a vowel).

hematin or (*esp. Brit.*) **haematin** **1** or **ferriheme hydroxide** the trivial name for ferriprotoporphyrin hydroxide (formerly called ferriprotoporphyrin IX hydroxide); hydroxo(proto-porphyrinato)iron(III); hydroxoiron(III) protoporphyrinate. Compare **heme** (def. 1), **hemin** (def. 1). **2** the generic name for any hydroxo(porphyrinato)iron(III) coordination complex.



hemato+ or (before a vowel) **hemat+** comb. form denoting blood. Also (*esp. Brit.*): **haemato+**, **haemat+**. See also **hemo+**.

hematocrit or (*esp. Brit.*) **haematocrit** **1** or **packed red cell volume** the proportion of the volume of a sample of blood that is represented by the red blood cells. **2** an apparatus, essentially a graduated centrifuge tube, used to determine the proportion of the volume of a sample of blood that is represented by the red blood cells.

hematocrit technique or (*esp. Brit.*) **haematocrit technique** a method for the rapid isolation of mitochondria or other particles, in which they are separated from the medium by centrifugation through a layer of silicone fluid. The latter is immiscible with water, and of a density intermediate between that of the particles to be separated and that of the medium. A high-density aqueous 'fixative' beneath the silicone layer can be used to stabilize labile compounds in the mitochondria or other particles. Compare **hematocrit** (def. 2).

hematology or (*esp. Brit.*) **haematology** the branch of medicine concerned with diseases of the blood or of the blood-forming tissues.

hematolysis or (*esp. Brit.*) **haematolysis** an alternative name for **hemolysis**.

hematopoiesis or (*esp. Brit.*) **haematopoiesis** an alternative name for **hemopoiesis**.

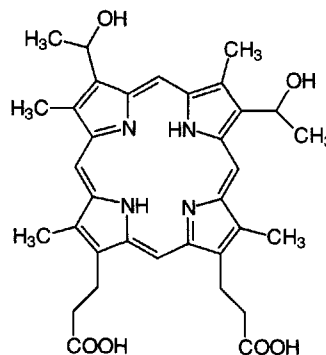
hematoporphyrin or (*esp. Brit.*) **haematoporphyrin** or (formerly) **hematoporphyrin IX** or (*esp. Brit.*) **haematoporphyrin IX**

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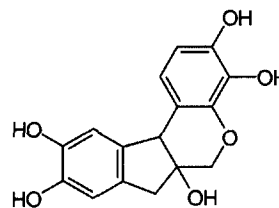
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the trivial name for 2,7,12,18-tetramethyl-3,8-bis(1-hydroxyethyl)porphyrin-13,17-dipropionic acid; the porphyrin formed from **heme** (def. 1) or a heme-containing protein on treatment with hydrogen bromide in glacial acetic acid.



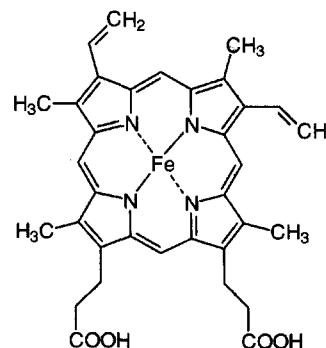
hematoside or (*esp. Brit.*) **haematoside** a **ganglioside**, NeuAc(α 2-3)Gal(β 1-4)Glc(β 1-1)Cer, found in red blood cells.

hematoxylin or (*esp. Brit.*) **haematoxylin** 7,11b-dihydrobenz[b]indeno[1,2-d]pyran-3,4,6a,9,10(6H)-pentol; a dye isolated from the heartwood of logwood, *Haematoxylon campechianum* L. Colourless or yellowish, it turns red on exposure to light, and is used as a stain in microscopy.



hematuria or (*esp. Brit.*) **haematuria** the presence of red blood cells in the urine. It can be due to lesions in the urinary tract, but also to glomerular disease of the kidney.

heme or (*esp. Brit.*) **haem** **1** or **ferroheme** or **ferrohaem** or **protoheme** the trivial name for ferriprotoporphyrin (formerly called ferriprotoporphyrin IX); 2,7,12,18-tetramethyl-3,8-divinylporphyrin-13,17-dipropionic acid iron(II) coordination complex. It occurs free and as the prosthetic group of a number of



heme (def. 1)

hemoproteins, e.g. hemoglobins, erythrocyruorins, myoglobins, some peroxidases, catalases, and cytochromes *b*. Compare



heme–heme interaction

hematin (def. 1), **hemin** (def. 1). **2** the generic name for any iron–porphyrin coordination complex irrespective of the valence state of the iron atom (which may be specified). The name of an individual heme is derived from that of the corresponding porphyrin; e.g. cytochrome (from cytoporphyrin) and ferroprothemo (from protoporphyrin). For biosynthesis, see **tetrapyrrole**.

heme–heme interaction or (*esp. Brit.*) **haem–haem interaction** the cooperative interaction between the various heme binding sites for dioxygen molecules that occurs when dioxygen binds with hemoglobin; as oxygenation proceeds, the combination of additional dioxygen molecules is made easier.

heme pocket or (*esp. Brit.*) **haem pocket** a hydrophobic crevice in the tertiary structure of myoglobin, or of a subunit of hemoglobin, into which the heme group fits. The nonpolar vinyl side chains of the heme are buried in the interior of the pocket, while the hydrophilic propionate side-chains project out of the pocket towards the surface of the molecule.

heme protein or (*esp. Brit.*) **haem protein** an alternative name for **hemoprotein**.

hemerythrin or (*esp. Brit.*) **haemerythrin** an oxygen carrier found in a few groups of invertebrates, e.g. sipunculid worms, certain molluscs, and crustaceans. It is a nonheme iron-containing protein, the subunits of which each contain about 113 amino-acid residues. Each monomer has an active site containing two atoms of Fe(II), 0.34 nm apart, between which a dioxygen molecule is thought to fit. The molecule is a homooctamer. Example from *Themiste dyscritum*: database code HEMT_THEDY, 113 amino acids (13.30 kDa); 3-D structure known.

heme synthetase see **ferrochelatase**.

heme–thiolate protein or (*esp. Brit.*) **haem–thiolate protein** see **P450**.

hemi+ prefix denoting half, or affecting one half. Compare **semi+**.

hemiacetal see **acetal**.

hemicellulose any or all of the cell-wall polysaccharides that are extractable from wood or other plant material with aqueous alkali (in which cellulose itself is insoluble). Hemicelluloses are chiefly xylans, but other homoglycans or heteroglycans containing hexose and/or pentose residues, uronic acid-containing xylans, and other **pectic substances** may also be present.

hemicystine an alternative name for **half-cystine**.

hemicystyl an alternative name for **half-cystyl**.

hemidesmosome a specialized cell junction between an epithelial cell and its underlying basal lamina. These resemble **desmosomes** morphologically but are chemically and functionally distinct. Between the cell membrane and the underlying connective tissue is a specialized mat of extracellular matrix sometimes known as the linker. The transmembrane linker proteins belong to the **integrin** family of extracellular matrix receptors rather than the **cadherin** family of cell–cell adhesion proteins found in desmosomes.

hemidiaphragm a muscle preparation widely used for metabolic studies *in vitro*, consisting of either of the two excised and separated left and right halves of the diaphragm, usually from a rat.

hemiketal see **acetal**.

Hemimetabola or **Exopterygota** one of the two divisions of the subclass Pterygota, comprising those insects, e.g. cockroaches, locusts, and grasshoppers, in which the juvenile forms resemble the adults, except for their lack of fully developed wings. Such insects undergo incomplete metamorphosis. Compare **Holometabola**.

hemin or (*esp. Brit.*) **haemin** **1** or **ferriheme chloride** or **protohemin** the trivial name for ferriprotoporphyrin chloride (formerly called ferriprotoporphyrin IX chloride); chloro(proto-porphyrinato)iron(III); chloroiron(III) protoporphyrinate. Compare **heme** (def. 1), **hematin** (def. 1). **2** the generic name for any chloro(porphyrinato)iron(III) coordination complex.

hemipore a model of translocation of solutes across mem-

branes, based on the possibility that there is interaction through the membrane between superficially disposed protein subunits. These subunits are envisaged as being embedded within the bimolecular lipid layer, anchored to the aqueous phase, and perhaps mobile in the plane of each face of the membrane. The subunits bind the solute in question and, when associated with the symmetrical protein at the opposite face of the membrane, transfer the solute across the membrane.

hemithioacetal see **acetal**.

hemithioketal see **acetal**.

hemizygous describing an organism in which a gene is present only once in a genotype, as a gene in a haploid cell or organism, a sex-linked gene in the heterogametic sex, or a gene in a segment of chromosome in a diploid cell or organism where its partner segment has been deleted. —**hemizygosity** *n*.

hemo+ or **hema+** or (*before a vowel*) **hem+** *comb. form* denoting blood. Also (*esp. Brit.*): **haemo+**, **haema+**, **haem+**. See also **hemato+**.

hemochromatosis or (*esp. Brit.*) **haemochromatosis** a disease of humans characterized by an abnormal, widespread, massive accumulation of tissue iron associated with characteristic pathological changes of tissue structure and function. **Primary hemochromatosis** is a specific genetically determined disease in which hemochromatosis occurs as a result of increased absorption of iron from a normal diet. **Secondary hemochromatosis** results from an increased intake and accumulation of iron secondary to other causes. Compare **hemosiderosis**.

hemochrome or (*esp. Brit.*) **haemochrome** any iron–porphyrin coordination complex with one or more strong-field axial ligands (e.g. pyridine).

hemochromogen or (*esp. Brit.*) **haemochromogen** any compound of heme with a nitrogenous base such as a protein (whether native or denatured) or pyridine, typically with two absorption bands in the green part of the visible spectrum. See also **chromogen**.

hemocuprein or (*esp. Brit.*) **haemocuprein** an alternative name for **superoxide dismutase**.

hemocyanin or (*esp. Brit.*) **haemocyanin** a blue, copper-containing oxygen carrier present in many molluscs and arthropods. It is a nonheme protein that binds one dioxygen molecule for two Cu(I) atoms; the dioxygen molecule is thought to form a bridge between the two copper atoms. The oxygenated compound is very bright blue with an intensity of light absorption 5–10 times that of other known copper complexes. Example from the lobster *Panulirus interruptus*: hexamer of a number of different chains, of which A, B (88% identity between these two), and C have been identified: chain A, database code HCYA_PANIN, 657 amino acids (75.61 kDa); this contains two copper-binding sites (His ligands); two CuA motifs; two other motifs form an oxygen-binding domain; one CuB motif.

hemocyte or (*esp. Brit.*) **haemocyte** any blood cell, especially of an invertebrate animal.

hemocytometer or (*esp. Brit.*) **haemocytometer** an apparatus for counting the number of cells in a known volume of blood. It consists of a slide with a chamber of known depth; the base of the chamber is ruled with a graticule over which the cells can be enumerated under a microscope.

hemodialysis or (*esp. Brit.*) **haemodialysis** the process of separating low-molecular-mass solutes from blood by **dialysis** through a semipermeable membrane, as in an artificial kidney machine.

hemoglobin or (*esp. Brit.*) **haemoglobin** *symbol*: Hb; any of a group of red, iron-containing, oxygen-carrying pigments of the blood of vertebrates and some invertebrates; hemoglobin also occurs in the root nodules of leguminous plants. All vertebrate hemoglobins consist of two pairs of associated **globin** polypeptide chains, each polypeptide chain carrying a heme prosthetic group bound noncovalently, the iron atom of which is in the ferrous state and forms a coordination complex with the pyrrole nitrogens. All normal human hemoglobins contain one pair of 15.7 kDa α chains, of 141 amino-acid residues, and



a pair of varying 16.5 kDa polypeptide chains, of 146 amino-acid residues and of similar amino-acid sequences. In adult human blood, 98% of the total hemoglobin is Hb A (or Hb A₁), containing a pair of β chains, i.e. it has the composition $\alpha_2\beta_2$, and 2% is Hb A₂, containing a pair of δ chains, i.e. $\alpha_2\delta_2$. In fetal blood, from about 10 weeks to 30 weeks of gestation, >90% of the hemoglobin is Hb F, containing a pair of γ chains, i.e. $\alpha_2\gamma_2$. In embryonic blood, the second pair of chains are ϵ chains, i.e. the structure is $\alpha_2\epsilon_2$. Each of the four heme groups in a hemoglobin molecule is able to combine reversibly with one dioxygen molecule; the oxygen-saturation curve of hemoglobin is sigmoidal, showing that there is positive cooperativity between the subunits. The hemoglobin molecule has a two-fold axis of symmetry, each half containing one α chain and one non- α chain; the overall shape of the molecule is globular with the heme groups buried in pockets in the polypeptide chains. There are eight helical regions, called A to G. The histidine at position F8 (i.e. the eighth residue in helix F) is the fifth ligand of the heme Fe(II). Upon oxygenation, dioxygen becomes the sixth ligand of Fe(II) and is also hydrogen bonded to HisE7 with an accompanying conformational change. In the absence of the correct bonding with the globin side chains, Fe(II) autooxidizes to Fe(III) and can no longer bind dioxygen. Examples: human and chimpanzee α chain, database code HBA_HUMAN, 141 amino acids (15.11 kDa); among α chain variants are Luxembourg, Ann Arbor, and Hirosaki; β chain, database code HBB_HUMAN, 146 amino acids (15.85 kDa). Among the **hemoglobinopathies** associated with β -chain variants is **sickle-cell anemia**. In those invertebrates that have hemoglobin (not **hemocyanin**) there is a polymeric hemoglobin. Example from earthworm, 12 subunits arranged in a hexagonal bilayer structure of 3.8 MDa; each one-twelfth subunit is composed primarily of disulfide-linked trimers (chains A, B, and C) and monomers (chain D); D chain, database code GLB1_LUTME, 142 amino acids (16.11 kDa). See also **hemoglobin A_{1c}**.

hemoglobin A_{1c} or (*esp. Brit.*) **haemoglobin A_{1c}** *abbr.*: Hb A_{1c}; the most abundant of the **glycated hemoglobins**, which are present in normal adult hemoglobin as minor components (together they represent 3–5% of the total hemoglobin of the erythrocyte). Hb A_{1c} is formed from Hb A₁ subsequent to biosynthesis by glycosylation of one of the two β chains, by a nonenzymic mechanism involving formation of an N-substituted aldime (Schiff's base) between the amino group of the peptide chain and the aldehyde group of glucose, followed by an **Amadori rearrangement** to the corresponding N-substituted ketoimine (aminodeoxyfructose derivative) (see also **Maillard reaction**). In inadequately controlled **diabetes mellitus** the steady-state concentration of Hb A_{1c} in serum is 2–3 times higher than normal, possibly reflecting an abnormally high mean blood glucose concentration during the preceding few months. Other glycosylated hemoglobins change very little in concentration in diabetes; thus measurement of serum levels of Hb A_{1c} is considered to be useful in assessing the degree of control of the condition.

hemoglobinopathy or (*esp. Brit.*) **haemoglobinopathy** any of many types of genetic disorder involving globin, the protein moiety of **hemoglobin**. These are divided into **thalassemia** syndromes, caused by defective synthesis of the component polypeptides of globin, and conditions that arise from mutations leading to amino-acid substitutions in the polypeptide chains. Of the latter there are over 400 known variants, including the classically important **sickle cell hemoglobin**. Many changes lead to increased oxygen affinity, e.g. Hb Rainier (β -145, Tyr $\rightarrow$ Cys), while in a few instances decreased oxygen affinity occurs, e.g. Hb Kansas (β -102, Asn $\rightarrow$ Thr). In some cases the substitution leads to an unstable hemoglobin molecule, as in Hb Hammersmith (β -42, Phe $\rightarrow$ Ser) where the important heme contact of the phenylalanine is lost, leading to the formation of **ferrihemoglobin**, or in Hb Bristol (β -67, Val $\rightarrow$ Asp) where replacement of the nonpolar valine by aspartic acid causes distortion of the E helix.

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hemolin or (*esp. Brit.*) **haemolin** an insect hemolymph protein that is of the immunoglobulin superfamily. It is induced by bacteria. It contains four internal immunoglobulin-type repeats. Example (precursor) from the moth *Hyalophora cecropia*: 413 amino acids (45.78 kDa).

hemolymph or (*esp. Brit.*) **haemolymph** a fluid occurring in the secondary body cavity (coelom) of some invertebrates, considered as functionally equivalent to the blood and lymph of higher animals.

hemolysate or (*esp. Brit.*) **haemolysate** a preparation obtained by lysis of erythrocytes.

hemolysin or (*esp. Brit.*) **haemolysin** 1 any substance, e.g. an antibody, that causes hemolysis. 2 any bacterial exotoxin that ruptures blood-cell membranes. Examples are the products of chromosomal genes of certain hemolytic *Escherichia coli* strains that cause predominantly nongastrointestinal (e.g. urinary tract) infections. There are four genes: *hlyA* encodes hemolysin (Ca²⁺-binding toxin): database code HLY1_EC01, 1023 amino acids (109.75 kDa); *hlyB* encodes an **ABC transporter** for HlyA export: database code HLY2_EC01, 707 amino acids (79.77 kDa); *hlyC* encodes a protein involved in hemolysin export: database code HLY3_EC01, 170 amino acids (19.70 kDa); *hlyD* encodes a transmembrane protein component of the HlyA export system: database code HLY4_EC01, 478 amino acids (54.53 kDa).

hemolysis or (*esp. Brit.*) **haemolysis** the lysis of red blood cells, either *in vivo* or *in vitro*. —**hemolyse** or **haemolyse** *vb*.

hemolytic or (*esp. Brit.*) **haemolytic** 1 of, or pertaining to **hemolysis**. 2 an agent or condition that causes hemolysis.

hemopexin or (*esp. Brit.*) **haemopexin** a 57 kDa β_1 -glycoprotein of human blood serum with one high-affinity binding site for **heme** (def. 1). The heme complex can be reduced with dithionite to give a three-band absorption spectrum characteristic of ferrohemochrome. Hemopexin may be necessary for heme to be taken up and degraded in the liver.

hemophilia or (*esp. Brit.*) **haemophilia** any of various hereditary disorders in which there is a deficiency or defect in certain of the **blood coagulation** factors resulting in prolonged bleeding following injury. Classical hemophilia, or **hemophilia A**, is a condition in which the blood has a prolonged clotting time due to an abnormally slow conversion of prothrombin to thrombin. It is caused by a functional deficiency of anti-hemophilic factor (factor VIII), inherited as an X-linked recessive trait. Other deficiencies in the blood coagulation system, e.g. **Christmas disease**, or **hemophilia B**, due to a deficiency of factor IX, produce similar clinical pictures.

hemopoiesis or **hematopoiesis** the process of blood formation, especially the formation of blood cells, occurring mainly in bone marrow. Also (*esp. Brit.*): **haemopoiesis**, **haematopoiesis**. —**hemopoietic**, **hematopoietic** or **haemopoietic**, **haematopoietic** *adj*.

hemoprotein or **heme protein** any protein to which an iron-porphyrin compound is linked in a stoichiometric manner. Hemoproteins include hemoglobins, myoglobins, cytochromes, catalase, and some peroxidases. Also (*esp. Brit.*): **haemoprotein**, **haem protein**.

hemorphin or (*esp. Brit.*) **haemorphin** any of a group of hemoglobin-derived peptides with affinity for opioid receptors.

hemorrhage or (*esp. Brit.*) **haemorrhage** bleeding; the escape of blood from a ruptured blood vessel, either externally or internally. —**hemorrhagic** or (*esp. Brit.*) **haemorrhagic** *adj*.

hemorrhagic toxin C and D see **atrolysin C**.

hemosiderin or (*esp. Brit.*) **haemosiderin** an insoluble, granular, ill-defined complex formed by the denaturation of **ferritin** with associated loss of apoferritin and micellar aggregation; about one third of its mass is iron. Hemosiderin occurs in liver, spleen, and red bone marrow, its amount increasing with increased iron content of the body. It is responsible for the histochemical staining of tissue by the Prussian Blue reaction, a deep-blue ferric ferrocyanide (Fe^{III}[Fe^{II}(CN)₆]). The distribution of stored iron is shifted from ferritin to hemosiderin in



hemosiderosis

direct correlation with ascorbic acid deficiency. *See iron overload.*

hemosiderosis or (*esp. Brit.*) **haemosiderosis** an abnormal increase in tissue iron content due to some defect in the control of iron turnover in the body or to increased iron ingestion. It is not associated with pathological changes in tissue structure and function. *Compare hemochromatosis.*

hemostasis or (*esp. Brit.*) **haemostasis** 1 the stopping of bleeding or the arrest of the circulation to an organ or part. 2 the stagnation of blood in an organ or part.

hemostat or (*esp. Brit.*) **haemostat** a device or a chemical agent that stops or retards bleeding.

hemostatic or (*esp. Brit.*) **haemostatic** 1 of, relating to, or functioning to cause **hemostasis** (def. 1). 2 a drug or other agent that retards or stops bleeding.

Hench, Philip Showalter (1896–1965), US physician notable for his demonstration (with E. C. Kendall) that rheumatoid arthritis could be treated with an adrenal hormone (later known as cortisone); Nobel Laureate in Physiology or Medicine (1950) jointly with E. C. Kendall and T. Reichstein 'for their discoveries relating to the hormones of the adrenal cortex, their structure and biological effects'.

Henderson-Hasselbalch equation an equation relating pH to the composition of buffer solutions:

$$\text{pH} = \text{pK}_A + \log([\text{conjugate base}]/[\text{acid}]),$$

where pK_A is the common logarithm of the acid dissociation constant of the buffer acid. The following approximation is often used:

$$\text{pH} = \text{pK}_A + \log([\text{salt}]/[\text{acid}]).$$

[After Lawrence Joseph Henderson (1878–1942), US physician and biochemist, and Karl A. Hasselbalch (1874–), Danish biochemist, who together formulated the equation in 1910.]

henna a preparation obtained from the dried and powdered leaves of *Lawsonia alba*, *L. inermis*, or *L. spinosa*, widely used for dyeing hair and fingernails. The active ingredient is **lawsone**.

henry (*pl. henrys or henries*) *symbol:* H; the SI derived unit of inductance, equal to the inductance of a closed circuit with a **magnetic flux** of 1 weber per ampere of current,

$$H = Wb A^{-1} s = m^2 kg s^{-2} A^{-2}$$

[After Joseph Henry (1797–1878), US physicist.]

Henry's function a function used in the theoretical treatment of the electrophoretic mobility of macromolecules that is dependent on the radius of the macromolecule, r , and its reciprocal ion-atmosphere radius, κ (derived from **Debye-Hückel theory**). It varies between 1.0 and 1.5 as κr goes from zero to infinity.

Henry's law the principle that the concentration of a gas in solution, at constant temperature, is directly proportional to the partial pressure of the gas in the vapour phase in equilibrium with that solution. It holds with high precision for gases of low solubility at pressures below approximately 10^5 Pa. It may be expressed by: $c_G = kp_G$, where c_G is the molar concentration of the gas in solution, p_G is the partial pressure of the gas in the gas phase, and k is a proportionality constant. [After William Henry (1774–1836), British chemist.]

heparan any polysaccharide derived by desulfation of **heparan sulfate** or **heparin**.

heparan sulfate any member of a group of glycosaminoglycans that have repeat units consisting of alternating $\alpha 1 \rightarrow 4$ -linked hexuronic acid and glucosamine residues, the former being a mixture of sulfated and nonsulfated D-glucuronic and L-iduronic acids, and the latter being either sulfated or acetylated on its amino group as well as sulfated on one of its hydroxyl groups. They usually occur attached to protein to form proteoglycans, the mode of attachment being through a xylose residue as in **chondroitin sulfates**. Heparan sulfate accumulates in a number of the mucopolysaccharidoses, e.g. Hunter syndrome, Hurler syndrome, and Sanfilippo syndrome. *See mucopolysaccharidosis. See also heparin.*

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heparan sulfate proteoglycan core protein *abbr.:* HSPG core protein; any of a group of modular heparan sulfate proteoglycan proteins with three heparan sulfate chains. Also known as **perlecans**, they have domains with homology to the low-density lipoprotein receptor, epidermal growth factor, laminin, and neural cell adhesion molecule, and have 15 immunoglobulin-like domains. Examples from *Mus musculus*: database code S18252, 3707 amino acids (397.85 kDa), and database code PGBM_MOUSE, 2481 amino acids (270.81 kDa).

heparin any member of a group of glycosaminoglycans found mainly as an intracellular component of mast cells. They are similar to **heparan sulfates** but are of somewhat higher average M_r (6000–20 000) and contain fewer *N*-acetyl groups and more *N*-sulfate and *O*-sulfate groups; they may be attached in the same manner to protein, forming proteoglycans. They consist predominantly of alternating $\alpha 1 \rightarrow 4$ -linked D-galactose and *N*-acetyl-D-glucosamine-6-sulfate residues. Heparins are secreted into the bloodstream by the mast cells of liver, lung, and other tissues; they act as inhibitors of blood clotting by activating **antithrombin III**.

heparinize or **heparinise** to treat with a preparation of **heparin**, usually as its sodium or lithium salt, in order to prevent clotting, either *in vivo* or *in vitro*, of blood or blood plasma. —**heparinization** or **heparinisation** *n*.

hepat+ a variant form of **hepat+** (before a vowel).

hepatectomy surgical removal of the liver. In **partial hepatectomy**, only some of the lobes of the liver are removed; this causes the remaining lobes to regenerate so that the liver *in toto* regains its former weight. —**hepatectomize** or **heparectomise** *vb*.

hepatic of, or pertaining to, the liver.

hepatitis any inflammatory disease of the liver. In humans the most common cause is infection with a virus or an amoeba. The three most common forms of viral infectious hepatitis are designated A, B, and C. Hepatitis A is caused by a single-stranded RNA **picornavirus**, for which there is now a killed virus vaccine, the virus having been grown in tissue culture. Hepatitis B is caused by a double-stranded DNA virus of the Hepadnaviridae family. The causative agent is a spherical virion with a diameter of 42 nm, often referred to as the **Dane particle**. This virus is especially virulent and its presence is associated with the development of primary liver tumours – the incidence of these is particularly high in China. The antigens may be in the form of 22 nm diameter, spherical particles designated HBsAg; principal components include gp27, which is a glycosylated form of p24, and a 42 kDa protein, database code VMSA_HPBVA, 389 amino acids (42.73 kDa). HBsAg is produced in recombinant yeast, the 22 nm particles being similar to those found in the plasma of infected individuals. These provide a very effective vaccine. Hepatitis C, formerly called 'non-A non-B', is caused by a single-stranded RNA virus, for which there is as yet no vaccine. Another form, hepatitis D, can be controlled by the vaccine to hepatitis B, and there are probably three other types: E, F, and G, of which G has recently been characterized.

hepat+ or (*before a vowel*) **hepat+** *comb. form* denoting of or pertaining to the liver.

hepatocarcinogen any **carcinogen** that acts specifically or primarily on liver.

hepatocellular of, or pertaining to **hepatocytes**.

hepatocyte or (*formerly*) **parenchymal liver cell**. the major (but not the only) cell type of the liver. They are arranged in folded sheets facing blood-filled spaces called sinusoids. Hepatocytes are responsible for the synthesis, degradation, and storage of a wide range of substances. They are the site of synthesis of all the plasma proteins, except for antibody, and are the site of storage of **glycogen**.

hepatocyte growth factor *abbr.:* HGF; *other name:* scatter factor; a **cytokine** produced by platelets, fibroblasts, macrophages, endothelial cells, and smooth muscle cells that is a po-



hepatocyte growth factor receptor

tent **mitogen** for hepatocytes, epithelial cells, keratinocytes, melanocytes, and hematopoietic precursor cells. It is a glycoprotein, with four **kringle** domains and no known protease activity. The molecule is a disulfide-linked dimer of α (463 amino acids) and β (234 amino acids) subunits, which are derived from a common precursor. Example from human: database code HGF_HUMAN, 728 amino acids (83.04 kDa).

hepatocyte growth factor receptor a type I membrane protein that binds **hepatocyte growth factor** and mediates its intracellular action by a protein tyrosine kinase (EC 2.7.1.112) activity on its intracellular domain. It is encoded by the *MET* protooncogene, and comprises a dimer of two subunits, α and β . Example (precursor) from human: database code KMET_HUMAN, 1390 amino acids (155.35 kDa). See also *tpo-met*.

hepatoflavin or **hepatoflavine** a former name for **riboflavin**.

hepatolenticular degeneration an alternative name for **Wilson's disease**.

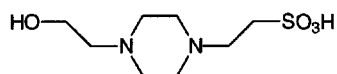
hepatoma any **carcinoma** derived from liver cells.

hepatopancreas a digestive gland, found in many invertebrates, that performs functions similar to those of the liver and the pancreas in vertebrates.

hepatosis any noninflammatory disease of the liver.

hepatotoxin any toxin that acts specifically or primarily on the liver.

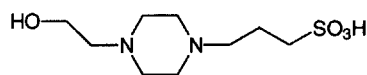
Hepes or **HEPES** *abbr. for (and trivial name of)* 4-(2-hydroxyethyl)-1-piperazine-ethanesulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 7.55.



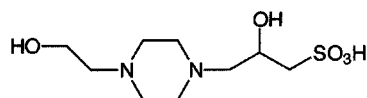
hexoxilin *abbr.:* Hx; any of various epoxyhydroxy metabolites of 12-hydroperoxyeicosatetraenoate, e.g. 13-hydroxy-14,15-epoxyeicosatetraenoate. A number of sites of formation have been reported, activities including modulation of neurotransmission in brain and release of insulin in pancreatic islets.

hexoxilin-epoxide hydrolase EC 3.3.2.7; an enzyme that converts (5Z,9E,14Z)-(8 ξ ,11R,12S)-11,12-epoxy-8-hydroxy-eicosa-5,9,14-trienoate **hexoxilin** A_3 to the trioxilin, (5Z,9E,14Z)-(8 ξ ,11 ξ ,12S)-8,11,12-trihydroxyeicosa-5,9,14-trienoate (trioxilin A_3).

Hepps or **HEPPS** or **EPSPS** *abbr. for (and trivial name of)* 4-(2-hydroxyethyl)-1-piperazinepropanesulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 8.0.



Heppso or **HEPPSO** *abbr. for (and trivial name of)* 4-(2-hydroxyethyl)-1-piperazine-(2-hydroxypropanesulfonic acid); a **Good buffer substance**, pK_a (25 °C) = 7.8.



hepsin EC 3.4.21.-; a serine protease, similar to other trypsin-like enzymes, that is a type II membrane protein. It is required for growth and maintenance of cell morphology in many tissues. Example from human: database code HEPS_HUMAN, 417 amino acids (44.96 kDa).

hepta+ or (before a vowel) **hept+** *comb. form* denoting seven.

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heptanoate 1 the anion, $\text{CH}_3-[\text{CH}_2]_5-\text{COO}^-$, derived from heptanoic acid (formerly enanthic acid). 2 any salt or ester of heptanoic acid.

heptanoyl *symbol:* Hpo; the univalent acyl group, $\text{CH}_3-[\text{CH}_2]_5-\text{CO}-$, derived from heptanoic acid.

heptitol any **alditol** having a chain of seven carbon atoms in the molecule.

heptose any **aldose** having a chain of seven carbon atoms in the molecule.

heptulose any **ketose** having a chain of seven carbon atoms in the molecule.

heptyl *symbol:* Hp; the alkyl group, $\text{CH}_3-[\text{CH}_2]_5-\text{CH}_2-$, derived from heptane.

heptyl glucoside *n*-heptyl β -D-glucopyranoside; a nonionic detergent, CMC 79 mM.

heptyl thioglucoside *n*-heptyl 1-thio- β -D-glucopyranoside; a nonionic detergent; the CMC, 30 mM, remains unchanged between 1 and 20 °C.

HER2 *see neu*.

herbicide any chemical substance that destroys plants or inhibits their growth, especially one used to control weeds.

hereditary of, pertaining to, or caused by **heredity**; inherited.

hereditary disease any pathological condition that is caused by the presence in the organism of an abnormal gene.

hereditary fructose-bisphosphatase deficiency an inherited autosomal recessive condition in humans characterized by less than 20% of normal activity of **fructose-bisphosphatase** (EC 3.1.3.11). It is characterized in infants by episodic spells of hyperventilation, apnoea, hypoglycemia, ketosis, and lactic acidosis. Surviving infants appear to become normal, but gluconeogenesis in their livers is severely impaired, and when glycogen stores are depleted, amino acids, lactate, and ketones accumulate. Subjects are not sensitive to fructose intake. *Compare hereditary fructose intolerance.*

hereditary fructose intolerance a condition in humans, inherited as an autosomal recessive character, in which there is a deficiency of the aldolase-B isoenzyme of **fructose-bisphosphate aldolase** (EC 4.1.2.13) (there is less than 15% normal activity with fructose 1-phosphate as substrate but activity is less depressed with fructose 1,6-bisphosphate as substrate) while the levels of aldolases A and C are normal (*see aldolase* (def. 2)). The condition is characterized by severe hypoglycemia and vomiting shortly after fructose intake. In this condition prolonged fructose ingestion in infants leads to poor feeding, vomiting, hepatomegaly, jaundice, hemorrhage, proximal-renal-tubule syndrome, and death. The hypoglycemia ensuing after fructose ingestion is caused by fructose 1-phosphate inhibition of glycogenolysis at the phosphorylase level and of gluconeogenesis at the mutant aldolase level. *Compare hereditary fructose-bisphosphatase deficiency.*

heredity 1 the transmission from one generation to another of genetic factors that determine the characteristics of an organism. 2 all of the inherited factors in an organism or all the characteristics determined by them.

heregulin *abbr.:* HRG; one of several glycoproteins that bind to the transmembrane tyrosine kinase ErbB2 p185 and stimulate its activity. They are secreted by human breast cancer cells in culture. Example heregulin- α precursor from human: database code A43273, 640 amino acids (70.29 kDa).

hormone an alternative term for **hormone** (suggested by E. Schäfer in 1913 as better expressing the concept of chemical messengers).

heroin *see* **diacetylmorphine**, **morphine**.

herpes virus any of a family (Herpesviridae) of enveloped, icosahedral, double-stranded DNA viruses, many of which infect animals. The family includes human herpesviruses types 1 and 2, Epstein-Barr virus, zoster (varicella) virus, pseudorabies virus, cytomegalovirus, Lucke virus, and Marek's disease virus.

herring any marine fish of the family Clupeidae, the flesh of



which is comparatively rich in oils with a high content of long-chain *n*-3 fatty acids; typical composition (range of major fatty acids, expressed as % of total fatty acids): 16:0, 12–18%; 16:1(*n*-7), 6–8%; 18:1(*n*-9), 11–25%; 20:1(*n*-9), 7–19%; 20:5(*n*-3), 11–15%; 22:6(*n*-3), 5–8%.

Hershey, Alfred Day (1908–97), US geneticist; Nobel Laureate in Physiology or Medicine (1969) jointly with M. Delbrück and S. E. Luria 'for their discoveries concerning the replication mechanism and the genetic structure of viruses'.

Hershey-Chase experiment a seminal experiment, reported in 1952, in which it was demonstrated that when an *Escherichia coli* cell is infected with T2 bacteriophage, the phage DNA is injected into the cell whereas the phage coat protein remains on the outside of the infected bacterial cell. This was shown by demonstrating that most of the ³⁵S-labelled phage protein could be removed from the surface of the bacterial cells by application of a shearing force without removing the ³²P-labelled nucleic acid and without interfering with the course of productive infection. The phage nucleic acid could be shown to have moved into the bacterial cell. It was therefore concluded that only the phage DNA was required for infection, indicating that DNA must be the hereditary material. This confirmed earlier experiments in 1944 by Avery and colleagues (see **transforming principle**), but was somewhat more readily accepted when Watson and Crick deduced the structure of DNA. [After A. D. Hershey and Martha Chase (1927–).]

Hershey circle a circular duplex DNA structure in which each strand contains one interruption and in which the interruptions are not opposite each other. It is formed by some linear, double-stranded DNA molecules (e.g. from *Escherichia coli* phage λ) containing short, complementary single strands (cohesive ends) at each end when these cohesive ends base-pair with each other.

Hers's disease an alternative name for type VI glycogen disease.

hertz *symbol*: Hz; the SI derived unit of frequency of a periodic phenomenon, equal to 1 cycle per second; i.e. 1 Hz = 1 s⁻¹. [After Heinrich Hertz (1857–94), German physicist.]

Hertz, Gustav (1887–1975), German physicist; Nobel Laureate in Physics (1926) jointly with J. Franck 'for their discovery of the laws governing the impact of an electron upon an atom'.

Hess's law of constant heat summation the statement that the enthalpy change in a chemical reaction is the same whether the reaction takes place in one stage or in more than one stage. [After Germain Henri Hess (1802–50), Genevese-born Russian chemist; known in Russia as German Ivanovich Gess.]

HETE *abbr.* for hydroxyeicosatetraenoate; see also **dihydroxy-eicosatetraenoate**.

hetero+ *comb. form* 1 other, different, unusual. *Compare* **allo+** (def. 1), **homo+** (def. 1), **iso+** (def. 1), **xeno+** (def. 1). 2 for, from, or directed towards a different species. *Compare* **allo+** (def. 2), **homo+** (def. 2), **iso+** (def. 2), **xeno+** (def. 2). 3 containing atoms, groups, linkages, residues, or subunits of different kinds. 4 being or pertaining to an atom other than the predominant or significant one, especially of a nitrogen, oxygen, or sulfur atom in a ring containing mainly carbon atoms. *Compare* **homo+** (def. 4).

heteroantibody any antibody (including any autoantibody) that is able to react with an antigen derived from another species. *Compare* **isoantibody**.

heteroantigen any antigen derived from one species that is able to stimulate an immune response in another species. *Compare* **isoantigen**.

heteroatom any atom in the chain or ring of an organic compound that is not a carbon atom. —**heteroatomic** *adj.*

heterobifunctional describing any reagent that carries two differing reactive groups. *Compare* **homobifunctional**.

heterocaryon a variant spelling of **heterokaryon**.

heterochromatic 1 relating to or possessing more than one colour; relating to light or other electromagnetic radiation of more than one frequency. *Compare* **monochromatic**. 2 of, relating to, or possessing **heterochromatin**.

heterochromatin a condensed form of **chromatin**, occurring in the nucleus at interphase, that stains strongly with basophilic dyes. The DNA of heterochromatin is replicated at a later stage in the cell-division cycle than **euchromatin**. 'Satellite' DNA is associated with heterochromatin, and the sex chromosomes of animals, especially the Y chromosome, contain large regions of heterochromatin. *Compare* **euchromatin**.

heteroconjugation the association between a base and the conjugate acid of a different base through a hydrogen bond. *See* **conjugate acid-base pair**. *Compare* **homoconjugation**.

heterocycle any compound or molecular structure that is **heterocyclic** (def. 1); a **heterocyclic** (def. 2).

heterocyclic 1 describing any cyclic molecular structure containing atoms of at least two different elements in the ring or rings; describing any compound having such a structure. *Compare* **carbocyclic** (def. 1), **homocyclic** (def. 1). 2 or **heterocycle** any such compound.

heterocyst any of the clear cells that occur at intervals on the filaments of certain cyanobacteria. They are thought to be important sites of nitrogen fixation.

heterodetic describing a cyclic peptide in which the ring consists of amino-acid residues only, but the linkages forming the ring are not solely eupeptide bonds; one (or, if nonproximate, more than one) is an isopeptide, disulfide, ester, or other bond. *Compare* **homodetic**.

heterodimer any molecular structure in which two nonidentical subunits are associated. *Compare* **dimer**, **homodimer**.

heteroduplex 1 any double-stranded DNA in which the two strands do not have completely complementary base sequences. In a **mutational heteroduplex** a mutational event has occurred in one of the polynucleotide chains; such a molecule does not produce uniform progeny on replication. A **recombinational heteroduplex** is a double-stranded DNA molecule that gives rise to one parental and one recombinant molecule of DNA in the reproductive cycle that immediately follows its production. Heteroduplex DNA may also arise from annealing DNA single strands *in vitro*. 2 a DNA-RNA hybrid. *Compare* **homoduplex**.

heteroduplex mapping the physical mapping of the base sequence homology of two DNA molecules. It is accomplished after the *in vitro* formation of a hybrid double-stranded molecule, in which one polynucleotide strand is from one of the DNA molecules and the complementary polynucleotide strand is from the other. Unpaired single-stranded regions are located and measured by electron microscopy.

heterogametic describing the sex whose cell nuclei contain a pair of dissimilar **sex chromosomes**. The heterogametic sex is usually male in mammals but is female in the Lepidoptera, birds, reptiles, some amphibians and fishes, and a few plants. *Compare* **homogametic**.

heterogeneous 1 differing in kind, as in a heterogeneous population. 2 describing a group of molecules that is not uniform with respect to size, charge, structure, or other property. 3 describing a system consisting of two or more phases. *Compare* **heterogenous**. —**heterogeneity** *n*.

heterogeneous catalysis any catalytic system in which the catalyst constitutes a different phase from that of the reactants. *Compare* **homogeneous catalysis**.

heterogeneous immunoassay any form of **immunoassay** that involves physical separation, at some stage of the procedure, of antibody-bound antigen from remaining free antigen. It includes **radioimmunoassay**. *Compare* **homogeneous immunoassay**.

heterogeneous nuclear ribonucleoprotein *abbr.*: hnRNP; particulate complexes of **heterogeneous nuclear RNA** (hnRNA) with proteins, which are cell-specific and themselves heterogeneous. The protein component may play a role in the processing of the hnRNA to mRNA.

heterogeneous nuclear RNA *abbr.*: hnRNA or HnRNA or H-RNA; a heterogeneous mixture of RNA molecules of high *M_r* with a rapid turnover rate that occurs in cell nuclei during protein synthesis. It is the form of RNA synthesized in eu-

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heterogenesis

karyotes by RNA polymerase II, i.e. that which is translated into protein. It represents the transcription products of both the exons and the introns of a gene or genes and is processed to form mRNA by the removal of the nontranslated portions (corresponding to the introns in the DNA). So-named because heterogeneity in size was one of the first characteristics used to distinguish it from other forms of nuclear RNA.

heterogenesis the production of offspring having different characteristics in successive generations; e.g. the alternation between generations that produce gametes and those that reproduce through agametes or by parthenogenesis.

heterogenetic describing chromosome pairing during meiosis when the pairing partners are derived from different original ancestors. *Compare* **homogenetic**.

heterogenic 1 describing a population or a gamete containing more than one allele of a particular gene. 2 describing two genetic elements that are not known to have a common ancestry. 3 *an alternative word for* **heterogenous**. *Compare* **homogenic**.

heterogenize or **xenogenize** to introduce a foreign, especially a more potent, antigen into a cell surface in order to enhance the response by T lymphocytes to other cell-surface antigens. — **heterogenization** *n*.

heterogenous or **heterogenic** not originating within the organism in question; of foreign origin. *Compare* **heterogeneous**.

heteroglycan or **heteropolysaccharide** any polysaccharide (i.e. glycan) that contains residues of two or more kinds of monosaccharide (i.e. glycose) molecule. *Compare* **homoglycan**.

heterograft or **heterologous graft** *an alternative name for* **xenograft**. *Compare* **homograft**.

heteroimmune describing a pair of bacteriophages in which each is sensitive to its own repressor but not to that of the other.

heterokaryon or **heterocaryon** any cell with more than one nucleus, in which the nuclei are not all of the same genetic constitution, or a tissue composed of such cells. *Compare* **homokaryon**. — **heterokaryotic** or **heterocaryotic** *adj*.

heterokaryosis or **heterocaryosis** the association of genetically different nuclei in a common cytoplasm.

heterolactic fermentation a type of fermentation of glucose in which the products include lactate, acetate and/or ethanol, and carbon dioxide. A key enzyme in the pathway is **phosphoketolase**, which cleaves xylulose 5-phosphate (produced by the pentose phosphate pathway) to D-glyceraldehyde 3-phosphate and acetyl phosphate, the former being then converted to lactate while the latter can form acetate and/or ethanol. Organisms that ferment in this manner include *Leuconostoc* spp. and some species of *Lactobacillus*; some produce D(–)-lactate while others produce DL-lactate. *Compare* **homolactic fermentation**.

heterolipid *a former name for* any complex lipid.

heterologous 1 consisting of different elements or of like elements in different proportions. 2 or **heterospecific** derived from another species, e.g. heterologous serum or tissue. *Compare* **homologous**, **isologous**.

heterologous association describing any association between two **protomers** in which the **domain of bonding** is made up of two different **binding sets**. *Compare* **isologous**.

heterologous desensitization *see* **desensitization**.

heterologous graft or **heterograft** *an alternative name for* **xenograft**.

heterolysin any antibody or other hemolysin formed in response to the introduction of erythrocytes from another species into the bloodstream of an animal. *Compare* **autolysin**, **homolysin**.

heterolysis 1 (*in chemistry*) the cleavage of a bond in such a manner that both bonding electrons remain within one of the two fragments between which the bond is broken; e.g. $A-B \rightarrow A^+ + B^-$. *Compare* **homolysis**. 2 (*in biochemistry*) the lysis of cells or tissues by the action of exogenous agents, e.g. enzymes, detergents, or lysins. *Compare* **autolysis**. — **heterolytic** *adj*.

heterolysosome any early **secondary lysosome** concerned with the digestion of exogenous material (i.e. **heterophagy**). *See* **heterophagosome**.

heteromorphic or **heteromorphous** 1 differing from the nor-

mal form, e.g. in size, shape, or function. 2 describing the members of any pair of homologous chromosomes that differ from each other in size or shape. 3 (*especially of insects*) having differing forms at different stages of the life cycle. 4 *an alternative term for* **polymorphic**. *Compare* **isomorphic**. — **heteromorphism** or **heteromorphy** *n*.

heterooligomer any **oligomer** made up of two or more kinds of constitutional repeating unit. — **heterooligomeric** *adj*.

heterophagosome any **phagosome** containing material exogenous to the cell. Fusion with a lysosome leads to the formation of a heterolysosome. *Compare* **autophagosome**.

heterophagy the process by which cells digest exogenous material that they have taken up. *Compare* **autophagy**. — **heterophagic** *adj*.

heterophilic binding (*of adhesion molecules*) binding of an adhesion molecule in one cell to a non-identical adhesion molecule in an adjacent cell. *Compare* **homophilic binding**.

heteroplasmy the presence of more than one type of mitochondrial DNA in a cell. A mutant mitochondrial DNA may be present in varying amounts so that phenotypic variation between cells is possible.

heteroploid describing a cell that contains any number of chromosomes except the **haploid** or **diploid** number. The term includes all **aneuploid** or **polyploid** numbers of chromosomes.

heteropolymer any polymer made up of two or more kinds of constitutional repeating unit. Proteins are typical heteropolymers. — **heteropolymetric** *adj*.

heteropolysaccharide *an alternative term for* **heteroglycan**.

heteroside *a former name for* any glycoside containing a non-carbohydrate moiety (aglycon).

heterospecific having specificity for or derived from a different species; **heterologous** (def. 2).

heterotetramer any molecular structure that can be dissociated into four (or two pairs of) nonidentical monomers. — **heterotetrameric** *adj*.

heterotope any of two or more chemical elements having atoms of different atomic number and occupying different places in the periodic table. The term includes **isobar** (def. 1). *Compare* **isotope**.

heterotopia or **heterotopy** the displacement, or unusual position, of a cell, organ, tissue graft, etc.

heterotopic 1 (*in chemistry*) of, relating to, or being a **heterotope** or heterotopes. *Compare* **isotopic**. 2 (*in biology and medicine*) or **heterotopous** displaced or in an unusual anatomical position.

heterotopic transplant any tissue graft that is transplanted from its normal position to another in the same organism. *Compare* **orthotopic transplant**.

heterotopous *a variant form of* **heterotopic** (def. 2).

heterotopy *a variant form of* **heterotopia**.

heterotrimer any molecular structure that is dissociable into three nonidentical monomers. — **heterotrimeric** *adj*.

heterotroph or **heterotrophe** or **organotroph** any organism whose nutritional requirements are not satisfied solely by simple inorganic substances and for which a supply of organic carbon is required for the synthesis of cellular constituents. Heterotrophs may be subdivided into **chemoheterotrophs** (or **chemoorganotrophs**) and **photoheterotrophs** (or **photoorganotrophs**) according, respectively, to whether they are **chemotrophs** or **phototrophs**. *Compare* **autotroph**. — **heterotrophic** *adj*; **heterotrophy** *n*.

heterotropic 1 describing an **allosteric** effect in which interaction occurs between nonidentical ligands; the effect may be either cooperative or antagonistic. The term is applied also to such an interaction, to an allosteric enzyme for which different substances act as the substrate and the effector, to the regulation of such an allosteric system, etc. *Compare* **homotropic**. 2 describing a chromosome that does not pair with another at meiosis, especially a sex chromosome in the heterogametic sex. **heterotypic** pertaining to, or being the first, reductive division in meiosis. *Compare* **homeotypic**.

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heterozygosis

heterozygosis 1 the union of genetically different gametes to form a **heterozygote**. 2 or **heterozygosity** the state or condition of being a heterozygote. *Compare* **homozygosis**.

heterozygosity *an alternative term for heterozygosis* (def. 2).

heterozygote any cell or organism having two different genes (alleles) at one or more corresponding loci on homologous chromosomes. With respect to such loci a heterozygote will produce two different types of gametes. *Compare* **homozygote**.

heterozygous or **heterozygotic** of, pertaining to, or being a **heterozygote**; hybrid.

HETP *abbr. for* height equivalent to a theoretical plate.

heuristic describing any method for solving mathematical or other problems for which no **algorithm** (or closed method) exists. It involves narrowing down the field of search for a solution by inductive reasoning from past experience of similar problems. *Compare* **stochastic**.

HEV *abbr. for* high-walled endothelium of the post-capillary venules.

hevein an *N*-acetyl-D-glucosamine and *N*-acetyl-D-neuraminic acid binding lectin; it is present in the latex of the rubber tree, *Hevea brasiliensis* and has 56% homology with wheat germ agglutinin. Example (precursor) from *H. brasiliensis*: database code HEMO_HYACE, 413 amino acids (45.78 kDa).

Hevesy, György (Károly) (1885–1966), *also styled, inter alia, as Georg (Karl) von Hevesy or George (Charles) von Hevesy or George (Charles) Hevesy or George (Charles) de Hevesy* Hungarian-born Swedish radiochemist, analytical chemist, and biochemist who (with F. A. Paneth in 1913) was the first to recognize the possibility of using natural radioactive elements as indicators and is remembered for his subsequent pioneering researches employing natural and artificial radioisotopes as indicators or tracers in chemical and biological systems and also for the discovery of hafnium; Nobel Laureate in Chemistry (1943) 'for his work on the use of isotopes as tracers in the study of chemical processes'.

hexa+ or (*before a vowel*) **hex+** *comb. form* denoting six, six-fold, six times.

hexabrachion a six-armed structure, characteristic of the **tenascin** molecule. Regions of the protein radiate outwards like spokes of a wheel. [From *L. brachium* arm.]

hexadecadlenoic acid any straight-chain fatty acid having sixteen carbon atoms and two double bonds per molecule. (*all-Z*)-Hexadeca-6,9-dienoic acid is found in small amounts in some animal lipids.

hexadecanoic acid *the systematic name for* palmitic acid; *see* **palmitate**.

hexadecenoic acid any straight-chain fatty acid having sixteen carbon atoms and one double bond per molecule. The most common are the 9Z-isomer (palmitoleic acid; *see* **palmitoleate**) and the 11Z-isomer, but the 3E-, 6E-, 6Z-, 9E-, and 10Z-isomers are also variously found in animals or plants; the 9E-isomer (i.e. palmitelaidic acid) is a pheromone.

hexaethyleneglycol mono-*n*-decyl ether a nonionic detergent; *see* **C₆E₇**.

hexaethyleneglycol mono-*n*-dodecyl ether a nonionic detergent; *see* **C₆E₈**.

hexagonal phase (of phospholipids) a phase in which the polar headgroups are orientated towards the interior of a phospholipid aggregate, with the fatty acyl carbon chains pointing outwards; a form of inverted micelle. It can occur in lipid bilayers, especially at higher temperatures. Certain phospholipids, such as phosphatidylethanolamine, have a greater tendency to adopt this phase than, e.g., phosphatidylcholine. A hexagonal phase with the polar headgroups oriented towards the exterior of the micelle can also exist. *See also* **liquid-crystalline phase**, **phase transition**.

hexamer 1 any oligomer consisting of six monomers. 2 the group of six capsomeres in an icosahedral virus capsid.

hexanoate or **caproate** 1 the anion, CH₃-[CH₂]₄-COO⁻, derived from hexanoic acid. 2 any salt or ester of hexanoic acid.

hexanolamino-PAF 1-*O*-hexadecyl-2-*O*-acetyl-*sn*-glycero-3-

phospho-(*N,N,N*-trimethyl)-hexanolamine; a substance that inhibits the effects induced by **platelet activating factor**, including platelet aggregation, secretion, and inositol phospholipid turnover; superoxide ion and H₂O₂ release from macrophages; hepatic glycogenolysis; and vasoconstriction.

hexanoyl or (*formerly*) **caproyl** *symbol*: Hxo; the univalent acyl group, CH₃-[CH₂]₄-CO-, derived from hexanoic acid. *See* **caproyl** (def. 2).

hexaric acid an **aldaric acid** formally derived from a hexose by oxidation at both C-1 and C-6.

hexitol any **alditol** having a chain of six carbon atoms in the molecule.

hexokinase EC 2.7.1.1; *systematic name*: ATP:D-hexose 6-phosphotransferase; *other name* (used for hexokinase IV): glucokinase; the phosphotransferase enzyme that catalyses the phosphorylation by ATP of D-hexose to form D-hexose 6-phosphate and ADP. In vertebrates there are four major isoenzymes, designated hexokinase types I, II, III, and IV, also called hexokinases A–D. All of these types are relatively nonspecific with respect to sugar, and phosphorylate a number of hexoses such as glucose, fructose, and mannose. Thus the name glucokinase for type IV may mislead. Types I–III all have *M_r* ≈ 100 kDa and are sensitive to the product, glucose 6-phosphate (*abbr.*: Glc-6-P). They differ from hexokinases found in yeast, which have *M_r* ≈ 50 kDa, and are not inhibited by Glc-6-P. Type III is inhibited by excess glucose. Type IV, despite being a monomer with only one active site, shows cooperativity with respect to glucose, and reaches physiologically significant activity at higher glucose concentrations than types I–III. In hepatocytes, at less than 5 mM glucose it is present as an inactive enzyme bound to a regulatory protein; a glucose concentration of 15 mM causes translocation and activation of the enzyme. Type II is the major isoenzyme of skeletal muscle and adipose tissue. There are strong sequence similarities between the different types. The mammalian enzymes could have evolved by duplication and fusion of an ancestral 50 kDa hexokinase. Examples: hexokinase IV from human mitochondrial outer membrane: database code HXK1_HUMAN, 917 amino acids (102.39 kDa); the N-terminal and C-terminal halves of the sequence are similar; regulatory function of this monomeric allosteric enzyme is associated with the N-terminal half; human brain hexokinase, type I: database code HXK1_HUMAN, 917 amino acids (102.39 kDa); human liver glucokinase isozyme (hexokinase D): database code HXKH_HUMAN, 466 amino acids (52.08 kDa); yeast hexokinase (3-D structure with glucose): database code NRL_1HKG, 457 amino acids (38.89 kDa); also, from yeast: database code HXKG_YEAST, 500 amino acids (55.31 kDa).

hexon a **capsomere** that is surrounded by six other capsomeres in an icosahedral virus capsid.

hexonic acid any monocarboxylic acid (i.e. aldonic acid) formally derived from a hexose by oxidation at C-1.

hexosamine any aminodeoxy sugar with six carbon atoms.

hexosaminidase EC 3.2.1.52; *recommended name*: β-*N*-acetylhexosaminidase; *systematic name*: β-*N*-acetyl-D-hexosaminide *N*-acetylhexosaminohydrolase; *other name*: β-hexosaminidase. An enzyme that catalyses the hydrolysis of terminal nonreducing *N*-acetyl-D-hexosamine residues in *N*-acetyl-β-D-hexosaminides. It is a lysosomal enzyme in mammals; there are two precursors (human): α, database code HEXA_HUMAN, 529 amino acids (60.62 kDa); and β, database code HEXB_HUMAN, 556 amino acids (63.04 kDa). The α chain, a glycoprotein, comprises amino acids 109–529 of its precursor; three glycoproteins (all similar to α) are derived from the β precursor: β (amino acids 122–556), β-B (amino acids 122–311), and β-A (amino acids 315–556). There are three forms of the enzyme: form A, responsible for degradation of GM₂ gangliosides, is a trimer composed of one α chain, one β-A chain, and one β-B chain; form B is a tetramer of two β-A and two β-B chains; form S is a homodimer of two α chains. **Tay–Sachs dis-**



hexosan

ease is due to defects of the α chain and Sandoff disease due to defects of the β chain.

hexosan a former name for any of a class of polysaccharides that yield only hexoses on hydrolysis.

hexose any aldose having a chain of six carbon atoms in the molecule.

hexose diphosphate pathway an alternative name for **Embden-Meyerhof pathway**.

hexose monophosphate pathway or **hexose monophosphate shunt** an alternative name for **pentose phosphate pathway**.

hexose phosphate transport protein an integral membrane transport protein responsible for sugar phosphate uptake in bacteria. Example from *Escherichia coli*: database code UHPT_ECOLI, 463 amino acids (50.55 kDa).

hexose-1-phosphate uridylyltransferase see **UDP-glucose-hexose-1-phosphate uridylyltransferase**.

hexulose any ketose having a chain of six carbon atoms in the molecule.

hexuronic acid 1 any monocarboxylic acid (i.e. uronic acid) formally derived from a hexose by oxidation at C-6. 2 a historical name originally used for ascorbic acid.

hexyl symbol: Hx; the alkyl group, $\text{CH}_3\text{-(CH}_2\text{)}_4\text{-CH}_2\text{-}$, derived from hexane.

hexyl glucoside hexyl β -D-glucopyranoside; a mild nonionic detergent, CMC 250 mM.

Heyrovský, Jaroslav (1890–1967), Czech physical chemist; Nobel Laureate in Chemistry (1959) 'for his discovery and development of the polarographic methods of analysis'.

Hf symbol for hafnium.

HFBA abbr. for heptafluorobutyric acid; a reagent used in automatic peptide sequence analysis. It renders the action of cyanogen bromide specific to the cleavage of tryptophanyl peptide bonds.

HFCS abbr. for high fructose corn syrup.

Hfr abbr. for high-frequency recombinant.

HFT abbr. for high-frequency transduction.

Hg symbol for mercury.

HGF 1 abbr. for hepatocyte growth factor. 2 abbr. for hyperglycemic-glycogenolytic factor; see **glucagon**.

HGF receptor see **hepatocyte growth factor receptor**.

HGG abbr. for human gamma-globulin.

hGH or **HGH** abbr. for human growth hormone (i.e. human somatotropin).

HGPRT abbr. for hypoxanthine/guanine phosphoribosyltransferase; an enzyme that is important for the metabolic activation of purines, e.g. 6-mercaptopurine. Its deficiency is the cause of **Lesch-Nyhan syndrome**.

HHT abbr. for hydroxyheptadecatrienoic acid.

5-HIAA abbr. for 5-hydroxyindoleacetic acid.

hidden (immunological) determinant or **cryptodeterminant** any antigenic determinant so positioned that it is cryptic, i.e. normally not accessible for recognition by lymphocytes or antibody, but that becomes accessible after some stereochemical change occurs, e.g. by breakage, decomposition, or denaturation of the antigen. See also **cryptotope**.

hide the skin of an animal, especially the tough, thick skin of larger mammals. Powdered, dried hide is used in assays of some proteinases.

high-density lipoprotein abbr.: HDL; one of the classes of lipoproteins found in blood plasma in many animals (data normally relate to humans). They are also known as α lipoproteins, from having the highest electrophoretic mobility of the lipoproteins. HDL particles are the smallest of the blood lipoproteins: diameter 7.5–10 nm; solvent density for isolation 1.063–1.21 g mL⁻¹; hydrated density 1.21 g mL⁻¹. Their approximate composition (% by weight) is 6% unesterified cholesterol, 13% esterified cholesterol, 28% phospholipid, 3% triacylglycerol, 50% protein. Their apolipoprotein composition (% by weight total apolipoprotein) is 67% A-I, 22% A-II, 5–11% C-I + C-II + C-III, 1–2% E-II + E-III + E-IV, trace of D. They are synthesized in liver as precursor molecules, which

undergo modification in plasma to the mature molecules, especially as a result of the action of **lecithin-cholesterol acyltransferase**. They appear to function in reverse transport of cholesterol from tissues to liver. See also **apolipoprotein** for individual apolipoproteins.

high-energy bond symbol: ~; a term formerly sometimes used by biochemists for a chemical bond whose hydrolysis is accompanied by an unusually large (>20 kJ mol⁻¹) negative standard free energy change. Although long used, the term has been recognized as incorrect and misleading, since it suggests that such a bond actually contains an unusually large amount of energy and fails to take into account the various factors that contribute to the large negative free energy change.

high-energy compound a term formerly sometimes applied by biochemists to any compound containing one or more so-called **high-energy bonds**, e.g. ATP, phosphoenolpyruvate, acetyl-coenzyme A, etc.

high-frequency recombinant any bacterial strain that has a high frequency of recombination because the **F plasmid** has become incorporated into the bacterial chromosome. See also **Hfr**.

high-frequency transduction abbr.: HFT; any transduction in which the transducing phage(s) constitute a large proportion of the total phage population.

high-fructose corn syrup abbr.: HFCS; a mixture of fructose and glucose used as a sweetener in soft drinks, etc.; it is sometimes called **high fructose syrup** or, especially in Europe, **isoglucose**. The breakdown of corn starch to glucose (plus oligosaccharides) is followed by the action of **glucose isomerase** yielding a mixture consisting mainly of fructose and glucose. The product thus manifests the greater sweetness of fructose compared with glucose. The content is typically 42% fructose, 51% glucose, 5% maltose, and 2% other oligosaccharides.

highly conserved see **conserved**.

highly repetitive DNA a form of DNA that consists of short tandemly repeated sequences. It is found outside coding regions, and its role is unknown. Such DNA forms a significant proportion of most mammalian genomes.

high mobility group proteins abbr.: HMG proteins; small nuclear proteins that bind to nucleosomal DNA and probably maintain genes in a transcribable conformation. The name derives from their behaviour on electrophoresis. Example, HMG 14 from human: database code HG17_HUMAN, 89 amino acids (9.25 kDa).

high-performance capillary electrophoresis electrophoresis through a long coiled capillary column that gives very high resolution of e.g. peptides. See also **capillary electrophoresis**.

high-performance liquid affinity chromatography abbr.: HPLAC; a technique that combines the use of bioaffinity supports and **high-pressure liquid chromatography**.

high-performance liquid chromatography abbr.: HPLC; an alternative name for **high-pressure liquid chromatography**.

high-potential iron-sulfur protein abbr.: Hipip or HiPIP or HiPiP; a type of **ferredoxin**, originally found in the photosynthetic bacterium *Chromatium vinosum*, that has a very much higher redox potential ($E_0' \approx 0.3$ V) and different magnetic properties compared with other ferredoxins. The term has been used increasingly, but inappropriately, for other ferredoxins that, like the *Chromatium* protein, are able to form a higher than usual oxidation level but do not have high redox potentials.

high-pressure liquid chromatography or **high-performance liquid chromatography** abbr.: HPLC; a technique of **column chromatography** that is rapid and provides high resolution. It can be used with the various modes of liquid chromatography, the liquid being forced at high pressure, normally up to 420 kg cm⁻² (6000 psi), through the column (which is usually of stainless steel) and thence through a detector into a fraction collector. Because of the ability to use high pressures, the particles of the support materials used for the stationary phase can be of very small diameter (3–10 μ m), which greatly improves resolution.



high-speed supernatant

high-speed supernatant a jargon term for the supernatant resulting from high-speed centrifugation ($\approx 100\,000\text{ g}$).

high-spin complex any of the complex transition-metal ions having a maximal number of unpaired electrons in the d- or f-shell. This state occurs when pairing energy is higher than the crystal field splitting energy. Compare **low-spin complex**.

high-voltage electron microscope a type of electron microscope that employs accelerating voltages of 1 MeV, ten times that of previous instruments, enabling images of thick sections and even intact cells to be produced without significant loss of resolution.

high-voltage electrophoresis *abbr.*: HVE; a technique of paper or thin-layer electrophoresis that results in better and more rapid separations (10–60 min) than the standard technique, which suffers from minimal diffusion of low- M_r components. The high-voltage technique employs currents of up to 500 mA and potential gradients of up to 200 V cm^{-1} . The high field strength requires cooling of the support and adequate protection devices to be incorporated into the apparatus.

Hildebrand solubility parameter *symbol*: δ ; a measure of what is commonly called the polarity of a solvent; nonpolar solvents have low δ values while polar solvents have large values. In liquid–liquid chromatography the Hildebrand parameter indicates the relative position of a solvent in an elutropic series. The parameter is the sum of a number of factors:

$$\delta = \delta_d + \delta_o + \delta_a + \delta_h,$$

where δ_d is a measure of interactions due to **dispersion forces**, δ_o is a measure of interactions due to dipole interactions, δ_a is a measure of the ability of the solvent to interact as a hydrogen acceptor, and δ_h is a measure of its interactions as a hydrogen donor. [After Joel Henry Hildebrand (1881–1983), US chemist.]

Hill 1 (Sir) Archibald Vivian Hill (1886–1977), British physiologist and biophysicist; Nobel Laureate in Physiology or Medicine (1922) 'for his discovery relating to the production of heat in the muscle' [prize shared with O. F. Meyerhof].
2 Robert ('Robin') Hill (1899–1991), British plant biochemist noted for his studies of the mechanism of photosynthesis.

Hill coefficient or **Hill constant** *symbol*: h or n_H ; see **Hill equation** and **Hill plot**.

Hill equation 1 an equation,

$$y = K[x]^h / (1 + K[x]^h),$$

used to express the binding of a ligand to a (macro)molecule. It may be rearranged to give:

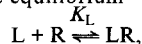
$$\log[y/(1-y)] = -\log K + h \log[x],$$

where y , the **fractional saturation**, is the fraction of the total number of binding sites occupied by ligand, $[x]$ is the free (unbound) ligand concentration, K is a constant, and h is the Hill coefficient. **2 (in pharmacology)** an equation derived from the above relating the effect of a drug to its concentration. For an effect, E , produced when an agent, A , is applied at a concentration $[A]$, then the relationship between E and $[A]$ may often be described empirically by the equation

$$E/E_{\max} = [A]^h / ([A]_{50}^h + [A]^h),$$

where $E_{\max}$ is the maximal value of E and $[A]_{50}$ is the concentration that produces an effect that is 50% of $E_{\max}$. [After A. V. Hill.]

Hill–Langmuir equation (in pharmacology) a variation of the **Langmuir absorption isotherm**; for a ligand, L , binding to a receptor, R , according to the equilibrium



$$p_{LR} = [L] / (K_L + [L]),$$

where p_{LR} is the fraction (proportion) of binding sites occupied by a ligand at equilibrium and K_L is the dissociation equilibrium constant.

Hill plot a plot of the **Hill equation** in the form $\log[y/(1-y)]$

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against $\log[x]$, where y is the fractional saturation with ligand and $[x]$ is the free (unbound) ligand concentration. The slope at any one point, h , is the Hill coefficient at that value of y . The values of h along the plot, which usually do not exceed the number of binding sites on each molecule and which tend to unity as y tends to 0 or 1, give an indication of the nature and degree of cooperativity (interaction) between the various binding sites on the molecule. [After A. V. Hill.]

Hill reaction the phenomenon in which isolated chloroplasts evolve dioxygen when illuminated in the presence of a suitable electron acceptor, e.g. ferricyanide or ferrioxalate, with a concomitant reduction of the electron acceptor. [After R. Hill.]

Hill reagent any electron acceptor that functions in a **Hill reaction**, e.g. ferrioxalate or ferricyanide.

hinge region 1 a flexible, proline-rich region of an **immunoglobulin** molecule, adjacent to the two disulfide bonds linking the two heavy chains together. It probably acts as a hinge on which the **Fab fragments** can rotate allowing the immunoglobulin molecule to take up a Y-shape when it reacts with antigen. The hinge region is near the sites of action of papain and pepsin. **2** the region in the helical, rod-shaped part of a **myosin** molecule that is susceptible to cleavage by trypsin to give light meromyosin and heavy meromyosin. **3** a region in any protein about which rigid body motions occur.

Hipip or **HiPIP** or **HiPiP** *abbr.* for high-potential iron–sulfur protein.

hippocalcin a neuron-specific Ca^{2+} -binding protein with **EF-hands**, similar to recoverin. It is expressed exclusively in the pyramidal layer of the hippocampus. Example from human: database code HIPP_HUMAN, 192 amino acids (22.30 kDa).

hirudin a dried extract obtained from leeches and possessing anticoagulant activity. The name is also applied to the purified active component, a 65-amino-acid polypeptide with three disulfide bridges; this has highly specific antithrombin properties, inactivating thrombin by blocking substrate-binding groups. Example from the leech *Hirudo medicinalis*: database code ITH1_HIRME, 65 amino acids (6.97 kDa).

hirudisin a recombinant **hirudin** variant engineered to inhibit α -thrombin and exhibit **disintegrin** activity. Hirudisins have antiplatelet aggregation activity. Replacement of the hirudin sequence Ser-Asp-Gly-Glu (residues 32–35) by Arg-Gly-Asp-Ser (RGDS) yields hirudisin, or by Lys-Gly-Asp-Ser yields hirudisin-1. The integrin-binding RGD motif of hirudisin is implicated in its antiplatelet action.

hirulog any synthetic peptide based on **hirudin** and designed as an inhibitor of the thrombin catalytic site, and exhibiting specificity for the anion-binding site of thrombin.

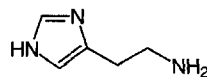
hirstasin an **antistatin**-like serine proteinase inhibitor isolated from the medicinal leech, *Hirudo medicinalis*.

His *symbol* for a residue of the α -amino acid **L-histidine** (alternative to H).

hist+ a variant form of **histo+** (before a vowel).

histaminase see **amine oxidase (copper-containing)**.

histamine 2-(4-imidazolyl)ethylamine; a compound formed by the decarboxylation of L-histidine and present in many mammalian tissues, with especially high concentrations in lung, skin, and intestine; it is stored in the granules of tissue mast cells and circulating basophil cells. Histamine is a potent vasodilator; it also increases capillary permeability, causes contraction of smooth muscle, plays a role in the regulation of gastric secretion, and acts as a mediator in allergic and anaphylactic conditions.



histamine receptor any of the receptors in mammalian tissues that interact specifically with **histamine**. Histamine receptors



histidase

may be divided into three categories. **H1 receptors** mediate the stimulating effects of histamine on smooth muscle in the gut and bronchi and most of the depressor effects on blood pressure. They are antagonized (blocked) by such classical **antihistamines** as pyrilamine (mepyramine) and promethazine. The **H1** receptor has seven transmembrane domains coupled to the phosphatidylinositol- Ca^{2+} second messenger system. Example: database code HHIR_HUMAN, 487 amino acids (55.78 kDa). **H2 receptors** mediate the stimulation of gastric acid secretion, the increase in the contraction frequency of cardiac muscle, the increase in cyclic AMP stores in basophil and mast cells, the inhibition of release of histamine by mast cells, and the remaining depressor effects on blood pressure. They are antagonized by, e.g., **cimetidine**. In humans the **H2** receptor has seven transmembrane domains, positively coupled to adenylate cyclase. Example: database code HH2R_HUMAN, 359 amino acids (40.1 kDa). **H3 receptors** have a receptor site characterized pharmacologically by radioligand binding; second messenger coupling and structure are unknown. They are associated with presynaptic nerve terminals and attenuate the release of neurotransmitters, e.g. histamine, acetylcholine, and norepinephrine. They are antagonized and blocked, e.g., by thioperamide and clobenpropit.

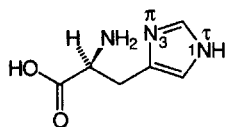
histidase an alternative name for **histidine ammonia-lyase**.

histidinal α -amino- β -4-imidazolepropionaldehyde; 2-amino-3-(1*H*-imidazol-4-yl)propanal; $(\text{C}_3\text{H}_3\text{N}_2)\text{CH}_2\text{CH}(\text{NH}_2)\text{CHO}$; the α -amino-aldehyde analogue of histidine. It is the immediate biosynthetic precursor of histidine.

histidinase an alternative name for **histidine ammonia-lyase**.

histidinate **1** histidine anion; the anion, $(\text{C}_3\text{H}_3\text{N}_2)\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}^-$, derived from histidine. **2** any salt of histidine anion. **3** any ester of histidine.

histidine the trivial name for β -4-imidazolylalanine; α -amino- β -imidazole-4-propionic acid; 2-amino-3-(1*H*-imidazol-4-yl)propanoic acid; $(\text{C}_3\text{H}_3\text{N}_2)\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$; a chiral α -amino acid. L-Histidine (symbol: H or His), (S)-2-amino-3-(1*H*-imidazol-4-yl)propanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: CAC or CAU. It is also a component of the isopeptide **carnosine**. In mammals it is an essential dietary amino acid and is glucogenic. D-Histidine (symbol: D-His or DHis), (R)-2-amino-3-(1*H*-imidazol-4-yl)propanoic acid, is not known to occur naturally. N-1 and N-3 of the imidazole ring are designated τ and π respectively.



L-histidine

histidine ammonia-lyase EC 4.3.1.3; *systematic name*: L-histidine ammonia-lyase; *other names*: histidase; histidine α -deaminase; histidinase. An ammonia-lyase (deaminase) enzyme that catalyses the deamination of L-histidine to urocanate. It is widely distributed in animals, plants, and microorganisms; in plants it is present in **glyoxysomes**. The activity of this enzyme is defective in **histidinemia**.

histidine (base) a histidine residue in which the imidazole group is unprotonated.

histidine decarboxylase *abbr.*: HDC; EC 4.1.1.22; *systematic name*: L-histidine carboxy-lyase. A pyridoxal phosphate enzyme that catalyses the decarboxylation of L-histidine to histamine. It is similar to dopa and glutamate decarboxylases. Example from human (homodimer): database code DCHS_HUMAN, 662 amino acids (74.10 kDa).

histidinemia or (*esp. Brit.*) **histidinaemia** a rare, hereditary,

human metabolic disorder in which abnormally high levels of histidine are found in the blood and urine. It is due to a defect in the activity of **histidine ammonia-lyase**.

histidine tail a sequence of (usually) about six histidine residues that is deliberately introduced at the C terminus of certain recombinant proteins. The purpose is to allow purification of the recombinant product by a type of affinity chromatography using a nickel chelating resin. Only the histidine-tailed proteins are retained by the column and these may then be eluted using a buffer containing imidazole.

histidinium **1** a term normally used to mean histidinium(1+); the (imidazolinium) monocation of histidine. In theory, the term denotes any ion or mixture of ions formed from histidine and having a net charge of +1, although the species $(\text{C}_3\text{H}_4\text{N}_2^+)\text{CH}_2\text{CH}(\text{NH}_3^+)\text{COO}^-$ generally predominates in practice. **2** the *systematic name* for histidinium(2+); the dication of histidine.

histidino **1** N^2 -histidino; the alkylamino group, $(\text{C}_3\text{H}_3\text{N}_2)\text{CH}_2\text{CH}(\text{COOH})\text{NH}-$, derived from histidine by loss of a hydrogen atom from its α -amino group. **2** either of the two possible groups, N^{τ} -histidino or N^{π} -histidino, $\text{HOOC-CH}(\text{NH}_2)\text{CH}_2\text{-(C}_3\text{H}_2\text{N}_2\text{)-}$, derived from histidine by loss of a hydrogen atom from its position on one or other of the nitrogen atoms of the imidazole ring.

histidinol 4-(2-amino-3-hydroxypropyl)imidazole; 2-amino-3-(1*H*-imidazol-4-yl)propanol; $(\text{C}_3\text{H}_3\text{N}_2)\text{CH}_2\text{CH}(\text{NH}_2)\text{CH}_2\text{OH}$; the α -amino-alcohol analogue of histidine. It is an intermediate, together with its phosphate ester, in the biosynthesis of histidine, and a powerful reversible inhibitor of protein synthesis.

histidinol dehydrogenase *abbr.*: HDH; EC 1.1.1.23; L-histidinol:NAD⁺ oxidoreductase. An enzyme that catalyses the oxidation of two molecules of NAD⁺ of L-histidinol to L-histidine with reduction of the NAD⁺ to NADH. Example from the *his* operon of *Salmonella typhimurium*: database code HISX_SALTY, 434 amino acids (45.85 kDa); the protein is a dimer and bifunctional as it also has an aldehyde dehydrogenase site, histidinol; being an intermediate. See also **Salmonella mutagenicity test**.

histidyl the acyl group, $(\text{C}_3\text{H}_3\text{N}_2)\text{CH}_2\text{CH}(\text{NH}_2)\text{CO}-$, derived from histidine.

histo+ or (*before a vowel*) **hist+** *comb. form* denoting tissue.

histochemistry the chemistry of the tissues, as studied with a combination of the methods of chemistry and histology.

histocompatibility the extent to which an organism will tolerate a graft of a foreign tissue. —**histocompatible** *adj.*

histocompatibility antigen any of the genetically determined **isoantigens** present on the surface of many animal cells that determine the immune reactions to tissue grafts. If tissue is grafted onto a recipient of the same species that does not carry the same histocompatibility antigens as the graft material, an immune response may be provoked in the recipient leading to graft rejection. See also **HLA histocompatibility system**, **major histocompatibility complex**.

histocompatibility gene any gene determining the formation of a **histocompatibility antigen**. See **HLA histocompatibility system**, **major histocompatibility complex**.

histoelectrofocusing a technique in which an unfixed frozen section of tissue, 25–40 μm thick, is applied to a support gel and then subjected to **electrofocusing**. The proteins and enzymes present in the tissue move into the gel and may be revealed by staining or zymographic techniques.

histogram a diagram representing a frequency distribution. It consists of a number of contiguous rectangles whose widths are proportional to the class interval under consideration and whose heights are proportional to the frequency associated with each class.

histohematin or (*esp. Brit.*) **histohaematin** a term originally used (1884) by Charles Alexander MacMunn (1852–1911) for the pigments now known as **cytochromes**. It included **myohematin**.



histone any of a group of evolutionarily highly conserved basic proteins, of molecular mass 11–21 kDa, that constitute about half the mass of the chromosomes of all eukaryotic cells, except spermatozoa. They comprise single polypeptide chains with a (lysine plus arginine) content of about 25%, and are concerned in the packing of DNA in **chromatin**. The principal histones may be classified into five types as shown in the table.

Type	Other names	Lys/Arg ratio	M_r	Location
H1	F1,Ia	20.0	21 000	linker
H2A	F2a2,IIB1	1.25	14 500	core
H2B	F2b,IIB2	2.25	13 800	core
H3	F3,III	0.72	15 300	core
H4	F2a1,IV	0.79	11 300	core

The octomer (H2A,H2B,H3,H4)₂ forms the **nucleosome** core. Examples from yeast: H2A.2, database code H2A2_YEAST, 131 amino acids (13.84 kDa); H2B.1, database code H2B1_YEAST, 130 amino acids (14.10 kDa); H3, database code H3_YEAST, 135 amino acids (15.21 kDa); H4, database code H4_YEAST, 102 amino acids (11.22 kDa). H1 is associated with nucleosomes, especially the linking DNA; example H1A (human): database code H1A_HUMAN, 222 amino acids (22.15 kDa). *See also* **H-NS**.

hitchhiking 1 a jargon term for the sequence of events in which a genetic element (or **transposon**) inserts into an active secondary site on a replicon which is then transferred to a recipient, the transposon then coming immediately to occupy its normal primary site in the DNA of the recipient organism. 2 the spread of a neutral allele through a population due to its linkage with a beneficial allele that is subject to selection.

Hitchings, George Herbert (1905–98), US biochemist and pharmacologist distinguished for his introduction, largely in collaboration with his colleague G. B. Elion, of a range of widely used synthetic drugs designed as antimetabolites, including the antifolate bactericidal agent co-trimoxazole, the immunosuppressants mercaptopurine and its derivative azathioprine, and the xanthine-oxidase inhibitor allopurinol; Nobel Laureate in Physiology or Medicine (1988) jointly with J. W. Black and G. B. Elion 'for their discoveries of important principles for drug treatment'.

HIV *abbr.* for human immunodeficiency virus; a **retrovirus**, the causative agent of **AIDS**. Two such viruses are known: HIV-1, found worldwide, and HIV-2, which is much less common, being found mainly in West Africa. In terms of the gene structure of the **viral envelope sequences** (*see* **HIV protein**) HIV-1 and HIV-2, although both members of the Lentivirus family of retroviruses, are not closely related. HIV-2 is closely related to a **simian immunodeficiency virus** (SIV) common in sooty mangabeys and probably originated there. HIV-1, which probably originated from a non-human primate in Africa, is closely related to SIV from chimpanzees. However, since the rate of SIV infection in chimpanzees is very low, it is not clear that they represent the natural host of these viruses. HIV is subclassified into eight alphabetical subtypes or clades, plus one orphan clade. The reverse transcriptase enzyme of HIV is highly error-prone, leading to a high mutation rate. Thus these viruses evolve very quickly even within single patients, and drug-resistant mutations appear rapidly. For HIV-1 to enter its target cells, it must not only bind to the cell-surface protein **CD4** but also to one or more chemokine receptors. One of these, named CCR5, is effective in the early stages of infection. Patients with defective CCR5 cannot be infected with HIV-1. During the later stages of infection, another chemokine receptor, CXCR4 (*former name*: fusin) dominates.

HIV-1 protease an aspartyl protease that is a homodimer of two subunits each containing 99 amino acids, with cysteines at positions 67 and 95. Its proteolytic function is essential for the life cycle of human immunodeficiency virus (HIV) and is, therefore, a target for the design of inhibitors as therapeutic drugs, several examples of which are in use.

HIV protein any of the proteins encoded by the human immunodeficiency virus (HIV). The HIV-1 genome (9.5 kb) is transcribed to give 12 different mRNAs generated by alternative splicing. The proteins are conventionally named for the gene and/or their size (kDa) with the convention that p = protein and gp = glycoprotein. The primary translation products are: Gag precursor (p53); Pol precursor; Env precursor (gp160); Nef (p27; negative regulator); Vif (p23); Tat (p14; regulatory protein); Rev (p19; regulatory protein); Vpr (p18); and Vpu (p15). The Gag precursor is cleaved to myristoylated Gag protein (p17), structural protein (p24), and p15; p15 is the precursor for RNA-binding protein (p7) and proline-rich protein (p9). The Pol precursor is cleaved to protease (p10), reverse transcriptase (p51), and endonuclease (p32). The Env precursor is cleaved to extracellular protein (gp120) and transmembrane protein (gp41). Protease, reverse transcriptase, and endonuclease are synthesized as part of a polyprotein: database code POL_HV1BR, 1015 amino acids (115.03 kDa). *See also* **retrovirus**.

HLA *abbr.* for human leukocyte-associated antigen. *See also* **histocompatibility antigen**, **major histocompatibility complex**.

HLA histocompatibility system a family of histocompatibility antigens encoded by the complex of genetic loci known as the **major histocompatibility complex** (MHC). It is the major histocompatibility system in humans, equivalent to the H-2 histocompatibility system in mice. The MHC genes control the presence of **HLA isoantigens** on cell surfaces. These antigens are of primary importance in tissue grafting since HLA-incompatible grafts are more likely to be rejected than incompatible grafts of other isoantigen groups.

HLH *abbr.* for helix-loop-helix (*see* **helix-turn-helix**).

hly genes *see* **hemolysin**.

HMC *abbr.* for 5-hydroxymethylcytosine.

H-meromyosin *abbr.* for heavy meromyosin.

hMG or HMG *abbr.* for human menopausal gonadotropin (i.e. human urogonadotropin).

HMG 1 *abbr.* for 2-hydroxy-2-methylglutaryl. 2 *abbr.* for high mobility group of nuclear proteins. 3 *see* **hMG**.

HMG-CoA *abbr.* for hydroxymethylglutaryl-CoA.

HMG-CoA lyase *abbr.* for hydroxymethylglutaryl-CoA lyase.

HMG-CoA reductase *abbr.* for hydroxymethylglutaryl-CoA reductase.

HMG-CoA synthase *abbr.* for hydroxymethylglutaryl-CoA synthase.

HMM *abbr.* for heavy meromyosin.

HMA *abbr.* for hydroxymethoxymandelate.

HMP *abbr.* for hexose monophosphate.

HMP pathway or HMP shunt *abbr.* for hexose monophosphate pathway (shunt) (i.e. **pentose phosphate pathway**).

HNPCC a human gene that may normally function in the repair of regions of mismatch of DNA. It is named from hereditary nonpolyposis colorectal cancer, a condition associated with mutations in this gene.

hnRNA or HnRNA *abbr.* for heterogeneous nuclear RNA (alternatives to H-RNA).

hnRNP *abbr.* for heterogeneous nuclear ribonucleoprotein.

hnRNP particles *abbr.* for heterogeneous nuclear ribonucleoprotein particles; particles in the nucleus containing **hnRNA** bound to protein.

H-NS bacterial proteins of several families that are basic DNA-binding proteins, possibly the prokaryotic counterparts of **histones** in eukaryotic chromatin. Their role remains unclear. They have the ability to bind to curved DNA and may have a role in transcriptional regulation. The N-terminal tetrapeptide, consensus Ser-Glu-Xaa-Leu, is characteristic. Example from *Escherichia coli*: protein H1, dimer, database code HNS_ECOLI, 136 amino acids (15.39 kDa). The name derives from an isolation procedure extracting them from the 30S ribosomal native subunit.

Ho symbol for holmium.

Hodgkin 1 (Sir) Alan Lloyd (1914–98) British physiologist

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renowned for his researches on nerve-impulse conduction; Nobel Laureate in Physiology or Medicine (1963) jointly with J. C. Eccles and A. F. Huxley 'for their discoveries concerning the ionic mechanisms involved in excitation and inhibition in the peripheral and central portions of the nerve cells membrane'. **2 Dorothy (Mary) Crowfoot-Hodgkin** née Crowfoot, (1910–94), British chemist and crystallographer renowned for her elucidation by X-ray crystallography of the molecular structures of benzylpenicillin (formerly termed penicillin G) and cyanocobalamin (a version of vitamin B₁₂) and in particular of the complete three-dimensional structure of crystalline zinc insulin (announced in 1969, having – as D. M. Crowfoot – first observed its diffraction pattern in 1934); Nobel Laureate in Chemistry (1964) 'for her determination by X-ray techniques of the structures of important biochemical substances', e.g. penicillin, vitamin B₁₂, and insulin.

Hoechst 33258 *see* bisbenzimidazole.

Hoechst 33342 *see* bisbenzimidazole.

Hoff *see* van't Hoff.

Hofmeister series or **lyotropic series** an arrangement of either the simple anions or the simple cations (or of the salts of such ions) in order of their ability to remove lyophilic substances from colloidal solutions. Each series is an index of, *inter alia*, the relative abilities of the ions to salt-out proteins from solution. The order for anions is: citrate³⁻ > tartrate²⁻ > acetate⁻ > NO₃⁻ > ClO₄⁻ > CNS⁻, and that for cations is: Mg²⁺ > Ca²⁺ > Sr²⁺ > Ba²⁺ > Li⁺ > Na⁺ > K⁺ > Rb⁺ > Cs⁺. [After Franz Hofmeister (1850–1922).]

Hofstee plot *see* Eadie-Hofstee plot.

Hogness box an alternative name for **TATA box**.

hol+ a variant form of (before a vowel) **holo+**.

holandric inherited solely in the male line. *Compare* **hologynic**.

holdback agent or **holdback carrier** any nonradioactive substance that is added to a sample to prevent either coprecipitation or adsorption of some soluble radioactive substance.

hold-up volume *symbol*: V_{M} (in chromatography) the volume of eluent required to elute a component whose concentration in the stationary phase is negligible relative to that in the mobile phase. In gas chromatography, the volume of carrier gas (eluent) is specified at the same temperature and pressure as the total retention volume.

hole (in physics) the absence of an electron in a normally filled valency structure of a crystalline **semiconductor** that behaves as a carrier of charge and is mathematically equivalent to a positron.

Holley, Robert William (1922–93), US biochemist and molecular biologist noted for his establishment of the structure of alanine transfer RNA, this being the first occasion on which the sequence of a nucleic acid had been determined; Nobel Laureate jointly with H. G. Khorana and M. W. Nirenberg (1968) for their 'interpretation of the genetic code and its function in protein synthesis'.

Holliday junction or **Holliday structure** one of the junctions between four strands of DNA that are important intermediates in **genetic recombination**. They are part of the mechanism for **crossing over** during the long prophase of meiotic cell division I, when four strands of DNA are aligned. The process involves breaking the DNA double helix in a maternal chromatid and in a homologous paternal chromatid and then exchanging fragments between the two non-sister chromatids in a reciprocal fashion. At this point, in each four strands, a paternal strand links across to a maternal strand, forming the junction known by the name of its proposer. [After Robin Holliday (1932–), British biologist.]

hollow-cathode discharge lamp a radiation source used in **atomic absorption spectrophotometry** to produce a spectrum of radiation specific to the element being assayed. A sample of the element in question is held in a metal cup-shaped cathode, which, together with a tungsten anode, is enclosed in a glass or quartz envelope filled with argon at low pressure. A high voltage is used to produce an arc spectrum of the element.

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hollow-fibre technique a technique in which bundles of semi-permeable, hollow fibres with pores of controlled dimensions that function as molecular sieves are used for concentrating, desalting, dialysing, and fractionating macromolecules in solution. *Compare* **capillary filtration**.

holo+ or (before a vowel) **hol+** *comb. form* whole, entire, complete; wholly, entirely, completely.

holocellulose the complex mixture of polysaccharides remaining after the removal of lignin from tree-wood by treatment with sodium chlorite solution. It consists of a mixture of **cellulose** and **hemicellulose**.

holocrine **1** describing a type of secretion that involves disintegration of the entire secretory cell to release the secretory products. **2** describing a gland made up of such cells. *Compare* **apocrine**, **eccrine**, **merocrine**.

holoenzyme a catalytically active complex comprising the protein part of an enzyme (**apoenzyme**) combined with the appropriate cofactor or cofactors.

hologram **1** the pattern produced when light, or other electromagnetic radiation, that has been reflected, diffracted, or transmitted by an object placed in a coherent beam, interferes with a reference beam related in phase to the first beam. **2** a photograph of such a pattern.

holograph **1** to make a **hologram** (def. 1, 2). **2** any book or other document handwritten by its author.

holography the process of making and using **holograms** (def. 1, 2). —**holographic** *adj.*

hologynic inherited solely in the female line. *Compare* **holandric**.

Holometabola or **Endopterygota** one of the two divisions of the subclass Pterygota, comprising those insects, e.g. butterflies, moths, flies, and beetles, in which the juvenile forms (larvae) are very different from the adults. Such insects undergo complete metamorphosis during a resting stage (pupa). *Compare* **Hemimetabola**.

holoprotein the functional form of a protein containing a protein part (**apoprotein**) together with any appropriate ligand or ligands.

holorepressor a functional repressor protein consisting of an **aporepressor** plus **corepressor** complex.

homeo+ or **homoeo+** or **homoio+** *comb. form* denoting like or similar.

homeobox a short (180 bp) conserved DNA sequence that encodes a DNA-binding motif famous for its presence in genes that are involved in orchestrating the development of a wide range of organisms. *See also* **homeotic gene**.

homeodomain the DNA-binding motif of 60 amino acids encoded by a **homeobox**.

homeomorphism or **homoeomorphism** (in chemistry) the property shown by certain substances of having the same crystal form but different compositions. —**homeomorphic**, **homeomorphous**, or **homoeomorphic**, **homoeomorphous** *adj.*

homeostasis or **homoeostasis** the maintenance of a relatively constant internal environment in the bodies of higher animals by means of a series of interacting physiological and biochemical processes. —**homeostatic** or **homoeostatic** *adj.*

homeotic gene or **homeodomain-encoding gene** or **homeobox-encoding gene** or **Hox gene**; any of the genes that are allelic for mutations resulting in the conversion of one body part into another. Homeotic genes were first discovered in *Drosophila*; a mutation in such a gene could, e.g., result in a leg replacing an antenna. While *Drosophila* has only two *Hox* clusters, in nematodes there is one, and in vertebrates four clusters of 9 to 11 genes each located on a separate chromosome and spanning more than 100 kb. The vertebrate *Hox* genes are expressed in specific patterns and at particular stages in embryogenesis. They are activated by **retinoic acid** and **sonic hedgehog protein**. In the limb these are active in concert with **fibroblast growth factor**. The degree of homology of the *Hox* genes in different species is remarkably high. The structure of some homeodomains has been determined, as has their interaction with DNA.



homeothermic the usual US spelling of **homoiothermic**.

homeotypic pertaining to, or being the second, reductive division in **meiosis**. Compare **heterotypic**.

homo+ *comb. form* 1 the same, similar, alike. Compare **allo+** (def. 1); **hetero+** (def. 1); **iso+** (def. 1); **xeno+** (def. 1). 2 from, or directed towards, the same species. Compare **allo+** (def. 2); **hetero+** (def. 2); **iso+** (def. 2); **xeno+** (def. 2). 3 containing atoms, groups, linkages, residues, or subunits of the same kind. Compare **hetero+** (def. 3). 4 denoting the next higher homologue of a specified chemical compound, especially one whose molecules contain, in a hydrocarbon chain or ring, one more methylene, $-CH_2-$, than those of the parent compound, e.g. homocysteine, homoserine, homo-5 α -pregnane. Compare **nort** (def. 2).

Homo a genus of primates that includes modern man (*Homo sapiens*) and a number of more primitive, fossil species.

homoarginine (sometimes) a trivial name for *N*⁶-amidino lysine.

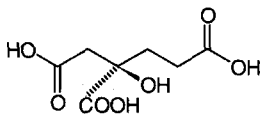
homobifunctional describing a chemical reagent that carries two identical reactive groups. Compare **heterobifunctional**.

homocarnosine (sometimes) a trivial name for *N*^α-(4-aminobutyl)histidine.

homochromatic a less common term for **monochromatic** (def. 1).

homochromatography a chromatographic technique for the separation of labelled oligonucleotides, in which a mixture of unlabelled oligonucleotides is used to displace the labelled components according to differences in size.

homocitrate 2-hydroxybutane-1,2,4-tricarboxylate; the (R) enantiomer is a component of the **aminoadipic pathway** and is formed by **homocitrate synthase**.

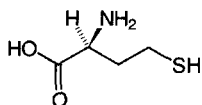


homocitrate synthase EC 4.1.3.21; *systematic name*: 2-hydroxybutane-1,2,4-tricarboxylate 2-oxoglutarate-lyase (CoA-acetylating). An enzyme that catalyses the formation of **homocitrate** from 2-oxoglutarate and acetyl-CoA and H_2O . See also **aminoadipic pathway**.

homoconjugation the association between a base and its conjugate acid through a hydrogen bond. See **conjugate acid-base pair**. Compare **heteroconjugation**.

homocyclic 1 describing any cyclic molecular structure containing atoms of a single element in the ring or rings; describing any compound having such a structure. The term is preferred to its synonym **isocyclic**; it includes **carbocyclic** (def. 1). Compare **heterocyclic** (def. 1). 2 any such compound.

homocysteine *symbol*: Hcy; α -amino- γ -mercaptobutyric acid; 2-amino-4-mercaptobutanoic acid; an important intermediate in the metabolic reactions of its *S*-methyl derivative, methionine. Its concentration in human serum is raised after stroke.



L-homocysteine

homocysteine methyltransferase one of several enzymes, e.g. the two alternative forms of enzyme synthesizing methionine: (1) 5-methyltetrahydrofolate-homocysteine *S*-methyltransferase. (EC 2.1.1.13); *other name*: tetrahydropteroylglutamate methyltransferase; it catalyses a reaction between

5-methyltetrahydrofolate and L-homocysteine to form tetrahydrofolate and L-methionine; cobalamin is a cofactor. Example, *metH* gene product in *Escherichia coli*: database code METH_ECOLI, 1226 amino acids (135.81 kDa). (2) 5-methyltetrahydropteroyltri-L-glutamate-homocysteine *S*-methyltransferase (EC 2.1.1.14); *other name*: tetrahydropteroyltri-L-glutamate methyltransferase; it catalyses a reaction between 5-methyltetrahydropteroyltri-L-glutamate and L-homocysteine to form tetrahydropteroyltri-L-glutamate and L-methionine. Example, *metE* gene product in *E. coli*: database code METE_ECOLI, 752 amino acids (84.91 kDa).

homocystine 2,2'-diamino-4,4'-dithiobis(butanoic acid); the disulfide oxidation product of **homocysteine**.

homocystinemia or (*esp. Brit.*) **homocystinaemia** inherited biochemical abnormality of humans, characterized by an abnormally high concentration of homocystine in the blood; this may be accompanied by a corresponding elevation of homocystine in the urine (**homocystinuria**); alternatively, homocystinuria may be the chief presenting sign. Either condition may be due to either: (1) a deficiency of cystathionine β -synthase (EC 4.2.1.22) activity (the deficiency preventing the condensation of homocysteine with serine to form cystathionine); or (2) a deficiency of the 5-methyltetrahydrofolate-dependent pathway for the remethylation of homocystine to methionine.

homodetic describing a cyclic peptide in which the ring consists solely of amino-acid residues in eupeptide linkage. Compare **heterodetic**.

homodimer any macromolecular structure in which two identical subunits (monomers or protomers) are noncovalently associated. See also **dimer**. Compare **heterodimer**.

homoduplex hybrid DNA involving two strands that are complementary. Compare **heteroduplex**.

homoeo+ a variant form of **homeo+**.

homogametic describing the sex whose cell nuclei contain a pair of similar **sex chromosomes**. The homogametic sex is usually female, but is male in the Lepidoptera, birds, some amphibians and fishes, and a few plants. Compare **heterogametic**.

homogenate any disrupted tissue preparation in which cell disruption is maximal while the destruction of intracellular organelles is kept to the minimum.

homogeneic an alternative term for **allogeneic**. Compare **heterogeneic**.

homogeneous 1 being uniform in kind, as in a homogeneous population. 2 (*describing a group of molecules*) uniform in structure or composition. 3 (*describing a system*), consisting of only a single chemical phase. Compare **heterogeneous**, **homogeneous**. —homogeneity *n*.

homogeneous catalysis any catalytic action or process in which the catalyst is uniformly distributed in the same chemical phase as the reactants. Compare **heterogeneous catalysis**.

homogeneous enzyme immunoassay or enzyme-multiplied immunoassay technique (*abbr.*: EMIT) a technique of **homogeneous immunoassay** in which an enzyme is covalently coupled to the substance to be assayed. The enzymic activity of the complex is inhibited by an antiserum specific for the substance and this inhibition is relieved by the presence of free molecules of the substance to be assayed. The enzyme acts as a molecular amplifier.

homogeneous immunoassay any **immunoassay** that does not involve physical separation of antibody-bound antigen from antigen molecules that remain free. Labels that may be attached to the antigen include: enzymes, cofactors, prosthetic groups or substrates of enzymes, chemiluminescent or fluorescent molecules, stable free radicals. Compare **heterogeneous immunoassay**.

homogeneously staining region a region in a chromosome that stains homogeneously, rather than with the banding pattern normally seen. The region contains massively amplified numbers of copies of a small segment of the genome. The amplified DNA may often consist of multidrug resistance gene DNA (see **MDR**).



homogenesis

homogenesis the production of offspring having the same characteristics in successive generations. *Compare heterogenesis.*

homogenetic describing chromosome pairing during meiosis when the pairing partners are derived from one of the original ancestors. *Compare heterogenetic.*

homogenic or **congenic** 1 describing a gamete that contains only one allele of a particular gene. 2 describing two genetic elements that are descended from a common ancestor by a known sequence of steps. 3 *an alternative word for homogenous.* *Compare heterogenic.*

homogenize or **homogenise** produce a **homogenate**. —**homogenization** or **homogenisation** *n.*

homogenizer or **homogeniser** any apparatus for producing a **homogenate**.

homogenous 1 of, pertaining to, or exhibiting **homogeny**. 2 *an alternative word for homogeneous.*

homogentisate or (formerly) **alcapton** 2,5-dihydroxyphenylacetate; an intermediate in the catabolism of tyrosine and in the biosynthesis of plastoquinone and tocopherol. It is excreted in abnormally large quantities in the urine in **alcaptonuria**. *See also homogentisate 1,2-dioxygenase.*

homogentisate 1,2-dioxygenase EC 1.13.11.5; *systematic name:* homogentisate: oxygen 1,2-oxidoreductase (deacylizing); *other names:* homogentisicase; homogentisate oxygenase; homogentisic acid oxidase. An enzyme that catalyses the oxidation by dioxygen of homogentisate to 4-maleylacetoacetate; iron(II) is a cofactor. In the catabolism of L-tyrosine, **homogentisate** is formed from 4-hydroxyphenylpyruvate; maleylacetoacetate is converted to fumarylacetoacetate, which is split to fumarate and acetoacetate.

homogeny (*in biology*) similarity in the structure of organisms or parts of organisms because of common ancestry.

homoglycan or **homopolysaccharide** any polysaccharide (i.e. glycan) that contains residues of only one kind of monosaccharide (i.e. glycoside) molecule. *Compare heteroglycan.*

homograft *an old term for allograft. Compare heterograft.*

homoio+ *a variant form of homeo+.*

homoiosmotic describing organisms with a constant internal osmotic pressure. *Compare poikilosmotic.*

homoiothermic or **homoiothermal** or (*esp. US*) **homeothermic** or **homeothermal** describing an organism (**poikilotherm**) that sustains a relatively constant body temperature, usually higher than that of its surroundings. *Compare poikilothermic.* —**homoiothermy** or (*esp. US*) **homeothermy** *n.*

homokaryon or **homocaryon** any cell with more than one nucleus, and in which the nuclei are all of the same genetic constitution; a tissue composed of such cells. *Compare heterokaryon.* —**homokaryotic** or **homocaryotic** *adj.*

homokaryosis or **homocaryosis** the condition of having a **homokaryon** or **homocaryons**. *Compare heterokaryosis.*

homolactic fermentation a type of fermentation of glucose to lactate via the **Emden-Meyerhof pathway**. *Compare heterolactic fermentation.*

homolog (*sometimes*) the *US* spelling of **homologue**.

homologous 1 having a related or similar position, structure, etc.; corresponding; exhibiting **homology**. 2 (*in chemistry*) describing compounds that form a series with successive constant differences in composition. 3 (*in biology*) of common ancestry; especially of organs and tissues that have a similar anatomical position and structure in different species by virtue of their common evolutionary origin, even though their functions may have come to differ; e.g. the wing of a bird and the forelimb of a reptile. *Compare analogous.* 4 (*in genetics*) describing chromosomes that pair during meiosis, each member of a homologous pair being a duplicate of one of the chromosomes contributed at syngamy by the mother or the father. 5 (*in biochemistry*) a (of sequences of residues in encoded macromolecules) having the same or similar residues at corresponding positions (*see also conserved*). With respect to proteins the term is used to imply a common evolutionary origin. Specifically this requires evidence, based on gene structure and

not merely a similarity of protein structure. **b** (of proteins from different species) having identical or similar functions.

Compare heterologous. —**homology** *n.*

homologous antibody the antibody elicited by a specified antigen.

homologous antigen the antigen that has elicited a specified antibody.

homologous desensitization *see desensitization.*

homologue or (*sometimes*) *US* **homolog** 1 (something) exhibiting homology. 2 (*in chemistry*) any member of a **homologous** (def. 2) series of compounds. 3 (*in biology*) any one of a series of **homologous** (def. 3) organs or structures. 4 (*in genetics*) **a** either members of a pair of **homologous** (def. 4) chromosomes. **b** either one of two genes in corresponding loci in homologous chromosomes.

homology domain any of the regions in an immunoglobulin involved in forming the **immunoglobulin fold**.

homolysis (*in chemistry*) the cleavage of a covalent bond in such a manner that each of the fragments between which the bond is broken retains one of the bonding electrons; e.g. A—B → A• + B•. —**homolytic** *adj.*

homooligomer any **oligomer** made up of only one kind of constitutional repeating unit. —**homooligomeric** *adj.*

homophilic binding (*of adhesion molecules*) binding of an adhesion molecule in one cell to an identical molecule in an adjacent cell. *Compare heterophilic binding.*

homopolymer any polymer made up of only one kind of constitutional repeating unit. For example, cellulose contains only glucose as the monomeric unit. —**homopolymeric** *adj.*

homopolymer tailing *a jargon term for a procedure useful in joining two types of duplex DNA molecules to form mixed dimeric circular DNA.* In this procedure, homopolymer sequences of one type, e.g. poly(dA), are added to the 3'-ends of one of the populations of DNA molecules, while complementary homopolymer sequences, e.g. poly(dT), are added to the 3'-ends of the other population of DNA molecules. The two types of molecules are then annealed to form mixed dimeric circles.

homopolysaccharide *an alternative name for homoglycan.*

homoserine *symbol:* Hse; α -amino- γ -hydroxybutyric acid; an intermediate in the biosynthesis of cystathionine, threonine, and methionine. It also occurs in bacterial peptidoglycans and, as its *O*-guanidino derivative, in canavanine. The term normally implies the *L* enantiomer.

homoserine deaminase *see cystathionine γ -lyase.*

homoserine dehydratase *see cystathionine γ -lyase.*

homoserine dehydrogenase EC 1.1.1.3; *systematic name:* L-homoserine:NAD(P)⁺ oxidoreductase; an enzyme of the pathways for the biosynthesis of methionine and threonine. It catalyses the formation of L-homoserine and NAD(P)⁺ from L-aspartate 4-semialdehyde and NAD(P)H. Example, a homotetrameric bifunctional enzyme from *Escherichia coli*: database code AK1H_ECOLI, 820 amino acids (89.02 kDa). Amino acids 1–249 form the aspartokinase I domain, amino acids 250–470 form a large 'interface', and 471–820 form homoserine dehydrogenase; regulation is by L-Thr. Another example is from *Corynebacterium glutamicum* and *Brevibacterium lactofermentum*: database code DHOM_CORGL, 445 amino acids (46.39 kDa). *See also aspartokinase/homoserine dehydrogenase.*

homoserine lactone *symbol:* Hsl; α -amino- γ -butyrolactone; a substance formed by the cleavage of methionine-containing peptides by cyanogen bromide.

homostерism or **homostery** the phenomenon in which a second molecule of normal substrate, or a structurally similar compound, combines at the catalytic site of an enzyme leading to a modification in the reaction of the bound intermediate. *Compare allosterism.* —**homosteric** *adj.*

homostery *an alternative term for homostерism.*

homothallic describing species, e.g. of certain fungi and algae, in which a sexual spore can result from the fusion of nuclei that are genetically distinct (i.e. not necessarily homozygous), but are derived from the same thallus; thus the species is self-fertile.



homothallic switching endonuclease

homothallic switching endonuclease a sequence-specific endonuclease, with a site on yeast chromosome III, that is involved in mating-type switching. It is a **zinc finger** protein. Database code HO_YEAST, 586 amino acids (66.06 kDa).

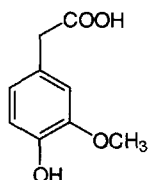
homotherm an alternative word for **homiotherm**.

homotope any residue that can take the place of another residue in a particular position in a polymer; e.g. the serine and glycine residues at position 9 of the A chain of bovine and ovine insulin, respectively, are homotopes.

homotransplant 1 an alternative term for **homograft**. 2 to effect such a transplant.

homotropic describing an **allosteric** effect in which interaction occurs between identical ligands; the effect appears to be always cooperative. The term is applied also to such an interaction in the case of an allosteric enzyme for which the same substance acts as both the substrate and the allosteric effector, etc. Compare **heterotropic** (def. 1).

homovanillic acid 4-hydroxy-3-methoxyphenylacetic acid; a metabolite of catecholamine metabolism, found in human urine. It is the principal urinary metabolite of dopa and dopamine. It is used as a reagent in the fluorometric determination of glucose oxidase and other oxidases.



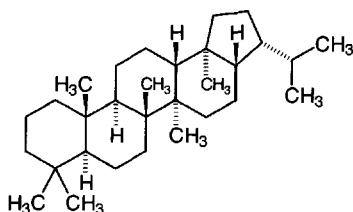
homozygosity 1 the union of gametes identical for one or more pairs of genes to form a **homozygote**. 2 or **homozygosity** the state or condition of being a homozygote. Compare **heterozygosity**.

homozygosity an alternative term for **homozygosity** (def. 2).

homozygote any cell or organism having identical genes (alleles) at one or more corresponding loci on homologous chromosomes. With respect to such loci, a homozygote will produce two identical gametes. Compare **heterozygote**. — **homozygotic** or **homozygous** adj.

Hoogsteen base pair a nucleic-acid base pair that differs from the Watson-Crick base pair. In the Hoogsteen adenine-thymine base pair the 6-NH₂ and N-7 of the adenine are hydrogen bonded respectively to the 4-O and H-1 of the thymine. The Hoogsteen guanine-cytosine pair requires that N-1 of the cytosine is protonated. In this base pair the 6-O and N-7 of the guanine are hydrogen bonded respectively to the 4-NH₂ and the protonated N-1 of cytosine. In Watson-Crick base pairs the glycosidic bond of both nucleotides has the *anti* conformation, but in the Hoogsteen pairs the glycosidic bond of the purine nucleotide has the *syn* conformation. Hoogsteen base pairs are particularly important in the structure of DNA triple helices, where the third strand forms Hoogsteen pairs to bases in the DNA duplex. [After Karst Hoogsteen (1923–), Dutch-American biochemist, who first described it.]

hopane the pentacyclic triterpene hydrocarbon parent of hopanoids and derivatives found in plant resins. See also **hopanoids**.

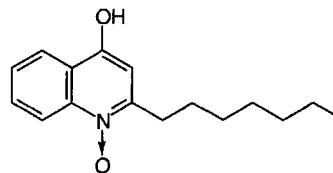


hopanoids pentacyclic sterol-like molecules based on the **hopane** nucleus. They are found in bacteria, some plants and lichens, and in sediments and crude oils, and are said to be the most abundant organic molecules in the biosphere (estimated total biomass 10¹¹ to 10¹² tonnes), thus outranking **cellulose**.

Hopkins, (Sir) Frederick Gowland (1861–1947), British biochemist renowned for his discoveries of essential amino acids and accessory food factors (later termed vitamins), for his isolation of glutathione and tryptophan, and for playing an important part in establishing biochemistry as an independent discipline; Nobel Laureate in Physiology or Medicine (1929) 'for his discovery of the growth-stimulating vitamins' [prize shared with Christiaan Eijkman].

Hopkins-Cole reaction a colour reaction for free or combined tryptophan, in which a violet colour is produced when a solution containing tryptophan is treated with glyoxylic acid (present in impure glacial acetic acid) and subsequently layered onto a solution of pure, concentrated sulfuric acid. [After F. G. Hopkins and Sydney William Cole (1877–1951).]

HOQNO or **HQNO** *abbr.* for 2-*n*-heptyl-4-hydroxyquinoline *N*-oxide; an inhibitor of the mitochondrial respiratory chain at cytochrome *bc*₁ and of photosynthetic electron flow immediately before cytochrome *b*₅₅₉LP.



hordein a type of **glutelin** found in barley.

hormone any substance formed in very small amounts in one specialized organ or group of cells and carried (sometimes in the bloodstream) to another organ or group of cells, in the same organism, upon which it has a specific regulatory action; an **endocrine** (def. 2). The term was originally applied to agents with a stimulatory physiological action in vertebrate animals (as opposed to a **chalone**, which has a depressant action). Usage is now extended to regulatory compounds in lower animals and plants, and to synthetic substances having comparable effects. Compare **autacoid**, **hormone**. See also **secretin**. — **hormonal** adj.

hormone cascade see **cascade sequence**.

hormone-sensitive lipase any enzyme of the sub-subclass EC 3.1.1 that, when activated in adipose tissue, catalyses the mobilization of fatty acids into the circulation. Such enzymes are activated by cyclic AMP-dependent phosphorylation under the influence of catecholamines, and are inactivated by insulin-dependent dephosphorylation. Example from human: database code LIPS_HUMAN, 786 amino acids (85.38 kDa).

hormonotoxin any conjugate of a peptide hormone and a peptide toxin, e.g. ovine luteinizing hormone-gelatin.

host 1 any organism in which another organism, especially a **parasite** or **symbiont**, spends part or all of its life cycle and from which it obtains nourishment and/or protection. 2 any organism that harbours a pathogenic or nonpathogenic infectious agent. 3 the recipient of a transplanted tissue or organ graft. 4 a cell or organism that contains recombinant DNA.

host cell any cell whose metabolism is used for the growth and reproduction of a virus.

host-cell reactivation *abbr.*: HCR; the restoration of activity in a UV-damaged DNA bacteriophage by the excision-repair mechanisms of the host cell in which it multiplies.

host-controlled modification or **host-controlled variation** or **host-induced modification** or **host-induced variation** any non-heritable change in the properties of a DNA bacteriophage re-



host-controlled modification

sulting from the phage DNA assuming the DNA **modification** pattern characteristic of the host bacterium. It is usually recognized by **restriction** in the efficiency of plating of the newly modified phage on its former host strain. Similar phenomena have been described for viruses infecting eukaryotic cells.

host-induced modification or **host-induced variation** *see* **host-controlled modification**.

host range the spectrum of host organisms that can be infected by a specified infectious agent or parasite.

host-vector system a compatible combination of host and **vector** (def. 3) that allows propagation of DNA.

hot (*in physics*) an *informal term* describing a substance or material that contains a radionuclide, especially if **carrier-free** or at a dangerously high activity or specific activity. *Compare* **cold**.

hot laboratory or **hot lab** an *informal term* for a laboratory reserved and equipped for the manipulation of radioactive substances or materials of (dangerously) high activity.

hot room 1 a room (or laboratory) maintained at a constant, higher-than-ambient temperature. 2 an *informal term* for a room (or laboratory) that is specially equipped to deal with high levels of radioactivity.

hot spot any site at which mutations are observed with an unusually high frequency. Such sites have a highly enhanced, inherent susceptibility to mutation that cannot be ascribed to the type of base pair involved, the way in which it is altered, or the resulting change in any protein encoded by the DNA molecule.

hour *symbol*: h or hr; a non-SI unit of time equal to 3600 s.

housekeeping genes *see* **constitutive genes**.

Houssay, Bernardo Alberto (1887–1971), Argentinian physiologist and endocrinologist noted for his researches on diabetes and on the internal secretions of the adrenal, anterior pituitary, and thyroid glands; Nobel Laureate in Physiology or Medicine (1947) 'for his discovery of the part played by the hormone of the anterior pituitary lobe in the metabolism of sugar' [prize shared with C. F. and G. T. Cori].

Houssay animal a pancreatectomized animal which has also undergone **hypophysectomy**. The latter ameliorates the diabetic state that results from pancreatectomy by abolishing the secretion of **somatotropin**. [After B. A. Houssay.]

Hox gene *see* **homeotic gene**.

Hp 1 *symbol* for the heptyl group, $\text{CH}_3\text{--}[\text{CH}_2]_5\text{--CH}_2\text{--}$. 2 *abbr.* for haptoglobin.

HP-1 *abbr.* for helper peak-1 (now called **interleukin 1**).

HPCE *abbr.* for high performance capillary electrophoresis.

HpETE *abbr.* for hydroperoxyeicosatetraenoate.

hPL or **HPL** *abbr.* for human placental lactogen.

HPLAC *abbr.* for high-performance liquid affinity chromatography.

HPLC *abbr.* for high-pressure liquid chromatography or high-performance liquid chromatography.

Hpo *symbol* for the heptanoyl group, $\text{CH}_3\text{--}[\text{CH}_2]_5\text{--CO--}$.

HpODE *abbr.* for hydroperoxyoctadecadienoate.

HPR *abbr.* for H protein; a low- M_r , heat-stable, cytoplasmic phosphocarrier protein found in some bacterial species and concerned in the phosphoenolpyruvate (PEP)-hexose transferase system for the transport of hexoses across the cell membrane. The phosphate of PEP is transferred transiently to HPR. Example from *Escherichia coli* (identical sequence for *Salmonella typhimurium*): database code PTHP_ECOLI, 85 amino acids (9.11 kDa). *See* **enzyme I**, **enzyme II**.

H⁺ pump *see* **hydrogen ion pump**.

HQNO *see* **HQNO**.

hr *abbr.* for hour (now superseded by h).

H1, H2, H3 receptors *see* **histamine receptor**.

H-RNA *abbr.* for heterogeneous nuclear RNA (alternative to hnRNA or HnRNA).

HRT *abbr.* for hybrid-released translation.

HSA *abbr.* for human serum albumin.

Hse *symbol* for a residue of the α -amino acid L-homoserine.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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HSE *abbr.* for heat-shock regulatory element or heat-shock response element.

α_2 -HS-glycoprotein or **fetuin** an α -globulin found in fetal calf serum, where it accounts for up to 45% of the total protein. It has a low isoelectric point and contains up to 35% carbohydrate, having three oligosaccharide chains each of M_r 3400; it comprises various glycoproteins that differ in M_r . It is a growth factor for a variety of cell types. It also promotes endocytosis, possesses opsonic properties, and influences the mineral phase of bone. The amino-acid sequence has **cystatin** features; two peptides are cleaved from a precursor to form chains held together by a single disulfide bond. Example precursor (human): database code A2HS_HUMAN, 367 amino acids (39.28 kDa).

Hsl *symbol* for a residue of the α -amino lactone L-homoserine lactone.

HSPG core protein *abbr.* for heparan sulfate proteoglycan core protein. *See also* **syndecan**.

hst/KS *see* **KS/hst**.

H strand *abbr.* for heavy strand.

5-HT *abbr.* for 5-hydroxytryptamine (i.e. **serotonin**).

HTH *abbr.* for helix-turn-helix.

HTLV *abbr.* for human T-cell lymphotropic virus; a **retrovirus** that causes a rare leukemia.

H⁺-transporting ATP synthase EC 3.6.1.34; *systematic name*: ATP phosphohydrolase (H⁺-transporting); *other names*: H⁺-transporting ATPase; mitochondrial ATPase; chloroplast ATPase; coupling factors F_0F_1 , C_0F_1 . The enzyme that is responsible for the phosphorylation of ADP to ATP in mitochondria, during **oxidative phosphorylation**. In *in vitro* preparations it can function as an ATPase. The 3-D structure is known for the major components of bovine heart mitochondrial enzyme (the following examples refer to this enzyme). The enzyme comprises an intrinsic membrane domain, F_0 , linked by a stalk to a globular catalytic domain, F_1 , and hence is sometimes termed F_0F_1 complex. Energy released by hydron flux through F_0 leads to ATP synthesis by F_1 ; the stoichiometry is approximately 3H^+ per ADP phosphorylated. F_1 (371 kDa) has the subunit composition: $3\alpha, 3\beta, \gamma, \delta, \epsilon$; the subunit sequences (precursors) are: α , database code ATP0_BOVIN, 553 amino acids (59.65 kDa); β , database code ATPB_BOVIN, 528 amino acids (56.22 kDa); γ , database code ATPG_BOVIN, 297 amino acids (32.92 kDa); δ , database code ATPD_BOVIN, 168 amino acids (17.59 kDa); ϵ (from rat), database code ATPE_RAT, 50 amino acids (5.63 kDa). The γ chain forms a core, the α and β (catalytic) subunits are arranged round it alternately; the three β subunits have different structures during the synthesis cycle. The N- and C-terminal regions of γ form an antiparallel coiled coil; the α and β proteins are essentially β barrels. There are several proteins in the F_0 component, the main subunits of which in eukaryotes are named A to G. The C chain is the protein that accumulates in ceroid lipofuscinosis (**Batten's disease**). It is a proteolipid encoded by two genes which code for two precursor proteins, P_1 and P_2 , having different signal peptides for locating the proteins in mitochondria, but the same structure in the mature protein. Example, subunit C precursor P_1 (bovine): database code ATPM_BOVIN, 143 amino acids (15.01 kDa). Other proteins of F_0 include the A chain, database code ATP6_BOVIN, 226 amino acids (24.8 kDa); the B chain, database code ATPF_BOVIN, 214 amino acids (24.64 kDa); and the D chain, database code ATPQ_BOVIN, 160 amino acids (18.56 kDa). (The last named chain has no apparent homologue in bacteria.)

huangosome a colloquial term for a bilayered **liposome**. *Compare* **bangosome**.

Huber, Robert (1937–), German biochemist notable for his crystallographic, immunological, and structural studies of proteins; Nobel Laureate in Chemistry (1988) jointly with J. Deisenhofer and H. Michel 'for the determination of the three-dimensional structure of a photosynthetic reaction centre'.



Hudson's convention

Hudson's convention a convention, no longer used, for designating anomeric configurations in monosaccharides. The anomer in the D-series with the highest positive rotation (or lowest negative rotation) was designated the α form and the anomer with the lowest positive rotation was called the β form. The reverse applied in the L-series, the anomer with the lowest positive rotation (or the highest negative rotation) being designated the α form. The α and β descriptors now refer to the absolute structural relationships. *See* **anomer**, **Haworth representation**.

Hughes press a device to disrupt microorganisms. It consists of a steel cylinder in which a paste of the microorganisms in question, mixed with abrasives, is subjected to sudden blows by a steel piston driven by a fly-press; the piston forces the crushed material out of the cylinder into a reservoir. At temperatures below -20°C no abrasives are needed, ice crystals taking their place. *See also* **French pressure cell**. [After D. E. Hughes (1922–), who published the design in 1951.]

HUGO *abbr.* for Human Genome Organization; an international organization of scientists involved in the **Human Genome Project**, the global initiative directed at mapping and sequencing the human genome. It was established in 1989, and carries out a coordinating role, supporting data collection for constructing genetic and physical maps of the human genome, organizing the Single Chromosome Workshops and other workshops and promoting consideration of ethical, legal, social, and intellectual property issues. It fosters exchange of data and biomaterials. The Human Genome Project is, however, not under the central control or direction of any individual, group, or organizing committee, the work being done in many countries being coordinated by the scientists carrying out the work.

human *abbr.* (in *bionomenclature*): h or (formerly) H; of, relating to, characteristic of, obtained from, or being a member of the species *Homo sapiens*. For names of substances or products beginning with the word human (e.g. human chorionadotropin, human serum albumin) see at the corresponding general entry.

Human Genome Project *see* **HUGO**

human immunodeficiency virus *see* **HIV**.

human T-cell lymphotropic virus *see* **HTLV**.

humectant 1 producing moisture. 2 any substance added to another substance or material to keep it moist.

humic acid any of the mixture of complex macromolecules having polymeric phenolic structures and extractable from soils and peat. It is produced by the oxidative degradation of lignin and has the ability to chelate metals, especially iron.

humidity cabinet or **humidity chamber** an enclosure in which the humidity is maintained at a constant, often high level. It is useful for keeping experimental samples or materials moist or at a predetermined moisture content.

humins the very dark-brown or black insoluble material that is produced when carbohydrate-containing proteins are subjected to acid hydrolysis.

humor the US spelling of **humour**.

humoral of, contained in, or involving the blood or other body fluid; consisting of, relating to, or involving some chemical agent or substance contained in a body fluid, especially the blood.

humoral immunity any specific immunity mediated by antibodies (i.e. **humoral antibodies**) present in blood plasma, lymph, and tissue fluids, and that may also become attached to cells.

humour or (US) **humor** a former name for any of the body fluids, e.g. blood, lymph.

humulin 1 human insulin made using cDNA technology. 2 the bitter aromatic principle of hops (lupulin).

humus the dark-brown or black complex mixture of colloidal organic material that results from the decomposition of plant and animal material in the soil. It is of great importance for plant growth.

hunchback a segmentation protein in insects. Similar to **ikaros**, it is a zinc-finger protein and is encoded by a **gap gene**. Example from *Drosophila melanogaster*: database code HUNB-DROME, 758 amino acids (83.02 kDa). *Compare* **Krüppel**.

Hunter syndrome an alternative name for mucopolysaccharidosis II. *See* **mucopolysaccharidosis**. [After Charles Hunter (1872–1955), British physician.]

Huntington's disease or **Huntington's chorea** an autosomal dominant genetic disease characterized by involuntary (choreic) movements and dementia over a 10–20 year course. It is inevitably fatal. Linking studies using a variety of RFLPs have localized the defective gene to a site on the short arm of chromosome 4. There is a selective loss of a group of neurons; the biochemical basis for their loss is unknown. The gene, which encodes a 3145-residue product, **huntingtin**, of unknown function, contains a polymorphic trinucleotide repeat of the codon for glutamine, CAG, within the coding sequence. Normals have 35–37 of these repeats. In no case has any individual with more than 41 repeats been found to be free of the disease. [After George Huntington (1850–1916), US physician.]

HU protein an abundant bacterial histone-like protein that has the ability to wrap and bend DNA *in vitro*, and is thought to play a role in chromosome DNA topology, probably by facilitating the action of gyrase. *See* **DNA topoisomerase** type II.

Hurler-Scheie compound syndrome an alternative name for **mucopolysaccharidosis I H/S**.

Hurler syndrome an alternative name for mucopolysaccharidosis I H; *see* **mucopolysaccharidosis**. [After Gertrude Hurler (1889–1965), Austrian physician.]

HUT (in *clinical chemistry*) *abbr.* for UDPglucose-hexose-1-phosphate uridylyl transferase (EC 2.7.7.12).

Huxley 1 (Sir) **Andrew Fielding** (1917–), British physiologist renowned for his researches on the ionic basis of contraction of muscle and of impulse conduction by nerve; Nobel Laureate in Physiology or Medicine (1963) jointly with J. C. Eccles and A. L. Hodgkin 'for their discoveries concerning the ionic mechanisms involved in excitation and inhibition in the peripheral and central portions of the nerve cell membrane'. 2 **Hugh Esmor Huxley** (1924–), British biophysicist noted for his study of the relation of ATP hydrolysis to the mechanics of muscle contraction.

HVA *abbr.* for homovanillic acid.

HVE *abbr.* for high-voltage electrophoresis.

Hvidt-Nielsen mechanism a proposed mechanism to account for the kinetics of hydrogen exchange between protein molecules in solution and solvent water molecules. It is considered that protein molecules exist in solution in more than one conformational state and that hydrogen atoms in the interior of the protein molecule in the most stable conformational state are inaccessible to exchange; these hydrogen atoms may become transiently externalized, and hence accessible to exchange, in one or more of the other conformational states. [After Aasa Hvidt and S. O. Nielsen.]

Hx *abbr.* for hepxilin.

Hx^c symbol for the cyclohexyl group (alternative to **cHx**).

hyal+ a variant form of **hyalo+** (before a vowel).

hyaladherin a name sometimes used for proteins in the extracellular matrix that bind to **hyaluronate**.

hyaline clear, transparent; with no inclusions.

hyalo+ or (before a vowel) **hyal+** *comb. form* denoting glassy, transparent.

hyaluronan an alternative name for **hyaluronate**.

hyaluronate or **hyaluronan** the naturally occurring anionic form of hyaluronic acid; any member of a group of glycosaminoglycans, the repeat units of which consist of $\beta 1 \rightarrow 4$ -linked D-glucuronyl-($\beta 1 \rightarrow 3$)-N-acetyl-D-glucosamine. Most preparations contain 1–2% by mass of protein, and have M_r values in the range 5×10^4 to 8×10^6 depending on source and methods of preparation and determination. Hyaluronates are widely distributed in many tissues, especially in the vitreous humour of the eye, synovial fluid, and umbilical cord. They

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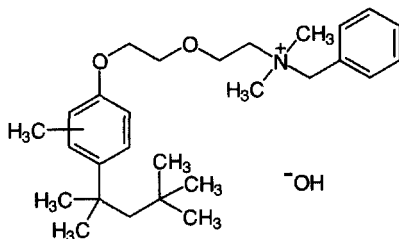


hyaluronidase

have an important role in the formation of proteoglycan aggregates. Hyaluronates form very viscous solutions and gels in water, a property that may be important in protecting animal organisms against invasion by bacteria.

hyaluronidase or **mucinase** or **spreading factor** a trivial name for either of the glycosidases hyaluronoglucosaminidase (EC 3.2.1.35) and hyaluronoglucuronidase (EC 3.2.1.36), and for the carbon-oxygen lyase hyaluronate lyase (EC 4.2.2.1). These enzymes have in common the ability to depolymerize hyaluronidate by cleavage of glycosidic bonds; e.g. hyaluronoglucosaminidase catalyses the random hydrolysis of 1,4-linkages between *N*-acetyl- β -D-glucosamine and D-glucuronate residues in hyaluronate. Hyaluronidases occur in certain animal tissues, e.g. sperm acrosome, and in some bacteria, e.g. *Clostridium* spp., and enhance the penetration of various substances and pathogens through the extracellular matrix (hence the name 'spreading factor'). Example, mu toxin from *Clostridium perfringens*: database code NAGH_CLOPE, 1042 amino acids (117.49 kDa).

hyamine hydroxide methylbenzethonium hydroxide; a water-soluble organic base, useful biochemically because it and some of its salts are soluble in organic solvents such as chloroform and toluene. It is therefore used to absorb radioactive CO_2 released during metabolic incubations; the resultant hyamine carbonate can then be dissolved in a scintillation cocktail for counting. It is also used as a topical anti-infective agent.



hybrid 1 something derived from heterogeneous sources or composed of different or incongruous elements. 2 (*in genetics*) the offspring of two animals or plants of different species, or (less strictly) different varieties. 3 (*in chemistry*) a bond or valence orbital obtained by the linear combination of two or more different atomic orbitals.

hybrid antibody any artificially produced immunoglobulin molecule in which the combining sites are of different specificities.

hybrid-arrested translation *abbr.*: HART; a method for analysing the relationship between a DNA sequence, its corresponding messenger RNA (mRNA), and the protein for which it codes, based on the observation that mRNA when hybridized with its complementary DNA is not translated in eukaryotic cell-free systems whereas heat-induced dissociation of the hybrid completely restores translational activity. The method has been used to map precisely specific protein-coding regions in **restriction fragments** of DNA. *See also* **hybrid-released translation**.

hybrid cell any cell containing the nuclei and cytoplasm from differing cells, formed artificially by **cell fusion**.

hybrid DNA 1 artificial duplex DNA produced by collinear (i.e. covalent) union of two (or more) fragments of duplex DNA (i.e. **recombinant DNA**; *see also* **plasmid chimera**). 2 artificial duplex DNA produced by collateral (hydrogen bonded) union of two incompletely complementary, single strands of DNA. *Compare* **DNA-RNA hybrid**.

hybridization or **hybridisation** 1 the act or process of forming a genetic **hybrid** (def. 2) by cross-breeding, or other unnatural recombination. 2 the act or process of forming a macromolecular **hybrid** (def. 1) by the artificial recombination of subunits.

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e.g. of polynucleotide strands or polypeptide chains. 3 the mathematical combination of atomic orbitals to form an orbital **hybrid** (def. 3).

hybridization analysis a method of testing for complementarity in the base sequences of two polynucleotides from different sources, based on the ability of complementary single-stranded DNA molecules to form duplex DNA molecules on being annealed together, and the ability of single strands of DNA and RNA similarly to form **DNA-RNA hybrids**.

hybridization probe *see* **probe** (def. 3).

hybrid myeloma *see* **hybridoma**.

hybridoma any permanent **cell line** derived from an artificial **somatic cell hybrid** formed between a cultured neoplastic lymphocyte and a normal primed B or T lymphocyte, all the cells of which express the specific immune potential of the normal parental cell. The term is derived from a contraction of 'hybrid' and 'myeloma'. Hybridomas are produced by the technique of **cell fusion**, using, e.g., a suspension of mouse myeloma or thymoma cells from a selected established cell line to which is added a preparation of spleen cells (predominantly normal lymphocytes) from mice or other animals that have been immunized against a specified antigen. After cell fusion, selection is aided by the use of **HAT** medium, which kills HGPRT⁻ myeloma cells, and lymphocytes that have not fused do not survive in culture. This treatment thus selects for the immortalized hybridoma cells that retain the lymphocyte HGPRT. Such hybrid cells are both permanently adapted to growth in culture and capable of forming specific antibody-producing tumours *in vivo*. A B-cell hybridoma continuously secretes pure **monoclonal antibody** of a predetermined specificity and may thus be used for the facile large-scale production of such antibody.

hybrid-released translation *abbr.*: HRT; a method in which cloned DNA is bound to a nitrocellulose filter and hybridized with unfractionated mRNA, or even total cellular RNA. After washing the filter, any mRNA that has hybridized with the DNA is eluted and translated in a cell-free translation system, thus revealing the product that is encoded by the DNA. *See also* **hybrid-arrested translation**.

hydantoin *generic name* for 2,4-imidazolidinedione or its derivatives. A hydantoin is formed by the reaction of an isocyanate, commonly phenylisocyanate, with the free amino group of an amino acid and the cyclization of the hydantoic acid derivative initially formed. When an isocyanate reacts with the N-terminal amino group of a peptide, the corresponding hydantoin is formed with simultaneous cleavage of the substituted N-terminal amino-acid residue from the peptide.

hydr+ a variant form of **hydro+** (sometimes before a vowel).

hydrase an obsolete term for **hydratase**.

hydratase *see* **dehydratase**.

hydrate 1 any adduct of water with another compound or element, usually in definite molecular proportions. 2 to form a **hydrate** (def. 1); to undergo **hydration**. 3 to cause to take up or to combine with water. 4 to maintain or restore the normal proportion of water in the body (of someone or something). —**hydrated** *adj.*

hydration 1 the act or process by which water molecules surround and bind to molecular entities. 2 the addition of H and OH to a chemical species. 3 the formation of a **hydrate** (def. 1). 4 the quality or state of being hydrated. 5 the introduction of additional water into the body (of something or someone). *Compare* **anhydrous**, **dehydration**, **solvation**.

hydration shell the layer of water molecules surrounding an ion or molecule in aqueous solution.

hydraulic 1 operated by pressure transmitted through a liquid, e.g. through water or oil. 2 of or concerning hydraulics (i.e. fluid mechanics).

hydrazine $\text{H}_2\text{N}-\text{NH}_2$, an oily liquid, fuming in air, that is very toxic and possibly a carcinogen. It is available as a hydrate and as various salts. *See also* **hydrazinolysis**.



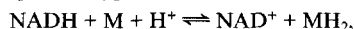
hydrazinolysis

hydrazinolysis the cleavage of amide and peptide bonds by heating at 100 °C with **hydrazine** in anhydrous conditions to form the corresponding acyl hydrazides. This reaction is useful in the determination of C-terminal amino-acid residues in peptides and proteins as all the other amino-acid residues will be converted to their hydrazides and only the C-terminal amino acid will be set free in an underivatized form.

hydrazone any condensation product of the type $R'R''C=N-NHR$, formed between carbonyl compounds ($R'R''=CO$) and hydrazine or a substituted hydrazine. See also **osazone**, **phenylhydrazine**.

hydride 1 the ion H^- , hydrogen anion. It has an independent existence only in the salt-like hydrides formed by strongly electropositive elements (see def. 2) but is also sometimes regarded as participating in the formation of complex hydrides (see def. 4) and in certain oxidation–reduction reactions (see **hydride-ion transfer**). 2 any stoichiometric binary compound of hydrogen with one of many (but not all) of the elements in groups 1, 2, and 3 of the IUPAC **periodic table**, e.g. NaH or CaH_2 . Such hydrides are salt-like ionic compounds, are solids at ambient temperatures, and on fusion can be electrolysed (dihydrogen then being liberated at the anode). They are readily decomposed by water and are potent reducing agents. 3 Any stoichiometric compound of hydrogen with one (or sometimes more than one) of many of the elements in other groups of the periodic table, and also with boron. Such hydrides are volatile covalent compounds containing one atom of the element per molecule, e.g. PbH_4 , or sometimes more than one, e.g. Si_3H_8 ; in the latter type certain of the hydrogen atoms may be present as bridges, as in B_2H_6 . 4 any salt-like compound containing a complex anion that may be considered as formed by coordination of a hydride ion to an uncharged hydride, e.g. KBH_4 (see **borohydride**) or $LiAlH_4$; such compounds are widely used in organic chemistry as reducing agents or for introduction of tritium, 3H . 5 any (frequently nonstoichiometric) product formed by uptake of dihydrogen by certain transition metals, especially palladium.

hydride-ion transfer term conventionally often applied to a chemical or enzyme-catalysed oxidation–reduction reaction involving the transfer from one molecule to another of a hydrogen atom together with its associated pair of bonding electrons. Such a reaction may notionally be considered to involve the transfer of a hydride ion without such an ion ever existing as a free entity. Examples include the **Cannizzaro reaction** and reactions of the general type



where M is a substrate molecule undergoing enzymic reduction, to which a hydride ion is transferred from NADH.

hydrindantin the reduced form of **ninhydrin**. It is sometimes used in conjunction with ninhydrin in the chromogenic determination or visualization of amino or imino groups.

hydrión a former name for **hydrogen ion**.

hydro+ or (sometimes before a vowel) **hydr+** comb. form 1 indicating or denoting water, liquid, or fluid. 2 indicating the presence of hydrogen in a compound.

hydrocarbon 1 any organic compound composed exclusively of hydrogen and carbon. 2 of, pertaining to, or being a **hydrocarbon** (def. 1) or a chemical group or side chain composed of hydrogen and carbon only.

hydrocarbyl any univalent group formed from a hydrocarbon by removal of a hydrogen atom; e.g. methyl (from methane).

hydrocolloid any colloidal substance for which the dispersing liquid is water or an aqueous solution.

hydrocortisone an alternative name for **cortisol**.

hydrodynamic 1 of or concerned with the mechanical properties of fluids. 2 of, pertaining to, or concerned with **hydrodynamics**.

hydrodynamic particle the entity formed by any (macro)molecule in solution together with its solvation shell.

hydrodynamic radius the radius of a (macro)molecule or

other particle in solution as determined by a hydrodynamic method, e.g. sedimentation equilibrium.

hydrodynamics the branch of physics concerned with the forces acting on or exerted by fluids.

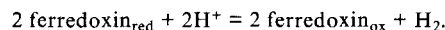
hydrodynamic shear the microscopic shear force caused by the motion of different layers in a fluid, where the layers move at different velocities. This acts to break any molecule sufficiently long to bridge the layers, e.g. DNA.

hydrodynamic volume (of a molecule in solution) the sum of the time-average of the molecular volume and the volume of the solvent molecules associated with it.

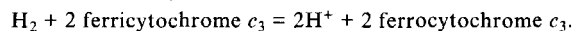
hydrogel any gel (def. 1) in which water is the liquid component. Compare **hydrosol**.

hydrogen symbol: H; the lightest of the elements and the most abundant in the universe. It exists as an odourless, colourless, flammable diatomic gas, dihydrogen, H_2 , and forms compounds with most of the elements, being present in water and all organic compounds. There are two naturally occurring isotopes, hydrogen-1 (**protium**) and hydrogen-2 (**deuterium**), and one artificial isotope, hydrogen-3 (**tritium**), which is radioactive.

hydrogenase 1 the recommended name for an enzyme, EC 1.18.99.1, also known as **hydrogenlyase**, that catalyses the reaction:



This enzyme recycles H_2 produced by **nitrogenase** to increase the production of ATP and to protect nitrogenase against inhibition or damage by O_2 . It is a heterodimer. Example from *Azotobacter vinelandii*: small subunit (precursor): database code MBHS_AZOVI, 358 amino acids (39.21 kDa); large subunit: database code MBHL_AZOVI, 602 amino acids (66.66 kDa). 2 an alternative name for **hydrogen dehydrogenase**. 3 EC 1.12.2.1; recommended name: cytochrome c_3 hydrogenase; an enzyme that catalyses the reaction:

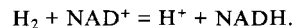


hydrogenate to undergo or cause to undergo reduction by addition of the hydrogen atoms of dihydrogen, usually with the aid of a catalyst such as finely divided forms of nickel, palladium, or platinum. The process is of industrial importance in petroleum refining, the petrochemical industry and the hardening of fats. —**hydrogenated** adj.; **hydrogenation** n.

hydrogen bond an association between an electronegative atom, e.g. fluorine, oxygen, nitrogen, or sulfur, and a hydrogen atom attached to another such electronegative atom. Although **hydrogen bonding** is due to interaction between dipoles, the force of attraction is large enough to permit formation of aggregates of small molecules or to stabilize the conformation of many macromolecules. The spatial relation of the donor and acceptor atoms is such that the hydrogen atom lies very close to the straight line between them.

hydrogen carrier a name sometimes used for any electron carrier that undergoes oxidation–reduction reactions by the loss or gain of hydrogen atoms, e.g. NADH, FADH.

hydrogen dehydrogenase the recommended name for the enzyme, EC 1.12.1.2, also known as **hydrogenase**, that catalyses the reaction:



It is a flavoprotein; iron and nickel are cofactors; the molecule is heterotetrameric. It may form part of a system involving **hydrogenase** (def. 1). Examples from *Alcaligenes eutrophus*: α subunit, database code HOXF_ALCEU, 602 amino acids (66.71 kDa); γ subunit, database code HOXU_ALCEU, 233 amino acids (26.01 kDa); the α and γ subunits form one dimer (FMN, 4Fe–4S cluster, NADH oxidoreductase); β subunit (contains Ni), database code HOXH_ALCEU, 487 amino acids (54.69 kDa); δ subunit, database code HOXY_ALCEU, 209 amino acids (22.85 kDa); the β and δ subunits form the other dimer which exhibits **hydrogenase** activity with artificial electron acceptors.

hydrogen–deuterium exchange see **hydrogen exchange**.

hydrogen electrode

hydrogen electrode an electrode that is reversible to hydrogen ions and dihydrogen molecules. It is formed by bubbling pure dihydrogen gas over a metal wire or a piece of foil whose surface is able to catalyse the reaction:



where e represents an electron. The most useful catalyst is platinum black on a platinum substrate. Hence the catalyst establishes an equilibrium between dihydrogen molecules (or hydrogen atoms) and the hydrogen ions (i.e. hydrons) in the solution in which the metal electrode is immersed. If the partial pressure (in atm) of dihydrogen gas at the electrode is p_{H_2} , the half-cell potential, E_{H} , of reaction (1) is given by:

$$E_{\text{H}} = E^{\circ}_{\text{H}} - (RT/F)\ln(a_{\text{H}^+}/p_{\text{H}_2}) \quad (2)$$

where R is the gas constant, F the Faraday constant, T the thermodynamic temperature, and a_{H^+} the hydrogen-ion activity. E°_{H} , the standard potential of the hydrogen electrode, is, by convention, zero at all temperatures, hence at $p_{\text{H}_2} = 1$, the usual standard state, equation (2) becomes:

$$E_{\text{H}} = -(RT/F)\ln a_{\text{H}^+}.$$

Thus the hydrogen electrode is the ultimate reference electrode for the determination of pH values. It should be noted that the potential of any single electrode cannot be determined by itself; another electrode must be connected to the solution.

hydrogen exchange or hydrogen-deuterium exchange a technique useful for demonstrating conformational differences between samples of the same protein and for providing insight into the dynamics of protein conformations in solution. It can be observed using nuclear magnetic resonance spectroscopy. The method is based on observation of the kinetics of the non-catalysed exchange of hydrogen atoms in peptides and proteins with protons or other hydron isotopes in aqueous solutions and the effects of pH (or p^{H}), temperature, etc. on the process.

hydrogen ion or (formerly) **hydron** the cation, H^+ , derived from the hydrogen atom. In aqueous solution it is mostly in the hydrated form, oxonium (H_3O^+). In nonaqueous solution it is likely, by analogy, that it is attached particularly to one solvent molecule, e.g. ROH_2^+ in an alcohol, R_2COH^+ in a ketone, RNH_3^+ in an amine, etc. See also **hydron**, **proton** (defs 1, 2).

hydrogen-ion concentration see **pH**.

hydrogen-ion pump any of a number of membrane proteins that transport hydrogen ions through membranes. The H^+ -transporting ATPase of fungi and plants (EC 3.6.1.35) has an action similar to that of the **H^+ -transporting ATP synthase** of mitochondria but without coupling factors; it is an integral membrane glycoprotein. The hydron gradient it generates drives the active transport of nutrients by H^+ -symport. Example from yeast: database code ATH1_YEAST, 918 amino acids (99.42 kDa). The **vacuolar ATPase** is another type.

hydrogenlyase see **hydrogenase** (def. 1).

hydrogen lamp or **hydrogen discharge lamp** a high-voltage discharge tube with a quartz window and containing dihydrogen at low pressure, in which the glow produced by excitation of the dihydrogen molecules is concentrated in a long capillary tube joining the two electrodes; this arrangement gives increased intensity. It produces a continuous emission spectrum over the range 180–350 nm. See also **deuterium (discharge) lamp**.

hydrogenosome any microbody-like particle occurring in the cytoplasm of some anaerobic flagellate protozoans. They are chromatid granules of high density containing enzymes, including hydrogenase, that catalyse the conversion of pyruvate to acetate, CO_2 , and H_2 . These organelles may substitute for mitochondria (which are absent from such organisms).

hydrolase any enzyme of class EC 3 that catalyses the hydrolysis of various bonds, e.g. C–O, C–N, C–C, phosphoric anhydride bonds, etc. Often the name of a hydrolase is formed from the name of the substrate and the suffix ‘-ase’.

hydrolysate or (sometimes US) **hydrolyzate** any substance or mixture of substances produced by hydrolysis.

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hydrolyse or (sometimes US) **hydrolyze** to subject to, to undergo, or to effect **hydrolysis**. —**hydrolysable** or (US) **hydrolyzable** *adj.*; **hydrolysat** or (US) **hydrolyzate** *n*.

hydrolysis (*pl. hydrolyses*) the rupture of one or more chemical bonds by reaction with, and involving the addition of, the elements of water; **solvolysis** by water.

hydrolytic of, pertaining to, producing, or produced by **hydrolysis**.

hydrolyzate an alternative US spelling of **hydrolysate**.

hydrolyze an alternative US spelling of **hydrolyse**.

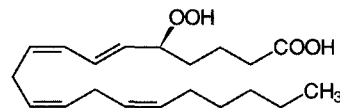
hydrometer an instrument used to measure the **density**, or **relative density**, of a liquid. It usually consists of a glass (or plastic) bulb, weighted at the bottom so that it floats upright, and fitted with a stem carrying a scale. The point on the scale that is level with the surface of the liquid indicates the (relative) density of the liquid.

hydron *symbol:* H^+ ; the recommended name for the cation derived from an atom of hydrogen in its natural isotopic abundance. See also **hydrogen ion**, **proton** (defs 1, 2).

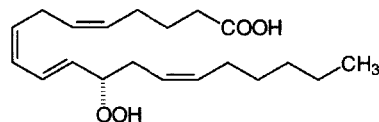
hydronate to combine with a **hydron** or hydrons. —**hydronation** *n*. **hydronium ion** an alternative name (now disfavoured) for **oxonium** (def. 1).

hydroperoxide any organic compound in which one hydrogen atom of a hydrocarbon is replaced by the –O–OH group.

hydroperoxyeicosatetraenoate *abbr.:* HpETE; any of various related products of the action of **lipoxygenase** on arachidonate, (all-*Z*)-eicosa-5,8,11,14-tetraenoate. The most notable are 5(*S*)-HpETE, (6*E*,8*Z*,11*Z*,14*Z*)-(5*S*)-5-hydroperoxyeicosa-6,8,11,14-tetraenoate; 12(*S*)-HpETE, (5*Z*,8*Z*,10*E*,14*Z*)-(12*S*)-12-hydroperoxyeicosa-5,8,10,14-tetraenoate; and 15(*S*)-HpETE, (5*Z*,8*Z*,11*Z*,13*E*)-(15*S*)-15-hydroperoxy-5,8,11,13-eicosatetraenoate. Their species distribution follows that of the lipoxygenases. HpETEs function as precursors of hydroxy-eicosatetraenoates, leukotrienes, dihydroxyeicosatetraenoates, heptoxilins, and lipoxins, but they also have biological activity in their own right, e.g. regulation of electrolyte flux, release of histamine, regulation of eicosanoid and corticosterone synthesis, and regulation of oocyte maturation. 12-HpETE inhibits platelet thromboxane synthase and prostaglandin-endoperoxide synthase, and acts as a second messenger in *Aplysia* sensory neurons.



5(*S*)-HpETE



12(*S*)-HpETE

hydroperoxyoctadecadienoate *abbr.:* HpODE; any of various related products of the action of **lipoxygenase** on linoleate, (all-*Z*)-octadeca-9,12-dienoate. The most notable are 9(*S*)-HpODE, (10*E*,12*Z*)-(9*S*)-octadeca-10,12-dienoate; and 13(*S*)-HpODE, (9*Z*,11*E*)-(13*S*)-octadeca-9,11-dienoate.

hydrophilic 1 tending to dissolve in water; having a strong affinity for water; readily mixing with or wetted by water. See also **lyophilic**. 2 **polar**. Compare **hydrophobic**. —**hydrophilicity** *n*.

hydrophobic

hydrophobic 1 tending not to dissolve in water; having a low affinity for water; not readily mixing with or wetted by water.

2 nonpolar. Compare **hydrophilic**. See also **lyophobic**. —**hydrophobicity** *n*.

hydrophobic bond a term sometimes used incorrectly for a **hydrophobic interaction**.

hydrophobic chromatography or **hydrophobic affinity chromatography** or **hydrophobic interaction chromatography** a technique for the separation and purification of proteins and other substances by **adsorption chromatography** on polysaccharide or other gels containing lipophilic substituents (which may or may not bear terminal ionizable groups). These hydrophobic centres on the gel interact with nonpolar amino-acid residues exposed on the surface of the protein molecules being separated.

hydrophobic interaction or (discouraged) **apolar bond** the tendency of hydrocarbons (or of lipophilic hydrocarbon-like groups in solutes) to form intermolecular or intramolecular aggregates in polar solvents. See also **apolar interaction**.

hydrophobic pocket (in a protein) a binding site (crevice, pocket) that contains mostly hydrophobic amino acids. For example, the binding site for the aromatic side chains of the specific substrates in **chymotrypsin** is a hydrophobic pocket, consisting of a slit in the enzyme, 100–120 nm deep and 35–40 nm by 55–65 nm in cross section, lined with nonpolar amino-acid side chains.

hydrophobic residue any building block of a polymer, the residue of which in that polymer is hydrophobic. The term is applied especially to amino-acid residues exhibiting such properties in polypeptides.

hydroquinone 1 1,4-benzenediol; *p*-dihydroxybenzene; quinol. Compare ***p*-benzoquinone**, **pyrocatechol**. See also **quinhydrone**. 2 any member of a class of aromatic *p*-diols derivable from *p*-quinones (see **quinone** (def. 2)) by reduction of two CO groups to COH groups with any necessary rearrangement of double bonds; any compound containing a quinol nucleus. See also **quinol** (def. 2), **semiquinone**. Compare **catechol** (def. 2).

hydrosol any sol in which water is the liquid component. Compare **hydrogel**.

hydrous (of a chemical compound) containing combined water (usually in stoichiometric proportions).

hydroxide 1 or (formerly) **hydroxyl** the ion OH⁻. 2 any inorganic or organic compound containing or dissociating to give this ion. Hydroxides that are soluble in water, and can dissociate, give aqueous solutions with an alkaline reaction (pH > 7).

hydroxo+ *comb. form* indicating the presence of a OH⁻ ligand in a coordination compound.

hydroxocobalamin or **vitamin B_{12b}** a form of vitamin B₁₂ in which a hydroxide ion is in coordinate linkage to the sixth (i.e. β) position on the cobalt atom in a molecule of **cobalamin**; the conjugate base of **aquacobalamin**.

hydroxonium ion an alternative name (now disfavoured) for **oxonium** (def. 1).

hydroxy denoting the presence in a compound of one or more **hydroxyl** groups.

hydroxy+ *comb. form* (in chemical nomenclature) denoting the presence of one or more **hydroxyl** groups; the number of hydroxyl groups may be specified, e.g. dihydroxy+ (two), trihydroxy+ (three), etc.

hydroxy acid any organic acid containing one or more **hydroxyl** groups.

2-hydroxy-acid oxidase EC 1.1.3.15; *systematic name*: (S)-2-hydroxy-acid: oxygen 2-oxidoreductase; *other name*: glycolate oxidase; an enzyme that catalyses the oxidation by dioxygen of (S)-2-hydroxy-acid to 2-oxo-acid and H₂O₂. It is a peroxisomal enzyme in plants that is involved in the conversion of glycolate to glycine in **photorespiration**. The coenzyme is FMN. Example from *Spinacia oleracea*: database code GOX_SPIOL, 369 amino acids (40.28 kDa).

3-hydroxyacyl-CoA dehydrogenase EC 1.1.1.35; *systematic name*: (S)-3-hydroxyacyl-CoA:NAD⁺ oxidoreductase; *other* *Bản đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:*

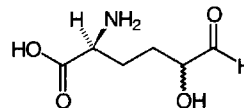
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

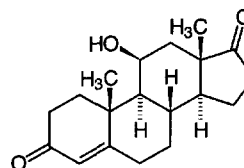
3-hydroxybutyrate dehydrogenase

names: β-hydroxyacyl dehydrogenase; β-keto-reductase; an enzyme of the **beta-oxidation system**. It catalyses the oxidation by NAD⁺ of S-3-hydroxyacyl-CoA to 3-oxoacyl-CoA with formation of NADH; example (rat liver): database code RATD2D3COA, 287 amino acids (31.88 kDa); for another example (*Escherichia coli*): see **fatty-acid oxidation complex**. See also **peroxisomal bifunctional enzyme**.

hydroxyallysine 5-hydroxy-6-oxonorleucine; 2-amino-5-hydroxyadipaldehyde. An α-amino acid not normally found in proteins but enzymically formed from 5-hydroxylysine *in situ* as an intermediate in the creation of covalent cross-links between adjacent polypeptide strands during the maturation of certain connective tissue proteins, e.g. collagen, elastin. See also **allysine**.

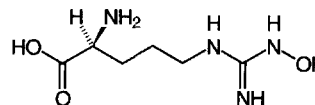


11β-hydroxy-4-androstene-3,17-dione 4-androsten-11β-ol-3,17-dione; a 17-oxo steroid formed from androstenedione by steroid 11β-monooxygenase (EC 1.14.15.4).



hydroxyapatite or **hydroxylapatite** a crystallized form of calcium orthophosphate hydroxide, Ca₁₀(PO₄)₆(OH)₂, virtually insoluble in water and in many aqueous solvents; it occurs as a mineral in phosphate rock. The main mineral component of bone, cartilage, and tooth approximates in composition to hydroxyapatite, small proportions of Ca²⁺ and OH⁻ being replaced respectively by Mg²⁺ and F⁻; trace amounts of other ions may be present as well. Hydroxyapatite may also occur in granules in mitochondria. Chemically prepared hydroxyapatite is useful in the separation of proteins or nucleic acids by **adsorption chromatography**. It is used in orthopedic and dental prostheses and in the prevention of osteoporosis.

N^ω-hydroxy-L-arginine *abbr.*: L-NHA; a derivative of L-arginine in which one of the guanidine nitrogens is oxidized to an N-hydroxy group. It is an intermediate in the formation of nitric oxide from L-arginine; it may function as an intercellular NO carrier and a vasodilator in its own right.



***p*-hydroxybenzoic acid** 4-hydroxybenzoic acid; an intermediate in the synthesis of **ubiquinone** from tyrosine in mammals.

3-hydroxybutyrate cycle or **3-hydroxybutyrate shuttle** an intracellular metabolic cycle that can transfer reducing equivalents from the cytoplasm to the mitochondria in liver. Extramitochondrial NADH is oxidized to NAD⁺ by acetoacetate, which is reduced to D(-)-3-hydroxybutyrate; the latter then passes into the mitochondria where it is reoxidized to acetoacetate by intramitochondrial NAD⁺. The acetoacetate passes out into the cytoplasm thereby completing the cycle.

3-hydroxybutyrate dehydrogenase EC 1.1.1.30; *other*

α -hydroxybutyrate (2-hydroxybutanoate)

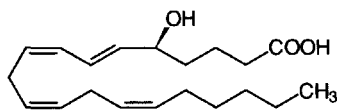
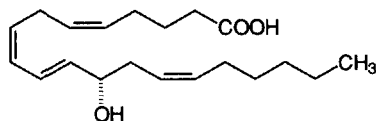
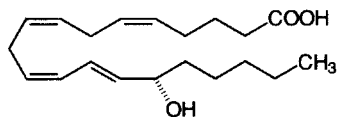
name: D- β -hydroxybutyrate dehydrogenase; an enzyme that catalyses the reaction:



Example (fragment) from human: database code BDH_HUMAN, 343 amino acids (38.09 kDa).

α -hydroxybutyrate (2-hydroxybutanoate) dehydrogenase an alternative name for the LD₁ isoenzyme of **lactate dehydrogenase** (EC 1.1.1.27).

hydroxyeicosatetraenoate *abbr.:* HETE or mono-HETE; any of various related products arising, either spontaneously or by the action of glutathione peroxidases, from reduction of **hydroperoxyeicosatetraenoates**. The main forms are the 5-, 12-, and 15-HETEs, resulting from the action of 5-, 12-, and 15-lipoxygenase enzymes, but 8-, 9-, 11-, 16-, 17-, 18-, 19-, and 20-HETEs are known. The hydroxy group in most HETEs is of the *S* configuration, but those of *R* configuration are known, e.g. 12(*R*)-HETE from psoriatic lesions; 12(*R*)-HETE and 11(*R*)-HETE are found in sea-urchin eggs. The three main HETEs are: (6*E*,8*Z*,11*Z*,14*Z*)-(5*S*)-5-hydroxyeicosa-6,8,11,14-tetraen-1-oate, 5(*S*)-HETE; (5*Z*,8*Z*,10*E*,14*Z*)-(12*S*)-12-hydroxyeicosa-5,8,10,14-tetraen-1-oate, 12(*S*)-HETE; and (5*Z*,8*Z*,11*Z*,13*E*)-(15*S*)-15-hydroxyeicosa-5,8,11,13-tetraen-1-oate, 15(*S*)-HETE. Their tissue distribution follows that of **lipoxigenases**. 15(*S*)-HETE can act as a precursor for 5(*S*),15(*S*)-dihydroxyeicosatetraenoate and **lipoxins**. 5-HETE and 12-HETE are chemoattractants, 12(*R*) being more potent than 12(*S*), but 5(*S*) being more potent than 5(*R*); 8(*R*)-HETE, synthesized by starfish oocytes, is orders of magnitude more potent than 8(*S*)-HETE in inducing oocyte maturation.

5(*S*)-HETE12(*S*)-HETE15(*S*)-HETE

hydroxy fatty acid any member of a class of fatty acids having a hydroxyl group, usually at the 2- or 3-position. 2-D(-)-hydroxy fatty acids are widespread minor constituents of plant and some animal (wool, skin) lipids; 3-L(+)-hydroxy fatty acids are found in low amounts in plant and bacterial lipids (e.g. in **mycolic acids**). Mid-chain hydroxy acids and polyhydroxy acids also occur in plant lipids.

5-hydroxyindoleacetaldehyde *see* **5-hydroxyindole metabolism**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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hydroxy-methoxymand

5-hydroxyindoleacetic acid *abbr.:* 5-HIAA; a metabolite of **5-hydroxytryptamine**, produced by oxidative deamination catalysed by the enzyme **monoamine oxidase**, followed by oxidation to 5-HIAA. *See also* **5-hydroxyindole metabolism**.

5-hydroxyindole metabolism the metabolism of 5-hydroxytryptophan and its derivatives; it is of particular interest because it is concerned with the synthesis and degradation of the neurotransmitter 5-hydroxytryptamine (i.e. **serotonin**). Serotonin is converted to 5-hydroxy-3-indoleacetaldehyde by **monoamine oxidase**, and thence to either 5-hydroxytryptophol (3%) or 5-hydroxy-3-indoleacetic acid (5-HIAA) (97%), and these are excreted in the urine. Urinary 5-HIAA is measured as a diagnostic aid in cases of carcinoid syndrome, as the tumour converts large amounts of dietary tryptophan to serotonin.

hydroxyl **1** the (covalently linked) -OH group in any chemical compound. **2** a former name for the hydroxide ion; *see* **hydroxide** (def. 1).

hydroxylamine **1** NH₂-OH; a chemical mutagen that causes the deamination of cytosine residues in DNA so that, during subsequent replication of the DNA, the altered cytosine pairs with adenine rather than with guanine, causing G-C to A-T transitions. Hydroxylamine is an intermediate in the oxidation of ammonium in nitrifying bacteria, and in the reverse reaction in denitrifying bacteria. It may be used to cleave specifically Asp-I-Gly bonds in peptides. **2** any organic compound of the type RNH-OH.

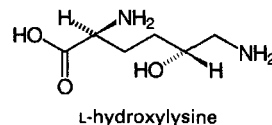
hydroxylapatite an alternative name for **hydroxyapatite**.

hydroxylase the common name for many monooxygenase and a few **dioxygenase** enzymes that catalyse hydroxylation reactions.

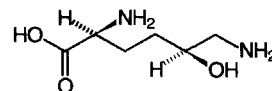
hydroxylate to introduce one or more **hydroxyl** (def. 1) groups into a compound, often by the replacement of hydrogen atom(s). —**hydroxylation** *n*.

hydroxylation-induced migration an alternative name for **NIH shift**.

hydroxylysine the trivial name for δ -hydroxylysine; δ -hydroxy- α,ϵ -diaminocaproic acid; 5-hydroxy-2,6-diaminohexanoic acid; H₂N-CH₂-CHOH-[CH₂]₂-CH(NH₂)-COOH; a chiral α -amino acid. Because the molecule of hydroxylysine possesses two chiral centres, four isomeric forms are possible, comprising an enantiomeric (D_s and L_s) pair of each of two (*erythro* and *threo*) diastereoisomers (the pair of *threo* enantiomers are sometimes known as **allohydroxylysine**). L-Hydroxylysine (symbol: 5Hyl or Hyl or Lys(5-OH) or (formerly) Hylys), *erythro*-5-hydroxy-L_s-lysine, is a **noncoded amino acid** found in peptide linkage in proteins (mainly collagen). In collagen 15–20% of the lysine residues are enzymically converted to hydroxylysine residues while the growing peptide chains are still attached to ribosomes; some of the hydroxyl groups of the hydroxylysine side chains are subsequently glycosylated.



L-hydroxylysine

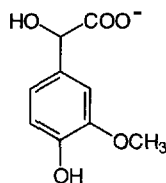


L-allohydroxylysine

hydroxy-methoxymandelate *abbr.:* HMMA; 4-hydroxy-3-methoxymandelate; often (incorrectly) known as vanillylmandelate (*abbr.:* VMA); the L enantiomer is the end product of

hydroxymethylbilane synthase

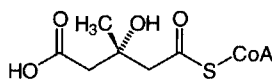
catecholamine catabolism, and is excreted in the urine. In clinical chemistry the measurement of HMMA in urine is used to detect excessive biosynthesis of catecholamines, as, e.g., in **pheochromocytoma**.



hydroxymethylbilane synthase EC 4.3.1.8; *systematic name*: porphobilinogen ammonia-lyase (polymerizing); *other names*: porphobilinogen deaminase; pre-uroporphyrinogen synthase. An enzyme that catalyses the conversion of four molecules of porphobilinogen to hydroxymethylbilane and four molecules of ammonia; dipyrromethane is a cofactor. Together with **uroporphyrinogen-III synthase** it is responsible for the step in heme biosynthesis in which four porphobilinogen molecules are condensed to form uroporphyrinogen III. Example from *Escherichia coli*: database code HEM3_ECOLI, 313 amino acids (33.81 kDa); 3-D structure known.

5-hydroxymethylcytosine *abbr.*: HMC; a minor base occurring in some types of DNA, e.g. in T4 bacteriophage.

hydroxymethylglutaryl-CoA *abbr.*: HMG-CoA; β -hydroxy- β -methylglutaryl-CoA; (S)-3-hydroxy-3-methylglutaryl-CoA; an important intermediate in the synthesis of cholesterol, farnesyl and geranyl derivatives, and ketone bodies. *See also* **hydroxymethylglutaryl-CoA reductase**, **hydroxymethylglutaryl-CoA lyase**.



hydroxymethylglutaryl-CoA lyase *abbr.*: HMG-CoA lyase; EC 4.1.3.4; *systematic name*: (S)-3-hydroxy-3-methylglutaryl-CoA acetoacetate lyase. An enzyme that catalyses the formation of acetyl-CoA and acetoacetate from (S)-3-hydroxy-3-methylglutaryl-CoA.

In mammals, a mitochondrial enzyme that converts hydroxymethylglutaryl-CoA to acetoacetate in the synthesis of **ketone bodies**. Example from *Pseudomonas mevalonii*: database code MVAB_PSEMV, 301 amino acids (31.57 kDa).

hydroxymethylglutaryl-CoA reductase (NADPH) *abbr.*: HMG-CoA reductase; EC 1.1.1.34; *systematic name*: (R)-mevalonate:NADP⁺ oxidoreductase (CoA-acylating). An enzyme that catalyses a reaction between (S)-3-hydroxy-3-methylglutaryl-CoA and two molecules of NADPH to form (R)-mevalonate, CoA, and two molecules of NADP⁺. Its function is to reduce HMG-CoA to (R)-mevalonate. In mammals it is an integral membrane glycoprotein of endoplasmic reticulum and peroxisomes. It catalyses a rate-limiting step and is an important regulatory enzyme in pathways for the synthesis of cholesterol and other sterols. It is the target for therapeutic strategies for reducing plasma cholesterol. Example from human: database code HMDH_HUMAN, 888 amino acids (97.36 kDa). *See also* **dephospho-[reductase kinase] kinase**.

[hydroxymethylglutaryl-CoA reductase (NADPH)] kinase EC 2.7.1.109; *other name*: reductase kinase; an enzyme that catalyses the phosphorylation by ATP of [3-hydroxy-3-methylglutaryl-CoA reductase (NADPH)] with release of ADP. It thereby inactivates the latter enzyme.

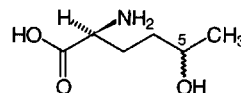
[hydroxymethylglutaryl-CoA reductase (NADPH)]-phosphatase EC 3.1.3.47; *other name*: reductase phosphatase; an

hydroxypr

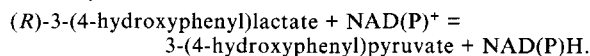
enzyme that hydrolyses phosphate from [hydroxymethylglutaryl-CoA reductase (NADPH)] phosphate. It thereby activates the latter enzyme.

hydroxymethylglutaryl-CoA synthase *abbr.*: HMG-CoA synthase; EC 4.1.3.5; *systematic name*: (S)-3-hydroxy-3-methylglutaryl-CoA acetoacetyl-CoA-lyase (CoA-acylating). An enzyme that catalyses a reaction between acetyl-CoA + H₂O and acetoacetyl-CoA to form (S)-3-hydroxy-3-methylglutaryl-CoA and CoA. In mammals it is a cytoplasmic and mitochondrial enzyme, catalysing the formation of hydroxymethylglutaryl-CoA from acetoacetyl-CoA and acetyl-CoA in cytoplasm for sterol biosynthesis, or in mitochondria for **ketone body** synthesis. Example from human: database code HMCS_HUMAN, 520 amino acids (57.29 kDa).

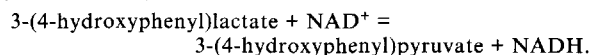
5-hydroxynorleucine *an older name (no longer recommended)* for 2-amino-5-hydroxyhexanoic acid; a nonprotein (non-coded) amino acid found in plants, e.g. *Crotalaria juncea*.



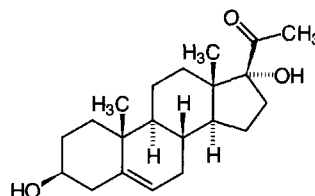
(R)-4-hydroxyphenyllactate dehydrogenase EC 1.1.1.222; *other name*: (R)-aromatic lactate dehydrogenase; an enzyme that catalyses the reaction:



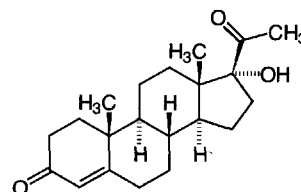
hydroxyphenylpyruvate reductase EC 1.1.1.237; an enzyme that catalyses the reaction:



17 α -hydroxypregnenolone 3 β ,17-dihydroxy-5-en-20-one; an intermediate in the synthesis of androgens, estrogens, and glucocorticoid hormones from cholesterol in the adrenal cortex. It is formed from **pregnenolone** by steroid 17 α -monooxygenase (EC 1.14.99.9).



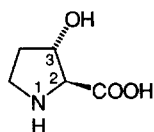
17 α -hydroxyprogesterone 17-hydroxypregn-4-one; an intermediate in the synthesis of androgens, estrogens, and glucocorticoid hormones from cholesterol in the adrenal cortex. It is formed from **progesterone** by steroid 17 α -monooxygenase (EC 1.14.99.9).



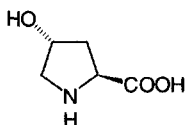
hydroxyproline *symbol*: Hyp or (formerly) Hypro; either of the two constitutional isomers 3-hydroxyproline and 4-hydroxyproline.

hydroxyprolinemia

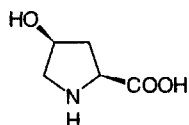
droxyproline, $\text{HO}-(\text{C}_4\text{H}_6\text{NH})-\text{COOH}$, both of which are chiral cyclic N-alkylated α -amino acids. Each constitutional isomer possesses two chiral centres; each can therefore exist as an enantiomeric (D and L) pair of each of two (*cis* and *trans*) diastereoisomers, making eight possible isomeric structures for hydroxyproline in all. Both *trans*-3-hydroxy-L-proline (symbol: 3Hyp or Pro(3-OH)), (2S,3S)-3-hydroxypyrrolidine-2-carboxylic acid, and *trans*-4-hydroxy-L-proline (symbol: 4Hyp or Pro(4-OH)), (2S,4R)-4-hydroxypyrrolidine-2-carboxylic acid, are **noncoded amino acids** found in peptide linkage in proteins. In collagen nearly 50% of the proline residues are hydroxylated to 4Hyp and a small proportion to 3Hyp; hydroxyproline residues are also found in elastin, enamel of the teeth, in the C1a moiety of the C1 component of **complement**, and in **extensin**. In collagen formation, enzymic hydroxylation of proline residues occurs while the growing peptide chains are still attached to ribosomes. Residues of both the *cis*-(2S,3R)-, and the *trans*-(2S,3S)-isomers of 3-hydroxy-L-proline, one of each per molecule, occur in the peptide antibiotic telomycin. One residue per molecule of *cis*-4-hydroxy-L-proline, the (2S,4S)-isomer, occurs in every member of the **amatoin** group of toxic fungal octapeptides. None of the D series of isomers is known to occur naturally.



trans-(2S,3S)-3-hydroxy-L-proline



trans-(2S,4R)-4-hydroxy-L-proline



cis-(2S,4S)-4-hydroxy-L-proline

hydroxyprolinemia or **hyperhydroxyprolinemia** a metabolic disorder of humans characterized by high levels of free 4-hydroxyproline in the blood plasma and urine. It is probably due to a deficiency in the activity of so-called hydroxyproline oxidase, i.e. the enzyme 4-oxoprolinase reductase (EC 1.1.1.104).

15-hydroxyprostaglandin dehydrogenase (NAD⁺) EC 1.1.1.141; an enzyme involved in the inactivation of prostaglandins E₂, F_{2α}, and B₁, by oxidizing the 15-hydroxyl group to a keto group. It catalyses a reaction between (5Z,13E)-(15S)-11α,15-dihydroxy-9-oxoprost-5,13-dienoate and NAD⁺ to form (5Z,13E)-11α-hydroxy-9,15-dioxoprost-5,13-dienoate (15-keto PGE₂) and NADH. Example from human: database code PGDH_HUMAN, 266 amino acids (28.94 kDa).

5-hydroxytryptamine *abbr.*: 5-HT; an alternative name for serotonin.

5-hydroxytryptamine receptor, any of several membrane

proteins that bind 5-hydroxytryptamine (i.e. **serotonin**) and mediate its intracellular effects. The receptors are classified pharmacologically into seven main subtypes or families; the members of these are designated 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, 5-HT_{1F}, 5-HT_{2A}, 5-HT_{2B}, 5-HT_{2C}, 5-HT₃, 5-HT₄, 5-HT_{5A}, 5-HT_{5B}, 5-HT₆, and 5-HT₇. The members of the 5-HT₁ family all have seven transmembrane domains and are negatively coupled to adenylate cyclase through G-proteins, but the 5-HT_{1A} receptor also increases K⁺ conductance; example, 5-HT_{1A} from rat: database code 5H1A_RAT, 422 amino acids (46.43 kDa). 5-HT_{1C} is now classified as a 5-HT₂ receptor. 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} have seven transmembrane domains and activate the phosphatidylinositol cycle; example, 5-HT_{2C} from rat: database code 5H2C_RAT, 460 amino acids (51.92 kDa). 5-HT₃ opens a Cl⁻ channel; 5-HT₄ has four transmembrane domains and is positively coupled to adenylate cyclase; 5-HT₅ has seven transmembrane domains but the main effector is unknown; example, 5-HT_{5A} from rat: database code 5H5A_RAT, 357 amino acids (40.67 kDa). 5-HT₆ and 5-HT₇ are both of the seven-transmembrane-helix type and are positively coupled to adenylate cyclase.

5-hydroxytryptophan a nonprotein amino acid found in plants; it is formed in legumes by hydroxylation of tryptophan. It is also formed in animals. *See also* **serotonin**.

hydroxyurea NH₂CONHOH; an analogue of **urea** with antineoplastic activity, possibly through inhibition of DNA synthesis.

hygro+ *comb. form* denoting moisture, especially that due to water.

hygrometer any instrument for measuring the relative humidity of a gas.

hygromycin B an aminoglycoside antibiotic from *Streptomyces hygroscopicus*; the inclusion of genes conferring resistance to hygromycin B in certain fungal recombinant DNA vectors allows selection of recombinants when these are plated out on medium containing this antibiotic.

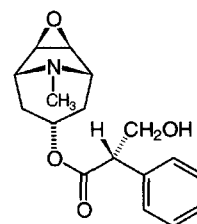
hygroscopic (*of a substance*) tending to absorb water from the surrounding atmosphere. *Compare* **deliquescent**.

Hyl or (formerly) **Hyls** symbol for a residue of any of the hydroxy derivatives of the α -amino acid L-lysine, especially 5-hydroxy-L-lysine. *See* **hydroxyllysine**.

5Hyl symbol for a residue of the α -amino acid 5-hydroxy-L-lysine (*preferred alternative to* Lys(5-OH) or Hyl). *See* **hydroxyllysine**.

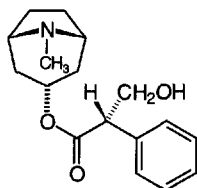
Hyls former symbol for a residue of (5-)hydroxy-L-lysine (5Hyl or Hyl now recommended).

hyoscine or **scopolamine** an alkaloid found, e.g., in *Datura stramonium* (thorn apple) and related to **atropine** in its structure and effects; it is a tertiary ammonium compound noted for its actions as a **muscarinic receptor** antagonist. It has found use as a sedative and preanesthetic and for prevention of motion sickness.



hyoscyamine 3α-tropanyl S-(-)-tropate; an anticholinergic material obtained from *Atropa belladonna* and other plants. It undergoes partial racemization on isolation; this process is completed by alkali treatment, and the racemic mixture is known as atropine. (Illustrated on p. 318.)

hypo+, a variant form of **hypo+** (*before a vowel*).



(-)-hyoscyamine

Hyp 1 or (formerly) **Hyp** symbol for a residue of either the 3-hydroxy or the 4-hydroxy derivative of the α -amino (strictly, α -imino) acid L-proline, usually 4-hydroxy-L-proline. See **hydroxyproline**. See also **3Hyp**, **4Hyp**. **2** symbol for a residue of the purine base **hypoxanthine**.

3Hyp symbol for a residue of the α -amino acid 3-hydroxy-L-proline (preferred alternative to Pro(3-OH) or Hyp). See **hydroxyproline**.

4Hyp symbol for a residue of the α -amino acid 4-hydroxy-L-proline (preferred alternative to Pro(4-OH) or Hyp). See **hydroxyproline**.

hyelcin A a 20-residue peptide of microbial origin containing 10 α -aminoisobutyric acid residues. The N terminus is acylated and the C terminus amidated with **leucinol**. It modifies the permeability of phospholipid bilayers.

hyper+ prefix indicating over, beyond, excessive, or in excess; above the normal range. Compare **hypo+**, **normo+**.

hyperaldosteronism an elevated level of **aldosterone** in the blood. In primary aldosteronism, there is overproduction of aldosterone due to a lesion of the adrenal(s), e.g. an adrenal tumour as in Conn's syndrome. In secondary aldosteronism the overproduction is caused by excessive levels of stimulatory hormones, especially increased renin secretion.

hyperammonemia or (esp. Brit.) **hyperammonaemia** the presence of excess ammonia in the blood (the normal range is 10–47 $\mu\text{mol L}^{-1}$). It occurs in a number of conditions including inherited disorders of the urea cycle, organic acidemias, liver disease, and severe systemic illness. See also **ornithine-urea cycle**.

hyperbaric of, relating to, or operating at higher than normal pressures.

hyperbilirubinemia or (esp. Brit.) **hyperbilirubinaemia** the presence of higher than normal concentrations of **bilirubin** in the blood; it may involve unconjugated or conjugated bilirubin. It occurs in liver disease or biliary obstruction, and is characteristic of neonatal hemolytic **jaundice**, where the bilirubin is mainly unconjugated. —**hyperbilirubinemic** or **hyperbilirubinaemic** *adj.*

hyperbola one of the conic sections; a plane curve in which the focal distance of any point bears to its distance from the directrix a constant ratio greater than unity. It has two foci and two asymptotes. Compare **rectangular hyperbola**. —**hyperbolic** *adj.*

hyperbolic function any one of a group of functions of an angle, analogous to the trigonometric functions, expressed as a relationship between the distances of a point on a hyperbola to the origin and to the coordinate axes. The group includes **sinh** (hyperbolic sine), defined as $\sinh x = (e^x - e^{-x})/2$; **cosh** (hyperbolic cosine), defined as $\cosh x = (e^x + e^{-x})/2$; and **tanh** (hyperbolic tangent), defined as $\tanh x = (\sinh x)/(\cosh x)$. The corresponding reciprocal functions are **cosec** (hyperbolic cosecant), **sech** (hyperbolic secant), and **coth** (hyperbolic cotangent), respectively.

hyperbolic kinetics curve a description sometimes used of the $v/[S]$ plot obtained for enzymes exhibiting **Michaelis-Menten kinetics**. The axes of the plot are not those of the rectangular hyperbola that results, but are rotated 45° (and displaced) from the true axes of the hyperbola.

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hypercalcemia or (esp. Brit.) **hypercalcaemia** the presence of higher than normal concentrations of **calcium** in the blood. Clinical features include tiredness, muscle weakness, anorexia, weight loss, cardiac arrhythmia, and hypertension. It arises from a multiplicity of causes including malignancy, primary hyperparathyroidism, hyperthyroidism, acute adrenal insufficiency, familial hypocalciuric hypercalcemia, vitamin D or A overdose, or sarcoidosis. —**hypercalcemic** or **hypercalcaemic** *adj.*

hypercapnia or **hypercarbia** the presence of greater than normal amounts of carbon dioxide in a vertebrate or in its blood.

Compare **hypocapnia**. —**hypercapnic** or **hypercarbic** *adj.*

hyperchlorhydria the presence of greater than normal amounts of hydrochloric acid in the gastric juice. **hyperchlorhydric** *adj.*

hypercholesterolemia or (esp. Brit.) **hypercholesterolaemia** the presence of higher than normal concentrations of cholesterol in the blood. See also **hyperlipidemia** —**hypercholesterolemic** or **hypercholesterolaemic** *adj.*

hyperchromatic 1 of, relating to, or exhibiting **hyperchromatism**. 2 an alternative word for **hyperchromic** (def. 1).

hyperchromatism the condition or property (of a cell or cell organelle) of staining more intensely than normal; the presence of a greater than normal amount of chromatin.

hyperchromic 1 or **hyperchromatic** more highly coloured than normal; intensely coloured. 2 of, pertaining to, exhibiting, or resulting from **hyperchromism**. 3 (of a substance) displaying a **hyperchromic effect**. 4 describing erythrocytes that contain, or appear to contain, more hemoglobin than normal.

hyperchromic effect the large increase in absorption of ultraviolet light, usually measured at 260 nm, shown by a solution of any natural or synthetic polynucleotide with a hydrogen-bonded structure when it is denatured or degraded. The effect is due to alterations in the electronic interactions between the initially stacked and hydrogen-bonded bases.

hyperchromicity a measure of the increase in absorption of electromagnetic radiation at a specific wavelength, usually that of the absorption maximum, due to **hyperchromism**, or to the **hyperchromic effect**. It is equal to the reciprocal of the **hypochromicity** minus unity.

hyperchromism the increased absorption of electromagnetic radiation exhibited by an ordered structure above that predicted on the basis of its constitution.

hyperfine splitting the splitting of a spectral line into multiplets of closely spaced lines. In electron spin resonance spectra hyperfine splitting is due to the interaction of unpaired electrons with neighbouring nuclei and can be used to determine the structure of a free radical or to identify the ligands of a paramagnetic ion and to measure the degree of covalent binding that exists between them. In nuclear magnetic resonance spectra hyperfine splitting, sometimes called **spin-spin splitting**, is due to the interaction of the nuclear magnetic moment with those of neighbouring nuclei and can be used to assign particular nuclear resonances and to determine molecular conformations.

hyperfine structure the presence in a spectrum of multiplets of closely spaced lines resulting from **hyperfine splitting**.

hyperglycemia or (esp. Brit.) **hyperglycaemia** an abnormally high blood glucose concentration, especially in relation to the fasting value. See also **glycemia**. —**hyperglycemic** or **hyperglycaemic** *adj.*

hyperglycemic factor or **hyperglycemic-glycogenolytic factor** a former name for **glucagon**.

hyperhydroxyprolinemia see **hydroxyprolinemia**.

hyperimmune describing the state of an animal in which very high concentrations of specific antibody are present in the serum, brought about by **hyperimmunization**.

hyperimmunization or **hyperimmunisation** any method of immunization designed to stimulate the production in the animal immunized of very large quantities of specific antibody. It typically involves the repeated administration of immunogen, often in increasing amounts.



hyperkalemia

hyperkalemia or (*esp. Brit.*) **hyperkalaemia** the presence in the blood of an abnormally high concentration of **potassium**. There are many causes including excessive K^+ intake (infusion, transfusion), increased transcellular movement (tissue damage, systemic acidosis), and decreased K^+ loss (renal failure, mineralocorticoid deficiency). —**hyperkalemic** or **hyperkalaemic** *adj.*

hyperlactatemia or (*esp. Brit.*) **hyperlactataemia** a clinical condition characterized by a persistent raised blood lactate concentration, usually below 5 mmol L^{-1} , and not accompanied by a lowered blood pH. The normal range is between 0.5 and 2.2 mmol L^{-1} . *Compare* **lactic acidosis**. —**hyperlactatemic** or **hyperlactataemic** *adj.*

hyperlipidemia or **hyperlipemia** or (*esp. Brit.*) **hyperlipidaemia** or **hyperlipaemia** the presence in the blood of an abnormally high concentration of fats (or lipids). Hyperlipidemias have been classified into various types according to which lipoprotein class is found to be elevated in the blood. The Fredrickson classification scheme designates: type I, elevated chylomicrons; type II, elevated low-density lipoprotein; type III, elevated intermediate-density lipoprotein; type IV, elevated very-low-density lipoprotein; type V, elevated chylomicrons and very-low-density lipoprotein. A simpler distinction, of use in treatment, is between elevated cholesterol levels (type II, **hypercholesterolemia**) and elevated triacylglycerol levels (types I, III–V, **hypertriglyceridemia**, an older term still often used). Hyperlipidemia may be secondary to other conditions; several drugs cause or exacerbate hyperlipidemia, including thiazides, **beta blockers** lacking intrinsic sympathomimetic activity, and **corticosteroids**. Estrogens may lower hypercholesterolemia, but may cause or exacerbate hypertriglyceridemia; hypothyroidism is commonly associated with hypercholesterolemia; high alcohol intake causes hypertriglyceridemia, but modest alcohol intake may have a beneficial effect on cholesterol status, as it tends to increase **high-density lipoprotein** concentration. —**hyperlipidemic**, **hyperlipemic** or **hyperlipidaemic**, **hyperlipaemic** *adj.*

hypermodified base any extensively modified nucleic-acid base occurring adjacent to the 3' end of the anticodon in many tRNA molecules. Examples are N^6 -(δ_2 -isopentenyl)adenosine (*symbol*: iA) and N^6 -(threonylcarbamoyl)adenosine (*symbol*: tA). Hypermodified bases are thought to be important for proper binding of tRNA molecules to ribosomes.

hypernatremia or (*esp. Brit.*) **hypernatraemia** the presence in the blood of an abnormally high concentration of sodium. It may be associated with conditions in which body water is depleted. —**hypernatremic** or **hypernatraemic** *adj.*

hyperoxaluria the presence of oxalate in the urine in abnormally high amounts. Primary hyperoxaluria is a rare inherited metabolic disorder of which there are two types: one results in increased oxalate synthesis, the other in increased oxalate excretion by the kidney. More usually, hyperoxaluria results from increased absorption of oxalate from the intestine. It is associated with the formation of renal calculi.

hyperoxia the presence of greater than normal amounts of dioxygen in a vertebrate or in its blood. *Compare* **hypoxia**. —**hyperoxic** *adj.*

hyperparathyroidism the increased production of **parathyroid hormone**. It may be primary, as a result of a tumour of the parathyroid gland, or secondary, as a result of a disturbance in calcium metabolism.

hyperphosphatemia or (*esp. Brit.*) **hyperphosphataemia** an excessive level of phosphate in the blood. It occurs most commonly in renal insufficiency, but may also accompany hypoparathyroidism, vitamin D intoxication, or excessive administration of phosphate. —**hyperphosphatemic** or **hyperphosphataemic** *adj.*

hyperplasia the increase in size of a tissue or organ resulting from an increase in the total number of cells present. The part thus affected retains its normal form. *Compare* **hypertrophy**. —**hyperplastic** *adj.*

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hyperprolactinemia or (*esp. Brit.*) **hyperprolactinaemia** the presence in the blood of abnormally high levels of **prolactin**, due to excessive secretion of the hormone. The most common example of pituitary hyperfunction, it results from pituitary adenoma, and also from treatment with dopamine receptor blockers. It can be treated with dopamine agonists, e.g. **bromocriptine**. —**hyperprolactinemic** or **hyperprolactinaemic** *adj.*

hypersecretion excessive secretion by any endocrine or exocrine gland.

hypersensitive having an abnormally great sensitivity, especially to an allergen, drug, or other agonist.

hypersensitivity **1** the state or condition of being **hypersensitive**. **2** (*in immunology*) an immune response that occurs in an exaggerated or inappropriate form. Such responses may result from agents including pollen or drugs, or from genuine pathogens. Also of importance are the different kinds of tissue damage seen in autoimmune diseases. In general terms the reactions may be classified into four types, although they do not occur in isolation from each other: type I, **immediate hypersensitivity**; type II, **antibody-mediated hypersensitivity**; type III, **immune complex-mediated hypersensitivity**; and type IV, **delayed hypersensitivity**.

hypersharping the formation of a 'sharper' boundary in sedimentation and moving boundary electrophoresis experiments than would be expected as a result of diffusion-broadening, because molecules in the trailing portion of the boundary tend to move at a greater velocity per unit field than molecules on the leading edge or in the plateau region.

hypertension the state or condition of having a higher than normal arterial blood pressure. —**hypertensive** *adj.*

hyperthyroidism or **thyrotoxicosis** a clinical state resulting from excessive secretion of thyroid hormones. The most common cause is Graves' disease; other causes include goitre and solitary toxic adenoma. The clinical manifestations include nervousness, irritability, fatigue, and heat intolerance; exophthalmia is present in some cases. Most symptoms derive from exaggeration of normal responses to thyroid hormones. *See also* **thyroxine**. —**hyperthyroid** *adj.*

hypertonic **1** (*of a solution*) having a higher osmotic pressure than that of some given solution, especially that of the fluid in or surrounding a given type of cell or a body fluid. **2** (*of a muscle*) showing or characterized by excessive tone or tension. *Compare* **hypotonic**. —**hypertonicity** *n.*

hypertriglyceridemia or (*esp. Brit.*) **hypertriglyceridaemia** *see* **hyperlipidaemia**.

hypertrophy the increase in the size of a tissue or organ resulting from an increase in the size of the cells present. *Compare* **hyperplasia**. —**hypertrophic** *adj.*

hyperuricemia or (*esp. Brit.*) **hyperuricaemia** the presence of abnormally high amounts of uric acid in the blood. This may be due to increased purine synthesis arising from a metabolic disorder, inherited or otherwise, or it may be due to reduced excretion – this may result from renal disease, the effect of drugs (e.g. salicylates) on the kidney, or the presence of organic acids (e.g. lactate) that competitively inhibit urate excretion. Hyperuricemia is associated with **gout**.

hypervariable regions regions in an **immunoglobulin** molecule where the amino-acid residues show a high degree of variation between one molecule and another. There are three hypervariable regions, two in the **V region** and one in the **D region**.

hypo a common name for sodium thiosulfate, especially when used in photographic processing.

hypo+ or (*before a vowel*) **hyp+** prefix denoting under, beneath, below, less than, or in deficit; below the normal range. *Compare* **hyper+**, **normo+**.

hypopalpalipoproteinemia *see* **hypolipoproteinemia**.

hypobaric of, relating to, or operating at pressures lower than normal.

hypobetalipoproteinemia *see* **hypolipoproteinemia**.

hypocalcemia or (*esp. Brit.*) **hypocalcaemia** the presence of a lower than normal concentration of **calcium** in the blood. It

hypocapnia

may be associated with disorders of vitamin D metabolism, **hypoparathyroidism**, magnesium deficiency, or pancreatitis. Clinical features include behavioural disturbances, stupor, numbness, paresthesia, muscle cramps or spasms, and convulsions. —**hypocalcemic** or **hypocalcaemic** *adj.*

hypocapnia or **hypocarbica** the presence of less than the normal amount of carbon dioxide in the vertebrate or in its blood. *Compare* **hypercapnia**. —**hypocapnic** or **hypocarbic** *adj.*

hypochlorhydria the presence of less than the normal amount of hydrochloric acid in the gastric juice. —**hypochlorhydric** *adj.*

hypcholesterolemia or (*esp. Brit.*) **hypcholesterolaemia** the presence of a lower than normal concentration of cholesterol in the blood. —**hypcholesterolemic** or **hypcholesterolemic** *adj.*

hypochromatic 1 of, relating to, or exhibiting **hypochromatism**. 2 an alternative word for **hypochromic** (def. 1).

hypochromatism the condition or property (of a cell or cell organelle) of staining less intensely than normal; the presence of an abnormally low amount of chromatin.

hypochromic 1 or **hypochromatic** less highly coloured than normal; weakly coloured. 2 of, pertaining to, exhibiting, or resulting from **hypochromism**. 3 (*of a substance*) displaying a **hypochromic effect**. 4 describing erythrocytes that contain, or appear to contain, less hemoglobin than normal.

hypochromic effect the observed decrease in absorption of certain frequencies of electromagnetic radiation by solutions of some macromolecules when the structure of the molecules becomes more ordered.

hypochromicity a measure of the decrease in absorption of electromagnetic radiation at a specific wavelength, usually that of the absorption maximum, due to **hypochromism**.

hypochromism the decreased absorption of electromagnetic radiation exhibited by an ordered structure below that predicted on the basis of its constitution.

hypoglycemia or (*esp. Brit.*) **hypoglycaemia** the presence of an abnormally low blood glucose concentration, especially in relation to the fasting value. *See also* **glycemia**. —**hypoglycemic** or **hypoglycaemic** *adj.*

hypoglycin or **hypoglycine** α -(methylenecyclopropyl)alanine; 2-amino-3-(methylenecyclopropane)propanoic acid. A non-protein α -amino acid identified as the hypoglycemic and toxic principle of the unripe fruit of the **ackee** (*Blighia sapida*), where it is present both in the fleshy aril and in the seeds. In the latter it occurs also as the *N*-L-glutamyl derivative, **hypoglycin B**. Certain other fruits also contain hypoglycin. It is degraded in animals to methylenecyclopropylacetate, either free or conjugated to glycine or coenzyme A. Methylenecyclopropylacetyl-CoA is a potent and specific inhibitor of the oxidation of short-chain acyl-CoAs, particularly of butyryl-CoA. In hypoglycin poisoning butyryl-CoA undergoes hydrolysis to yield free butyrate, which accumulates in the blood to abnormal concentrations. *See also* **Jamaican vomiting sickness**.

Hyp(3-OH) *symbol* for a residue of the α -amino acid 3-hydroxy-L-proline (alternative to 3Hyp or Hyp). *See* **hydroxyproline**.

Hyp(4-OH) *symbol* for a residue of the α -amino acid 4-hydroxy-L-proline (alternative to 4Hyp or Hyp). *See* **hydroxyproline**.

hypokalemia or (*esp. Brit.*) **hypokalaemia** the presence in the blood of an abnormally low concentration of potassium. It may result from alkalosis, insulin administration, primary or secondary aldosteronism, or administration of drugs that are aldosterone antagonists. —**hypokalemic** or **hypokalaemic** *adj.*

hypolipoproteinemia the partial or total absence of any lipoprotein, or the presence of a defective lipoprotein, in the plasma. Several types of this condition are recognized. **Hypobetalipoproteinemia** is arbitrarily defined as a low density lipoprotein (LDL) cholesterol level below the fifth percentile for a given sex- and age-matched population. Such patients also have low triacylglycerol levels. In this condition, LDL cholesterol is always detectable, in contrast to homozygous **abetalipoproteinemia**, in which it is entirely absent. Hypobetalipoproteinemia appears to be due to a low rate of synthesis of

LDL from very-low-density lipoprotein. **Abetalipoproteinemia** is a rare autosomal recessive disorder. It probably results from an abnormality in the synthesis or secretion of apo B, and it is accompanied by fat malabsorption, as chylomicron formation requires apo B. In the homozygote, levels of triacylglycerols, cholesterol, and phospholipids are extremely low. Heterozygotes may present with hypobetalipoproteinemia. **Hypoalphalipoproteinemia** is in some cases familial, and may result from defective apo A-I. In another disorder, known as fish-eye disease (due to severe corneal opacity) the structure of apo A-I is normal, but high-density lipoprotein (HDL) levels are only 10% of normal. In **analphalipoproteinemia**, or **Tangier disease**, HDL is virtually absent.

hyponatremia or (*esp. Brit.*) **hyponatraemia** the presence in the blood of an abnormally low concentration of sodium. It can occur as a result of incorrect administration of fluid intravenously or in parenteral feeding, after administration of diuretics, in congestive heart failure, or in protein imbalance. —**hyponatremic** or **hyponatraemic** *adj.*

hypoparathyroidism underactivity of the **parathyroid glands**.

hypophase the lower layer of a two-phase system. *Compare* **epiphase**. —**hypophasic** *adj.*

hypophosphatemia or (*esp. Brit.*) **hypophosphataemia** the presence of an abnormally low level of phosphate in the blood. It occurs in vitamin D deficiency, primary **hyperparathyroidism**, incorrect parenteral nutrition, and during recovery from diabetic ketoacidosis.

hypophysectomy surgical removal of the pituitary gland (i.e. hypophysis). —**hypophysectomize** or **hypophysectomise** *vb.*

hypophysiostrophic factor or **hypophysiostrophic hormone** an alternative name for **hypothalamic (regulatory) factor**.

hypophysis or **hypophysis cerebri** an alternative name for **pituitary gland**. *See also* **adenohypophysis**, **neurohypophysis**. —**hypophyseal** or **hypophysial** *adj.*

hypopituitarism total or (more often) partial loss of pituitary function, usually due to destructive lesions of the pituitary.

hypoplasia underdevelopment or defective formation of an organ or tissue. —**hypoplastic** *adj.*

hypoprothrombinemia or (*esp. Brit.*) **hypoprothrombinaemia** the presence of abnormally low (inadequate) levels of plasma prothrombin (factor II). It is commonly linked with vitamin K deficiency.

hyposecretion lesser than normal secretion by any endocrine or exocrine gland.

hypotension the state or condition of having a lower than normal arterial blood pressure. —**hypotensive** *adj.*

hypothalamic (regulatory) factor or **hypothalamic hormone** or **hypophysiostrophic factor** or **hypophysiostrophic hormone** any peptidic substance that is synthesized in the mammalian hypothalamus and released into the hypophyseal-portal circulation in response to neural and/or chemical stimuli, and that regulates the secretion (and perhaps also the synthesis) of a specific polypeptide hormone by the anterior lobe of the pituitary gland (i.e. **adenohypophysis**). These specific factors may be either releasing or release-inhibiting factors (or hormones); their recommended names are now formed by replacing the word-ending 'tropin' (or '-in' in prolactin) in the name of the corresponding pituitary hormone by the word-ending '-liberin' (for a releasing factor) or '-statin' (for a release-inhibiting factor).

hypothalamus (*pl.* **hypothalami**) the region of the vertebrate brain lying below the thalamus and around the floor of the third ventricle, just posterior to the attachment of the cerebral hemispheres. It lies just above the **pituitary gland**, to which it supplies various **hypothalamic (regulatory) factors**. —**hypothalamic** *adj.*

hypothyroidism deficient secretion by the thyroid gland, for whatever reason. In infants it leads to cretinism; in adults the consequences include low metabolic rate, lethargy, menstrual disorders, and in extreme cases **myxedema**. —**hypothyroid** *adj.*

hypotonic 1 (*of a solution*) having a lower osmotic pressure



hypovolemia

than that of some given solution, particularly the fluid in or surrounding a given type of cell or a body fluid. **2** (of a muscle) showing or characterized by diminished tone or tension. *Compare hypertonic. —hypotonicity n.*

hypovolemia or (*esp. Brit.*) **hypovolaemia** a reduction in the volume of circulating blood. It may be due to hemorrhage or to redistribution of fluid from the plasma to the extravascular tissues and spaces.

hypox *abbr. for* hypophysectomized. *See hypophysectomy.*

hypoxanthine *symbol:* Hyp; purin-6(1*H*)-one; 6-hydroxy purine; an intermediate in the degradation of adenylate. Its ribonucleoside is known as **inosine** and its ribonucleotide as **inosinate**. *Compare xanthine. See also hypoxanthine phosphoribosyltransferase.*

hypoxanthine-guanine phosphoribosyltransferase *abbr.:* HGPRT; *see hypoxanthine phosphoribosyltransferase.*

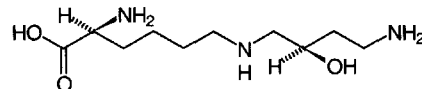
hypoxanthine phosphoribosyltransferase EC 2.4.2.8; *systematic name:* IMP: pyrophosphate phospho-D-ribosyltransferase; *other names:* hypoxanthine-guanine phosphoribosyltransferase (*abbr.:* HGPRT); IMP pyrophosphorylase. An enzyme that catalyses a reaction between 5-phospho- α -D-ribose 1-diphosphate and either hypoxanthine or guanine to form IMP or GMP respectively, with release of pyrophosphate. It is an enzyme of the purine salvage pathway, and a deficiency is associated with **Lesch-Nyhan syndrome** and **gout**. Example from human: database code HPRT_HUMAN, 217 amino acids (24.45 kDa); many mutations leading to deficiency diseases have been identified.

hypoxia the presence of less than normal amounts of dioxygen in a vertebrate or in its blood. *Compare hyperoxia. —hypoxic adj.*

Hypro (*formerly*) *symbol for* a residue of (4-)hydroxy-L-proline (4Hyp or Hyp now recommended).

hypsochromic describing the shift of an absorption band in the direction of shorter wavelengths. *Compare bathochromic.*

hypusine *N*^ε-[4-amino-2-hydroxybutyl]lysine; an amino acid uniquely formed by post-translational modification of one of the lysyl residues of eukaryotic translation initiation factor 5A (eIF-5A). It is formed by deoxyhypusine synthase, an enzyme that transfers the butylamine portion of spermidine to the ϵ -NH₂ group of a specific lysine of the eIF-5A precursor, followed by addition of the hydroxyl group by deoxyhypusine hydroxylase.



hysteresis the delay between the effect on a system and the cause producing the effect. *—hysteretic adj.*

hysteretic enzyme any enzyme that responds slowly (in terms of some kinetic characteristic) to a rapid change in ligand (either substrate or modifier) concentration. Such slow changes, defined in terms of their rates relative to the overall catalytic reaction, result in a lag in the response of the enzyme to changes in the ligand level.

Hz *symbol for* hertz.

H zone *see sarcomere.*

I

i *symbol for 1* iso (def. 3); (as a prefix or superscript in symbols and abbreviations for iso compounds or groups; e.g. isopropyl can be symbolized as iPr or Prⁱ). **2** the chemical group 2-isopentenyl; 3,3-dimethylallyl (as a prefix to a single-letter symbol for a nucleoside residue to designate substitution of the base, as in **IA**). **3** the square root of -1.

i *symbol for 1* electric current (alternative to *I*). **2** van't Hoff factor.

I *symbol for 1* iodine. **2** an iodo group in an organic compound. **3** a residue of the α -amino acid L-isoleucine (alternative to Ile). **4** a residue of the ribonucleoside inosine (alternative to Ino).

I *symbol for 1* electric current (preferred alternative to *i*). **2** intensity; the subscripts e (for energetic), p (for photon), or v (for visible) may be added to distinguish between radiant intensity, photon intensity, and luminous intensity, respectively. **3** ionic strength; the subscripts c or m may be added to indicate whether it is expressed based on concentration or molality respectively. **4** the inductive effect of a particular chemical group in an organic compound.

IA *symbol for* N⁶-(2-isopentenyl)adenosine, a **hypermodified base**.

IAA *abbr. for* indole-3-acetic acid or **indoleacetate**.

IAN *abbr. for* indoleacetonitrile.

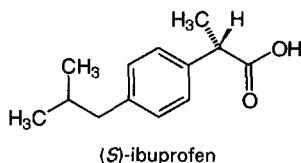
IAP *abbr. for* islet-activating protein; *see* **pertussis toxin**.

iatrogenic describing a condition or disease induced unintentionally by a physician through his or her diagnosis, manner, or therapy. —**iatrogenicity** *n*.

I-band *abbr. for* isotropic band. The I-bands of striated muscle contain the **thin filaments** and correspond to the light bands. The name derives from the fact that they are isotropic in polarized light. *See also* **sarcomere**.

IBMX *abbr. for* isobutylmethylxanthine.

ibuprofen 2-(4-isobutylphenyl)propionic acid; a nonsteroidal anti-inflammatory agent of the substituted propionic acid type; others of this type are flurbiprofen, **ketoprofen**, and naproxen. They inhibit the cyclooxygenase activity of **prostaglandin-endoperoxide synthase**, reversibly over short time intervals, followed by time-dependent irreversible inactivation, due probably to conformational rather than covalent changes. The racemic mixture is present in many over-the-counter drugs, but the (*S*)-enantiomer is the active form. *Proprietary names include:* Advil; Brufen; Motrin.



ic or **i.c.** *abbr. for* intracutaneous or intracutaneously.

IC *abbr. for* internal conversion.

IC₅₀ **1** the median inhibitory concentration (in mol L⁻¹) of an antagonist, i.e. the concentration that reduces a specified response to 50% of its former value; *compare* **EC₅₀**. **2** the median inhibitory concentration (in mol L⁻¹) of an agent (agonist or antagonist), i.e. the concentration that causes a 50% reduction in the specific binding of a radioligand.

ICAM *abbr. for* intercellular adhesion molecule; any of several type I membrane glycoproteins of the immunoglobulin superfamily. They act as ligands for leukocyte adhesion to target cells, in conjunction with **LFA-1**; in fact LFA-1-ICAM links mediate adhesion between many cell types. There are three subclasses. **ICAM-1** (or CD54), of molecular mass 90–115

kDa, are expressed on B and T cells, endothelial, epithelial, and dendritic cells, fibroblasts, keratinocytes, and chondrocytes. They are inducible in 12–24 hours by cytokines such as gamma interferon, interleukin-1 β , and tumour necrosis factor- α . Example (human, precursors): **ICAM-1** major group rhinovirus receptor: database code ICA1_HUMAN, 532 amino acids (57.76 kDa). **ICAM-2** (or CD102), of molecular mass 55–65 kDa, are constitutively expressed on endothelial cells, some lymphocytes, monocytes, and dendritic cells. Example (human): database code ICA2_HUMAN, 275 amino acids (30.62 kDa). **ICAM-3** (or CD50), of molecular mass 116–140 kDa, are constitutively expressed on monocytes, granulocytes, and lymphocytes; upon physiological stimulation they become rapidly and transiently phosphorylated on serine residues. Example (human): database code ICA3_HUMAN, 547 amino acids (59.32 kDa).

ICD (*in clinical chemistry*) *abbr. for* isocitrate dehydrogenase (EC 1.1.1.41).

ice the solid form of water that can exist at temperatures below the triple point of water (273.16 K at 101 325 Pa).

iceberg a metaphor used to describe an interpretation of the anomalous entropies of solution of noble gases and other non-polar substances in water, suggesting that water tends to organize itself into quasi-solid supramolecular structures around the molecules of such substances. In the case of alkyl compounds, this tendency increases markedly with the length of the alkyl chain.

ICE-like protease any of a family of endopeptidases that structurally resemble **interleukin-1 β convertase** (ICE). They are involved in **apoptosis**, being implicated in the proteolysis that causes cell death.

I cell 1 or **CCK cell** any of a group of cells, widely distributed in the duodenal and jejunal mucosa, that produce **cholecystokinin**. So named because their histological features are intermediate between those of **S cells** and **L cells**. **2** *abbr. for* inclusion cell. *See also* **I-cell disease**.

I-cell disease or **inclusion-cell disease** or **mucopolidosis II** an autosomal recessive disease in which most of the lysosomes in the connective tissue (fibroblasts) contain large inclusions of glycoaminoglycans and glycolipids as a result of the absence of several lysosomal hydrolases. These enzymes, which are synthesized in the endoplasmic reticulum, are secreted into the extracellular medium rather than being directed to the lysosomes. This is due to the absence of a mannose 6-phosphate marker on the carbohydrate moieties of these hydrolases because of a deficiency in an enzyme required for mannose phosphorylation. The failure of the phosphorylation in the *cis* Golgi network means that the enzymes are not segregated by the **mannose 6-phosphate receptors** into the appropriate transport vesicles in the *trans* Golgi network. The patients have an abnormally high level of lysosomal enzymes in their sera and body fluids. A milder form of I-cell disease is Hurler's polydystrophy (**Hurler syndrome**).

ice point the temperature at which ice melts. It is taken as the temperature (273.15 K) at which ice and water are in equilibrium at standard pressure (101 325 Pa). It was used as a reference temperature on the Celsius scale, but the kelvin is based instead on the temperature at the **triple point** of water (273.16 K).

ICF *abbr. for* intracellular fluid.

ichthyotocin *an alternative name for* **isotocin**.

icos+ or (*before a vowel*) **icos+** *comb. form* recommended for denoting twenty or twenty times. *Also (formerly):* **eicosa+**, **eicos+**. (*Note:* the eicos(a)+ variant is still always used for the C₂₀ fatty acids and skeletally related compounds, e.g. the eicosanoids.)

icosadeltahedron (*pl.* **icosadeltahedra**) any solid geometrical



icosahedral symmetry

figure obtained by dividing the triangular faces of an **icosahedron** into smaller triangles. If divided into more than four smaller triangles, the original faces will not be flat, in which case the 12 pentameric vertices will be supplemented by hexameric vertices. —**icosadeltahedral** *adj.*

icosahedral symmetry a form of **cubic symmetry**, being the crystallographic symmetry displayed by an **icosahedron** and by the capsid of some types of virus.

icosahedron (*pl.* **icosahedra**) any solid geometrical figure having 20 triangular faces, 30 edges, and 12 pentameric vertices. A regular icosahedron has faces that are congruent plane equilateral triangles; there is twofold rotational symmetry about an axis through the midpoint of each edge, threefold rotational symmetry about an axis through the centre of each face, and fivefold rotational symmetry about an axis through each vertex. —**icosahedral** *adj.*

icosanoate or (formerly) **eicosanoate** the systematic name for **arachidate**.

icosanoid the recommended name for **eicosanoid**, but not generally used.

ICRO *abbr.* for International Cell Research Organization.

ICSH *abbr.* for interstitial cell-stimulating hormone (i.e. **luteinizing hormone**).

ICSU *abbr.* for International Council of Scientific Unions.

icterus (*in pathology*) an alternative name for **jaundice**.

icterogenic causing **jaundice**.

id or **i.d.** *abbr.* for **1** intradermal, or intradermally (alternative to **ID**). **2** inside diameter.

ID **1** (*in clinical chemistry*) *abbr.* for iditol dehydrogenase (EC 1.1.1.14). **2** *abbr.* for intradermal, or intradermally (alternative to **id**).

ID₅₀ or **ID₅₀** *abbr.* for median infectious dose.

idaric acid the **aldaric acid** derived from idose.

IDDM *abbr.* for insulin-dependent diabetes mellitus (*see* **diabetes mellitus**).

+ide *suffix* (*in chemical nomenclature*) denoting a **lactide**. It replaces the terminal '+ic acid' of the trivial name of the parent hydroxy acid, and a multiplying prefix indicates the number of molecules of hydroxy acid involved; e.g. dilactide (formed from two molecules of lactic acid).

ideal describing any object, process, or system that is perfect, or that behaves exactly in accordance with some simple law or theory. *See also* **ideal gas**, **ideal solution**. —**ideality** *n.*

ideal gas any gas whose behaviour is accurately described by the **gas laws**.

ideal solution any solution in which, for each component,

$$d\mu_i = RT \ln x_i,$$

where μ_i is the chemical potential of the *i*th component and x_i is its mole fraction, R is the universal gas constant, and T is the thermodynamic temperature. Integration of the above equation gives:

$$\mu_i = \mu_i^\circ + RT \ln x_i,$$

where μ_i° is the standard chemical potential of the *i*th component.

idio+ *comb. form* denoting **1** peculiar, distinct, one's own, separate. **2** self-produced, arising within.

idiogram or **karyogram** a diagrammatic representation of the **karyotype** of a cell, based on measurement of the chromosomes in several or in many cells.

idiom (*in immunology*) the group of **idiotypes** to which the **epitopes** carried by components of one individual animal belong.

idiopathic **1** (*in pathology*) describing a disease arising by itself, not consequent upon, or symptomatic of, another disease; of the nature of a primary morbid state; essential. **2** of the nature of a particular affection or susceptibility. —**idiopathy** *n.*

idiopathic pentosuria *see* **pentosuria** (def. 1).

idiotope **1** the group of **epitopes** carried by components of one individual animal. **2** as originally defined by **Jerne**, an individual. *Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:*

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ual determinant (i.e. site) on an antibody molecule; together, the idiotopes constitute the **idiotype**.

idiotype **1** an antigenic specificity, particularly of antibodies directed against a single antigen. **2** any class or group of related antigenic characteristics or **idiotopes**; an **epitope** that is characteristic of a particular animal. —**idiotypic** *adj.*

iditol the hexitol derived formally by reduction of the aldehyde group of idose. *L*-Iditol is a rare natural product accompanying sorbitol in the fermented juice of berries of mountain ash (*Sorbus aucuparia*).

IDL *abbr.* for intermediate-density lipoprotein.

IDNA *abbr.* for intercalary DNA.

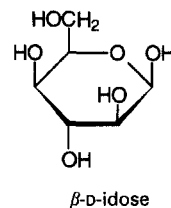
IDNPAP *abbr.* for isethionyl-3-(*N*-2,4-dinitrophenyl)amino propionimide; a reagent useful in the isolation of plasma membranes. It reacts with, but does not penetrate, the membranes and leaves the net charge on the proteins unaltered.

ido- *prefix* (*in chemical nomenclature*) indicating a particular configuration of a set of four (usually) contiguous >CHOH groups, as in the acyclic form of *D*- or *L*-idose. *See* **monosaccharide**.

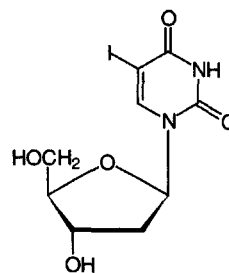
Ido *symbol* for a residue (or sometimes a molecule) of the aldohexose idose.

IdoA *symbol* for a residue of iduronic acid.

idose *symbol:* **Ido**; *ido*-hexose; an aldohexose whose *D*- and *L*-enantiomers are epimeric at C-2, C-3, and C-4 with *D*- and *L*-glucose, respectively.



idoxuridine 5-iodo-2'-deoxyuridine; a pyrimidine analogue, metabolized to the triphosphate, that inhibits DNA replication by substituting for thymidine in viral and mammalian DNA incorporation. It is used as an antiviral agent.



IDP *abbr.* for **1** the ribonucleotide inosine (5')-diphosphate. **2** isopentenyl diphosphate (*formerly called* Δ^3 -isopentenyl pyrophosphate).

IDU *abbr.* for 5-iodo-2'-deoxyuridine (*see* **idoxuridine**).

IdUrd *symbol* for 5-iodo-2'-deoxyuridine (*see* **idoxuridine**) (*preferred* to **IDU**, **IUDr**, or **IUDR**).

iduronate **1** the anion of **iduronic acid**. **2** any salt or ester of iduronic acid.

iduronate-2-sulfatase EC 3.1.6.13; *other name:* chondroitin-sulfatase; an enzyme that hydrolyses the 2-sulfate groups of the *L*-iduronate 2-sulfate units of dermatan sulfate, heparan sulfate, and heparin. Example (precursor) from human: database code IDS_HUMAN, 550 amino acids (61.80 kDa).

iduronic acid *symbol:* **IdoA**; the **uronic acid** formally derived



L-iduronidase

from idose by oxidation to a carboxyl group of the hydroxymethylene group at position 6. In nature α -linked L-iduronic-acid residues, sulfated at position 2, occur in terminal positions in dermatan sulfate and heparan sulfate; these iduronic-acid residues are formed by epimerization of the configuration at C-5 of D-glucuronic-acid residues in these heteropolysaccharides.

L-iduronidase EC 3.2.1.76; an enzyme that hydrolyses α -L-iduronosidic linkages in desulfated dermatan. Example (precursor) from human; database code IDUA_HUMAN, 653 amino acids (72.58 kDa).

iduronosyl the glycosyl group formed by removal of the anomeric hydroxyl group from **iduronic acid** or iduronate.

IE *abbr.* for immunoelectrophoresis (alternative to IEP).

IEC *abbr.* for ion-exchange cellulose.

IED *abbr.* for individual effective dose. IED₅₀ denotes the median of the IEDs within a group of subjects.

IEF *abbr.* for isoelectric focusing.

IEP *abbr.* for 1 immunoelectrophoresis (alternative to IE). **2** or **i.e.p.** isoelectric point (pI preferred).

IF *abbr.* for initiation factor. Suffixed numerals are added to designate individual factors, e.g. **IF-1**, **IF-2**, **IF-2a**, **IF-3**. Eukaryotic initiation factors are generally denoted by the prefix 'e', e.g. eIF-2, eIF-3.

IF-1 an *Escherichia coli* initiation factor.

IF-2 an *Escherichia coli* initiation factor.

IF-3 an *Escherichia coli* initiation factor.

IFCC *abbr.* for International Federation of Clinical Chemistry.

IFN *abbr.* for interferon. Suffixed Greek letters are added to differentiate the main classes, e.g. IFN- α , IFN- β , IFN- γ .

Ig *symbol* for immunoglobulin.

IgA *symbol* for immunoglobulin A.

IgA protease another name for **IgA-specific serine endopeptidase**.

IgA-specific serine endopeptidase EC 3.4.21.72; *other name*: IgA protease; an enzyme that catalyses the cleavage of immunoglobulin A molecules at certain Pro-I-Xaa bonds in the hinge region. No small molecule substrates are known. It is an excreted bacterial serine proteinase. A signal peptide guides the enzyme precursor to the periplasmic space, and the C-terminal helper domain associates with the outer membrane to form a pore for excretion of the protease domain. The helper domain is then released by autolysis. Example from *Neisseria gonorrhoeae*; database code IGA_NEIGO, 1532 amino acids (168.79 kDa).

IG cell the intestinal gastrin cell, a small, granulated, gastrin-producing cell occurring in the human upper small intestine. IG cells react both with gastrin-specific antisera and with antisera recognizing the C-terminal pentapeptide sequence common to gastrin and cholecystokinin (CCK). IG cells are smaller than, and ultrastructurally different from, CCK-producing **I cells**, gastrin-producing **G cells** of the gastric pylorus, and **TG cells**.

IgD *symbol* for immunoglobulin D.

IgE *symbol* for immunoglobulin E.

igensin a neutral cytosolic serine proteinase, activated by sodium dodecyl sulfate and found in a number of mammalian tissues.

IGF *abbr.* for insulin-like growth factor.

IGFBP *abbr.* for insulin-like growth factor binding protein.

IgG *symbol* for immunoglobulin G. Suffixed numerals are added to designate individual subclasses, e.g. IgG1, IgG2, IgG3, IgG4.

Ig-like domain *see* immunoglobulin fold.

IgM *symbol* for immunoglobulin M.

β IG-M1 *see* connective tissue growth factor.

β IG-M2 *see* connective tissue growth factor.

ignose the name first proposed for ascorbic acid by A. Szent-Györgyi, before he knew its structure. *See also* **godnose**.

IGT *abbr.* for impaired glucose tolerance.

Ii system a blood group system in which the antigenic determinant is an oligosaccharide structure. It is structurally related to

the ABH system (*see* **ABH antigens**), anti-I and anti-i antibodies being inhibited by ABH and Lewis glycoproteins. Anti-I and anti-i are the major **cold autoantibodies**; the anti-I factors occur in the transient anemia accompanying viral or *Mycoplasma pneumoniae* infections; the i-specific antibodies are characteristic of the so-called chronic cold agglutinin disease, accompanying chronic lymphoproliferative diseases of the non-Hodgkin type. The structures of the determinants are: i determinant, GlcNAc(β 1-3)Gal(β 1-4)GlcNAc β 1-R; I determinant, Gal(β 1-4)[Gal(β 1-4)GlcNAc(β 1-6)]GlcNAc(β 1-3)Gal(β 1-4)GlcNAc β 1-R.

ikaros a protein that binds and activates the **enhancer** of a gene *CD3- δ* involved in T-lymphocyte specification and maturation. Similar to **hunchback**, it is a zinc-finger protein. Example from mouse; database code IKAR_MOUSE, 431 amino acids (48.13 kDa).

IL *abbr.* for interleukin. Various distinct interleukins are distinguished by suffixed numerals, e.g. IL-1, IL-2, IL-3, etc.

ILA *abbr.* for insulin-like activity.

Ile *symbol* for a residue of the α -amino acid L-isoleucine (alternative to I).

ileum the lower part of the small intestine, extending from the jejunum to the large intestine. —**ileal** *adj.*

illegitimate transcription transcription at a low level of a normally tissue-specific transcribing gene in nonspecific cells.

IM or **i.m.** *abbr.* for intramuscular, or intramuscularly.

imide **1** any salt or ester of an **imidic acid**. **2** to convert an oxo diacid into its corresponding **imide**. —**imidation** *n.*

imidazole or (formerly) **glyoxaline** or **iminazole** 1,3-diazole; 1,3-diaza-2,3-cyclopentadiene; a weak base, $pK_a \approx 6.95$. The numbering and tautomerism of the imidazole ring are described at **histidine**.



imidazole 4-acetate a product of histamine metabolism; it results from the action of monoamine oxidase on histamine to form imidazole 4-acetaldehyde, which is then oxidized by an NAD⁺-dependent aldehyde dehydrogenase to imidazole 4-acetate. *See also* **histidine**.

imidazoline receptor any of a class of mitochondrial or plasma membrane binding sites for compounds from the generic groups imidazoles, imidazolines, imidazolidines, guanidines, and oxazolines, and having a putative endogenous ligand, **agmatine**. They occur throughout the brain, where they are possibly associated with glial cells, and in peripheral tissues. Two subtypes (at least) have been characterized pharmacologically: **I₁ receptors**, which have high affinity for *p*-aminoclonidine and are localized in plasma membranes, and **I₂ receptors**, which have high affinity for idazoxan and are localized in mitochondrial membranes. The latter are known alternatively as nonadrenergic imidazoline binding sites (NAIBSs). Photoaffinity labelling to at least three proteins occurs: two, of *M_r* 55 000 and 61 000, are found in rat liver and differ in their binding affinity for the guanidium, amiloride; a third, of *M_r* 61 000, is found in PC-12 cells and has high affinity for amiloride but low affinity for idazoxan. These proteins may represent different **I₂ receptors**. Subtypes of **I₂ receptors** are also suggested pharmacologically. The function of imidazoline receptors is not yet known, but many of the effects of imidazoline compounds, formerly attributed to their binding to α_2 receptors (e.g. hypotension) are now attributed to NAIBSs.

imidazolium **1** the cation formed by addition of a proton (hydron) to a molecule of **imidazole**. **2** the cationic chemical group formed by addition of a proton (hydron) to an **imidazolyl** group.

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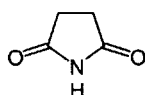
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imidazolyl the chemical group derived formally by removal of one hydrogen atom from **imidazole**. The position from which the hydrogen atom has been removed may be indicated, e.g. 2-imidazolyl. The imidazolyl group is present in **histidine** and its derivatives.

imide any secondary organic amide (i.e. any diacyl derivative of ammonia or a primary amine), especially any that is a cyclic compound in which both of the acyl groups (which may be the same or different) are derived formally or actually from a diacid. (Note: Noncyclic symmetrical secondary organic amides are generically termed **diacylamines**.)

+imide suffix (*in chemical nomenclature*) designating any organic compound that is an **imide**. The nitrogen atom may be substituted with a named alkyl or other group. An example is succinimide:



succinimide

imidic acid any derivative of a carboxylic acid in which the carbonyl oxygen atom of the carboxyl group has been replaced by an **imino** (def. 1) or alkylimino group; i.e. any containing the $-C(=NR)-OH$ group. *Distinguish from imino acid.*

imido+ *comb. form* designating any inorganic oxo acid (including the anions, salts, and esters) in which $=O$ has been replaced by $=NH$ or $=NR$ (where R is a named substituent), or any derived di-, tri-, etc. acid in which a bridging oxygen atom, $-O-$, has similarly been replaced by $-NH-$ or $-NR-$; e.g., **adenosine 5'-[β , γ -imido]triphosphate**.

Imigran *see* **sumatriptan**.

iminazole a former name for **imidazole**.

imine 1 any organic compound having the general structure $R_2C=NR$ where R may be any organyl group or H. *See also* **aldimine**, **azomethine**, **ketimine**, **Schiff base**. 2 an obsolete term for **azacycloalkane**.

iminium the bivalent quaternary nitrogen-containing cationic group $=^+NR_2$ attached to carbon in a chemical compound, where the R groups may each be hydrogen or any organyl group.

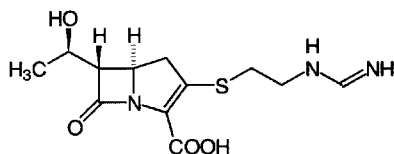
imino the bivalent group $=NH$ attached to carbon in a chemical compound.

imino+ *comb. form* denoting the presence in a chemical compound of one or more **imino** groups; the number of imino groups in the molecule may be specified, e.g. diimino+.

imino acid any carboxylic acid in which an imino substituent replaces two hydrogen atoms. In biochemistry the term is commonly applied also to certain cyclic alkylamino (especially α -alkylamino) derivatives of aliphatic carboxylic acids, e.g. **proline**, although such compounds are now preferably classed as **azacycloalkane carboxylic acids**. *Compare* **imidic acid**.

iminodipeptidase *see* **Pro-X dipeptidase**.

imipenem a carbapenem β -lactam antibiotic that is a semisynthetic derivative of thienamycin produced by *Streptomyces cattleya*. Its broad-spectrum antibiotic actions resemble those of penicillins, i.e. inhibition of bacterial peptidoglycan synthesis.



immediate gene or **immediate-early gene** *see* **early gene**.

immediate hypersensitivity or **type I hypersensitivity** a type of **hypersensitivity** responsible for asthma, hayfever, and some types of eczema. It occurs within minutes of exposure to antigen and is dependent on the activation of **mast cells** and the release of mediators of acute inflammation. The mast cells bind **immunoglobulin E** (IgE) via their surface **Fc receptors** and when antigen cross-links the IgE, the mast cells degranulate releasing vasoactive amines that mediate the symptoms.

immobilize or **immobilise** to render any agent, whether a micro- or macromolecule, a particle, or an intact cell, nondispersible in an aqueous medium with retention of its specific ligating, antigenic, catalytic, or other properties. Immobilization may be achieved by, e.g., encapsulation, entrapment in a small-pore gel, adsorption on, or covalent linkage to, an insoluble supporting material (matrix), or through formation of aggregates by cross-linkage.

immobilized enzyme *see* **solid-phase technique**.

immortalization or **immortalisation** the transformation of a cell population with a finite life span to one possessing an infinite life span. It is a characteristic of cancer cells, and is of practical importance in, e.g., the creation of **monoclonal antibodies**. Immobilization of eukaryotic cells may be effected experimentally by infection with certain viruses, or by fusion of the cell with a neoplastic cell to form a **hybridoma**. —**immortalize** or **immortalise** *vb*.

immune 1 relating to or possessing **immunity**. 2 regarding or describing an animal that has been immunized. 3 an alternative word for **immunological**.

immune adherence the **complement**-dependent adherence of antibody-antigen complexes, or of antibody-coated bacteria (or other particles), to primate erythrocytes, causing the agglutination of the latter.

immune clearance an alternative term for **immune elimination**.

immune complex an alternative term for **antigen-antibody complex**.

immune complex-mediated hypersensitivity or **type III hypersensitivity** a type of **hypersensitivity** caused by the deposition of antigen-antibody complexes in tissue and blood vessels. The complexes activate **complement** and attract polymorphs and macrophages to the site.

immune cytolysis the **complement**-dependent lysis of cells by antibody molecules. When the cells concerned are erythrocytes the process is termed **immune hemolysis**.

immune elimination or **immune clearance** the accelerated, exponential removal of antigen from an immune animal as a result of complexing with antibody molecules.

immune globulin (*sometimes*) an alternative name for **immunoglobulin**.

immune hemolysis *see* **immune cytolysis**.

immune response the total immunological reaction of an animal to an immunogenic stimulus. It includes antibody formation, the development of hypersensitivity, and immunological tolerance.

immune serum an alternative term for **antiserum**.

immunity freedom and protection from 'infection'. Two forms of immunity can be recognized: (1) innate, meaning non-specific anti-microbial systems (e.g. phagocytosis) that are innate in that they are not intrinsically affected by prior contact with the infectious agent; and (2) the state of an animal that has an enhanced ability, above the nonimmune state, to respond to some (specific) antigen in which the antigen is bound and rendered inactive or eliminated from the body. *See also* **active immunity**, **passive immunity**.

immunization or **immunisation** any procedure carried out on an animal, e.g. the administration of antigen or antibody, that leads to an increased reactivity of the animal's immune system towards an antigen or antigens. —**immunize** or **immunise** *vb*.

immuno+ *comb. form* indicating immune or immunity.

immunoabsorbent any insoluble preparation of an antigen



immunoabsorption

(or antibody) suitable for use in **immunoabsorption**. *See also* **immunosorbent**.

immunoabsorption the use of an insoluble antigen (or antibody) to remove specifically unwanted antibodies (or antigens) from a mixture so as to make the antibody (or antigen) more specific. *Compare* **adsorption** (def. 2). *See also* **immunosorption**.

immunoaffinity chromatography a type of **affinity chromatography** in which one of the two components of an antigen-antibody system is coupled to an insoluble matrix and used for the separation and purification of the other component.

immunoaffinity column a chromatographic column used in **immunoaffinity chromatography**.

immunoassay 1 any of a group of techniques for the measurement of specific biochemical substances, commonly at low concentrations and in complex mixtures such as biological fluids, that depend upon the specificity and high affinity shown by suitably prepared and selected antibodies for their complementary antigens. A substance to be measured must, of necessity, be antigenic – either an immunogenic macromolecule or a haptenic small molecule. To each sample a known, limited amount of specific antibody is added and the fraction of the antigen combining with it, often expressed as the bound:free ratio, is estimated, using as indicator a form of the antigen labelled with a radioisotope (**radioimmunoassay**), fluorescent molecule (**fluoroimmunoassay**), stable free radical (**spin immunoassay**), enzyme (**enzyme immunoassay**), or other readily distinguishable label. The amount of the antigen in the sample is found by comparison with standards containing known amounts. *See also* **heterogeneous immunoassay**, **homogeneous immunoassay**, **saturation analysis**. 2 an alternative name for **radioimmunoassay**. 3 to carry out or to estimate (something) by an **immunoassay** (def. 1, 2).

immunobiology the branch of biology dealing with the activities of the cells of the immune system and their relationship to each other and to their environment. —**immunobiological** *adj.*

immunochemical 1 of or relating to **immunochemistry**. 2 any specific immunological reagent that consists of or incorporates an antigen or an antibody, especially one of commerce.

immunochemistry the branch of biochemistry dealing with the chemical nature of antigens, antibodies, and their interactions, and with the chemical methods and concepts as applied to immunology.

immunochromatogram the pattern of precipitation bands formed in **immunochromatography**.

immunochromatography any of various techniques for separating and identifying soluble antigens. In one method, analogous to **immuno-electrophoresis**, the antigens are first fractionated through a thin-layer chromatogram, e.g. of Sephadex; the fractions are then diffused against a trough containing a solution of antibodies (antiserum). In another method a small drop of antigen-antiserum mixture is applied to filter paper, or resin-impregnated paper, and the chromatogram developed with buffer; any antigen-antibody complex remains at the point of application and can be detected with a protein stain, while the other proteins are washed away from the origin.

immunocompetent cell an immunologically competent cell.

immunocyte any immunologically competent cell.

immunocytochemistry **cytochemistry** using appropriately labelled antibody preparations to detect specific cellular components. —**immunocytochemical** *adj.*

immunodeficiency any condition in which there is a deficiency in the production of humoral and/or cell-mediated immunity. —**immunodeficient** *adj.*

immunodeterminant an alternative name for **antigenic determinant**.

immunodiagnosis the diagnosis of disease by immunological methods.

immunodiagnostic 1 of or relating to **immunodiagnosis**. 2 any specific immunological reagent used in diagnosis.

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immunodiffusion any of various analytical methods by which components of soluble antigen or antibody mixtures may be distinguished. Essentially, the antigens and antibodies are allowed to diffuse towards each other in a translucent gel, wherein they react to give lines or bands of precipitation in characteristic positions. Such methods include: **double immunodiffusion**, **electroimmunodiffusion**, **immunochromatography**, **immuno-electrophoresis**, **immunorheophoresis**, and **single radial immunodiffusion**. —**immunodiffuse** *vb.*

immunodominance the attribute of a part of an **epitope** that contributes a disproportionately great portion of the binding energy; e.g. the immunodominance shown by a monosaccharide residue in determining the antigenic specificity of a polysaccharide. —**immunodominant** *adj.*

immuno-electrofocusing or **immunoisoelectric focusing** a technique that is analogous to **immuno-electrophoresis** but with the preliminary fractionation of one of the reactants being effected by electrofocusing instead of electrophoresis.

immuno-electron microscopy a form of electron microscopy in which structures are stained with specific antibodies labelled with an electron-dense material.

immuno-electropherogram (*sometimes*) the record of an immuno-electrophoresis experiment, either the electrophoretic support itself or a tracing derived therefrom.

immuno-electrophoresis *abbr.*: IE or IEP; a technique for separating and identifying soluble antigens. It consists of **zone electrophoresis**, in a reasonably transparent gel, in one direction followed by **immunodiffusion** against a solution of antibodies (antiserum) placed in a trough parallel to the direction of electrophoresis. In suitable conditions each antigen shows up as an arc of precipitation in a characteristic position. —**immuno-electrophoretic**, **immuno-electrophoretic** *adj.*

immuno-enzymatic of or pertaining to the particular techniques used in **immuno-enzymology**.

immuno-enzymology the subspeciality of immunochemistry in which the activity of enzymes coupled to antigens or antibodies is utilized as a molecular amplifier of antigen-antibody reactions.

immunofixation or **immunofixation electrophoresis** a variant of **immuno-electrophoresis**, or **immuno-electrofocusing**, in which proteins of a single immunological species are anchored to the support by treatment with monospecific antibodies, allowing the other proteins to be washed out of the support. It is useful in the identification of proteins in complex mixtures or of minor proteins obscured in electrophoretic patterns.

immunofluorescence or **immunofluorescence microscopy** a technique in which an antigen or antibody is made fluorescent by conjugation to a fluorescent dye and then allowed to react with the complementary antibody or antigen in a tissue section or smear. The location of the antigen or antibody can then be determined by observing the fluorescence by microscopy under ultraviolet light. —**immunofluorescent** *adj.*

immunogelfiltration a variant of **radial single immunodiffusion** in which antigen is first fractionated according to its molecular size by permeation, in a buffer solution, into a thin layer of an appropriate cross-linked dextran gel. The dextran layer is then covered with a layer of agarose gel containing antibody. Rings or spots of antigen-antibody precipitate develop in the agarose gel over the complementary antigens.

immunogen 1 any substance that, when introduced into the body, elicits humoral or cell-mediated immunity, but not immunological tolerance. 2 any substance that is able to stimulate an immune response, as distinct from substances (antigens) that can combine with antibody. 3 any substance that is able to stimulate protective immunity against a pathogen, as distinct from an immune response that is of no intrinsic value to the animal immunized. —**immunogenic** *adj.*

immunogenetics (*functioning as sing.*) the branch of biology that combines immunology and genetics. It includes the use of immunological methods and knowledge in the study of genet-



ics, and the use of genetics in the study of immunological phenomena and substances. —**immunogenetic** *adj.*

immunogenicity *an alternative term for antigenicity.*

immunoglobulin *symbol:* Ig; any member of a group of proteins occurring in higher animals as major components of the immune system. They are produced by cells of the lymphocyte series, and virtually all possess specific antibody activity. Each immunoglobulin molecule essentially comprises four polypeptide chains, two identical **heavy chains** and two identical **light chains**, linked together by disulfide bonds; all contain in addition differing amounts of attached oligosaccharide. There are five classes, IgA, IgD, IgE, IgG, and IgM. The primary structures of the heavy chains differ among the various classes, being designated α , δ , ϵ , γ , and μ , respectively for the classes listed above, and there are two types of light chain, κ and λ . IgA and IgM molecules contain multiples of the four-chain unit. The homology units form a characteristic motif, the **immunoglobulin fold**. *See also* immunoglobulin A, immunoglobulin D, immunoglobulin E, immunoglobulin G, immunoglobulin M.

immunoglobulin A *symbol:* IgA; an immunoglobulin found in human plasma, where its role is uncertain. It occurs mostly as a 170 kDa monomer of the basic four-chain immunoglobulin structure or, as a 400 kDa dimer, together with an extra 15 kDa polypeptide, the **J chain**, linked to **secretory piece**. It is the major immunoglobulin of seromucous secretions, in which it is concerned in the defence of external body surfaces against attack by microorganisms. It has the structure $(\alpha_2)_n\lambda_2$, where α is the heavy chain, λ is the light chain (κ or λ), J is the J chain, and n is 1 or 2. Example, α_1 C region: database code ALC1_HUMAN, 353 amino acids (37.61 kDa). *See also* IgA-specific serine endopeptidase.

immunoglobulin D *symbol:* IgD; a 185 kDa immunoglobulin that is very susceptible to proteolytic degradation and has a very short half-life in plasma. It is found in large amounts on the surface of B lymphocytes together with immunoglobulin M; it is probable that these two immunoglobulins function as mutually interacting antigen receptors and play a part in the control of lymphocyte activation and suppression. It has the structure $\delta_2\lambda_2$ where δ is the heavy chain and λ is a light chain (κ or λ). Example, heavy chain (C region): database code DTC_HUMAN, 383 amino acids (42.08 kDa).

immunoglobulin E *symbol:* IgE; a 200 kDa immunoglobulin of high carbohydrate content, normally found in very low concentrations in plasma, although its concentration there is elevated in a number of allergic conditions. IgE is synthesized mainly in lymphoid tissue of gut and respiratory tract. It binds strongly to mast cells and contact with antigen leads to degranulation of the mast cells and release of inflammatory mediators. It has the structure $\epsilon_2\lambda_2$ where ϵ is a heavy chain and λ is a light chain (κ or λ). Example, heavy chain C region: database code EPC_HUMAN, 428 amino acids (46.97 kDa).

immunoglobulin fold a series of β strands, arranged as two antiparallel β sheets packed tightly against each other, that is highly characteristic of immunoglobulin structure. In the constant region, the two β sheets are formed from four and three β strands respectively. In the variable region there are nine β strands (five and four) rather than seven. This type of structure is found in many other proteins, especially cell surface receptors, and is often referred to as an **Ig-like domain**.

immunoglobulin G *symbol:* IgG; the principal immunoglobulin of human plasma and other internal body fluids. It is a 150 kDa protein and is the major immunoglobulin synthesized during the period of secondary response to an antigen. In the tissue fluids it neutralizes bacterial toxins and binds to microorganisms thereby enhancing their phagocytosis. Complexes of IgG with microorganisms activate **complement** by the classical pathway. IgG binds to macrophages and polymorphonuclear lymphocytes and can cross the placenta. It has the structure $\gamma_2\lambda_2$ where γ is a heavy chain and λ is a light chain (κ or λ). There are subclasses of IgG depending on different types of γ chain. Example, IgG γ_1 C region: database code

GC1_HUMAN, 330 amino acids (36.06 kDa); 3-D structure known.

immunoglobulin M *symbol:* IgM; a 900 kDa immunoglobulin consisting of a star-shaped polymer of five basic four-chain structures, each heavy chain bearing an extra constant domain. Polymerization is dependent on the presence of **J chain**, which may act to stabilize the sulfhydryl groups in the constant portion of the heavy chains during IgM synthesis. IgM is produced early in the immune response and is mainly found intravascularly. Because of its high antigen-combining valency it is very effective in binding microorganisms and mediating **complement**-dependent cytotoxicity. It has the structure $(\mu_2)_5\lambda_5$ where μ is a heavy chain, λ is a light chain (κ or λ), and J is a J chain. Virgin B lymphocytes have an altered sequence at the C terminus of μ and the switch is effected by alternative splicing. Example, the protein from **heavy chain disease**, which has no C and C_H1 regions due to deletion: database code MUCB_HUMAN, 391 amino acids (43.00 kDa).

immunoglobulin superfamily a superfamily of proteins that have similarities of sequence and structure to immunoglobulins. The superfamily has been divided into three sets, based on immunoglobulin domains: one with variable-like domains, the V set, and two with different variants of the constant-like domains, the C1 and C2 sets.

immunogram (*sometimes*) the pattern of bands or lines of precipitation formed in a gel by **immunodiffusion**.

immunohistochemistry a form of **histochemistry** in which appropriately labelled antibody preparations are used to detect specific structures in tissues. —**immunohistochemical** *adj.*

immunoliposome a **liposome** bearing a chemically coupled monoclonal antibody.

immunolocalization or **immunolocalisation** the use of appropriately labelled antibody (or antigen) preparations to detect the position of specific structures or components that contain or are the complementary antigens (or antibodies).

immunological or (*esp. US*) **immunologic**, concerned with, or pertaining to immunity or immunology. —**immunologically** *adv.*

immunologically competent (*of cells, tissues, etc.*) qualified for and capable of giving an **immune response**.

immunological paralysis *an alternative name for acquired tolerance.*

immunological rejection the destruction, by a specific immune reaction, of foreign cells or tissues transplanted or inoculated into a recipient animal from a donor animal.

immunological response any specific response to an antigen. It includes **cell-mediated immunity**, **humoral immunity**, and **immunological tolerance**.

immunological surveillance the postulated monitoring of the cells of large, long-lived animals by their immunological systems so that aberrant cells arising from somatic mutations, and so containing new antigens, are destroyed.

immunological tolerance a type of immunological response in which there develops a specific nonreactivity of the lymphoid tissues towards a given antigen, one that in other circumstances is able to induce cell-mediated or humoral immunity. Immunological tolerance may follow contact with the antigen in fetal or early postnatal life or the administration of the antigen to adults (**acquired tolerance**). The reaction towards other, unrelated antigens is unaffected.

immunology the study of **immunity** and related phenomena. The science of immunology grew out of the study of resistance to infectious disease, and now encompasses the study of antigens, antibodies, and their interactions both *in vivo* and *in vitro*, and the cellular phenomena of recognition of, and responsiveness to, foreign substances. Related disciplines include **immunobiology** and **immunochemistry**.

immunometry the measurement of amounts of substances by the use of specific antigen-antibody reactions. —**immunometric** *adj.*; **immunometrically** *adv.*

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immunomodulation

immunomodulation the alteration of an immune response by an agent other than the antigen.

immunomodulator any agent that alters the extent of the immune response to an antigen, by altering the antigenicity of the antigen or by altering in a nonspecific manner the specific reactivity or the nonspecific effector mechanisms of the host. Immunomodulators include adjuvants, immunostimulants, and immunosuppressants.

immunoosmophoresis an alternative term for counter-(immuno)electrophoresis.

immunophenotyping the typing of cells with immunological markers such as monoclonal antibodies.

immunophilin any member of a family of receptors that includes the major FK506 binding protein FKBP and cyclophilin. These two proteins are unrelated in amino-acid sequence, but both possess peptidyl-prolyl isomerase activity which is blocked by immunosuppressants that block signal-transduction pathways leading to T-cell activation such as FK506 and rapamycin, which block FKBP, or cyclosporin A, which blocks cyclophilin. Example, the rapamycin-selective 25 kDa human immunophilin: database code FKB2_HUMAN, 224 amino acids (25.15 kDa).

immunoprecipitate 1 the precipitate formed in an antigen-antibody reaction. 2 to precipitate (something) by reaction with a specific antibody or antigen. —**immunoprecipitation** *n*.

immunoradiometric assay *abbr.*: IRMA; an alternative method to radioimmunoassay for the measurement of very small amounts of nonradioactive material. An excess amount of a specific antibody labelled with a radioactive isotope is added to the sample containing the substance to be assayed. After equilibration, unreacted antibody is removed and the amount of radioactive material remaining is measured.

immunoreactive capable of reacting with a specific antibody. The term is used especially of substances that react with an antibody directed against a particular peptide hormone (or part of such a hormone) but that do not necessarily exhibit the expected physiological or pharmacological activity. —**immunoreactivity** *n*.

immunosorbent any sorbent used in, or useful for, immunosorption. *See also* immunoadsorbent.

immunosorption the use of antigens (or antibodies) immobilized on a solid matrix to remove specifically the complementary antibodies (or antigens) from a mixture and their subsequent elution from the solid phase. It is used in, and is useful for, purification of antibodies (or antigens). *See also* immunoadsorption.

immunostimulant any agent that nonspecifically enhances the immunologically specific reactivity of an animal to an antigen and also the animal's nonspecific effector mechanisms. —**immunostimulatory** *adj*.

immunosuppressant or (sometimes) **immunosuppressive** any agent that causes immunosuppression.

immunosuppressed describing an organism that is incapable of showing an immune response to an antigen.

immunosuppression the suppression of immune responses to antigens. This can be achieved by various means, including physical (e.g. X-irradiation), chemical (e.g. antimetabolic drugs), or biological (e.g. antilymphocyte serum). After transplant surgery azathioprine is commonly used. This drug has a preferential effect on T-cell-mediated reactions. It is degraded in the body first to mercaptopurine and is then converted to the active agent, the ribotide, which inhibits nucleic-acid synthesis. Methotrexate, cyclophosphamide, cyclosporin A, and FK506 are also highly effective immunosuppressants, the latter two being used extensively. Infection by HIV-1 eventually leads to a reduction in the population of helper T-cells and macrophages. This immunosuppression of the system is associated with occurrence of Kaposi's sarcoma and allows infection by agents such as *Pneumocystis carini*, cytomegalovirus, and tuberculosis. *Compare* immunodeficiency. *See also* suppressor cell.

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immunosuppressive 1 able to cause immunosuppression. 2 an alternative term for immunosuppressant.

immunosympathectomy the destruction of some or all of the neurons of the sympathetic nervous system by injections into a newborn animal of an antiserum to the appropriate nerve growth factor.

immunotoxin any of a class of therapeutic drugs consisting of a monoclonal antibody linked to a toxic protein (e.g. ricin, diphtheria toxin).

IMP *abbr.* for 1 inosine monophosphate; i.e. inosine phosphate. 2 ion-moderated partition.

impaired glucose tolerance *abbr.*: IGT; the term recommended to denote a clinical state or condition of humans that is intermediate between the normal one and the overtly diabetic. Characteristically, in a standard oral glucose-tolerance test in non-pregnant adults, the glucose concentration in capillary whole blood or venous plasma taken at 2 hours is in the range 8–11 mmol L⁻¹ (1.4–2.0 g L⁻¹); the corresponding values for venous whole blood are 7–10 mmol L⁻¹ (1.2–1.8 g L⁻¹). In addition, the fasting plasma glucose must be below 8 mmol L⁻¹, and plasma glucose during the test must be greater than 11 mmol L⁻¹ at 30, 60, and 90 min.

IMP cyclohydrolase EC 3.5.4.10; *other names*: IMP synthetase; inosinase; an enzyme of the pathway for *de novo* purine biosynthesis that cyclizes 5-formamido-1-(5'-phosphoribosyl)imidazole-4-carboxamide to IMP with elimination of H₂O. For an example, *see* phosphoribosylaminoimidazolecarboxamide formyltransferase.

IMP dehydrogenase EC 1.1.1.205; *other name*: inosine-5'-monophosphate dehydrogenase; an enzyme that catalyses the conversion of inosine 5'-phosphate, NAD⁺, and H₂O to xanthosine 5'-phosphate and NADH. The reaction involves the introduction of an oxo group at the purine C-2. Xanthosine 5'-phosphate is then converted to guanosine 5'-phosphate by GMP synthase (glutamine hydrolysing). Examples from human, in which there are two isoenzymes: database code IMP1_HUMAN, 514 amino acids (55.39 kDa); database code IMP2_HUMAN, 514 amino acids (55.74 kDa).

impedance *symbol*: Z; the quality determining the amplitude of the current flowing in a circuit for a given applied alternating electric potential. It depends upon the electric resistance, self-inductance, and capacitance of the circuit.

impeller 1 the vaned, rotating disk of a centrifugal pump. 2 any device incorporating an impeller (def. 1).

impermeable (of a substance or structure) not permitting the passage through it of a fluid or of a particular solute in a fluid; not permeable. —**impermeability** *n*.

impermeant (of a fluid or a solute in a fluid) incapable of passing through a particular substance or structure.

imprinting (*in genetics*) the differential degree to which the effects of maternally and paternally derived DNA are exerted. For example, the onset of Huntington's disease is earlier if the defective allele is inherited from the father rather than the mother.

impulse 1 (*in physics*) a vector quantity equal to the integral of the force, *F*, acting on a body with respect to the time, *t*, over which the force acts: $\int F dt$. If the force is constant the expression reduces to the product *Ft*. It also equals the total change of momentum of the body induced by the applied force. 2 (*in physics*) an electrical surge of unidirectional polarity. 3 (*in physiology*) or **nerve impulse** an all-or-none signal propagated along the axons of neurons by which information is transmitted rapidly and precisely through the nervous system. It consists of a sequence of changes in ionic permeability of the neuronal membrane that gives rise to ionic currents and associated changes in transmembrane electrical potential.

impure not pure, lacking purity; (of a substance or material) containing contaminants.

impurity 1 the state or quality of being impure. 2 any or all of the admixed components in an impure substance or material.

impurity quenching

impurity quenching chemical quenching due to an **impurity** (def. 2).

In symbol for indium.

inappropriate secretion a clinical syndrome in which the secretion of a specified hormone, commonly **antidiuretic hormone**, is abnormally high relative to the state of the normal regulatory mechanisms. It is so termed because it results from an unknown or unmistakably abnormal cause and serves no recognizable homeostatic function.

inborn error of metabolism any of the genetically determined biochemical variations occurring in humans. They are highly specific and represent many diverse metabolic phenomena, resulting in very varied effects on the viability of the individual.

inbred strain or **inbred line** a group of organisms obtained by repeated inbreeding over several successive generations, e.g. by self-pollination in plants or by repeated brother × sister matings in animals. Eventually, all individuals will possess identical sets of **autosomes**, and all the chromosome pairs will become homozygous.

inclusion (*in cytology*) 1 any discrete body or particle within a cell, especially a passive product of cell activity such as a starch or volutin granule. 2 *an alternative (less common) name for inclusion body.*

inclusion body or **inclusion** 1 any discrete assembly of virions and/or viral particles that may be visualized in virus-infected cells, either in the nucleus or in the cytoplasm. 2 (*in bacteriology*) any one of a number of bodies found within bacterial cells, e.g. storage granules, gas vacuoles, carboxysomes.

inclusion cell *abbr.*: I cell; a type of cell that is seen in tissues from patients with mucopolipidosis II (**I-cell disease**) and that contains inclusions consisting of very large lysosomes whose contents are heterogeneous, comprising mucopolysaccharide, whorls of membrane, and other material.

Inclusion-cell disease or **mucopolipidosis II** *see* **I-cell disease**.

inclusion complex or **inclusion compound** any chemical complex in which one component (the host molecule) forms a crystal lattice containing tunnel- or channel-shaped spaces in which molecular entities of a second species (the guest molecule) are located. There is no bonding between the host and the guest molecules.

incompatibility 1 a criterion for classifying bacterial **plasmids**; two plasmids of the same incompatibility group cannot coexist in one host cell. The molecular basis is that the incompatible plasmids share sites during plasmid **segregation** (def. 3). 2 the state of being **incompatible**.

incompatibility group a group to which plasmids are said to belong if they cannot coexist in the same bacterial host over several generations without selection.

incompatible 1 (*in immunology*) having antigenic nonidentity between a donor and a recipient, e.g. in blood transfusion or tissue transplantation. 2 (*in therapeutics*) (*of two drugs*) incapable of being used together or combined; antagonistic.

incomplete antibody 1 any antibody whose activity in an agglutination test can only be demonstrated indirectly, e.g. by an antiglobulin test. 2 (*sometimes*) a univalent antibody fragment produced enzymically, e.g. a **Fab fragment**.

incomplete Freund's adjuvant *see* **Freund's adjuvant**.

incorporation 1 the act or process of including (something) as a part of a whole. 2 (*in biochemistry*) the inclusion of (isotopically labelled) metabolites, by covalent linkage, into biopolymers by living cells or cell-free preparations; often taken as an index of biosynthesis. —**incorporate** *vb*.

incretin a putative insulinogenic principle secreted from the duodenum and jejunum in response to the presence of glucose. Its effects are now generally considered to be attributable to **glucose-dependent insulinotropic peptide**.

incubate 1 to maintain at an appropriate temperature so as to favour growth, development, or continued survival (e.g. of cells); by extension, to maintain under specified conditions in a controlled or artificial environment (e.g. in studies of en-

zymic or antigen-antibody reactions), especially at a particular ambient temperature (which may be above or below that of the environment). 2 a material or preparation that has been incubated. —**incubation** *n*.

incubation mixture any reaction mixture that is maintained at a controlled temperature and/or in a controlled, artificial environment.

incubation period 1 (*in pathology*) the time interval between the invasion of an organism by a pathogenic virus or microorganism and the overt manifestation of disease. 2 (*in microbiology*) the period of development of a culture of bacteria or other microorganisms.

incubator any closable, heat-insulated cabinet that is maintained at a constant internal temperature (and, sometimes, humidity and/or atmosphere) and that is used for the growth or maintenance of cells, cultures, tissues, or organisms, or for the hatching of eggs.

Inderal *see* **propranolol**.

index of refraction *see* **refractive index**.

Indian tobacco *see* **lobelia**.

indican 1 metabolic **indican** indol-3-yl hydrogensulfate; 3-indoxylsulfuric acid; a so-called ethereal sulfate, formed in the gut by bacterial action on tryptophan and excreted in the urine, where its level may be taken as an index of intestinal stagnation. 2 **plant indican** indol-3-yl- β -D-glucopyranoside; indoxyl- β -D-glucoside; a substance isolated from *Indigofera* spp., members of the Leguminosae, and *Polygonum tinctorium*. The glucoside is hydrolysed during extraction by water or dilute acid, and the indoxyl is liberated spontaneously, oxidizing to the blue dye indigo.

indicated hydrogen *symbol*: *H-* (*or sometimes H*) (with a prefixed locant); (*in chemical nomenclature*) an atom of hydrogen whose position in the molecular structure of an unsaturated cyclic organic compound or group requires to be explicitly stated in order to make the name of the compound or group specific to a particular structural isomer (or tautomer) if otherwise the name would apply equally to two or more isomers (or tautomers). The compound or group in question commonly consists of or is derived from a condensed ring system with the maximum number of non-cumulative double bonds, although the notation is applied also to certain monocyclic systems. The symbol (with its locant) ordinarily precedes the name or the relevant part of the name, e.g. 1*H*-purin-6-amine or 6-amino-1*H*-purine (alternative semisystematic names for one of the tautomers of adenine); however, where the insertion of a substituent into a cyclic structure has necessitated the addition of a hydrogen atom at another position the symbol is placed in parentheses after the locant for the substituent, e.g. purin-6(1*H*)-one (*a semisystematic name for one of the tautomers of hypoxanthine*).

indicator 1 any substance used in a chemical operation to indicate by a colour change the completion of a reaction or the attainment of a desired state. 2 any substance that by a characteristic colour change indicates the presence of another particular substance. 3 (*in saturation analysis*) the labelled substance whose distribution between the reactants of the system is used to determine the amount of analyte present. 4 an isotope, often a radioactive one, that is used as a tracer.

indicator enzyme *an alternative name for marker enzyme*.

indicator organism or **indicator species** any species of organism whose presence or population characteristics in a particular habitat are used to provide information about the nature of that habitat, especially regarding its degree of pollution.

indicator yellow *the original name for N-retinylidene opsin*. It was so called because its colour changes between yellow and orange according to the pH of its aqueous solution. It spontaneously hydrolyses into opsin and (all-*trans*)-retinal. *See* **rhodopsin**.

indigo a blue dye prepared from the same plants as plant **indican**. Its major blue component is **indigotin**.

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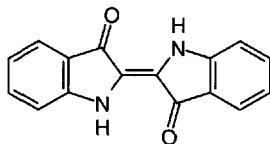
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indigotin

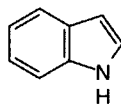
indigotin [*A*^{2,2'}-biindoline]-3,3'-dione; the major blue component of **indigo**.



indirect antagonism a form of **antagonism** (def. 2) arising as a consequence of competition by the antagonist for the binding site of an intermediate that links receptor activation to the observed effect.

INDO-1 1-[2-amino-5-(6-carboxyindol-2-yl)phenoxy]-2-(2-amino-5-methylphenoxy)ethane-*N,N,N',N'*-tetraacetic acid; a fluorescent calcium chelator, named from the indolyl moiety, that can be used as an indicator of Ca^{2+} concentration; it has a large shift in fluorescence emission, from 480 to 400 nm, on binding Ca^{2+} .

indole 2,3-benzopyrrole; a compound formed by the degradation of tryptophan.



indoleacetate *abbr.*: IAA; the anion of indole-3-acetic acid. Formed from L-tryptophan, it is a plant hormone, and the main **auxin** of higher plants.

indoleacetoneitrile *abbr.*: IAN; an **auxin**, second in abundance to indoleacetate.

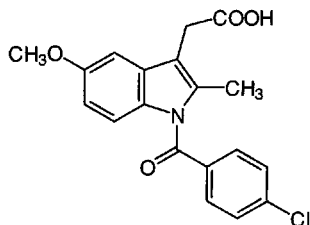
indole-3-glycerol-phosphate synthase *see* *trp* genes.

indole test a test used for the identification of certain microorganisms, e.g. members of the Enterobacteriaceae. It determines the ability of the organism to produce indole from tryptophan.

indolicidin a bactericidal and fungicidal 13-residue peptide amide from the cytoplasmic granules of bovine neutrophils. It contains five tryptophan residues, hence its name. Some therapeutic success has been achieved in animals against the fungus, *Aspergillus fumigatus*, using liposomally encapsulated indolicidin, indicating that neutrophil-derived antimicrobial peptides may have therapeutic value.

indoluria the presence in (human) urine of compounds containing an indole ring.

indomethacin 1-(*p*-chlorobenzoyl)-5-methoxy-2-methylindole-3-acetic acid; an inhibitor of **prostaglandin-endoperoxide synthase**. It is useful as an anti-inflammatory drug.



INDOR *abbr.* for internuclear double resonance spectrometry.

induced enzyme or **inducible enzyme** any enzyme that is synthesized in a cell only in very small amounts except when induced by the presence of its substrate or a closely related compound. *Compare* **constitutive enzyme**.

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induced-fit theory a possible explanation, first proposed by D. E. Koshland, of enzyme specificity. It suggests that reaction between an enzyme and its substrate can occur only following a change in the enzyme's structure, induced by the substrate itself. It is proposed that: (1) a precise orientation of catalytic groups is required for enzyme action; (2) the substrate may cause an appreciable change in the three-dimensional relationship of the amino-acid residues at the active site; and (3) the changes in enzyme structure caused by a substrate will bring the catalytic groups into the proper orientation for reaction, whereas a nonsubstrate will not. It is similar to the **zipper model** of hormone-receptor interaction. *Compare* **lock-and-key model**.

inducer any compound that causes cells to produce larger amounts of the particular enzymes involved in its uptake and/or metabolism (or in the uptake and/or metabolism of a structurally similar compound), than are found in cells when the inducer is absent.

inducible capable of being induced.

inducible enzyme *an alternative name for induced enzyme*.

induction 1 *see* **Jacob-Monod model**. 2 (*in physics*) any physical effect brought about (i.e. induced) in an object by the action of a field without any direct physical contact with the inducer. It may be electrostatic, magnetic, or electromagnetic. *See also* **induced enzyme**.

induction coil a device for producing a rapid succession of large pulses of induced emf in a secondary coil from an intermittent low emf in a primary coil.

induction motor an electric motor in which an alternating current is supplied to the winding of the stator, arranged in such a manner as to produce in effect a rotating magnetic field. This in turn induces electric currents in the winding of the rotor, and interaction between these currents and the magnetic flux exerts a torque on the rotor. Induction motors are useful in situations where sparks caused by moving electric contacts are undesirable.

inductive effect (*in chemistry*) *symbol*: *I*; an experimentally observable effect (on reaction rates, etc.) caused by the transmission of charge through a chain of atoms by electrostatic induction. It is symbolized by +*I* or -*I*, according to whether the group in question is left with a small positive or a small negative charge.

+ine *suffix* designating 1 an organic compound that is either an amine (including any monoamino monocarboxylic acid other than tryptophan), or a nitrogen-containing heterocycle other than one containing four or five atoms in the ring. 2 a halogen. 3 any of a group of certain inorganic binary hydrogen compounds, namely arsine, bismuthine, phosphine, and stibine; also hydrazine.

inert 1 having little or no ability to react chemically or physiologically. 2 having no inherent ability to move or resist being moved. —**inertness** *n*.

inertia 1 (*in physics*) the tendency of an object to resist changes to its state of rest or uniform motion; by analogy extended to other physical qualities that resist change. 2 the state of being inert or inactive. —**inertial** *adj*.

inertial mass *see* **mass**.

infect 1 to cause infection in (an organism, wound, etc.), especially with pathogenic microorganisms. 2 to affect or become affected with a communicable disease.

infection 1 the invasion of a cell or organism by other (especially pathogenic) organisms. 2 the act or process of infecting; the state of being infected. 3 any agent or process that infects.

infectious 1 (*of a disease*) caused by pathogenic (micro)organisms. 2 causing or transmitting infection. 3 (*of a disease*) capable of being transmitted to another organism. —**infectiously** *adv.*; **infectiousness** *n*.

infectious mononucleosis *see* **Epstein-Barr virus**.

infectious viral nucleic acid a purified viral nucleic acid that, when it infects a host cell, causes the production of progeny viral particles.

infective 1 capable of causing infection. 2 *an alternative* (*less*



infinite

common) word for **infectious**. —**infectively** *adv.*; **infectiveness** or **infectivity** *n.*

infinite having no limits in extent, magnitude, space, or time; immeasurably numerous or great. —**infiniteness** *n.*; **infinitely** *adv.*

infinite dilution a hypothetical state (of a solution) in which the solute concentration is considered to be zero and the activity coefficient unity.

infinitely thick (of a radioactive sample) having a thickness no less than **infinite thickness** for the particular radionuclide to be measured.

infinitely thin (of a radioactive sample) having a thickness no greater than **infinite thinness** for the particular radionuclide to be measured.

infinite thickness the thickness of a sample containing a particular radionuclide, especially a soft beta-emitter, beyond which, because of self-absorption, any increase results in no further increase in recordable disintegrations. Under these conditions the (radio)activity is proportional to the specific (radio)activity of the substance.

infinite thinness the thickness of a sample containing a particular radionuclide at or below which self-absorption is negligible. Under these conditions the (radio)activity is proportional to the amount of the radionuclide present.

inflammable very liable to catch fire and produce flame; flammable.

inflammation the immediate defensive reaction of vertebrate tissue to infection or to injury by chemical or physical agents. The part affected is characterized by pain, heat, redness, swelling, and loss of function; there is local vasodilatation, extravasation of plasma into the intercellular spaces, and accumulation of white blood cells and other macrophages in the injured part. Plasma enzyme systems are important sources of inflammatory mediators. These include the **complement**, **blood coagulation**, fibrinolytic, and **kinin** systems. Also active are the mediators released by mast cells, basophils, and platelets, as well as the eicosanoids generated by many cells at inflammatory sites. —**inflamm** *vb.*

informatics the science concerned with the structure and properties of scientific information, its theory, history, methodology, and organization. *Compare* **information science**. —**informatical** *adj.*; **informatician** *n.*

informatin *see* **informofer**.

information 1 communicated or communicable factual knowledge; data; news. 2 **genetic information**. 3 (in computing) the results obtained by processing raw data according to programmed instructions.

informational 1 of, pertaining to, giving, or carrying information. 2 (in biochemistry) describing any biological macromolecule that contains or transmits **genetic information** in the form of specific sequences of nucleotides or amino acids.

information retrieval the selective tracing and recovery of information stored in printed works, computers, etc.

information science the science concerned with the procedures by which information, especially scientific or technical information, is stored, retrieved, and disseminated. *Compare* **informatics**.

information theory the quantitative theory of the coding and transmission of signals and information.

informofer a specific, 30S macroglobular particle consisting of the protein **informatin**. Informofers are present in the cell nucleus, tightly packed along molecules of **heterogeneous nuclear RNA**.

infososome a cytoplasmic mRNA-carrying ribonucleoprotein particle of relatively low buoyant density. Infosomes are believed to function in the protection of mRNA from nuclease attack and in the transport of mRNA from the nucleus to the cytoplasm.

infra+ prefix denoting below, beneath, coming after, within. *Compare* **sub+**, **supra+**, **ultra+** (def., 1).

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infradian describing any biological activity that occurs, or varies cyclically, more often than once every 24 hours. *Compare* **circadian**, **ultradian**.

infranatant 1 lying below. 2 a solid or liquid lying below a supernatant liquid. *Compare* **subnatant**.

infrared of, pertaining to, or being **infrared radiation**.

infrared dichroism the dichroism of polarized infrared radiation. It is used in studies of polypeptide structure.

infrared radiation *abbr.*: IR or i.r.; electromagnetic radiation of wavelengths in the range 730 nm to 1 mm (frequencies 300 GHz to 410 THz). This range lies between those of visible light and microwaves.

infrared spectrophotometer any instrument for **infrared spectrophotometry**.

infrared spectrophotometry the measurement of the absorption or emission of infrared radiation at particular wavelengths or frequencies. *See* **infrared spectrum**.

infrared spectrum any absorption spectrum or emission spectrum of infrared radiation. It includes photons that are absorbed or emitted during vibrational and rotational transitions in a molecule. Infrared spectra are widely used in organic chemistry to identify particular groups of atoms by their characteristic vibrational frequencies.

infuse 1 to introduce gradually, as of a solution into a vein or body cavity. 2 to make an **infusion** (def. 2).

infusion 1 the process of infusing. 2 an extract, e.g. of meat or leaves, obtained by soaking in water.

ingest to take material, e.g. food or liquid, into a cell or organism. —**ingestible** *adj.*; **ingestion** *n.*; **ingestive** *adj.*

inhibin one of several polypeptides (another is **activin** of the transforming growth factor β family) that participate in differentiation and growth of diverse cell types. Inhibin and activin are glycoproteins, secreted by the gonads. Inhibin inhibits secretion of **follicle-stimulating hormone** by the pituitary. It is present in seminal plasma and follicular fluid. It has an M_r of 31 000, and consists of two subunits crosslinked by disulfide bridges. There are two isoforms, A and B, with the same α subunit but different β subunits. Inhibin A is a dimer of α and β_A subunits, and inhibin B is a dimer of α and β_B subunits. Example of the α chain, from *Mus musculus*: database code IHA_MOUSE, 366 amino acids (39.54 kDa). The inhibin/activin precursor is a protein generated by alternative splicing of a gene for subunits for both activin and inhibin. Examples of β chains, inhibin β_A chain (precursor) from human (*other name*: erythroid differentiation protein; *abbr.*: EDF): database code IHBA_HUMAN, 426 amino acids (47.39 kDa); amino acids 311–426 are the inhibin β_A chain; inhibin β_B chain (precursor) from *Bos taurus*: database code IHBB_BOVIN, 408 amino acids (44.89 kDa).

inhibit 1 to restrain or hinder. 2 to stop, prevent, or reduce the rate of (some process, e.g. the growth or functioning of an organism or part of an organism, or a chemical or enzymic reaction).

inhibition constant *symbol*: K_i ; the equilibrium (dissociation) constant for the reaction: $EI \rightleftharpoons E + I$. It is given by $K_i = [E][I]/[EI]$, where E is the enzyme and I the inhibitor, the square brackets indicating concentrations.

inhibition index an inverse measure of the potency of an antimetabolite. It is the ratio of the concentration of antimetabolite that is required to inhibit the biological effect of unit concentration of corresponding metabolite.

inhibitor 1 any agent that inhibits. 2 (in biochemistry) any substance that inhibits an enzymic reaction. *See also* **competitive inhibition**, **noncompetitive inhibition**, **uncompetitive inhibition**.

initial velocity or **initial steady-state rate** *symbol*: v_0 or v_i ; the reaction velocity in the earliest stage of an enzyme-catalysed reaction. It is given by the tangent at the origin to the curve of reaction velocity as a function of time. In practice it is often measured over the period when the initial substrate concentration has diminished by <5%.



initiation

initiation 1 the formation of **initiation complex(es)** at the start of ribosome-mediated translation of mRNA into polypeptide. 2 the first stage in carcinogenesis, in which a normal cell is converted to a precancerous one. 3 the first step in a chain reaction. —**initiate** *vb.*; **initiator** *adj.*

initiation codon or **initiator codon** the codon AUG (or, sometimes in prokaryotes, GUG) that binds *N*-formylmethionyl-tRNA^{Met} in prokaryotes or initiator methionyl-tRNA^{Met} in eukaryotes, and serves as a signal for the start of polypeptide synthesis at the ribosome.

initiation complex any of the complexes formed at the start of ribosome-mediated translation of mRNA into polypeptide. They contain mRNA, initiation factors, initiator fMet-tRNA_f^{Met} or Met-tRNA_f^{Met}, one or two ribosomal subunits, and sometimes GTP. *See* **initiation factor**.

initiation factor *abbr.*: IF (in prokaryotes) or eIF (in eukaryotes); any of a group of soluble proteins that function in the initiation of ribosome-mediated translation of mRNA into polypeptide. Among prokaryotic initiation factors, those of *Escherichia coli* are well characterized. This organism has three principal initiation factors, each consisting of a single polypeptide chain. **IF-1** is a basic, stable protein of *M_r* 9000. It is a stimulator of IF-2 and IF-3. Example: database code IF1_ECOLI, 71 amino acids (8.11 kDa). There are two forms of **IF-2**: IF-2a, which is acidic, unstable, and of *M_r* 115 000 and the almost equally active artefactual product, IF-2b, formed by limited proteolysis of IF-2a. It is a GTP-binding protein, protects formylmethionyl-tRNA_f^{Met} from spontaneous hydrolysis, and promotes its binding to the 30S ribosomal subunits; it is also involved in the hydrolysis of GTP during the formation of the 70S ribosomal complex. By using alternative initiation codons in the same reading frame, the gene is translated to give three isozymes, α (example), β, and β', having different N termini. Example: database code IF2_ECOLI, 894 amino acids (97.74 kDa). **IF-3** is a basic, stable protein of *M_r* 22 000. It binds to the 30S ribosomal subunit and shifts the equilibrium between 70S ribosomes and their subunits in favour of the free subunits, thus enhancing the availability of 30S subunits on which protein synthesis initiation begins. It is also a translational repressor of its own gene. Example: database code IF3_ECOLI, 180 amino acids (20.54 kDa). IF-1, IF-2, and IF-3 bind cooperatively to a 30S ribosomal subunit to form a 30S-IF-1-IF-2-IF-3 particle in which the three initiation factors are bound contiguously near the 3' end of 16S RNA in a 30S ribosomal subunit. This particle subsequently binds mRNA at a site that includes the initiation codon AUG (or GUG), initiator fMet-tRNA_f^{Met}, and GTP with concomitant release of IF-3 to form a 30S initiation complex, 30S-IF-1-IF-2-mRNA-GTP-fMet-tRNA_f^{Met}. Joining of a 50S ribosomal subunit with concomitant release of IF-1, IF-2, GDP, and P_i forms a 70S initiation complex, mRNA-70S-fMet-tRNA_f^{Met}.

In eukaryotes there are more initiation factors and many have been characterized in lysates of rabbit reticulocytes, they include: eIF-1 (single polypeptide chain, *M_r* 15 000); eIF-2 (three nonidentical polypeptide chains, *M_r* 150 000); eIF-3 (nine nonidentical polypeptide chains, *M_r* 700 000); eIF-4A (single chain, *M_r* 49 000); eIF-4B (single chain, *M_r* 80 000); eIF-4C (single chain, *M_r* 17 500); eIF-4D (single chain, *M_r* 16 500); eIF-5 (single chain, *M_r* 190 000); Co.eIF-2 (single chain, *M_r* 22 000); cap-binding protein (single chain, *M_r* 24 000). eIF-2, initiator Met-tRNA_f^{Met}, and GTP form a ternary complex; eIF-3 promotes ribosomal dissociation and stabilizes binding of ternary complex to a 40S ribosomal subunit to form the complex 40S-eIF-2-eIF-3-Met-tRNA_f^{Met}-GTP. Then eIF-1, eIF-4A, eIF-4B, eIF-4C, and cap-binding protein promote binding of mRNA and ATP-dependent formation of a **40S initiation complex**, 40S-eIF-2-eIF-3-Met-tRNA_f^{Met}-mRNA. Factor eIF-5 is required for formation of an **80S initiation complex**, 80S-Met-tRNA_f^{Met}-mRNA, from a 40S complex, together with hydrolysis of the bound GTP and release

of initiation factors. Co.eIF-2 stimulates ternary complex formation.

initiation signal 1 any of the purine-rich regions of mRNA on the 5' side of the initiation codon that serve to distinguish the latter from other similar codons. 2 any of the promoter regions of DNA on the 5' side of the start of transcription to form mRNA.

initiator 1 a gene product that interacts specifically with the replicator site on a replicon resulting in the initiation of a new round of DNA replication. 2 a starting point for either nucleic acid transcription or translation. 3 a carcinogen that initiates the first stage of carcinogenesis. 4 any compound that, when added to monomer, initiates a polymerization reaction.

initiator codon *see* **initiation codon**.

initiator transfer RNA *abbr.*: tRNA_f^{Met}; any of the tRNA molecules that, when aminoacylated, specifically binds to an initiation codon (usually AUG) in mRNA at the start of ribosome-mediated translation of mRNA into polypeptide. In prokaryotes it is *N*-formylmethionyl-tRNA_f^{Met}, and in eukaryotes it is methionyl-tRNA_f^{Met}; neither is able to promote the insertion of methionine residues at internal sites.

innate being an essential characteristic of an organism or thing; inborn.

innervation the nerve supply to an organ or part.

ino *symbol* for a residue of the ribonucleoside inosine (alternative to I).

inoculate 1 to introduce cells, microorganisms, or viruses into or onto a culture medium. 2 to introduce causative agents of disease (alive, dead, or attenuated) into an animal or person for the purpose of immunization. —**inoculation**, **inoculator** *n.*; **inoculative** *adj.*

inoculum (*pl.* **inocula**) a quantity or suspension of cells, microorganisms, or viruses used to start a new culture or to infect another culture; something used to inoculate.

Ino-3',5'-P *symbol* for inosine 3',5'-phosphate.

Ino5'P *symbol* for inosine 5'-phosphate.

Ino5'PP *symbol* for inosine 5'-diphosphate (alternative to ppI).

Ino5'PPP *symbol* for inosine 5'-triphosphate (alternative to pppI).

inorganic not having the characteristics of living organisms; not **organic**.

inorganic biochemistry the branch of biochemistry concerned with the role of inorganic substances in biochemical systems. *Compare* **bioinorganic chemistry**.

inorganic chemistry the branch of chemistry dealing with the elements and all their compounds other than ones containing carbon, except for carbon itself and some simple compounds of carbon, e.g. oxides, sulfides, and cyanides.

inorganic diphosphate *symbol*: PP_i; any of the anions formed from diphosphoric acid, (HO)₂PO-O-PO(OH)₂, or any mixture of the acid and its anions. The older term **inorganic pyrophosphate** may be retained but is now less preferable.

inorganic phosphate or **inorganic orthophosphate** *symbol*: P_i; any of the anions formed from [ortho]phosphoric acid, PO(OH)₃, or any mixture of the acid and its anions.

inorganic pyrophosphatase an enzyme, EC 3.6.1.1, often less precisely termed pyrophosphatase; *systematic name*: pyrophosphate phosphohydrolase. It catalyses the hydrolysis of pyrophosphate by two molecules of orthophosphate. Example from *Escherichia coli*: database code IPYR_ECOLI, 175 amino acids (19.55 kDa).

inorganic pyrophosphate another (less preferred) term for **inorganic diphosphate**.

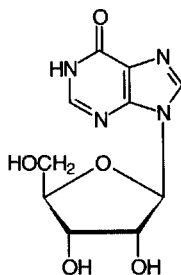
INOS *see* **nitric-oxide synthase**.

inosinate 1 either the mono- and/or the di-anion of **inosinic acid**. 2 any mixture of the acid and its anions. 3 any salt or ester of inosinic acid.

inosine *symbol*: I or Ino; hypoxanthosine; hypoxanthine riboside; 9-β-D-ribofuranosylhypoxanthine; a nucleoside found free, especially in meat, sugar beet, and yeast, but not in combination in nucleic acids except in the anticodons of some

inosine 3',5'-(cyclic)phosphate

transfer RNAs. It may be formed by the action of adenosine deaminase (EC 3.5.4.4) in the first step of the breakdown of adenosine to uric acid, and it occurs as an intermediate in the recycling of both adenosine and inosine 5'-phosphate by **salvage pathways**.



inosine 3',5'-(cyclic)phosphate or **inosine 3',5'-cyclophosphate** *see* **inosine 3',5'-phosphate**.

inosine 5'-diphosphate *symbol:* Ino5'PP or ppi; *recommended name for inosine diphosphate* (*abbr.:* IDP), 5'-diphosphoinosine, 5'-inosinyl phosphate, inosine 5'-(trihydrogen diphosphate). It is an intermediate in the formation of inosine 5'-triphosphate from inosine 5'-phosphate.

inosine monophosphate *abbr.:* IMP; *an alternative name for any inosine phosphate*, but particularly for **inosine 5'-phosphate**, especially when its distinction from inosine (5'-)diphosphate and inosine (5'-)triphosphate requires emphasis.

inosine 5'-monophosphate dehydrogenase *see* **IMP dehydrogenase**.

inosine phosphate inosine monophosphate (*abbr.:* IMP); any phosphoric monoester or diester of inosine. Of the three possible monoesters – the 2'-phosphate, the 3'-phosphate, and the 5'-phosphate – only **inosine 5'-phosphate** is of any importance (the locant being omitted if no ambiguity may arise). The only diester of significance is **inosine 3',5'-phosphate**.

inosine 3',5'-phosphate *symbol:* Ino-3',5'-P; *the recommended name for cyclic inosine 3',5'-monophosphate* (*abbr.:* 3',5'-cyclic IMP or cyclic IMP or cIMP), inosine 3',5'-cyclophosphate, inosine 3',5'-(cyclic)phosphate; a monophosphoric diester of inosine. It is a cyclic nucleotide similar in structure and metabolism to **adenosine 3',5'-phosphate** (i.e. cyclic AMP) but only rarely encountered.

inosine 5'-phosphate or **5'-inosinic acid** or **5'-phosphoinosine** or **5'-O-phosphonoinosine** *symbol:* Ino5'P; *alternative recommended names for inosine monophosphate* (*abbr.:* IMP), inosine 5'-(dihydrogen phosphate), hypoxanthine ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) It is the initial biosynthetic product containing an intact purine nucleus, its immediate precursor being 4-carbamoyl-5-formamidoimidazole riboside 5'-phosphate. Hence IMP is a key intermediate in the biosynthesis of all other purine nucleotides and in the formation of purine nucleosides and free purines. In addition, resynthesis of IMP from hypoxanthine can be effected via a **salvage pathway** by the action of hypoxanthine phosphoribosyltransferase (EC 2.4.2.8) (*see* **phosphoribosyltransferase**).

inosine 5'-triphosphate *symbol:* Ino5'PPP or pppI; *the recommended name for inosine triphosphate* (*abbr.:* ITP), 5'-triphosphoinosine, 5'-inosinyl diphosphate, inosine 5'-(tetrahydrogen triphosphate). ITP can substitute for adenosine 5'-triphosphate or guanosine 5'-triphosphate in certain enzymic reactions. It is formed from inosine 5'-diphosphate by transfer of the terminal phosphoric residue from adenosine 5'-triphosphate through the action of nucleoside-diphosphate kinase (EC 2.7.4.6).

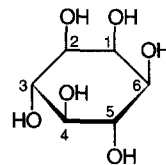
inosinic acid *the trivial name for any phosphoric monoester of inosine*. The position of the phosphoric residue on the ribose

inositol-1,4,5-trisphosphate 5-phospho-

moiety of a given ester may be specified by a prefixed locant (*see* **inosine phosphate**). However, 5'-inosinic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Inosinic acid is also an alternative recommended name for **inosine 5'-phosphate**.

inosinyl the inosine [mono]phospho group; the acyl group derived from **inosinic acid**.

inositol **1** *symbol:* Ins; *the common name for the cyclitol myo-inositol* (formerly called *meso-inositol* or *i-inositol* or *bios I* or *muscle sugar*); 1,2,3,5/4,6-cyclohexanexol; a growth factor for animals and microorganisms, widely distributed in nature. Inositol occurs free, as various phosphates, or as a constituent of certain phospholipids (*see* **phosphatidylinositol**); in plants it also occurs as its hexaphosphate phytate. Note: whereas *myo-inositol* itself is achiral, its phosphorylated derivatives are chiral and the numbering of their carbon atoms can be confusing. For a proposed solution of the problem, *see* **Agranoff's turtle**. **2** *generic name for any member of a subclass of cyclitols comprising the nine possible 1,2,3,4,5,6-cyclohexanexols, of which inositol (def. 1) is the most prevalent and most important.*



inositol-1,4-bisphosphate 1-phosphatase EC 3.1.3.57; *systematic name:* D-*myo*-inositol-1,4-bisphosphate 1-phosphohydrolase; *other name:* inositol polyphosphate 1-phosphatase; an enzyme involved in the degradation of inositol phosphates that hydrolyses D-*myo*-inositol 1,4-bisphosphate to D-*myo*-inositol 4-phosphate and orthophosphate. Example (bovine): database code INPP_BOVIN, 400 amino acids (43.88 kDa).

inositol phosphate any phosphoric ester of an inositol, but normally referring to the esters of *myo*-inositol (*abbr.:* Ins). Of particular interest are those formed as a result of the hydrolysis of membrane phosphoinositides following activation of phosphoinositide-specific phospholipase C (*see* **phospholipase**). The abbreviation Ins is reserved for *myo*-inositol, and the generalized abbreviation for phosphorylated *myo*-inositols follows the convention Ins(numbers of the carbons bearing phosphorylated hydroxyls) P_x , where P_x indicates the number of phosphates. They include inositol 1,4,5-trisphosphate (*abbr.:* Ins(1,4,5) P_3), formed by phospholipase C action on phosphatidylinositol 4,5-bisphosphate. Ins(1,4,5) P_3 has a number of physiological actions related to the **phosphatidylinositol cycle**. It can be phosphorylated to Ins(1,3,4,5) P_4 , which is also physiologically active in calcium metabolism. Other inositol phosphates are intermediates in the degradation of Ins(1,4,5) P_3 and Ins(1,3,4,5) P_4 . Ins(1,4,5) P_3 is hydrolysed to Ins(1,4) P_2 , which is hydrolysed to Ins(4) P and thence to Ins. Ins(1,3,4,5) P_4 is hydrolysed to Ins(1,3,4) P_3 which is hydrolysed to either Ins(1,3) P_2 or Ins(3,4) P_2 these being hydrolysed to Ins(1) P or Ins(3) P respectively and thence to Ins.

inositol-1,3-bisphosphate 3-phosphatase EC 3.1.3.65; an enzyme that hydrolyses D-*myo*-inositol 1,3-bisphosphate to D-*myo*-inositol 1-monophosphate and orthophosphate.

inositol-1,3,4,5-tetrakisphosphate 3-phosphatase EC 3.1.3.62; an enzyme that hydrolyses D-*myo*-inositol 1,3,4,5-tetrakisphosphate to D-*myo*-inositol 1,4,5-trisphosphate and orthophosphate.

inositol-1,4,5-trisphosphate 1-phosphatase EC 3.1.3.61; an enzyme that hydrolyses D-*myo*-inositol 1,4,5-trisphosphate to D-*myo*-inositol 4,5-bisphosphate and orthophosphate.

inositol-1,4,5-trisphosphate 5-phosphatase EC 3.1.3.56; *systematic name:* D-*myo*-inositol-1,4,5-trisphosphate 5-phosphatase



inosyl

phosphatase; *other names*: inositol trisphosphate phosphomonoesterase; inositol polyphosphate 5-phosphatase; 5PTase; an enzyme that hydrolyses D-myoinositol 1,4,5-trisphosphate to D-myoinositol 1,4-bisphosphate and orthophosphate. The enzyme has two types, I and II, with different activities; type I (but not type II) hydrolyses inositol 1,3,4,5-tetrakisphosphate to inositol 1,3,4-trisphosphate; neither enzyme acts on the latter or on inositol 1,4-bisphosphate. Example (precursor, fragment) from human: database code IT5P_HUMAN, 672 amino acids (76.96 kDa).

inosyl any chemical group formed by the loss of a 2', 3', or 5'-hydroxyl group from the ribose moiety of **inosine**.

inotropic influencing contractility of muscles, especially cardiac muscle.

Ins symbol for a residue of the cyclitol 1-D-myoinositol (see **inositol** (def. 1)).

insect any small air-breathing arthropod of the class Insecta, typically having a segmented body with an external, chitinous covering, three pairs of legs, and in most groups two pairs of wings.

insecticide any substance or agent that kills insects. —**insecticidal** *adj.*

(insect) moulting hormone (or **US**) (**insect**) molting hormone a common name for **ecdysterone**.

insectorubin any member of a group of red or red-brown eye pigments of insects derived from oxidation products of tryptophan. See also **ommin**.

insectoverdin any member of a group of green pigments of insects formed of mixtures of carotenoids and biliverdin.

insert (*in genetics*) a fragment of DNA that has been linked into a cloning vector. The term can also be used as a modifier as, e.g., in **insert DNA**.

insertional inactivation the inactivation of a gene in a plasmid cloning vector by insertion of foreign DNA at endonuclease-sensitive sites. It is useful for distinguishing DNA chimeras from cleaved plasmid molecules that have self-annealed.

insertional mutagenesis the induction of a mutation by the insertion of DNA into a gene using recombinant DNA techniques. One approach, known as restriction enzyme-mediated integration (REMI), utilizes a plasmid that is cut with a restriction enzyme. Then the cut plasmid, together with the restriction enzyme, is introduced into cells (e.g. *Dictyostelium*) by electroporation. The restriction enzyme cuts the genomic DNA at sites into which the plasmid can integrate through its **sticky ends**. If the plasmid contains a suitable marker, transformants can be isolated. The gene affected can be identified because it is effectively tagged with the plasmid. If the plasmid also contains an additional marker conferring drug resistance, the gene from the transformant can be readily cloned.

insertion mutation any mutation caused by insertion of an extranumerary nucleotide residue between two successive nucleotide residues in a DNA molecule.

insertion sequence *abbr.*: IS; see **transposon**.

insertion vector (*in recombinant DNA technology*) any **vector** (def. 3) that has been cut with a restriction enzyme having a single target site in the centre of the vector, so that two 'arms' are generated between which foreign DNA may be inserted.

insertosome an 800–1400 bp stretch of DNA that can insert itself randomly into the chromosome of *Escherichia coli*, thereby causing polar mutations analogous to those caused by phage Mu-1.

inside-out vesicle any vesicle formed from disrupted cell or other membranes that has sealed up so that the face of the membrane that was originally on the outside is now on the inside.

in situ *Latin* in the normal, natural, original, or appropriate position.

in situ hybridization a technique of **hybridization analysis** in which chromosomes *in situ* are used as the target for hy-

bridization with radioactive or fluorescent putative complementary DNA or RNA.

insolubilize or **insolubilise** to render **insoluble** (def. 1), especially by coupling covalently to an insoluble matrix. —**insolubilization** or **insolubilisation** *n.*

insoluble 1 incapable of being dissolved (in water unless otherwise specified); of extremely low solubility. 2 incapable of being solved; having no solution.

Ins (3,4)P₂ symbol for 1D-myoinositol 3,4-bisphosphate.

Ins (1,3,4)P₃ symbol for 1D-myoinositol 1,3,4-trisphosphate.

Ins (1,4,5)P₃ symbol for 1D-myoinositol 1,4,5-trisphosphate.

Ins (1,3,4,5)P₄ symbol for 1D-myoinositol 1,3,4,5-tetrakisphosphate.

instructive theory of immunity a theory ascribing to an antigen the function in immunity of instructing or directing uncommitted cells to form new antibodies that reflect the nature of the antigen itself. Compare **selective theory of immunity**.

insulate to prevent, or reduce, the passage of electricity, heat, or sound to or from (something) by interposition of a (relevant) nonconducting material.

insulin or (*formerly*) **insuline** the hypoglycemic, antidiabetic, and anabolic polypeptide hormone of the endocrine portion of the mammalian pancreas or of homologous organs of lower vertebrates. In mammals, it is elaborated in the B cells of the islets of Langerhans and stored within them in the B granules prior to release by exocytosis. Insulin is a slightly acidic protein that readily aggregates to form dimers, hexamers, and higher multimers. It is very poorly soluble in aqueous media around its isoelectric point (pH ≈ 5.4), especially in the presence of certain divalent metal ions, typically Zn²⁺, with which it can form crystals, usually containing two or four metal ions per insulin hexamer, depending on conditions. It does, however, dissolve without denaturation in organic solvents in an acid medium, a property that can be exploited in its isolation. The monomer of insulin is one of the smallest known proteins, with *M_r* values such as 5733 (bovine), 5777 (porcine), or 5807 (human). Bovine insulin was the first protein to have its primary sequence determined, by **Sanger** in 1955. The sequences of a large number of insulins are now known, all of which consist of two polypeptide chains, commonly termed the **A chain** and the **B chain**, with three disulfide bridges formed between half-cystine residues; two of these bridges are interchain, forming A7–B7 and A19–B20 links, and the third is intrachain, linking positions A6 and A11 to form a heterodetic ring in the A chain. The numbers of amino-acid residues vary slightly with the species of origin, commonly being 21 and 30 in the A and B chains, respectively; six amide groups are present per molecule of the monomer if insulin is carefully isolated. The three-dimensional structure is known: chains A and B have two motifs each. Insulin is structurally related to **relaxin** and **insulin-like growth factor**. Insulin is biosynthesized by post-translational modification of preproinsulin via its single-chain precursor **preproinsulin**. In preproinsulin, residues 1–24 are the signal sequence. When this is removed, proinsulin is formed. In the Golgi apparatus, prohormone convertases PC2 and PC3 cleave proinsulin on the carboxyl side of a pair of basic amino-acid residues (Arg⁵⁵-Arg⁵⁶ and Lys⁸⁸-Arg⁸⁹). This excises the C peptide; (residues 57–87, leaving the A and B chains (two Arg residues (55 and 56) must first be cleaved from the B chain by carboxypeptidase H for the mature B chain to be formed). Residues 55–87 are known as the connecting peptide. Example (human): database code INS_HUMAN, 110 amino acids (11.97 kDa); the A chain is derived from residues 90–110, and the B chain from residues 25–54.

Insulin stimulates glucose uptake by muscle and adipose tissue, and promotes glycogenesis, lipogenesis, and synthesis of protein and nucleic acid. Deficiencies in the secretion or action of insulin in humans result in the overproduction and increased urinary excretion of glucose (see **glucose-tolerance test**, **diabetes mellitus**); continuing therapeutic administration of insulin is necessary in type I (insulin-dependent) diabetes



insulinase

mellitus. Commercial insulin for therapeutic use is extracted mostly from bovine or porcine pancreas; human insulin is now also being made by recombinant DNA technology. The international unit for insulin is defined as the activity contained in 0.041 67 mg of the Fourth International Standard for Insulin, Bovine and Porcine, for Bioassay (which is defined as crystalline zinc insulin derived from a mixture of 52% bovine insulin and 48% porcine insulin).

insulinase EC 3.4.99.45; *other names*: insulin-degrading enzyme; insulin protease; an enzyme that catalyses the degradation of insulin and glucagon, progressively cleaving a number of peptide bonds in insulin, but only Arg¹⁷-I-Arg¹⁸ in glucagon.

insulin-dependent diabetes mellitus *abbr.*: IDDM; *see* diabetes mellitus.

insulin-like activity *abbr.*: ILA; *a provisional name given to material thought to be present in plasma or serum that showed apparently higher insulin levels when measured by the rat epididymal fat-pad assay than when other in vitro assay methods were used; it was found to be due to insulin-like growth factor.*

insulin-like growth factor *abbr.*: IGF; any member of a group of polypeptides that are structurally homologous to **insulin** and share many of its biological activities but are immunologically distinct from it; they are mitogens of the insulin/relaxin family. They include substances now named insulin-like growth factor-I (IGF-I), which is a monomer of 70 amino acids, and insulin-like growth factor-II (IGF-II), which is a monomer of 67 amino acids; together these were formerly known as **nonsuppressible insulin-like activity** (NSILA). The insulin-like growth factors also include **somatomedin C**, which is IGF-I, and possibly also somatomedins A₁ and A₂, and **multiplication-stimulating activity**. They appear to be more potent than insulin as growth-promoting factors but less potent in stimulating glucose utilization. Brain IGF is a truncated form of IGF-I, in which the N-terminal tripeptide sequence is lacking. Example, (precursor) human IGF-II: database code IGF2_HUMAN, 180 amino acids (20.12 kDa). *See also* **insulin-related growth factor**, **somatomedin**.

insulin-like growth factor binding protein *abbr.*: IGFBP; any of several proteins that bind insulin-like growth factors (IGFs) and are found in serum, amniotic and other body fluids, and in conditioned media from a variety of cell types. They are believed to prolong the half-life and regulate the endocrine effects of IGFs. Six IGFBPs (IGFBP-1 to -6) have been identified in humans, their complementary DNA having been cloned and sequenced. IGFBP-1 was originally identified as a placental protein, placental protein-12 (PP-12) and as α -1 pregnancy-associated globulin; it is a nonglycosylated protein of M_r 25 000; IGFBP-2 has been identified in human and rat central nervous systems; it is a nonglycosylated protein of M_r 31 000. IGFBP-1 and IGFBP-2 both have the RGD motif, a recognition site for **integrins**. IGFBP-3 is the principal IGFBP in human serum; it is a glycoprotein of M_r 28 500, and circulates as a complex with IGF peptide (γ subunit). For IGFBP-5 *see* **skeletal growth factor**.

insulinogenesis the formation of **insulin**.

insulinogenic resulting from the administration of **insulin**.

insulinoma any neoplasm (benign or malignant), originating in the B cells of the pancreatic islets (or sometimes from non-islet tissue), that autonomously secretes insulin and/or biosynthetic precursors of insulin. Such lesions are frequently recognized by their metabolic consequences.

insulinotropic stimulating **insulin** secretion.

insulinotropic effect *see* **gastric inhibitory peptide**.

insulinotropin a peptide containing residues 7–37 of glucagon-like peptide I, coencoded with glucagon in preproglucagon. It is a potent stimulator of insulin release from the perfused rat pancreas. *See* **glucagon-like insulinotropic peptide**.

insulin receptor a type I membrane protein with tyrosine kinase (EC 2.7.1.112) activity. The protein is derived from a precursor; after being transported from the endoplasmic reticulum

to the Golgi apparatus, the single glycosylated precursor is further glycosylated and then cleaved to α and β chains. In the plasma membrane the receptor is a disulfide-bonded heterotetramer, $\alpha_2\beta_2$; the α chains are entirely extracellular, while each β chain has one transmembrane domain. The ligand binds to the α -subunit extracellular domain and the kinase is associated with the β -subunit intracellular domain; it is partly regulated by autophosphorylation. Associated diseases include various forms of insulin resistance, including type 2 **diabetes mellitus** and **leprechaunism**. Example (human precursor): database code INSR_HUMAN, 1382 amino acids (156.10 kDa); amino acids 28–762 form the α subunit, 763–1382 form the β subunit.

insulin-related growth factor (*sometimes*) any of a group of vertebrate and invertebrate protein or polypeptide hormones or growth factors (including **relaxin**, **nerve growth factor**, **insulin-like growth factors**, silkworm prothoracicotropic hormone, and possibly others) that share with **insulin** certain features of primary and three-dimensional structure and may also display some similarities to **insulin** as regulators of growth and/or metabolism.

insulin-tolerance test a sensitive and accurate test of the integrity of the hypothalamic-pituitary-adrenal axis and its ability to respond to stress, and also of somatotropin responsiveness. The test is performed after an overnight fast with measurement of plasma cortisol, glucose, and somatotropin levels before and periodically after intravenous administration of insulin. The hypoglycemia induced by insulin elicits a potent stress response via the central nervous system, stimulating the release of **corticotropin releasing hormone** and consequently, of **corticotropin**.

Int the product of phage gene *int*, involved in the integration of phage DNA into host DNA during transition from the lytic to the lysogenic states of the phage. *Compare* **Int-2**. *See also* **retroregulation**.

Int-2 an oncogene first identified as a transforming gene following the integration of mouse mammary tumour virus into the mouse DNA. Its product has been shown to be a **fibroblast growth factor** (FGF3). Example (precursor): database code FGF3_HUMAN 239 amino acids (26.89 kDa). *Compare* **Int**.

integer any rational whole number.

integral 1 of, or being an essential part of a whole; entire, complete. 2 (*in mathematics*) of, or denoted by, an **integer**; of or involving an **integral** (def. 3). 3 (*in mathematics*) the sum of a large number of infinitesimally small quantities, denoted $\int$, summed either between specified limits (**definite integral**) or within indeterminate limits (**indefinite integral**).

integral (cell membrane) protein or **intrinsic protein** an operational term for any protein that can only be dissociated from a cell membrane by drastic treatment with a reagent such as detergent, bile salt, protein denaturant, or organic solvent. Such proteins have one or more domains that traverse the cell membrane. *Compare* **peripheral (cell membrane) protein**.

integrase a protein of lambdoid phages that catalyses integration of phage DNA during establishment, probably by forming a transient DNA-protein link. Example from phage lambda: database code VINT_LAMBD, 356 amino acids (40.26 kDa). The same enzyme is one subunit of the **excisase** activity that reverses this; the other subunit is the Xis protein, known as **excisionase**. Example from phage lambda: database code VXIS_LAMBD, 72 amino acids (8.60 kDa).

integrate 1 to combine or be combined into a whole; to complete (something imperfect) by the addition of parts. 2 (*in mathematics*) to find the **integral** (def. 3) of (quantities, expression, etc.). —**integrable**, **integrative** *adj.*; **integrability** *n*.

integrated rate equation a general expression for the interpretation of a single enzyme-catalysed reaction. The expression for the velocity of a reaction at time t is:

$$v = d[P]/dt = V([S]_0 - [P]) / \{K_m + ([S]_0 - [P])\}.$$

This can be integrated with respect to time to give:

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$$Vt = [P]_t + K_m \ln \{ [S]_0 / ([S]_0 - [P]_t) \},$$

or

$$\ln \{ [S]_0 / ([S]_0 - [P]_t) \} / t = V / K_m - [P]_t / K_m t,$$

where $[S]$ and $[P]$ are the concentrations of substrate and product, v is the velocity of the reaction, V is the limiting rate, and K_m is the Michaelis constant. The subscripts indicate the concentrations at zero time and at time t . A plot of $\ln \{ [S]_0 / ([S]_0 - [P]_t) \} / t$ against $[P]_t / t$ will have a slope of $-1/K_m$ and an intercept of V/K_m .

integration **1** the act or process of uniting or adding parts to make a whole. **2** the process of determining and/or indicating the mean value or total sum of (some physical quantity, area, etc.). **3** (in mathematics) the process of determining the **integral** (def. 3) of a function or a variable. *Compare* **differentiation**. **4** (in genetics) the act or process of forming a **cointegrate**.

integrator any device (mechanical or electronic) or any person that determines the value of an **integral** (def. 3).

integrin any member of the large family of transmembrane proteins that act as receptors for **cell-adhesion molecules**. Integrins are heterodimeric molecules in which the α and β subunits are noncovalently bonded. The subunits are named from α_1 to α_8 , α_L (for **LFA-1**), α_M (for **Mac-1**), α_v (for **vitronectin receptor**), and from β_1 to β_7 . Both subunits penetrate the cell-membrane lipid bilayer; the α subunit has four Ca^{2+} -binding domains on its extracellular chain, and the β subunit bears a number of cysteine-rich domains extracellularly. The integrins are grouped into families: the VLA family, having the β_1 subunit; the LEUCAM family, which includes LFA-1 and Mac-1, having the β_2 subunit; and a group of other integrins having subunits β_3 – β_8 . The type of integrin expressed on the cell surface determines which adhesion molecules (and thus which other cells) the cell will bind, and can be varied in different circumstances. For example, transforming growth factor- β increases expression of $\alpha_1\beta_1$, $\alpha_2\beta_1$, $\alpha_3\beta_1$, and $\alpha_5\beta_1$ integrins on fibroblasts and the $\alpha_v\beta_3$ integrin on fibroblasts and osteosarcoma cells; interleukin- 1β enhances β_1 expression on osteosarcoma cells; and in response to wounding, the keratinocyte, which normally expresses integrin $\alpha_6\beta_4$, will express $\alpha_5\beta_1$ (VLA-1, the fibronectin receptor) so that the keratinocyte will then migrate over fibronectin in the cell matrix, thus covering the wound. Example, integrin α_1 precursor from *Rattus norvegicus*: database code ITA1_RAT, 1180 amino acids (130.81 kDa); residues 1–28 are the signal peptide, 29–1180 integrin α_1 . *See also* **adhesion molecules** (for details of the integrins to which they bind).

intein a proposed term for an internal peptide sequence of a protein precursor that is spliced out by transpeptidation during processing to form the mature protein. The peptide sequences that are spliced together are termed **exteins**. The terms are derived, respectively, by analogy from **introns**, the nucleotide sequences that are removed during mRNA processing in eukaryotes, and **exons**, the coding sequences that are spliced together to form the mature mRNA transcript.

intensify (in photography) to increase the opacity (of a film or plate). —**intensification**, **intensifier** *n*.

intensity (in physics) the magnitude of an energy flux, field strength, or force. *See* **luminous intensity**, **radiant intensity**.

intensity millicurie an obsolete name for **sievert** (def. 2).

intensive property any of the properties of a substance or object that describe a specific characteristic of the substance or object in a given state. Intensive properties are independent of the amount of the substance or the size of the object being considered, e.g. density, specific volume, temperature, etc. *Compare* **extensive property**. *See also* **intensive**.

inter+ *comb. form* signifying between, within, or among. *Compare* **intra+**.

interaction **1** any reciprocal or mutual action or effect. **2** (in physics) the transfer of energy between elementary particles, between particles and a field, or between two or more fields. —**interact** *vb*.

interaction broadening the concentration-dependent broadening of lines in an electron spin resonance spectrum caused by magnetic interactions between paramagnetic molecular or atomic species.

interactive (of two or more forces, fields, processes, etc.) acting reciprocally or in close and/or continuing relation with each other; interacting.

interallelic complementation interaction between products of two mutant alleles to form a functional protein. It can occur in the case of multimeric proteins, for which subunits are encoded by two alleles. If products from only one mutant allele compose the protein, an inactive protein results, but in certain cases, if the products from the two alleles (each of which codes for an inactive protein) are combined in the protein, an active protein results. This was first shown in *in vitro* experiments by taking two inactive proteins of alleles encoding glutamate dehydrogenase (NADP^+), and recombining them by freezing and thawing in the presence of NaCl to form an active enzyme. There is some evidence that it can also occur *in vivo*.

interband any DNA-poor region between two (DNA-rich) bands in a polytene chromosome.

intercalary deletion *see* **deletion**.

intercalary DNA *abbr.*: iDNA; moderately repetitive DNA sequences occurring between structural genes in chromosomal DNA.

intercalation (in molecular biology) the insertion of additional material between the parts of an existing series, especially the insertion of molecules of antibiotic substances (e.g. actinomycin D) or acridine dyes (e.g. proflavine) between adjacent base pairs in a DNA duplex, thereby interfering with replication, etc. —**intercalate** *vb*; **intercalation** *n*; **intercalary** or **intercalative** *adj*.

intercalator or **intercalating agent** (in molecular biology) any substance capable of being intercalated.

intercellular between cells. *Compare* **intracellular**.

intercept (in mathematics) a point at which two figures intersect; the point at which a line, curve, or surface cuts a coordinate axis.

interchain (of any interaction or linkage) between two chains of a biopolymer.

interconvertible enzyme any enzyme that exists in at least two well-defined reversibly convertible forms, produced by covalent modifications of amino-acid side chains under biological conditions (covalent modifications occurring as intermediates in the catalytic process are excluded). *Compare* **isoenzyme**.

intercrine an alternative name for **chemokine**.

interdomain (of any interaction or linkage) between two domains of a protein or other biopolymer.

interface **1** any surface forming a common boundary between two bodies, molecules, liquids, or phases. **2** any device, instrument, or method connecting two systems or processes; used especially of connections between computer systems. **3** to connect with or by an **interface** (def. 2). —**interfacial** *adj*.

interfacial tension the surface tension occurring at the interface between two mutually saturated liquids that tends to contract the area of the interface.

interference **1** (in physics) the process in which two or more wave trains of the same frequency and phase are superimposed to form a wave train in which there are alternate bands (called **interference fringes**) of no energy (due to **destructive interference**) or higher energy than that of the individual wave trains (due to **constructive interference**). **2** (in virology) the partial or complete prevention of the replication of one virus in a cell or tissue by the concomitant interaction between the given cell or tissue and another virus.

interference (light) filter a type of (light) filter with which a transmitted band-width of 10–20 nm can be obtained with a peak transmission of about 40%. It is possible to make inter-

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interference fringe

ference filters in which the position of the centre of the transmitted band can be set within 1–2 nm.

interference fringe *see* **interference** (def. 1).

interference optics a system of optics, based on the Rayleigh interferometer, that is sometimes fitted to analytical ultracentrifuges. A double-sector ultracentrifuge cell is used, one sector containing only solvent and the other the solution being examined. The pattern of interference fringes produced by light passing through both sectors is a function of the refractive index at each point in the cell, each fringe tracing a curve of the refractive index against distance. Interference optics allows smaller changes in concentration to be detected than is possible by the **Schlieren method**.

interferogram a photographic record of a pattern of interference fringes.

interferometer an instrument that divides a beam of electromagnetic radiation into a number of beams and reunites them to produce **interference** fringes. It is useful for accurate measurement of the wavelength of the radiation, examination of the hyperfine structure of spectral lines, the testing of the flatness of surfaces, etc. — **interferometric adj.**; **interferometrically adv.**; **interferometry n.**

interferon *abbr.*: IFN; any member of a group of proteins that form a closely related group of nonviral proteins, of M_r in the range 15 000 to 30 000, that are produced and liberated by animal cells following exposure to a variety of inducing agents; they are not normally present in uninduced cells. They exert nonspecific, antiviral activity, at least in homologous cells, through cellular metabolic processes involving the synthesis of both RNA and protein. Viruses are the most potent inducing agents, but interferons can also be induced by exposure of cells to a wide variety of microorganisms, bacterial endotoxins, phytohemagglutinins, and other substances. Interferons do not prevent viral penetration into cells but induce within the cells a complex of inhibitors of protein synthesis that block translation of viral mRNA, but not the cell's own mRNA. Among the interferon inhibitors are a protein kinase, which phosphorylates an initiation factor in protein synthesis, and an oligonucleotide synthetase, which catalyses the synthesis of pppA2'p5'A2'p5'A (*abbr.*: 2,5-A) — this trinucleotide activates a latent cellular endonuclease that degrades the viral mRNA. Interferons are classified into three types. **IFN- α** (*formerly* leukocyte interferon) comprises many examples; they are produced by macrophages, have antiviral activities, and stimulate production of two enzymes: a protein kinase and an oligoadenylate synthetase; they exist as monomers. Example, (human) α -4B precursor: database code INA4_HUMAN, 189 amino acids (21.78 kDa). **IFN- β** (*formerly* fibroblast interferon) has antiviral, antibacterial, and anticancer activities. It is related to the α family (three motifs). Example (human, precursor): database code INB_HUMAN, 187 amino acids (22.27 kDa). **IFN- γ** (*formerly* immune interferon) is produced by lymphocytes activated by specific antigens or mitogens. In addition to antiviral activity it has important immunoregulatory functions: it is a potent activator of macrophages, it has antiproliferative effects on transformed cells, and can potentiate the antiviral and antitumour effects of the type I interferons (α and β). It is a glycoprotein and exists as a homodimer. Example (human, precursor): database code ING_HUMAN, 166 amino acids (19.33 kDa). Interferons are produced commercially either by *in vitro* culture of human leukocytes or by recombinant DNA or RNA technology, γ interferon requiring a system that effects glycosylation.

intergenic suppression *see* **suppression**.

interjacent RNA a 26S single-stranded RNA found in cells infected with certain togaviruses, e.g. Semliki forest virus. It codes for all the virus coat proteins.

interleukin *abbr.*: IL; any member of a heterogeneous group of **cytokines** that have the ability to act as signalling molecules between different populations of leukocytes. Several classes

have been defined; each is designated by a suffixed arabic numeral, e.g. **interleukin 1** (IL-1), **interleukin 2** (IL-2), etc.

interleukin 1 *abbr.*: IL-1; *former names*: lymphocyte-activating factor (LAF); mitogenic protein (MP); helper-peak-1 (HP-1); T-cell-replacing factors III and m (TRF-III; TRF-m); B-cell-activating/differentiation factor (BAF; BDF). An interleukin of M_r 12 000–18 000, produced mainly by activated macrophages. It is not able to promote and maintain *in vitro* long-term cultures of T cells, but stimulates thymocyte proliferation by inducing **interleukin 2** release. It is involved in the inflammatory response, and is identified as an endogenous pyrogen. Example, IL-1 β (human): database code NRL1_6IIB, 153 amino acids (17.36 kDa); 3-D structure data obtained by NOESY.

interleukin 2 *abbr.*: IL-2; *other names*: thymocyte-stimulating factor (TSF); thymocyte mitogenic factor (TMF); T-cell growth factor (TCGF); co-stimulator; killer-cell helper factor (KHF); secondary cytotoxic T-cell-inducing factor (SCIF). An interleukin of M_r 30 000–35 000, produced by T cells in response to antigenic or mitogenic stimulation. Translocation involving the IL-2 gene causes one form of acute lymphoblastic leukemia. IL-2 promotes and maintains *in vitro* long-term cultures of T cells. Example, human (precursor): database code IL2_HUMAN, 153 amino acids (17.61 kDa).

interleukin 3 *abbr.*: IL-3; *other names*: multipotential colony-stimulating factor; hematopoietic growth factor; P-cell-stimulating factor; mast-cell growth factor. A class of interleukins that are granulocyte/macrophage colony-stimulating cytokines; they act in hematopoiesis by controlling the production, differentiation, and function of granulocytes and the monocytes/macrophages. They are glycoprotein monomers. Example, human (precursor): database code IL3_HUMAN, 152 amino acids (17.21 kDa).

interleukin 4 *abbr.*: IL-4; *other names*: B-cell stimulatory factor 1; lymphocyte stimulatory factor 1. An interleukin that participates in at least several B-cell activation processes as well as those of other cell types. It is a co-stimulator of DNA synthesis and is a glycoprotein. Example: database code NRL1_RBC, 129 amino acids (14.95 kDa); 3-D structure known.

interleukin 5 *abbr.*: IL-5; *other names*: T-cell-replacing factor; eosinophil differentiation factor. An interleukin that induces terminal differentiation of late-developing B cells to immunoglobulin-secreting cells. It is a homodimer, with subunits linked by a disulfide bond. Example, human (precursor): database code IL5_HUMAN, 134 amino acids (15.22 kDa).

interleukin 6 *abbr.*: IL-6; *other names*: B-cell stimulatory factor 2; interferon β -2; hybridoma growth factor. An interleukin that induces myeloma and plasmacytoma growth, and nerve-cell differentiation. In hepatocytes it induces acute-phase reactants. It is a glycoprotein. Example, human (precursor): database code IL6_HUMAN, 212 amino acids (23.69 kDa).

interleukin 7 *abbr.*: IL-7; *other name*: hematopoietic growth factor. An interleukin that stimulates proliferation of lymphoid progenitor cells. It is produced by monocytes, T cells, and NK cells. Example, human (precursor): database code IL7_HUMAN, 177 amino acids (20.16 kDa).

interleukin 8 *abbr.*: IL-8; *other names*: monocyte-derived neutrophil chemotactic factor (MDNCF); T-cell chemotactic factor; neutrophil-activating protein 1 (NAP-1); lymphocyte-derived neutrophil-activating factor; protein 3-10C. An interleukin that acts as a chemotactic factor, attracting neutrophils, basophils, and T cells, but not monocytes. It is released from several cell types in response to an inflammatory stimulus. Example, human (precursor): database code NRL_311, 68 amino acids (7.96 kDa); 3-D structure obtained from a combination of crystallography and NMR.

interleukin 9 *abbr.*: IL-9; *other names*: T-cell growth factor; p40 precursor. An interleukin that supports IL-2/IL-4-independent growth of helper T cells. Example, human (precursor): database code IL9_HUMAN, 144 amino acids (15.89 kDa).

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interleukin 10 *abbr.*: IL-10; *other names*: cytokine synthesis inhibitory factor (CSIF); cytokine glycoprotein. An interleukin that appears to be a suppressor factor for Th1 immune responses. Example, human (precursor): database code IL10_HUMAN, 178 amino acids (20.48 kDa).

interleukin 11 *abbr.*: IL-11; *other name*: adipogenesis inhibitory factor (AGIF). An interleukin that acts as both cytokine and growth factor. It stimulates plasmacytoma proliferation, and the T-cell-dependent development of immunoglobulin-producing B cells. Example, human (precursor): database code IL11_HUMAN, 199 amino acids (21.40 kDa).

interleukin 12 *abbr.*: IL-12; *other name*: cytotoxic lymphocyte maturation factor. An interleukin that can act as a growth factor for activated T and natural killer (NK) cells, enhance the lytic activity of NK/lymphokine-activated killer cells, and stimulate the production of interferon- γ by resting PMBC. It is a disulfide-linked heterodimer of 40 kDa and 35 kDa subunits. NK stimulatory factor (NKSf) is essentially a complex of cytokine and soluble receptor. Examples, human; precursor for the 35 kDa (α) subunit: database code I12A_HUMAN, 219 amino acids (1–22 signal) (24.82 kDa); precursor for the 40 kDa β subunit: database code I12B_HUMAN, 328 amino acids (1–22 signal) (37.13 kDa). Both subunits are glycoproteins; α has some similarity to **interleukin 6**; β has an immunoglobulin-like domain.

interleukin 13 *abbr.*: IL-13; an interleukin that inhibits inflammatory cytokine production. It synergizes with **interleukin 2** in regulating interferon- γ synthesis. Example, human (precursor): database code IL13_HUMAN, 199 amino acids (21.40 kDa).

interleukin 1 β convertase EC 3.4.22.36; *other name*: interleukin -1 β converting enzyme (*abbr.*: ICE); a proteolytic enzyme that cleaves the interleukin-1 β precursor specifically at Asp¹¹⁶-|-Ala¹¹⁷ and Asp²⁷-|-Gly²⁸ in the precursor, producing the mature cytokine. It is a cysteine endopeptidase. Example from human: database code I1BC_HUMAN, 404 amino acids (45.16 kDa).

intermediary metabolism the chemical pathways by which molecules and structures found in living cells are synthesized from molecules that are taken into the cells, and subsequently broken down to molecules that are eliminated from the cells.

intermediate 1 occurring, or situated between two things in time, place, or order. 2 (*in chemistry or biochemistry*) any compound that takes part in a reaction, or sequence of reactions, occurring between the starting materials and the products. In metabolism, intermediates occur between the nutrients taken in and cellular material synthesized, or between cellular material and excretory products.

intermediate body a former name for **amboceptor**.

intermediate-density lipoprotein *abbr.*: IDL; one of the classes of lipoproteins found in blood plasma in many animals (data normally relate to humans). These lipoprotein particles have diameter 19.6–22.7 nm, solvent density for isolation (g mL^{-1}) 1.006–1.019, hydrated density (g mL^{-1}) 1.003. Their approximate composition (% by weight) is 8% unesterified cholesterol, 22% esterified cholesterol, 25% phospholipid, 30% triacylglycerol, 15% protein. Their apolipoprotein composition (% by weight total apolipoprotein) is 50–70% B-100, 5–10% C-I + C-II + C-III, 10–20% E-II + E-III + E-IV. They are formed in the plasma from very-low-density lipoproteins as a result of the action of **lipoprotein lipase**, and represent an intermediate stage in the conversion of very-low-density lipoproteins to low-density lipoproteins. *See also* **apolipoprotein** for individual apolipoproteins.

intermediate filament a distinct elongated structure, characteristically 10 nm in diameter, that occurs in the cytoplasm of higher eukaryotic cells. Intermediate filaments form a fibrous system, composed of chemically heterogeneous subunits and involved in mechanically integrating the various components of the cytoplasmic space. Intermediate filaments may be divided into five chemically distinct classes: **keratin** filaments of

mammalian epithelial cells; **desmin** filaments found in cells of muscles of all types; **vimentin** filaments found in mesenchymal cells; neurofilaments of neurons; and glial filaments found in all types of glial cells. *Compare* **microfilament**, **microtubule**.

intermediate lobe or **intermediate part** or **pars intermedia** (**hypophyseos**) the part of the **pituitary gland** lying between the anterior and the posterior lobes. It is absent in adult humans but is present in fetal humans and in certain other species.

intermedin a former name for β -melanocyte-stimulating hormone (so-called because of its occurrence in the intermediate lobe of the pituitary gland). *See* **melanocyte-stimulating hormone**.

internal compensation the phenomenon of the existence of **internally compensated** molecular entities in general, or the property of such a molecular entity in a particular instance.

internal conversion *abbr.*: IC; a mode of radioactive decay of an excited atomic nucleus (formed by a nuclear transformation) in which, in a proportion of instances, excess energy released from the nucleus is not radiated from the atom as a quantum of low-energy gamma radiation but is imparted to one of the inner orbital electrons, which is thereupon ejected as a **conversion electron**; the electron emission is homoenergetic. The filling of the vacant orbital in the resultant excited ion causes emission also of either an Auger electron or a quantum of the characteristic X-radiation of the daughter atom.

internal energy *symbol*: U or E ; a function of state that describes the sum of the kinetic and potential energy of a system. For closed systems, a change in internal energy, ΔU , usually results in a transfer of work or heat between the system and the surroundings. Thus $\Delta U = q + w$, where q is the heat gained by the system and w is the work done on the system.

internal indicator any indicator substance used to follow a reaction, or other process, that is placed in the reaction mixture, or system.

internalize or **internalise** to render internal; especially to take a substance into a cell or organism. —**internalization** or **internalisation** *n*.

internally compensated describing any molecular entity that possesses equal numbers of structurally identical groups of opposite chirality. If no other chiral groups are present the overall entity is achiral and is designated by the prefix *meso*-.

internal secretion a former name for **endocrine** (def. 2).

internal standard any standard used as a reference in an assay that is added to the samples and subjected to the whole of the procedures applied to them.

internal-standard method (of quench correction) or **internal standardization** a method of correcting for quenching in **liquid scintillation counting**. After the sample has been counted a measured small amount of a (nonquenching) standard preparation, of known activity, of the radionuclide in question is added to it and the sample is recounted. From the increment in the count rate the counting efficiency within the sample, and hence the degree of quenching, may be calculated. The method is simple and reliable and corrects adequately for all types of quenching but is more time-consuming than the **external-standard method** and has other disadvantages.

International Biological Standard any of the numerous internationally agreed standards of activity or potency of antisera, hormones, vitamins, antibiotics, and other biologicals. Each has been defined in terms of a mass of an appropriate standard preparation, prepared and stored under specified conditions.

International Practical Temperature Scale *abbr.*: IPTS; *see* **temperature**.

International System of Units the English translation of *Système International d'Unités*; *see* **SI**.

International Unit *abbr.*: IU (preferred) or i.u. or u; any unit of activity or potency of an antiserum, hormone, vitamin, antibiotic, or other biological defined in terms of the activity of an appropriate **International Biological Standard** or **International Biological Reference Preparation**. International Units are useful for quantifying a specific biological activity of a

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Internet

material before the chemical nature of the substance responsible for the activity has been fully characterized, or before pure preparations with reproducible activity are available.

Internet a protocol for allowing communication between computers, including those at distant sites. More loosely 'the internet' is used to refer to the worldwide network of such computers, including the providers of database services, discussion forums, and other communication media. *See also* **Appendix D**.

internuclear double resonance spectrometry *abbr.*: INDOR; a technique used in **nuclear magnetic resonance spectrometry** as an aid to assignment of resonances. It involves monitoring a single frequency, f_1 (usually one corresponding to a single transition), while sweeping a perturbing (decoupling) frequency, f_2 , through a selected range of frequencies. Signals, negative or positive, will only appear in the INDOR spectrum when f_2 passes through a transition that has an energy level in common with the transition at f_2 , making it possible to determine otherwise inaccessible chemical shifts and coupling constants.

interphase the state of a eukaryotic cell when not undergoing mitosis or meiosis. It covers the G_1 , S, and G_2 phases of the **cell-division cycle**.

interpolate (*in mathematics*) to estimate a value of a function between data values already obtained or known either graphically or by calculation. *Compare* **extrapolate**. —**interpolation** *n*.

intersegment transfer a proposed mechanism for the transfer of a repressor of gene expression in a nonspecifically bound state to its target site. It involves the rapid and direct transfer of repressor from one segment of a DNA molecule to another as a consequence of the relative diffusion of these segments within the domain of the molecule.

interstice any minute space or crevice that intervenes between structures, e.g. between cells, organs, or tissues, or between atoms in a crystal lattice. —**interstitial** *adj*.

interstitial cell a connective tissue cell lying in the interstices between other cells in a tissue and serving to bind the tissue cells. Examples include the neuroglial cells and the testosterone-producing Leydig cells lying between the seminiferous tubules in the testis of vertebrates.

interstitial cell-stimulating hormone *abbr.*: ICSH; *a former name for luteinizing hormone*.

interstitial collagenase EC 3.4.24.7; *other names*: vertebrate collagenase; matrix metalloproteinase 1; an enzyme that cleaves preferentially one bond in native collagen and catalyses the cleavage of the triple helix of collagen at about three-quarters of the length of the molecule from the N terminus, at Gly⁷⁷⁵-Ile⁷⁷⁶ in the α 1(I) chain. It cleaves synthetic substrates and α -macroglobulins at bonds where Pl' is a hydrophobic residue. Zinc is a cofactor. Example (precursor) from human fibroblasts: database code COG1_HUMAN, 469 amino acids (53.95 kDa).

interstitial deletion the loss of DNA or part of a chromosome that occupies an interstitial (i.e. nonterminal) position.

interstitial fluid the portion of the **extracellular fluid**, consisting mainly of water, that occurs outside the blood vessels and the lymphatics. In a 70 kg man its volume is approximately 12 L and it represents approximately 17% of body mass. It has a higher content of sodium, chloride, and bicarbonate and a lower content of potassium, magnesium, sulfate, and phosphate than **intracellular fluid**.

intertrabecular space the space within cells lying between the elements of the **microtrabecular lattice**.

intervening sequence *an alternative name for intron*.

intimin an actin-polymerization-inducing protein produced by enteropathogenic and enterohemorrhagic strains of enteric bacteria. It is an outer membrane protein of *Escherichia coli*, necessary for attachment of the bacterium to epithelial cells. Example from *Hafnia alvei*: database code HAFEAEA, 280 amino acids (30.11 kDa).

intoxication 1 any metabolic process that converts a nontoxic

xenobiotic to a toxic product. 2 poisoning, especially with ethanol, manifested in behavioural change of the subject.

intra+ *prefix* signifying within, on the inside, or during. *See also* **endo+**. *Compare* **inter+**.

intracellular within a cell or cells.

intracellular fluid *abbr.*: ICF; the portion of the total fluid content of a multicellular organism that occurs within the cells, especially as distinct from **extracellular fluid**. In a 70 kg man the volume of ICF is approximately 26.5 L, and it represents approximately 38% of body mass. It has a higher content of potassium, magnesium, sulfate, and phosphate, and a lower content of sodium, chloride, and bicarbonate than **interstitial fluid**.

intrachain (*of any interaction or linkage*) between different parts of a single chain of a biopolymer.

intradermal *abbr. (in medicine)*: ID or i.d.; within the layers of the skin.

intradomain (*of any interaction or linkage*) within a single domain of a protein or other biopolymer.

intrinsic suppression *see* **suppression**.

intramolecular within the same molecule.

intramolecular chaperone a domain of a protein that mediates conformation (i.e. exercises a **chaperone** function) of the protein within which it resides. **Trypsinogen**, e.g., contains such a domain.

intramuscular *abbr. (in medicine)*: IM or i.m.; within a muscle or muscles; used especially of an injection.

intrapertitoneal *abbr.*: IP or i.p.; within the peritoneum; used especially of an injection, infusion, or aspiration.

intrapleural within the pleura or pleurae; used especially of an injection or aspiration.

intraathelial within the intradural or subarachnoid space; used especially of an injection or aspiration.

intravenous *abbr. (in medicine)*: IV or i.v.; within a vein; used especially of an injection, infusion, transfusion, or aspiration.

intrinsic 1 of or pertaining to the essential nature of something; inherent. 2 (*in morphology*) peculiar to, or situated in a part. —**intrinsically** *adv*.

intrinsic association constant or **intrinsic binding constant** *see* **intrinsic constant**.

intrinsic activity *an alternative name for maximal agonist effect*.

intrinsic birefringence the optical anisotropy of an individual molecule.

intrinsic constant any association, binding, or dissociation constant that describes the association, binding, or dissociation of a ligand to (or from) a particular site on a polyvalent molecule, when all the sites of this type are identical, present in the same molecule, and noninteracting.

intrinsic dissociation constant *see* **intrinsic constant**.

intrinsic efficacy *symbol*: ε ; an expression introduced by Furchgott that denotes the notional efficacy of a drug associated with a single receptor. It is given by $\varepsilon = e/[R_1]$, where e is the **efficacy** and $[R_1]$ indicates the total concentration of active receptors of a certain type.

intrinsic factor a glycoprotein, secreted by the parietal cells of the gastric mucosa, that is necessary for the adequate absorption of dietary **cobalamins**. Intrinsic factor binds cobalamins in the intestinal lumen to form a complex that is subsequently bound by a specific receptor on the mucosa of the ileum. Cobalamin is then dissociated from the complex by a releasing factor and actively transported across the mucosa into the bloodstream. Congenital pernicious anemia seems to be due to a defect of intrinsic factor. Example (precursor) from human: database code IF_HUMAN, 417 amino acids (45.37 kDa).

intrinsic pathway (of blood coagulation) *see* **blood coagulation**.

intrinsic protein *less common term for integral (cell membrane) protein*.

intrinsic viscosity *symbol*: η_{int} ; the limiting value of the **spe-**

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intron

cific viscosity, η_{sp} , of a solution as the concentration of the solute tends to zero. The units are $\text{cm}^3 \text{g}^{-1}$. Its value depends on the specific volume of the (solvated) solute, V , and on a shape-dependent factor, ν , as follows: $\eta_{\text{int}} = \nu V$. The value of ν is 2.5 for a sphere and its value increases for solutes of other shapes. The name **limiting viscosity number** has been recommended for this attribute.

intron or intervening sequence any intragenic region of DNA in eukaryotes that will not ultimately be expressed in a mature RNA molecule. Such sequences of deoxyribonucleotide residues intervene between coding sequences (**exons**); this arrangement of introns alternating with exons constitutes a **transcription unit**. By customary usage, the term is extended to the corresponding regions of the **primary transcript** of RNA. In RNA, four general types of introns are distinguishable: (1) yeast nuclear tRNA; (2) group I introns; (3) group II introns; and (4) nuclear mRNA precursor. Group I and II introns were originally described for fungal mitochondrial genes; group I introns are characterized by a lack of conserved sequences at splicing junctions but have short conserved sequences internally, while group II introns have conserved sequences at the splicing junctions. The introns of different RNA types are spliced by different mechanisms (*see* **splicing**). With respect to the function of introns it has been customary to describe them as derived from 'junk DNA', i.e. DNA for which no function can be found, but this concept must be modified. In the case of 'self-splicing introns' in the pre-rRNA of *Tetrahymena* many of the self-splicing introns double as mRNAs from which proteins are expressed that assist self-splicing and/or mediate intron translocation to new DNA sites. In the expression of a gene coding for small, stable RNAs that accumulate in the nucleolus, certainly of humans and mouse, it is the introns that give rise to RNA rather than the exons. Hence the terms intron and exon should simply signify RNA sequences that become physically separated during the process of RNA splicing. —**intronic adj.**

inulin a slightly soluble polysaccharide, of ≈ 6 kDa, found in members of the Compositae and Campanulaceae, in which it partially or completely replaces starch as a reserve carbohydrate. It is a linear glycan composed of about 35 ($\beta 2 \rightarrow 1$)-linked D-fructofuranose residues terminating in an ($\alpha 1 \rightarrow 2$)-D-glucopyranosyl residue. Inulin is useful in the measurement of renal clearance, as inulin is neither reabsorbed nor secreted by the renal tubular cells. It is also useful in the measurement of extracellular fluid volume (**inulin space**), since it can pass through the blood capillary walls but is not taken up by cells.

inulobiase the disaccharide 2-O- β -D-fructofuranosyl-D-fructose.

inulosucrase EC 2.4.1.9; an enzyme concerned in inulin biosynthesis. *See also* **sucrase** (def. 3).

in utero Latin within the uterus.

in vacuo Latin in a vacuum.

invaginate 1 to infold, especially to form a sheathlike or saclike enclosure. 2 describing an organ or part that is folded back upon itself. —**invagination n.**

invariant residue any amino-acid residue that always occurs in equivalent positions in the polypeptide chains of homologous proteins from different species, or in genetic variants of the protein from a given species.

invasin an alternative name for **hyaluronidase**.

invasive (*in pathology*) tending to penetrate into healthy tissue and grow within the host away from the site of entry or origin. It is used especially of a pathogenic organism, cell, or group of cells, e.g. cancer cells. —**invasiveness n.**

inverse agonist any ligand that by binding to receptors reduces the fraction of them in an active conformation, thereby having biological effects opposite to those produced by an **agonist** (as in **benzodiazepine inverse agonist**). This can occur if some of the receptors are in the active form even in the absence of a conventional agonist. *Compare* **partial inverse agonist**.

inverse-square law a law according to which the magnitude of a physical quantity, e.g. the intensity of electromagnetic or other radiation, is proportional to the reciprocal of the square of the distance from the source of the property.

inversion 1 **configurational inversion** the process of changing the **configuration** about a chiral centre into that with the opposite configuration. *See also* **Walden inversion**. 2 the hydrolysis of sucrose to an equimolar mixture of glucose and fructose. It is so-called because of the accompanying reversal of direction of optical rotation by the solution. 3 **chromosomal inversion** the deletion of a chromosomal segment followed by its reinsertion at the same position after having been turned through 180° .

inversion isomer an alternative name for **invertomer**.

invert 1 to turn or cause to turn inward, or upside down. 2 to reverse in direction, sequence, effect, etc. *See also* **inversion**.

invertase or saccharase EC 3.2.1.26; *recommended name*: β -fructofuranosidase; *systematic name*: β -D-fructofuranoside fructohydrolase. An enzyme from yeast and other sources that catalyses the **inversion** (def. 2) of sucrose. *See also* **saccharase**.

invertase another name for **β -fructofuranosidase**.

invertebrate 1 any animal lacking a vertebral column (backbone) and internal skeleton. 2 of, pertaining to, or designating any such animal.

inverted repeat or indirect repeat *abbr.*: IR; either of a pair of sequences in duplex DNA that exhibit two-fold rotational symmetry around a centre, the two sequences being complementary to each other. The sequence of the two repeats together is called a **palindrome**, e.g.

GGTACC
CCATGG.

The repeats need not be contiguous, e.g.

GGTNNACC
CCANNTGG.

Inverted repeats promote the formation of **cruciform** structures. Their function is unknown. However, they are found flanking transposable elements, which may indicate an associated role. *See also* **cruciform**, **dyad symmetry**, **hairpin**.

invertin a ferment (enzyme preparation) from yeast used by Fischer in work that led to his **lock-and-key model**. The major component is an α -glucosidase activity. *See also* **emulsin**.

invertomer or inversion isomer either of two **conformers** that interconvert by inversion at an atom possessing a nonbonding electron pair.

invert sugar an equimolar mixture of D-glucose and D-fructose formed by the hydrolysis, i.e. **inversion** (def. 2), of sucrose. It is sweeter than its parent disaccharide. *See also* **high-fructose corn syrup**.

in vitro Latin (*of any biological process, reaction, or experiment*) occurring or made to occur outside an organism, e.g. in extracts or cultures; literally it means 'in glass'. *Compare* **in vivo**.

in vivo Latin (*of any biological process, reaction, or experiment*) occurring or made to occur within a living organism; literally it means 'in life'. *Compare* **in vitro**.

involucrin a protein of keratinocytes and other stratified squamous epithelia that first appears in the cell cytosol, but ultimately becomes cross-linked to membrane proteins by the action of transglutaminase. Example from human: database code INVO_HUMAN, 585 amino acids (68.39 kDa).

involute 1 involved, intricate. 2 spiral, having whorls wound closely around the axis. 3 having edges that roll under or inwards. 4 to be subjected to a process involving the inward rolling of the surrounding tissue

iod+ a variant form of **iodo+** (sometimes before a vowel).

iodate 1 the anion, IO_3^- , derived from iodic(v) acid. 2 any salt of iodic(v) acid.

iodide 1 the anion, I^- , of the element iodine. 2 any salt of

iodimetry

hydriodic acid. **3** any compound containing an iodine atom in organic linkage.

iodimetry a volumetric method of analysis applicable to substances with a redox potential appreciably lower than that of the iodine-iodide system ($E^\circ = +0.535$ V at 25 °C). Such substances are oxidized by iodine and may therefore be titrated in acid solution with a standard solution of iodine. *Compare iodometry.* —**iodimetric** *adj.*; **iodimetrically** *adv.*

iodinatable capable of being converted into an iodo derivative. The term may be used of an organic compound or a particular part of one, such as a residue in a peptide.

iodinate **1** to treat or react (a substance) with iodine. **2** to introduce one or more iodo groups into (an organic compound), whether by addition or substitution. —**iodination** *n.*

iodine **1** *symbol:* I; a nonmetallic element of group 17 of the IUPAC periodic table and one of the halogens; proton number 53; it exists naturally as a single stable nuclide, **iodine-127**, of relative atomic mass 126.904. It is widely distributed in nature, although never found as the free element, and is an essential dietary element. Numerous (about 29 unstable and seven metastable) artificial radioactive nuclides are known. Of these, the most widely used in biochemistry and molecular biology are **iodine-125** and **iodine-131**. **2** *symbol:* I₂; the common name for the substance correctly known as diiodine, a bluish-black solid that sublimes at room temperature into a violet-coloured irritating vapour. An ethanolic solution is useful as a mild topical antiseptic.

iodine-125 the artificial radioactive nuclide ¹²⁵₅₃I. It decays to the stable nuclide tellurium-125 by capture to the nucleus of an orbital electron from the K (i.e. innermost) shell, emitting low-energy gamma radiation (35.5 keV, 7% emitted and 93% internally converted) and characteristic tellurium K X-radiation (27–32 keV, 140%), but no beta radiation. It has a half-life of 59.6 days.

iodine-127 the stable nuclide ¹²⁷₅₃I. Its relative abundance in natural iodine is 100 atom percent.

iodine-131 the artificial radioactive nuclide ¹³¹₅₃I. It decays to the stable nuclide xenon-131, emitting a moderate-energy electron or beta particle of various energies (0.25–0.81 MeV max., mainly 0.61 MeV max.) plus gamma radiation of various energies (0.08–0.72 MeV, mainly 0.36 MeV), 1.3% of nuclides decaying via the metastable nuclide xenon-131m. It has a half-life of 8.04 days.

iodine-labelling a widely used procedure for labelling a compound or material by introducing into it, directly or indirectly, one or more atoms per molecule of iodine, especially radioactive isotopes (i.e. radioiodine). *See also radioiodinate.*

iodine number or **iodine value** a measure of the degree of unsaturation of a fat or fatty acid, defined as the number of grams of iodine capable of reacting with 100 g of the sample. It is usually determined indirectly by measuring the amount of an iodine-containing reagent, e.g. iodine monobromide, that remains after excess has been added and reaction is complete. The iodine number of saturated fatty acids is zero, that of oleic acid is 90, of linoleic acid 181, and of linolenic acid 274.

iodo of, relating to, or describing a compound that contains an iodine atom in organic linkage.

iodo- or (sometimes before a vowel) **iod+** *comb. form* denoting **1** an iodine atom in organic linkage. **2** violet-coloured. *See also iodo.*

iodoacetamide I-CH₂-CONH₂; a compound that acts as an inhibitor of an enzyme containing -SH groups, Enz-SH, by reacting irreversibly to give Enz-S-CH₂-CONH₂, with elimination of HI.

iodo-Gen a proprietary name for 1,3,4,6-tetrachloro-3 α ,6 α -diphenylglycouril; a solid-phase reagent that oxidizes low concentrations of inorganic iodide to iodine and hence is useful for the rapid and efficient labelling of proteins with radioactive iodine. Being very stable and virtually insoluble in water, it can be used in the form of a prior coating on the walls of a

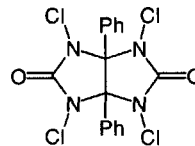
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ion-exclusion chromatography

reaction vessel containing aqueous solutions of iodide ion and protein.



iodogorgoic acid a former name for 3,5-diiodotyrosine.

iodometry a volumetric method of analysis applicable to substances with a redox potential appreciably higher than that of the iodine-iodide system ($E^\circ = +0.535$ V at 25 °C). Such substances are reduced by iodide, which is added in excess to acidified solutions and the liberated iodine titrated with a standard solution of thiosulfate. *Compare iodimetry.* —**iodimetric** *adj.*; **iodimetrically** *adv.*

iodopsin any of three light-sensitive visual pigments in the cone cells of the retinas of vertebrates other than freshwater and migratory fishes and some amphibians. They consist of different cone-type opsins combined (as Schiff's bases) with 11-*cis*-retinal. Each individual cone cell contains one of the three pigments, which have absorption maxima at ≈ 430 nm, ≈ 540 nm, and ≈ 575 nm. The iodopsins were once thought to be a single pigment, termed visual purple or visual violet. *See also cyanopsin, retinal.*

ion any atom or other molecular entity that has acquired an electric charge by loss or gain of one or more electrons. An anion is negatively charged, and a cation is positively charged.

ion atmosphere the region surrounding a charged molecular entity in which there is a statistical preference for ions of opposite charge.

ion chamber another name for ionization chamber.

ion channel any of several types of channels in cell membranes that allow the passage of specific inorganic ions through the membrane. They may be **ligand-gated**, or **voltage-gated**. In other cases, an ion channel may be opened by mechanical stress (**mechanically gated channels**). For ligand-gated channels, the ligand may be an extracellular mediator such as a neurotransmitter (**transmitter-gated channel**), or an intracellular mediator such as an ion (**ion-gated channel**) or a nucleotide (**nucleotide-gated channel**). Ion channels are common direct or indirect targets for the action of drugs, which may either potentiate or inhibit channel opening.

ion exchange the process in which ions of like charge, which may be the same or different chemically, are exchanged between two phases, such as two solutions separated by a semi-permeable membrane or a solution and an insoluble material, e.g. a resin.

ion-exchange cellulose *abbr.*: IEC; any derivatized cellulose containing cationic or anionic substituents used in **ion exchange**, especially for the separation of charged biopolymers.

ion-exchange chromatography any form of chromatography that depends on the process of **ion exchange** to effect the separation.

ion exchanger any **anion exchanger** or **cation exchanger**.

ion-exchange resin any insoluble polymer of high molecular mass and carrying groups with like charges that is used in **ion exchange**.

ion-exclusion chromatography a chromatographic technique by which strong electrolytes can be separated from non-electrolytes and weak electrolytes by passage of the mixture through an ion-exchange resin in a buffer of ionic strength and pH that can be varied in the course of the separation. At low salt concentrations particularly, ionic repulsive forces tend to exclude from the resin those ions that have the same charge as the functional groups on the resin so that these ions, together with an equivalent number of counterions, pass rapidly



ion-gated channel

through the column while nonelectrolytes and weak electrolytes enter the resin and thus pass more slowly through the column at a rate determined in part by the pK of their ionizable groups.

ion-gated channel *see* **ion channel**.

ionic 1 of, pertaining to, existing as, or characterized by ions. 2 using or operated by ions.

ionic atmosphere the cloud of ions of one charge type that tend to accumulate by interionic attraction around an ion of the other charge type in a solution of an electrolyte. The cloud is symmetrical about the charge on the central ion unless the solution is subjected to a shearing force conducive to flow or is exposed to an applied electric field. In the latter event the central ion and its surrounding counterions migrate in opposite directions, the resultant asymmetric disposition of the ionic atmosphere effecting retardation of the central ion in a concentration-dependent manner.

ionic bond any bond formed by attraction between a positively and a negatively charged group on different parts of the same molecule or on different molecules.

ionic conduction the passage of electricity due to the movement of ions.

ionic detergent any **anionic detergent** or **cationic detergent**. *Compare* **nonionic detergent**.

ionic double layer the region surrounding a charged molecular entity together with the layer of **counterions** that surrounds it.

ionic mobility a measure of the velocity at which an ion moves through a liquid under the influence of an applied electric potential.

ionic product *see* **ion product**.

ionic strength *symbol:* I or I_m (molality basis) or I_c (concentration basis); a physical quantity equal to half the summation over all ions in a solution of the concentration of each ion multiplied by the square of its charge. Hence,

$$I_m = 0.5(\sum m_i z_i^2)$$

and

$$I_c = 0.5(\sum c_i z_i^2),$$

where m_i and c_i are the molal and molar concentrations of the i th ion, respectively, and z_i is its charge.

ionizable group or **ionisable group** any uncharged group in a molecular entity that is capable of dissociating by yielding an ion (usually an H^+ ion) or an electron and itself becoming oppositely charged, or of associating with an ion (usually an H^+ ion) and itself assuming the ion's charge. The extent to which an ionizable group becomes an ionizing group depends on the constitution of the molecular entity and upon the environment.

ionization or **ionisation** any of the processes by which one or more ions are generated. It may occur, e.g., by loss or gain of a hydron by a solute in response to changes in pH, or by the loss of an electron from a neutral chemical species by unimolecular heterolysis of such a species into two or more ions. Further, a gas may become ionized by the passage through it of fast charged particles or electromagnetic radiation.

ionization chamber or **ionisation chamber** or **ion chamber** a chamber containing a gas and two electrodes between which an electric potential difference is maintained. It is used to detect ionizing particles or electromagnetic radiation by virtue of the current that flows following ionization of the gas.

ionization constant or **ionisation constant** the **dissociation constant** for a reversible reaction in which the products are ionic.

ionization energy or **ionisation energy** *symbol:* E_i ; the minimum energy (enthalpy) required to remove an electron from an isolated molecular entity (in its vibrational ground state) in the gaseous phase. It was formerly, and misleadingly, termed **ionization potential**.

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ionization potential or **ionisation potential** a former term for **ionization energy**.

ionize or **ionise** to convert, or to become converted, wholly or in part, into ions; to cause **ionization** of (something). —**ionizable** *adj.*

ionizing particle or **ionising particle** any charged particle of sufficient energy to enable it to dislodge an orbital electron from an atom or other molecular entity. *See* **ionizing radiation**.

ionizing power or **ionising power** or (sometimes) **polarity** a qualitative term used to denote the tendency of a particular solvent to promote ionization of an uncharged solute.

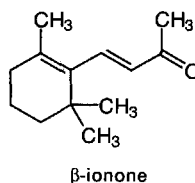
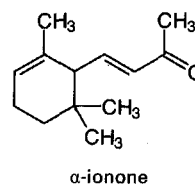
ionizing radiation or **ionising radiation** any radiation, corpuscular or electromagnetic, that is capable of causing **ionization** (directly or indirectly) of matter through which it passes. For ionization to occur, the kinetic or photon energy, respectively, must be greater than the **ionization energy** of the irradiated substance. Ultraviolet radiation, X-radiation, alpha particles, and high-energy electrons (including beta particles) are more effective than gamma radiation or neutrons. It is commonly known simply as radiation.

ionogen 1 any compound that can dissociate into ions in a suitable environment; an electrolyte. 2 any atom or group that is capable of becoming ionized chemically. —**ionogenic** *adj.*

ionography an alternative name for **zone electrophoresis**.

ionomycin a nonfluorescent antibiotic ionophore derived from *Streptomyces conglobatus*. It is highly specific for divalent cations and exhibits greater selectivity for Ca^{2+} over Mg^{2+} than **A23187** (calcimycin). It complexes with Ca^{2+} between pH 7 and 9.5 with intense UV absorption, and thus may be used as a carrier of Ca^{2+} and for measurement of cytoplasmic Ca^{2+} concentration.

ionone either of two terpene compounds, α -ionone and β -ionone, found in a number of plants. β -Ionone is an intermediate in the synthesis of vitamin A; its odour is reminiscent of cedarwood or, in dilute alcohol solution, of violets.



ionophore or **ionophorous agent** any member of a varied group of both natural and synthetic substances, cyclic or linear and of greatly differing relative molecular masses, that are capable of binding metal ions in solution and transporting them across lipid barriers in natural or artificial membranes. They may be classed as **cage carriers**, when the ionophore surrounds the metal ion during transport, or as **channel carriers**, when the ionophore forms an ion-conducting pore in the membrane. —**ionophorous** or **ionophoric** *adj.*

ionophoresis **electrophoresis**, especially of relatively small ions. *See also* **iontophoresis**.

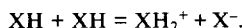
ionotropic describing a type of receptor that mediates its effects by regulating ion channels, in distinction from receptors that act by regulating enzymes such as adenylate cyclase or in-



osotide-specific phospholipase C. *Compare* **metabotropic**. *See also* **excitatory amino-acid receptor**.

ion pair a pair of oppositely charged ions held together by coulombic attraction. *See also* **ionic bond**.

ion product or (sometimes) **ionic product** *symbol*: K_i ; a measure of the tendency of an amphiprotic solvent, XH , to take up or to lose a hydron in the reaction:



In such a reaction, $K_i = a_{XH_2^+} \times a_{X^-}$ where $a_{XH_2^+}$ and a_{X^-} are the activities of XH_2^+ and X^- , respectively. The ion product of water, K_w , is given by

$$K_w = a_{H_3O^+} \times a_{OH^-}; K_w \approx 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6} \text{ at } 25^\circ \text{C}.$$

It has a large positive temperature coefficient.

ion-selective electrode or **ion-selective microelectrode** any electrode (or microelectrode) whose potential responds in a reproducible manner to changes in the concentration (strictly, in the activity) of a specific kind of ion in the solution in which the electrode is immersed and that is insensitive, or essentially so, to changes in the concentrations (strictly, in the activities) of all other kinds of ion in the solution. The electrode has at its surface a thin membrane of electrically conducting material, separating the internal and external solutions, across which an electric potential difference develops. The membrane consists of or contains an electroactive material responsive to the ions to be determined.

iontophoresis 1 the introduction of drugs and other substances through plasma membranes of cultured cells by the transfer of ions brought about by the application of a small electrical potential through electrodes placed on the membrane. 2 a less common word for **ionophoresis**.

iota *symbol*: ι (lower case) or I (upper case); the ninth letter of the Greek alphabet. For uses *see* **Appendix A**.

IP or **i.p.** *abbr.* for intraperitoneal or intraperitoneally.

IP₂ *abbr.* for 1D-myo-inositol 3,4-bisphosphate.

IP₃ *abbr.* for 1D-myo-inositol 1,3,4- (or 1,4,5-) triphosphate.

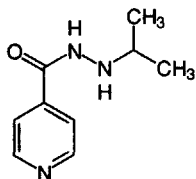
IP₄ *abbr.* for 1D-myo-inositol 1,3,4,5-tetrakisphosphate.

ipecacuanha or **ipecac** the dried root of the South American shrub, *Cephaelis ipecacuanha*; it contains the alkaloids **cephaeline** and **emetine**. Ipecacuanha preparations are used as emetics and expectorants.

IPP *abbr.* for Δ^3 -isopentenyl pyrophosphate. This compound is now preferably called isopentenyl diphosphate (*abbr.*: IDP).

iPr *symbol* for the isopropyl group, $(CH_3)_2CH-$ (alternative to Pr^i).

iproniazid 1-isonicotinoyl-2-isopropylhydrazine; a derivative of **isoniazid** that is a **monoamine oxidase** inhibitor.



ips *symbol* for the pipsyl group.

IPTG *abbr.* for isopropyl β -D-thiogalactoside.

IPTS *abbr.* for International Practical Temperature Scale.

Ir *symbol* for iridium.

IR 1 or **i.r.** *abbr.* for infrared (radiation). 2 *abbr.* for inverted repeat.

IRE *abbr.* for iron response element.

I₁, I₂ receptors *see* **imidazoline receptor**.

IRG *abbr.* for immunoreactive glucagon.

IRMA *abbr.* for immunoradiometric assay.

iron *symbol*: Fe; a metallic element of group 8 of the IUPAC periodic table; proton number 26; natural iron has a relative atomic mass of 55.847 and is a mixture of stable nuclides of nucleon numbers 54 (5.8 atom per cent), 56 (91.7 atom per cent), 57 (2.2 atom per cent), and 58 (0.3 atom per cent). About 12 artificial radioactive nuclides are known; of these, **iron-59** is the one most widely used in biochemistry and molecular biology. Iron is a **transition element** and hence forms ionic salts and numerous complexes in the oxidation states +2, i.e. iron(II) (ferrous iron), or +3, i.e. iron(III) (ferric iron). Iron is one of the most important elements in biology; it is essential in substantial quantities to all cellular organisms from bacteria to mammals, particularly for the synthesis of hemoproteins such as **cytochromes**, **hemoglobins** and related proteins, **myoglobin**, **catalase**, and some **peroxidases**. Important iron-transport proteins include **ferritin** and **transferrin**; iron is taken up into mammalian cells, bound to transferrin, by the **transferrin receptor**. There is no physiological excretory mechanism for iron, but it is lost from females during menstruation.

iron-59 the artificial radioactive nuclide $^{59}_{26}\text{Fe}$. It decays to the stable nuclide cobalt-59 with emission of a beta particle (i.e. an electron) in two moderate-energy ranges (0.27 MeV max., 48%, and 0.48 MeV max., 51%) plus gamma radiation of various energies (mainly 1.10 MeV, 56%, and 1.29 MeV, 43%). It has a half-life of 44.5 days.

iron-molybdenum cofactor *abbr.*: FeMo-co; *see* **nif**, **nitrogenase**.

iron overload a condition of humans in which iron is taken up in excess over a prolonged period. As there is no excretory mechanism for iron, it accumulates in Kupffer cells in the liver as **hemosiderin**, a form of **ferritin** complexed with other proteins and iron. The condition is known as **hemosiderosis**. It is sometimes referred to as **Bantu siderosis**, being very prevalent in these peoples owing to their use of iron cooking pots. It also occurs as a result of repeated blood transfusions, e.g. in treatment for the hemoglobinopathies.

iron response element or **iron regulatory element** *abbr.*: IRE; a base sequence in eukaryotic genomes responsible for regulating gene expression of proteins involved in iron metabolism.

ironophore an alternative name for **siderochrome**.

iron-sulfur protein *abbr.*: Fe-S protein; any member of a group of proteins to which iron is bound via sulfur-containing ligands (excluding hemoproteins containing iron linked only through axial sulfur ligands). Iron-sulfur proteins are characterized by a relatively low molecular mass and the possession of nonheme bound iron, as well as one or more of the following: (1) acid-labile sulfur; (2) absorption bands between 300 and 600 nm; (3) the capability of transferring electrons at low redox potential; and (4) paramagnetism in one or more oxidation states. They may be classified as **simple** if they contain one or more Fe-S clusters only (*see* **ferredoxin**, **rubredoxin**), and **complex** where additional active groups such as flavin, heme, or another metal are present.

irradiate to subject to or treat with electromagnetic radiation or beams of particles. —**irradiation** *n*.

irreversible competitive antagonist (*in pharmacology*), an antagonist for which the dissociation rate is so low that antagonist molecules cannot be replaced by increasing the concentration of an agonist for the same receptor.

irreversible process any process occurring in an isothermal system without change of internal energy but with an increase in entropy.

IS *abbr.* for insertion sequence.

islet any small group of cells that is structurally distinct from the surrounding tissue, especially the **islets of Langerhans**.

islet-activating protein *abbr.*: IAP; a former name for **pertussis toxin** (so-called because it stimulates release of insulin from the islets of Langerhans).

islets of Langerhans or (sometimes) **islands of Langerhans** or **pancreatic islets** the numerous small, spherical or oval, highly vascular clusters of endocrine cells that lie scattered throughout the acinar (i.e. exocrine) tissue of the pancreas of gnathostomes (i.e. jaw-bearing) vertebrates and are readily distinguished from the surrounding exocrine tissue.



iso+

guishable microscopically from it. The normal adult human pancreas contains about 2 million islets, which together comprise under 2% of its total volume. Islets commonly consist of three main types of cells, all granule-containing: the glucagon-secreting **A cells**; the insulin-secreting **B cells**; and the somatostatin-secreting **D cells**. Some other types of hormone-secreting cells may be present also. [After Paul Langerhans (1847–88), German physician and anatomist, who first described such clusters of cells (1869).]

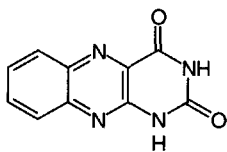
iso+ or (sometimes before a vowel) **is+** *comb. form* 1 equal, the same, uniform, homogeneous. Compare **allo+** (def. 1), **homo+** (def. 1), **hetero+** (def. 1), **xeno+** (def. 1). 2 (in immunology and genetics) **a** in or of some individual or ones that are genetically identical (e.g. **isogenic**). **b** in or of another, genetically differing individual of the same species (e.g. **isoantigen**); **allo+** (def. 2) is now usually preferred. **c** in or of all normal individuals of the same species (e.g. **isotypic**). Compare **allo+** (def. 2), **hetero+** (def. 2), **homo+** (def. 2), **xeno+** (def. 2). 3 (in chemical nomenclature) **a** signifying isomer of (formerly 'iso' was italicized and often hyphenated). **b** *symbol*: i (as a prefix or superscript); signifying the isomer of a straight-chain alkane or alkane derivative that differs only in having a terminal (CH₃)₂CH– group (leucine and isoleucine are exceptions). Compare **allo+** (def. 3), **nor+** (def. 1). 4 (in names of certain classes of macromolecules) signifying irrespective of location or origin, one of two or more structural variants with the same specificity of activity or action (e.g. **isoacceptor tRNA**, **isohormone**; see also **isoenzyme**).

ISO *abbr.* for International Standards Organization.

isoacceptor tRNA or **isoaccepting tRNA** any of two or more transfer RNA species obtainable from a given organism that can be acylated by the same amino acid; they may differ in their anticodons.

isoagglutinin any antibody formed in one individual that causes agglutination of cells derived from another individual of the same species.

isoalloxazine benzo[g]pteridine-2,4(1H,3H)dione; the hypothetical tricyclic parent compound from which is derived any N¹⁰-substituted alloxazine. The 7,8-dimethyl derivative is the heterocyclic ring structure in riboflavin and flavin mononucleotide, and one of the heterocyclic ring structures in flavin-adenine dinucleotide.



isoamylase EC 3.2.1.68; *other name*: debranching enzyme; an enzyme that catalyses the hydrolysis of 1,6- α -D-glucosidic branch linkages in glycogen, amylopectin, and their β -limit dextrins. It does not attack pullulan, and is limited in action on α -limit dextrins. Example (precursor) from *Pseudomonas amyloclavata*: database code ISOA_PSEAY, 771 amino acids (83.24 kDa).

isoantibody any antibody formed in one individual in response to an antigen derived from another individual of the same species.

isoantigen any antigen occurring in an individual that is capable of eliciting an immune response in genetically different individuals of the same species but not in the individual itself.

isoasparagine the trivial name for the α -amide of aspartic acid, aspartic 1-amide; β -aminosuccinamic acid; 3-amino-3-carbamoylpropanoic acid; H₂N–CO–CH(NH₂)–CH₂–COOH; a chiral α -aminoacylamide. L-Isoasparagine (*symbol*: Asp-NH₂) can occur as an N-terminal amidated residue of proteins. Compare **asparagine**.

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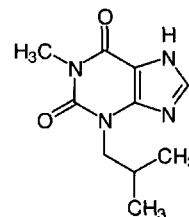
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
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isoasparaginy the acyl group, H₂N–CO–CH(NH₂)–CH₂–CO–, derived from **isoasparagine**. Compare **asparaginy**.

isoaspartyl the former name for β -aspartyl.

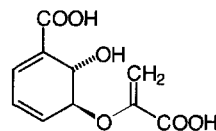
isobar 1 any of two or more nuclides having the same mass number (nucleon number) but different atomic numbers (proton numbers) and hence of differing chemical properties. See also **heterotope**. 2 any curve or equation relating quantities measured at the same pressure; any line joining points on a graph or map that have equal pressures. —**isobaric** *adj.*

isobutylmethylxanthine *abbr.*: IBMX; 3-isobutyl-1-methylxanthine; an inhibitor of cyclic AMP phosphodiesterase; it raises cyclic AMP levels in cells and tissues.



isocaloric (of foodstuffs) having the same calorific value.

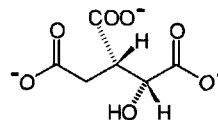
isochorismate 5-[(1-carboxylatoethenyl)oxy]-6-hydroxy-1,3-cyclohexadiene-1-carboxylate; the dianion of isochorismic acid; an isomer of **chorismate** and an important precursor to some compounds in the **shikimate pathway**.



isochorismic acid

isochromosome any chromosome having homologous, genetically identical arms that are mirror images of each other.

isocitrate 1 any of the anionic forms of isocitric acid, 1-hydroxy-1,2,3-propanetricarboxylic acid. (1R,2S)-1-hydroxypropane-1,2,3-tricarboxylate (also termed (2R,3S)-isocitrate or *threo*-D₃-isocitrate) is an important intermediate in the **tricarboxylic-acid cycle** and the **glyoxylate cycle**. 2 any mixture of free isocitric acid and its ionized forms. 3 any (partial or full) salt or ester of isocitric acid.



(1R,2S)-1-hydroxypropane-1,2,3-tricarboxylate

isocitrate dehydrogenase 1 isocitrate dehydrogenase (NAD⁺) EC 1.1.1.41; *other names*: isocitric dehydrogenase; β -ketoglutaric-isocitric carboxylase; an enzyme that catalyses the oxidation of isocitrate to 2-oxoglutarate and CO₂, with reduction of NAD⁺ to NADH. The isomer of isocitrate involved is (1R,2S)-1-hydroxypropane-1,2,3-tricarboxylate. Examples from yeast, in which there are two mitochondrial isozymes (precursors): database code IDH1_YEAST, 360 amino acids (39.28 kDa); database code IDH2_YEAST, 369 amino acids (39.69 kDa). In



mammalian tissues, this enzyme is localized exclusively in the mitochondria. **2 Isocitrate dehydrogenase (NADP⁺)** (EC 1.1.1.42); an enzyme that is localized in both mitochondria and cytosol.

[isocitrate dehydrogenase (NADP⁺)] kinase EC 2.7.1.116; *other names*: isocitrate dehydrogenase kinase/phosphatase (*abbr.*: IDH kinase/phosphatase); an enzyme that catalyses the phosphorylation by ATP of isocitrate dehydrogenase (NADP⁺) with formation of ADP. The phosphorylated protein is the inactive form of the enzyme. Example from *Escherichia coli* in which a bifunctional enzyme contains this and isocitrate dehydrogenase kinase/phosphatase (EC 3.1.3.-) activities: database code ACEK_ECOLI, 578 amino acids (67.51 kDa).

isocitrate lyase EC 4.1.3.1; *systematic name*: isocitrate glyoxalate-lyase; an enzyme, present in the glyoxysomes of seeds, that converts (1*R*,2*S*)-1-hydroxypropane-1,2,3-tricarboxylate (the isomer of isocitrate involved) to succinate and glyoxylate. *See glyoxylate cycle*.

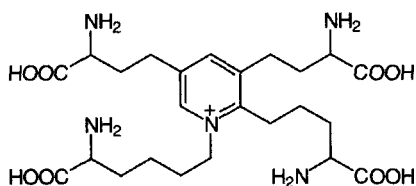
isocratic (*in chemistry*) describing any chromatographic procedure or separation in which the composition of the eluent is maintained constant. *Compare gradient elution*.

isocyanate any organic compound containing the univalent group —N=C=O . *Compare cyanate, isothiocyanate*.

isocyclic *see* **homocyclic** (def. 1).

isodalt a type of very-high-resolution two-dimensional electrophoresis system for the analysis of small (1–2 mg) samples of complex mixtures of proteins. It employs isoelectric focusing in cylindrical gels in the first dimension, followed by slab-gel electrophoresis in the presence of sodium dodecyl sulfate in the second dimension. Such systems are considered to be capable of resolving mixtures containing 10^4 or more individual proteins within the range 10–200 kDa.

isodesmosine the 2-(4-amino-4-carboxybutyl)-1-(5-amino-5-carboxypentyl)-3,5-bis(3-amino-3-carboxypropyl)pyridinium ion; a tetra(α -amino acid) isolated from hydrolysates of the fibrous protein elastin. *See desmosine*.



isodiametric **1** (*of cells*) having similar diameters in all planes. **2** (*of a crystal*) having three equal axes.

isodynamic (*in physics*) having equal force or strength.

isoelectric **1** having the same electrical potential; containing or indicating no potential difference. **2** (*of particles or molecular entities*) having no net electric charge.

isoelectric focusing or **electrofocusing** *abbr.*: IEF; a technique in which solutes of different isoelectric points are caused to form stationary bands in an electric field, which is superimposed on a (stable) pH gradient, the pH increasing from the anode to the cathode. In practice the pH gradient is most conveniently formed by electrolysis of a solution containing a mixture of **carrier ampholytes** of low molecular mass and slightly differing isoelectric points, each of which will move to its isoelectric region in the electric field and remain there.

isoelectric-focusing electrophoresis a separation technique in which electrophoresis is effected in a gel slab at right angles to a preformed, stationary pH gradient generated by isoelectrically focused carrier ampholytes (*see isoelectric focusing*).

isoelectric fractionation any technique for fractionating proteins or other ampholytes using **isoelectric focusing**.

isoelectric pH an alternative term for **isoelectric point**.

isoelectric point or **isoelectric pH** *abbr.*: IEP or i.e.p.; *symbol*: pI ; the pH of the solution in which a protein or other ampholyte has zero net charge; hence the pH at which the protein or other ampholyte has zero net charge, i.e. no charges or an equal number of positive and negative charges including those due to any extraneous ions bound to the ampholyte molecule. The pH value of the isoelectric point may depend on other ions, except hydrogen and hydroxide ions, present in the solution. *Compare isoionic point*.

isoelectric precipitation the precipitation of a protein at its **isoelectric point**, at which proteins are generally least soluble. It is useful in the fractionation of mixtures of proteins.

isoelectronic describing two or more chemical entities having the same number of electrons, especially valence electrons. *Compare isosteric*.

isoenergetic taking place at constant energy.

isoenzyme or **isozyme** any one of the multiple forms of an enzyme arising from a genetically determined difference in primary structure. Isoenzyme is an operational term to denote any one of a group of enzyme proteins with similar catalytic properties but separable by suitable methods, e.g. by electrophoresis, where knowledge of the nature of the multiplicity may be lacking. If the different enzyme proteins are derived from a common gene, then the term 'multiple forms' rather than 'isoenzymes' should be used. The term embraces **allelzyme** and **multilocus enzyme**. *Compare interconvertible enzyme*.

isoflavone 7-O-glucosyltransferase EC 2.4.1.170; an enzyme that catalyses the reaction between UDPglucose and isoflavone to form isoflavone 7-O- β -D-glucoside and UDP.

isoflurane 2-chloro-2-(difluoromethoxy)-1,1,1-trifluoroethane; a widely used nonflammable volatile anesthetic agent.

isoform any of several multiple forms of the same protein that differ in their primary structure, but retain the same function. They can be produced by different genes or by **alternative splicing** of RNA transcripts from the same gene.

isogenic or **isogenetic** describing two or more individuals that possess exactly the same genotype. *See also syngenic*.

isoglucose *see* **high-fructose corn syrup**.

isoglutamine a trivial name for the α -amide of glutamic acid; glutamic 1-amide; γ -aminoglutaric acid; 4-amino-4-carbamoylbutanoic acid; $\text{H}_2\text{N—CO—CH(NH}_2\text{)—[CH}_2\text{]}_2\text{—COOH}$; a chiral α -amino acylamide. L-Isoglutamine (*symbol*: Glu-NH₂) can occur as an N-terminal amidated residue of proteins. Residues of D-isoglutamine (*symbol*: D-Glu-NH₂) are found in *Staphylococcus aureus* peptidoglycan. *Compare glutamine*.

isoglutaminyl the acyl group, $\text{H}_2\text{N—CO—CH(NH}_2\text{)—[CH}_2\text{]}_2\text{—CO—}$, derived from **isoglutamine**. *Compare glutaminyl*.

isoglutamyl a former name for γ -glutamyl.

isograft an alternative term for **syngenic graft**.

isohormone any of two or more structural variants of a polypeptide hormone that have the same hormonal action and are produced within the same organism, or at least within the same species of organism. —**isohormonal** *adj.*

isohydric describing solutions that have the same hydrogen-ion concentration, hydrogen-ion activity, or pH.

isohydric shift the opposing and counterbalancing changes in hydrogen-ion concentration occurring in peripheral tissues when dioxygen leaves the blood and carbon dioxide enters it. The hydration of CO_2 to H_2CO_3 , catalysed by erythrocyte carbonic anhydrase, has a tendency to lower the blood pH. This tendency is counteracted in some degree by the simultaneous deoxygenation of hemoglobin; deoxyhemoglobin binds hydrogen ions more strongly than oxyhemoglobin, thus tending to raise the blood pH.

isoionic pH an alternative name for **isoionic point**.

isoionic point or **isoionic pH** the pH of a solution of a protein or other ampholyte from which all other ions have been removed, except hydrogen and hydroxide ions formed by dissociation of water or of the protein or ampholyte itself. *Compare isoelectric point*.

isoionic solution any solution of a protein or other ampholyte

isokinetic gradient

that contains no other ions except those arising from dissociation of the solute and the solvent.

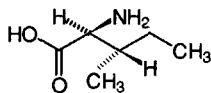
isokinetic gradient any concentration and viscosity gradient used in centrifugation that allows particles (molecules) of the same density to sediment at a constant velocity at all distances from the centre of rotation.

isolate 1 to separate (a pure substance, a cell type, or a sample of a subcellular component) from a mixture or from naturally occurring material and then (usually) to characterize it; to separate and put into pure culture (a particular species or strain of a microorganism) from a mixture, sample, or biological specimen. 2 to remove a viable tissue or organ for the purpose of study or experiment. 3 (*in physics*) to disconnect (part of a circuit), or to prevent interaction between (circuits or components). 4 a group of similar microorganisms, especially a pure culture, obtained by isolation for study, experiment, or as an aid to diagnosis; something isolated. —**isolable** or **isolatable** *adj.*; **isolability**, **isolator** *n.*

isolectin any of two or more molecular forms of lectins, of the same origin, apparently with the same biological properties, and of very similar molecular masses, but of differing electrophoretic mobilities. They may be the products of closely related genes or formed prior to or during isolation. In glycoprotein lectins, their differences may lie in the carbohydrate side chains.

isoleucinate 1 isoleucine anion; the anion, $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}(\text{NH}_2)\text{-COO}^-$, derived from **isoleucine**. 2 any salt containing isoleucine anion. 3 any ester of isoleucine.

isoleucine the trivial name for α -amino- β -methylvaleric acid; (2*R**,3*R**)-2-amino-3-methylpentanoic acid; $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}(\text{NH}_2)\text{-COOH}$; an α -amino acid with two chiral centres. Because molecules of isoleucine possess a second chiral centre, at C-3, in addition to the chiral centre at C-2 common to all α -amino acids other than glycine, the enantiomers L-isoleucine, (2*S*,3*S*)-2-amino-3-methylpentanoic acid (*symbol*: I or Ile), and D-isoleucine, (2*R*,3*R*)-2-amino-3-methylpentanoic acid (*symbol*: D-Ile or DIle), are diastereoisomeric with those of **alloisoleucine**, (2*R**,3*S**)-2-amino-3-methylpentanoic acid. L-isoleucine is a coded amino acid found in peptide linkage in proteins; codon: AUA (not in mitochondria of fruit fly, mammals, or yeast), AUC, or AUU. In mammals it is an essential dietary amino acid, and is both glucogenic and ketogenic. D-Isoleucine is not known to occur naturally.



L-isoleucine

isoleucine biosynthesis a pathway that commences with formation of α -aceto- α -hydroxybutyrate (I) from pyruvate and α -ketobutyrate (2-oxobutanoate, formed by removal of ammonia from threonine by serine dehydratase). I is converted via α,β -dihydroxy- β -methylvalerate to α -keto- β -methylvalerate (II). Isoleucine is then formed from II as a result of transamination with glutamate.

isoleucinium isoleucine cation; the cation, $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}(\text{NH}_3^+)\text{-COOH}$, derived from **isoleucine**.

isoleucino the alkylamino group, $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}(\text{COOH})\text{-NH-}$, derived from **isoleucine**.

isoleucyl the acyl group, $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}(\text{NH}_2)\text{-CO-}$, derived from **isoleucine**.

isoliquivitigenin *see* **chalcone**.

isolog the US spelling of **isologue**.

isologous 1 of, relating to, or being two or more closely related chemical compounds in a series whose successive mem-

isomeric tran:

bers differ in composition in a regular manner, other than by a difference of one methylene group. The term is now applied especially to compounds that have different atoms of the same valency at one or more positions in the molecule but are otherwise structurally identical, e.g. methionine and selenomethionine. 2 an alternative term for **isogeneic**.

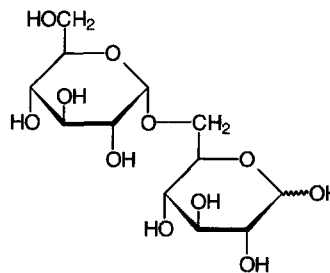
isologous association or **isologous binding** any association between two identical protein subunits (i.e. **protomers**) in which the **domain of bonding** is made up of two identical **binding sets**.

isologue or (*US*) **isolog** any of two or more **isologous** (def. 1) compounds.

isology the state or condition of being **isologous** (def. 1); chemical kinship among molecules or compounds.

isomaltase EC 3.2.1.10; *recommended name*: oligo-1,6-glucosidase; *systematic name*: dextrin 6- α -D-glucanohydrolase. An enzyme that hydrolyses 1,6- α -D-glucosidic bonds in dextrans and isomaltose produced after α -amylase degradation of starch and glycogen. The enzyme from intestinal mucosa also catalyses the reaction of EC 3.2.1.48 (*see* **sucrase**). Example from *Bacillus* sp.: database code O16G_BACSP, 508 amino acids (59.78 kDa).

isomaltose the disaccharide 6-*O*- α -D-glucopyranosyl-D-glucose; it exists as the branching unit in **amylopectin**, **glycogen**, and some bacterial **dextrans**. *Compare* **maltose**.



isomaltulose the disaccharide 6-*O*- α -D-glucopyranosyl-D-fructofuranose.

isomer 1 (*in chemistry*) **molecular isomer** any of two or more compounds that have identical molecular formulas but differ in the nature of bonding of their atoms (**constitutional isomers**) or in the arrangement of their atoms in space (**stereoisomers**). 2 (*in physics*) **nuclear isomer** any of two or more nuclides that have the same proton number and the same mass (nucleon) number but different energy states.

isomerase any enzyme of EC class 5 that catalyses geometric or structural changes within one molecule. According to the type of isomeric change they catalyse, they may be called racemases or epimerases (EC subclass 5.1); *cis-trans*-isomerases (EC subclass 5.2); intramolecular oxidoreductases (EC subclass 5.3); intramolecular transferases (mutases) (EC subclass 5.4); or intramolecular lyases (EC subclass 5.5); other isomerases are placed in EC subclass 5.99.

isomeric of or pertaining to an **isomer**; pertaining to the phenomenon of, or displaying, **isomerism**.

isomeric state any of the states of an atomic nucleus having a different energy and observable half-life from those of other states of the same nucleus; the condition of being an **isomer** (def. 2).

isomeric transition *abbr.*: IT; the change of one nuclear isomer (*see* **isomer** (def. 2)) into another without change in either proton number (atomic number) or nucleon number (mass number), energy being lost in the form of gamma radiation; e.g. the decay of the metastable nuclide technetium-99m (half-life 6 h) to the unstable nuclide technetium-99 (half-life 2.1×10^5 years), which in turn decays to ruthenium-99 with emission of an electron.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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isomerism

isomerism the phenomenon of the existence of **isomers** in general, or the occurrence of molecular or nuclear isomers in a particular instance.

isomerization or **isomerisation** the act of changing or the process of change of one molecular isomer into another one. —**isomerize** or **isomerise** *vb.*

isometric **1** having identical dimensions. **2** (*in physiology*) of or pertaining to a contraction of a muscle that does not cause shortening of the muscle. **3** (*in crystallography*) having three mutually perpendicular axes of equal length; cubic.

isomorph a substance or organism that is **isomorphic** (def. 1) to another substance or organism.

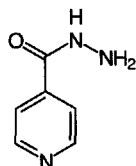
isomorphic **1** or **isomorphous** of identical or similar form, shape, or structure. **2** describing a type of alternation of generations in which morphologically identical generations occur in both diploid and haploid phases. It occurs mainly in algae. *Compare heteromorphic.*

isomorphism **1** (*in biology*) similarity of form in organisms of different ancestry. **2** (*in chemistry*) similarity of crystalline form and structure between substances of similar composition or substances that can form a more or less continuous series of solid solutions.

isomorphous **1** an alternative term for **isomorphic** (def. 1). **2** displaying **isomorphism** (esp. def. 2).

isomorphous replacement method a method used to solve the phase problem in **X-ray diffraction** analysis of macromolecular structures. It involves combining one or more **heavy atoms** (def. 2) (which are strong scatterers) in a specific manner with the macromolecule in question and investigating their effects on the diffraction pattern. The heavy atom must not modify either the conformation or the crystal form of the macromolecule, i.e. the replacement must be isomorphous.

isoniazid 4-pyridinecarboxylic acid hydrazide; an antitubercular drug that may act by inhibiting the synthesis of the mycolic acids that are vital for the construction of the cell wall of the pathogen, *Mycobacterium tuberculosis*. Like related hydrazides it is highly hepatotoxic and no longer used. Its euphoric side effects led to the discovery of monoamine oxidase inhibitors.

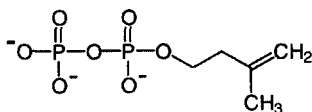


isoosmotic a variant spelling of **isosmotic**.

isopartitive (*in chromatography*) describing two or more solvents that bring about similar partitions of a given substance between the solvent and the stationary phase and hence result in R_f values that are the same or nearly so.

isopentenyl the trivial name for 3-methylbut-3-enyl, $\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-$, the alkenyl group formally derived from isopentenyl alcohol (i.e. 3-methylbut-3-en-1-ol) by loss of its hydroxyl group. *Compare prenyl.*

isopentenyl diphosphate *abbr.*: IDP; formerly called Δ^3 -isopentenyl pyrophosphate (*abbr.*: IPP); an isomer of dimethylallyl diphosphate and the key precursor of all **isoprenoids** (def. 2). *See also* **dimethylallyltransferase**, **isopentenyl-diphosphate Δ -isomerase**, **isoprene rule**.



isopentenyl-diphosphate Δ -isomerase EC 5.3.3.2; *systematic name*: isopentenyl diphosphate Δ^3 - Δ^2 -isomerase; *other name*: IPP isomerase. An enzyme in the pathway for the synthesis of cholesterol, terpenes, and prenyl groups; it catalyses the reversible reaction:

isopentenyl diphosphate = dimethylallyl diphosphate.

Example from *Saccharomyces cerevisiae*: database code IPPI_YEAST, 287 amino acids (33.18 kDa).

isopeptidase any peptidase catalysing the cleavage of an **isopeptide bond**.

isopeptide bond any peptide bond other than one that is a **peptide bond**, i.e. any amide bond formed between a carboxyl group of one amino-acid molecule or residue and an amino group of another where either group occupies a position other than α (or both do so). Examples include peptide bonds formed from the β -carboxyl group of aspartic acid, the γ -carboxyl group of glutamate from the ϵ -amino group of lysine, and from either the carboxyl group or the amino group of β -alanine.

isopiestic of, pertaining to, or characterized by equal pressures.

isopiestic vapour pressure method a method for the determination of the activity of water in a solution of a nonvolatile solute. The solution in question is equilibrated with a reference solution containing a nonvolatile solute for which the activity of water at different solute concentrations has been determined with high precision. The two solutions are placed in an enclosure at constant temperature until their vapour pressures are equal, i.e. the activity of water in both solutions is equal. They are then analysed for their solute contents, from which the activity of water in both of them can be calculated.

isoprenaline an alternative name for **isoproterenol**.

isoprene 2-methylbuta-1,3-diene; $\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}=\text{CH}_2$; a liquid hydrocarbon hemiterpene isolated from products of the pyrolysis of natural rubber, turpentine, etc., and widespread in plants although generally found only in extremely small amounts.

isoprene residue the branched-chain bivalent group, 3-methyl-2-butenylene; 3-methylbut-2-en-1,4-diyl; $-\text{CH}_2-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2-$. It is formally derived by bond rearrangement from **isoprene** and constitutes the repeating unit of the molecular structures of prenyls and many other isoprenoids.

isoprene rule a rule, formulated by Wallach in 1887, that stated that many natural products are built up from isoprene carbon skeletons. It was modified to the **biogenetic isoprene rule** by Ružička, who proposed that all terpenoids, steroids, etc. are synthesized from a precursor called 'active isoprene'. This precursor is now known to be **isopentenyl diphosphate**.

isoprene unit (*in an isoprenoid*) an **isoprene residue** or any branched five-carbon-atom moiety formally derived from an isoprene residue by chemical modification, e.g. by hydrogenation, hydroxylation, etc. It represents isopentenyl alcohol (i.e. 3-methylbut-3-en-1-ol), the diphosphate of which is the universal biogenetic precursor of isoprenoids.

isoprenoid **1** of, pertaining to, or being **isoprene** and its chemical relatives; having a molecular structure composed of, containing, or derived from linked **isoprene residues**. **2** any isoprenoid compound. Isoprenoids comprise a large and varied family of natural products, including carotenoids, phytol, prenyls, rubber, steroids, terpenoids, and tocopherols. Many substances contain both isoprenoid and nonisoprenoid components, e.g. the vitamins K.

isoprenyltransferase any of the enzymes within subclass EC 2.5.1, existing as α - β heterodimers, that add C_{15} or C_{20} carbon chains by **prenylation** near the carboxyl terminus of proteins. Examples include the nuclear lamins and Ras.

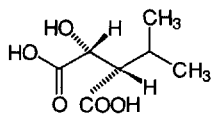
isopropyl *symbol*: iPr or Prⁱ; the alkyl group $(\text{CH}_3)_2\text{CH}-$, derived from propane. *Compare propyl.*

isopropylmalate either of the isomeric substances 2-isopropylmalate (i.e. 3-carboxy-3-hydroxy-4-methylpentanoate) or 3-



3-isopropylmalate dehydratase

isopropylmalate (i.e. 3-carboxy-2-hydroxy-4-methylpentanoate); they are enzymatically interconvertible intermediates in the biosynthesis of leucine in plants. *See also* **3-isopropylmalate dehydratase**.



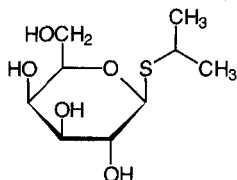
(2R,3S)-3-isopropylmalic acid

3-isopropylmalate dehydratase EC 4.2.1.33; *systematic name*: 3-isopropylmalate hydro-lyase; an enzyme that catalyses the isomerization of 3-isopropylmalate to 2-isopropylmalate, by dehydration to 2-isopropylmaleate and rehydration to generate the isomer, in the pathway of L-leucine biosynthesis in plants. *See* **leucine and valine biosynthesis**.

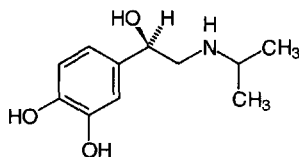
3-isopropylmalate dehydrogenase EC 1.1.1.85; *systematic name*: 3-carboxy-2-hydroxy-4-methylpentanoate:NAD⁺ oxidoreductase; an enzyme that catalyses the oxidation of 3-isopropylmalate by NAD⁺ (followed by loss of CO₂) to 4-methyl-2-oxopentanoate, in the pathway of L-leucine biosynthesis in plants. *See* **leucine and valine biosynthesis**.

2-isopropylmalate synthase EC 4.1.3.12; *systematic name*: 3-carboxy-3-hydroxy-4-methylpentanoate 3-methyl-2-oxobutanoate-lyase (CoA-acetylating); an enzyme that condenses acetyl-CoA with 3-methyl-2-oxobutanoate to form 2-isopropylmalate, in the pathway of L-leucine biosynthesis in plants. *See* **leucine and valine biosynthesis**.

isopropyl β-thiogalactoside *abbr.*: IPTG; isopropyl β-D-1-thiogalactopyranoside; an inducer of β-galactosidase in bacteria; it also induces the *lac* operon by binding and inactivating the *lac* repressor. It is used in cloning techniques to enhance expression of DNA cloned into vectors containing the *lac* operon.



isoproterenol or **isoprenaline** 3,4-dihydroxy-α-[(isopropylamino)methyl]benzyl alcohol; the synthetic isopropyl analogue of epinephrine. It is a selective agonist of β adrenoceptors, but does not discriminate between β₁ and β₂ subtypes.



isoproterenol

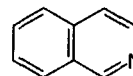
isopycnic of, pertaining to, or characterized by equal densities.

isopycnic banding the formation during **isopycnic centrifugation** of one or more bands composed of molecules that have the same density as that of the solvent at that position in the

gradient; the process of separating or purifying substances by forming such bands.

isopycnic centrifugation or **band centrifugation** or **density-gradient centrifugation** or **density-gradient sedimentation equilibrium** a separation technique in which macromolecules are subjected to a centrifugal field in a solvent containing a density gradient. The macromolecules will sediment or float and arrange themselves in bands (isopycnic bands) in the column of solvent where the density of the solvent is the same as that of the particular macromolecule. *Compare* **isokinetic gradient**.

isoquinoline 2-benzazine; benzo[c]pyridine; a bicyclic, aromatic, nitrogen-containing compound. Many alkaloids contain an isoquinoline ring system.



isoquinolinesulfonylmethylpiperazine *see* H7.

isosbestic point any wavelength at which the molar **absorption coefficients** of two or more particular chemical species are equal, i.e. their absorption spectra intersect. All mixtures (or solutions) of two or more such species, having the same total concentration, will then exhibit absorption spectra that pass through the isosbestic point, provided the species do not interact.

isoschizomer any of two or more restriction endonucleases from different sources that recognize the same sequence in a DNA molecule and display identical specificity of cleavage within that sequence. The term is sometimes extended to include those enzymes that cleave at different sites within the sequence recognized, or that respond to methyl groups on different bases in the sequence. Examples are *MspI* and *HpaII*, both of which cleave the unmethylated sequence CCGG. —**isoschizomeric** *adj.*

isosemantic substitution the incorporation of an amino-acid residue into its normal position in a polypeptide chain in response to a mutated but synonymous codon.

isosmotic or **isoosmotic** (of two or more solutions) having equal osmotic pressure; having the same osmotic pressure as another solution. *See also* **isotonic** (def. 2).

isostere or **isoster** any of two or more **isosteric** chemical species.

isosteric (of two or more chemical species) having the same, or similar, shape and size, e.g. L-threonine and L-valine; having the same number of valence electrons arranged in a similar manner. *Compare* **isoelectronic**. —**isosterism** *n.*

isotachophoresis or **isotachoelectrophoresis** or **displacement electrophoresis** a method for separating ion species in which all ions under observation are given the same migration velocity; moreover, they all have the same sign and a common counterion. The sample ions are introduced as a zone between a leading electrolyte, whose ions have a higher mobility than any of the sample ions, and a terminating electrolyte, whose ions have a lower mobility than any of the sample ions.

isotactic of, relating to, or having stereochemical regularity in the repeating units of a polymer. *Compare* **syndiotactic**.

isotherm or **isothermal** any line on a graph, or diagram, that connects points of equal temperature.

isothermal 1 (of a process or reaction) occurring at constant temperature. **2** of or pertaining to an isotherm. **3** an alternative name for isotherm.

isothiocyanate any organic compound containing the monovalent group —N=C=S. *Compare* **thiocyanate**.

isotocin or (formerly) **isotocine** [Ser⁴, Ile⁸]oxytocin; the somewhat less potent structural variant of **oxytocin**. It is present in many bony fish in place of oxytocin, hence it is sometimes known as **ichthyotocin**. *See also* **mesotocin**.

isotone any of two or more nuclides having the same number

isotonic

of neutrons but, usually, different numbers of protons and therefore differing in atomic number.

isotonic 1 of, or having the same tension; used especially of muscular contraction occurring in the absence of significant resistance and accompanied by marked shortening of the muscle fibres. *Compare isometric* (def. 2). 2 having the same osmotic pressure as some particular solution, especially one in a cell or body fluid; **isosmotic**. *Compare hypertonic, hypotonic*. 3 of, relating to, or being an **isotone** or isotones.

isotonically under **isotonic** (def. 1) conditions.

isotonicity the state or quality of being **isotonic** (def. 2); the degree of osmotic pressure.

isotope any of two or more **nuclides** that have the same **proton number**, i.e. that are nuclides of the same chemical element and thus display qualitatively the same chemical properties (for quantitative differences, see **isotope effect**). Isotopes with a given proton number usually have different **nucleon numbers** and thus characteristically usually have different relative atomic masses. In some instances, however, two isotopes of an element are **isomeric**, i.e. they have the same nucleon number (and hence the same relative atomic mass) but differ instead in their nuclear energy levels, the isotope with the higher level being **metastable** (def. 3). For the majority of elements, isotopes occur naturally and at more or less constant relative abundances; additional isotopes can be produced artificially. In general, any given element possesses both stable and unstable isotopes, although there are no stable ones for elements of high proton number; conversely, the naturally occurring isotopes of elements of lower proton number are, with a few exceptions, all stable. Metastable and unstable isotopes undergo radiative decay and are collectively termed **radioactive isotopes**, or **radioisotopes**. (*Note:* In common parlance the term isotope is often used loosely as a synonym of nuclide irrespective of its proton number.) *Compare heterotope, isobar, isotone*. —**isotopic**, *adj.*

isotope abundance the number of atoms of a particular isotope in a mixture of the isotopes of an element in a sample, expressed as a percentage of all the atoms of that element in the sample. *Compare isotope ratio*.

isotope dilution analysis an accurate method for measuring the concentration of an element, or a compound of that element, in a system by adding a known amount of a different isotope, or a labelled compound, and then measuring its concentration in a sample withdrawn from the system after mixing.

isotope effect any variation in some chemical, physicochemical, or biochemical property of a molecular entity that is detectable on isotopic substitution of an atom or atoms of the entity, e.g. variation in reaction rate, in chromatographic mobility, or in binding to macromolecules. For a given pair of isotopes, 'light' and 'heavy', the effect may be expressed quantitatively either (in the **kinetic isotope effect**) as the ratio of the rate constants, $k_{\text{light}}/k_{\text{heavy}}$, or (in the **equilibrium isotope effect**) as the ratio of the equilibrium constants, $K_{\text{light}}/K_{\text{heavy}}$, for two chemical reactions that differ only in the isotopic composition of one or more of their otherwise chemically identical components. The magnitude of the effect depends on the relative difference in atomic mass of the isotopes in question; thus it is the greatest by far for isotopes of hydrogen.

isotope exchange any chemical reaction in which the reactants and products are chemically identical but of different isotopic composition. In such a reaction the isotope distribution tends towards equilibrium.

isotope farming an informal term sometimes used to describe biological techniques for the large-scale preparation, usually for sale, of radioisotope-labelled biochemicals, especially when a number of such compounds are obtainable in the same operation, e.g. the simultaneous preparation of a number of ^{14}C -labelled compounds from $^{14}\text{CO}_2$ by photosynthesis in whole leaves or whole plants.

isotope ratio or **isotopic ratio** the ratio between the numbers of

atoms of any two isotopes of a given element in a particular preparation, sample, or substance. *Compare isotope abundance*.

isotope-ratio mass spectrometer a mass spectrometer designed specifically for the determination of **isotope ratios**.

isotopic 1 of, relating to, or being an **isotope** or isotopes. 2 being or containing a less common or artificial isotope, especially as a label or tracer. 3 (*especially of techniques*) using or depending on isotopes. —**isotopically** *adv.*

isotopy the phenomenon of the existence of isotopes in general; or the fact or condition of being isotopic in a particular instance.

isotropic or isotropous 1 (*in physics*): exhibiting uniform properties when measured along axes in all directions. *Compare anisotropic*. 2 (*in biology*) not having predetermined axes of growth. —**isotropically** *adv.*; **isotropy** or **isotropism** *n.*

isotropic band or **I-band** see **sarcomere**.

isotropic motion the random tumbling of a chemical species about its three axes.

isotropous an alternative word for **isotropic**.

isotypic specificity any antigenic specificity that is the same in a given protein in all normal individuals of the same animal species. For example anti-rabbit gamma globulins prepared in another animal species do not distinguish between gamma globulins from different rabbits.

isotypic variation structural variability of any one antigen that is common to all members of the same species.

isovaleric acidemia a hereditary metabolic disease of humans, due to defects in **isovaleryl-CoA dehydrogenase** i.e. $\text{Leu} \rightarrow \text{Pro}$ at position 13 and $\text{Gly} \rightarrow \text{Val}$ at position 170 (of the mature enzyme). It is of variable severity.

isovaleryl-CoA dehydrogenase *abbr.*: IVD; EC 1.3.99.10; a flavoprotein enzyme of the pathway for degradation of valine. It catalyses the oxidation of 3-methylbutanoyl-CoA to 3-methylbut-2-enoyl-CoA. The flavin is FAD. Example (precursor) from human: database code IVD_HUMAN, 423 amino acids (46.27 kDa).

isozyme an alternative word for **isoenzyme**.

isozymogen any precursor of an isozyme.

IT *abbr.* for isometric transition.

itaconate methylenesuccinate; methylenebutanedioate; the dianion $^-\text{OOC}-\text{CH}_2-\text{C}(\text{=CH}_2)-\text{COO}^-$; a constitutional isomer of **citraconate**. It is a metabolite of strains of *Aspergillus terreus*, which can yield it in large amounts by decarboxylation of **cis-aconitate**. It is a comparatively reactive compound, with properties useful in the manufacture of synthetic resins and plastics.

iteron any sequence of nucleotides in the duplex DNA of certain lambdoid phages that is reiterated several times with only minor changes. These repeated sequences, of 13–19 base pairs, are usually separated by short spacer sequences, each having internal dyad symmetry.

+itis *suffix forming nouns* denoting inflammation of the (specified) organ, tissue, etc.

ITP *abbr.* for 1 inosine (5'-)triphosphate. 2 isotachopheresis.

itraconazole a complex derivative of triazole, useful as an antimycotic. It inhibits cytochrome P450-dependent enzymes, thereby impairing the synthesis of ergosterol.

+ity *suffix forming nouns* meaning a property of a substance; e.g. absorptivity, density, solubility.

IU *abbr.* for International Unit (*preferred alternative to i.u. or u.*).

IUB *abbr.* for International Union of Biochemistry; renamed (1992) International Union of Biochemistry and Molecular Biology (IUBMB).

IUBMB *abbr.* for International Union of Biochemistry and Molecular Biology. See **Appendix C**.

IuDr or **IUDR** *abbr.* for 5-iododeoxyuridine (IdUrd preferred).

IUPAB *abbr.* for International Union of Pure and Applied Bio-
physics.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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IUPAC

IUPAC *abbr. for* International Union of Pure and Applied Chemistry.

iupaciubin alanyl-lysyl-glutamyl-tyrosyl-leucine; a pentapeptide containing one residue each of the five main types of amino acid found in peptides and proteins, i.e. small neutral, basic, acidic, aromatic, and lipophilic. It exists only on paper and was designed as a model to illustrate nomenclature recommendations relating to the modification of named peptides. Its name symbolizes the cooperation between IUPAC and IUB.

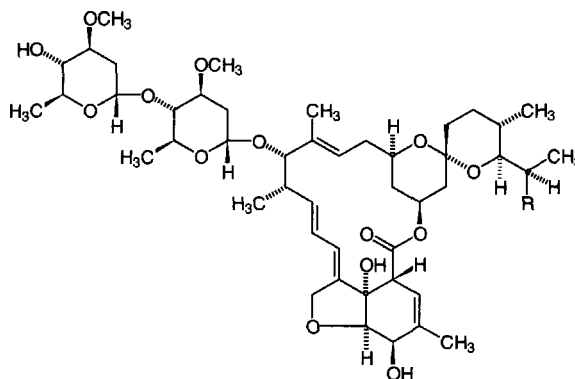
IUPAP *abbr. for* International Union of Pure and Applied Physics.

IV or **i.v.** *abbr. for* intravenous or intravenously.

IVD *abbr. for* isovaleryl-CoA dehydrogenase.

ivermectin 22,23-dihydroabamectin; 22,23-dihydroavermectin B₁; a semisynthetic material derived from **abamectin** and used as an anthelmintic in both clinical and veterinary practice, especially against *onchocerciasis*. It is used extensively as a once-a-month pill against heartworm in dogs.

iVi *symbol for* the isovaleryl group, (CH₃)₂CH-CH₂-CO- (alternative to VIⁱ).



component B_{1a}, R = C₂H₅
component B_{1b}, R = CH₃

ivermectin

Jj

j symbol for electric current density (alternative to *J*).

J symbol for 1 joule. **2 J** chain (of an immunoglobulin).

J symbol for 1 coupling constant. **2** flux (def. 2). **3** electric current density (preferred alternative to *j*).

jack bean 1 an annual tropical American leguminous plant, *Canavalia ensiformis*, grown in the USA for forage. **2** the seed of this plant. It is the source of the globulin **canavalin**, the toxic amino acid L-canavanine, the lectin **concanavalin A**, and the enzyme **urease**.

Jacob, François (1920–), French molecular biologist notable, *inter alia*, for coining the term **allosteric**, for postulating the existence of **messenger RNA**, and for studies of genetic regulatory mechanisms in protein synthesis (all with J. L. Monod); Nobel Laureate in Physiology or Medicine (1965) jointly with J. L. Monod and A. M. Lwoff 'for their discoveries concerning genetic control of enzyme and virus synthesis'.

Jacob-Monod model a model of genetic regulation of protein synthesis in prokaryotes whereby the structural genes that determine the organization of the proteins are controlled by other regions of DNA upstream from the structural genes. The latter consist of a regulator gene and a **control site**, which comprises the **promotor** (*p*) and **operator** (*o*). The regulator gene produces a **repressor** (*r*), which interacts with the operator. The operator and its associated genes are called an **operon**. The structural genes are transcribed into a single mRNA coding for the several proteins. An mRNA coding for more than one protein is known as a **polygenic** or **polycistronic** transcript. The first example was the **lac operon** but the model has been used to elucidate many other such phenomena, e.g. the **tryptophan** and **arabinose** operons. The repressor can be either inactivated (**induction**) or activated (**repression**) by specific metabolites. While the model has been helpful in explaining the mode of expression of genes in eukaryotes, operons have not been described in them. [After F. **Jacob** and J. **Monod**.]

Jamaican vomiting sickness a syndrome of sickness, vomiting, and poisoning of the nervous system, associated with profound hypoglycemia, induced by consumption of the unripe fruit of the **ackee**, the toxic principle of which is **hypoglycin**.

jaundice or **icterus** a yellowing of the skin and/or of the whites of the eyes indicating an excess of bilirubin in the blood. Jaundice may be classified into three types depending on cause: **obstructive jaundice**, in which the passage of bile from the liver to the intestine is obstructed; **hepatocellular jaundice**, in which there is disease and/or inflammation of liver cells thus diminishing their ability to secrete bile pigments into the bile passages; and **hemolytic jaundice**, in which there is an excessive breakdown of erythrocytes in the body and hence an excess of bile pigments.

JCBN abbr. for (IUPAC-IUB or IUPAC-IUBMB) Joint Commission on Biochemical Nomenclature. See also **CBN**.

J chain or **J piece** abbr. for joining chain or joining piece; a 15 kDa, cysteine-rich polypeptide chain occurring in **immunoglobulin A** (IgA) and **immunoglobulin M** (IgM) molecules. It is concerned in cross-linking the monomeric immunoglobulin moieties to form a dimer or pentamer in IgA and IgM, respectively. It is not to be confused with the J segment of an immunoglobulin gene (see **joining gene**). Example from human: database code IGH_HUMAN, 137 amino acids (15.58 kDa).

jejunum the part of the small intestine of mammals between the duodenum and the ileum. —**jejunal adj.**

jelly roll a motif in protein structure consisting of two antiparallel β sheets placed one above the other to form a sandwich structure. The polypeptide chain follows a route in which successive β strands alternate between the two sheets. If the β strands are numbered β_1, β_2 , etc. and the β sheets numbered in

parentheses, as (1), (2), etc. the structure of the motif can be represented as loop- β_1 (1)-loop- β_2 (2)-loop- β_3 (1)-loop- β_4 (2)-loop- β_5 (1), etc. The jelly-roll motif was first observed in viral coat proteins but has since been found in other proteins, the first of these to be determined being tumour necrosis factor.

Jerne, Niels Kaj (1911–94), Danish microbiologist and immunologist distinguished for his pioneering theories of antibody generation; Nobel Laureate in Physiology or Medicine (1984) jointly with G. J. F. Köhler and C. Milstein 'for theories concerning the specificity in development and control of the immune system and the discovery of the principle for production of monoclonal antibodies'.

Jerne's theory of immunity an alternative name for **natural-selection theory of immunity**.

J gene abbr. for joining gene.

Job's method an alternative name for **continuous variation**.

Johnston-Ogston effect a phenomenon occurring in sedimentation velocity experiments when two or more solutes are present in a mixture and their sedimentation coefficients are mutually concentration-dependent. It follows that the sedimentation coefficient of the more slowly sedimenting component is less in the plateau region containing both species than it is in the plateau region between the boundaries, resulting in a higher concentration of the more slowly moving component in the plateau region between the boundaries. As a consequence, the apparent concentration of the faster moving species, based on the concentration difference between the two plateaus, is too low, while the apparent concentration of the more slowly moving species is too high. [After Alexander George Ogston (1911–96) and Joseph Percy Johnston (1923–76), British biochemists.]

joinase or **joining enzyme** a former name for DNA ligase (ATP and DNA ligase (NAD⁺). See **DNA ligase**.

joining gene or **J segment** abbr.: J gene; any member of a group of approximately five germ-line genes concerned in coding for **immunoglobulin** synthesis. The J genes code for the fourth framework region of both light and heavy immunoglobulin chains and for part of the third hypervariable region of the light chain. Compare **J chain**.

joining peptide an acidic 18-residue peptide situated between γ_3 -MSH and ACTH in **proopiomelanocortin**.

Joliot, Jean Frédéric (1900–58), latterly adopting the surname Joliot-Curie, French nuclear physicist noted particularly for the production, in collaboration with his wife, I. Joliot-Curie, of the first artificial radioactive nuclides by bombarding stable elements, e.g. boron or aluminium, with fast alpha-particles; Nobel Laureate in Chemistry (1935) jointly with I. Joliot-Curie 'in recognition of their synthesis of new radioactive elements'.

Joliot-Curie 1 Irène Joliot-Curie (1897–1956), daughter of M. and P. Curie (see **Curie**), French nuclear physicist noted particularly for the production, in collaboration with her husband, J. F. Joliot, of the first artificial radioactive nuclides; Nobel Laureate in Chemistry (1935) jointly with J. F. Joliot. 2 surname latterly adopted by J. F. Joliot.

Jones-Mote sensitivity a weak, delayed-type hypersensitivity of the skin seen on challenge a few days after priming with soluble protein in aqueous solution or in incomplete Freund's adjuvant. The cells infiltrating the lesion characteristically include a high proportion of basophils.

joule symbol: J; the SI derived unit of energy, work, or heat, equal to a force of one newton acting over a distance of one metre; i.e. $1 \text{ J} = 1 \text{ N m} = 1 \text{ W s}$. The joule is equivalent to 10^7 ergs and is the energy dissipated by one watt in one second. [After James Prescott Joule (1818–89), British physicist.]

J piece an alternative name for **J chain**.



J segment

J segment *an alternative name for joining gene.*

jumping gene *an informal term for transposon.*

JUN a gene family encoding nuclear transcription factors; *v-jun* is the oncogene from avian sarcoma virus ASV17 (named from *junana*, Japanese for seventeen). *JUN* encodes transcription factor AP1; it belongs to the *FOS/JUN* family, and is rapidly and transiently expressed on stimulation of cells by mitogens. The product, Jun, forms heterodimers with Fos, and these, together with Jun homodimers, bind to the AP1 consensus site. Jun is a leucine zipper protein, which also contains basic sequences essential for DNA binding. In resting cells Jun is phosphorylated on four serines and one threonine; removal of one or more of these phosphates leads to increased binding to the AP1 promoter site. Jun has five motifs. Database code TAPI_HUMAN, 331 amino acids (35.63 kDa).

junction potential *see liquid-junction potential.*

'junk' amino acid any noncontributing amino acid.

'junk' DNA *see selfish DNA hypothesis.*

juvenile hormone any insect hormone, produced in the corpora allata glands, that functions to keep the insect in the larval (juvenile) stage. At least three are known; all are derivatives of farnesoic acid. They have potential as insect control agents.

juvenile-hormone esterase *see methylesterase.*

juxtacrine stimulation a mode of intercellular communication established by the binding of a membrane-anchored growth factor on one cell to its receptor on an adjacent cell, e.g. in the case of pro-transforming growth factor- α (*see transforming growth factor*). Unlike endocrine, paracrine, or autocrine modes there is no diffusible factor.

Kk

k symbol for kilo+ (SI prefix denoting 10^3).

k symbol for 1 rate constant; the various associated terms are distinguished by subscripts and/or superscripts (see individual entries below). **2 Boltzmann constant**. See also **k(T)**.

$\tilde{k}$ symbol for pH-independent value of the rate constant **k**.

k_0 symbol for catalytic constant (of an enzyme); alternative to k_{cat} .

K_A , K_B symbols for the specificity constants of substances A and B.

k_{app} symbol for the apparent value of the rate constant **k**.

k_{cat} symbol for catalytic constant (of an enzyme); alternative to k_0 .

k_i , k_{-i} symbols for the forward and reverse rate constants, respectively, for the *i*th step in an enzyme reaction.

k_{ij} symbol for the rate constant for the step from E_i to E_j in an enzyme reaction.

K symbol for 1 potassium. **2** kelvin. **3** a residue of the α -amino acid L-lysine (alternative to Lys). **4** a residue of an incompletely specified base in a nucleic-acid sequence that may be guanine, or either thymine (in DNA) or uracil (in RNA). **5** phylloquinone. **6** (in computing) indicating multiplied by $1024 (= 2^{10})$.

K symbol for 1 (thermodynamic) equilibrium constant (of a chemical reaction); the various associated terms are distinguished by subscripts and/or superscripts (see individual entries below). **2** kinetic energy. **3** kerma (kinetic energy released in matter). **4** luminous efficacy.

K' symbol for apparent (concentration) equilibrium constant.

K^0 or $K^\ominus$ symbol for standard equilibrium constant.

K_1 , K_2 , ... K_n symbols for the first, second, ... *n*th equilibrium (dissociation or association) constants, respectively, of the same substance.

K_a , K'_a symbols for the thermodynamic and apparent acid dissociation constants, respectively.

K_A symbol for activator constant.

K_b , K'_b symbols for the thermodynamic and apparent basic dissociation constants, respectively.

K_c symbol for equilibrium constant with respect to concentration (concentration equilibrium constant).

K_i symbol for inhibition constant (inhibition type unspecified); the reactant to which it relates may be indicated by an appropriate subscript, e.g. K_{iA} .

K_{ic} symbol for competitive inhibition constant.

K_{iu} symbol for uncompetitive inhibition constant.

K_m symbol for Michaelis constant (also called **Michaelis concentration** or, more appropriately **Michaelis–Menten constant**); the reactant to which it relates may be indicated by an appropriate subscript, e.g. K_{mA} .

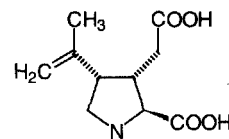
K_s symbol for substrate dissociation constant; the reactant to which it relates may be indicated by an appropriate subscript, e.g. K_{sA} .

K_s' symbol for salting out constant.

K_w symbol for the ion product of water.

kainate receptor a type of excitatory amino-acid receptor characterized by its specific agonist ligand, kainate (see **glutamate receptor**), and for which glutamate may be the endogenous ligand. The receptor is coupled to Na^+/K^+ channels in the cell membrane, which are opened when activated by agonist.

kainic acid (2*S*,3*S*,4*S*)-2-carboxy-4-(1-methylethenyl)-3-pyrrolidineacetic acid; digenic acid; a cyclic analogue of L-glutamic acid obtained from *Digenea simplex* (a red alga). It is an amino-acid agonist for the **kainate receptor**, is a neurotoxin, and has anthelmintic properties.



kainic acid

kairomone any of a heterogeneous group of chemical messengers that are emitted by organisms of one species but benefit members of another species; they are commonly nonadaptive or maladaptive to the emitter. They include attractants, phagostimulants, and other substances that mediate the positive responses of, e.g., predators to their prey, herbivores to their food plants, and parasites to their hosts. Compare **allomone**, **pheromone**.

kalinin **1** another name for laminin **5** (see **laminin**). **2** a filament-associated protein of the **laminin** family. Example (precursor) human kalinin B1, also known as nicein B1 or (most commonly) laminin B1k: database code A53612, 1170 amino acids (129.27 kDa).

kallidin an alternative name for lysylbradykinin; see **bradykinin**.

kallikrein any of two groups of serine endopeptidases, EC 3.4.21.34 and EC 3.4.21.35; the active site provides a **charge-relay system**. They are widely distributed in mammalian tissues and body fluids, including blood. **Plasma kallikrein** (EC 3.4.21.34; also called kininogenin or Fletcher factor or serum kallikrein) cleaves Arg-[Xaa and Lys]-Xaa bonds, including Lys-[Arg and Arg]-Ser bonds in kininogen to produce bradykinin, and activates blood coagulation factors XII and VII; it is formed from prekallikrein by factor XIIa. Example (precursor): database code KAL_HUMAN, 638 amino acids (71.37 kDa). **Tissue kallikrein** EC 3.4.21.35; other names: glandular kallikrein; kininogenin; a serine endopeptidase that catalyses the preferential cleavage of Arg-[Xaa bonds in small molecule substrates. Its highly selective action to release kallidin (lysyl-bradykinin) from **kininogen** involves hydrolysis of Met-[Xaa or Leu]-Xaa. The rat enzyme is unusual in liberating bradykinin directly from autologous kininogens by cleavage at two Arg-[Xaa bonds. Examples: human glandular kallikrein 1 precursor (kidney/pancreas/salivary gland kallikrein): database code KLK1_HUMAN, 262 amino acids (28.86 kDa); human glandular kallikrein 2 precursor (prostate): database code KLK2_HUMAN, 261 amino acids (28.64 kDa). 3-D structures of the protein with basic proteinase inhibitor from pig have been obtained: chain a, database code NRL_2KA1A, 80 amino acids (9.10 kDa); chain b, database code NRL_2KA1B, 152 amino acids (16.48 kDa). They were formerly classed as EC 3.4.4.21 and EC 3.4.21.8, respectively.

kallistatin or **protease inhibitor 4** an acidic protein (in plasma) of the **serpin** family, expressed widely in human tissues. It inhibits tissue **kallikrein**, forming a 1:1 complex with it. Example (precursor) from human: database code A55560, 427 amino acids (48.50 kDa).

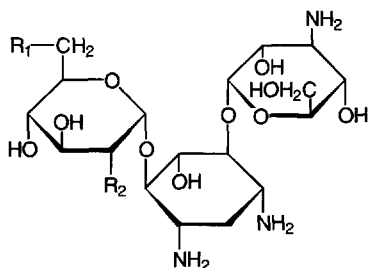
kanamycin an aminoglycoside antibiotic complex produced by *Streptomyces kanamyceticus*. It is made up of three components: kanamycin A (in which $R_1 = NH_2$, $R_2 = OH$; see structure), the major component, often designated as kanamycin; and two minor congeners, kanamycin B (R_1 and $R_2 = NH_2$) and kanamycin C ($R_1 = OH$, $R_2 = NH_2$). Kanamycin causes misreading during protein synthesis in bacteria, but its target

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site on the ribosome is apparently different from that of streptomycin. *See also* **kanosamine**.



kanosamine 3-amino-3-deoxy-D-glucose; an amino sugar present in **kanamycin** and other antibiotics, and as free carbohydrate in fermentation broths of *Bacillus aminoglucosidicus*.

K antigen *see* **capsular antigen**.

kaolin or **china clay** a finely powdered hydrated aluminium silicate; it is useful as an adsorbent.

Kaposi's sarcoma a skin cancer endemic to equatorial Africa and often occurring in AIDS patients. [After Moritz Kaposi (1837–1902), Austrian dermatologist.]

kappa *symbol*: κ (lower case) or K (upper case); the tenth letter of the Greek alphabet. For uses *see* **Appendix A**.

kappa B or **kB** an element common in eukaryotic promoters. The consensus sequence is GGRNYYCC, e.g. GGGGACTTCC in mouse and human.

kappa chain or **κ chain** *symbol*: κ ; one of the two types of **light chain** found in human immunoglobulins, the other type being a **lambda chain**. *See also* **Bence-Jones protein**.

kappa factor a protein factor that causes termination of transcription of various bacteriophage DNAs, including those of T4, T5, and T7, by *Escherichia coli* DNA-dependent RNA polymerase in a DNA site-specific manner. It is a dimer of two apparently identical 17 kDa peptide chains.

kappa particle a cell of any of several species of *Caedibacter* present in the cytoplasm of killer strains of *Paramecium aurelia*. (*Caedibacter* is a genus of Gram-negative bacterial obligate endosymbionts of this protozoan.) When kappa particles (and/or toxic constituents or products of them) are released by killer paramecia they may be lethal to sensitive paramecia (which lack both the endosymbiont and a protective genetic factor).

Karplus curve a curve describing the relation of the nuclear magnetic resonance spin-spin coupling constant and the dihedral angle. [After Martin Karplus (1930–).]

Karplus relation an equation describing the relation between the magnitude of the nuclear magnetic resonance spin-spin coupling constant, J , across three bonds and the dihedral angle, θ , about the central bond. It is:

$$J = A + B \cos \theta + C \cos^2 \theta,$$

where A , B , and C are coefficients that depend on the electronegativity of the substituents.

Karrer, Paul (1889–1971), Swiss organic chemist distinguished, *inter alia*, for his structural characterization of vitamin A, for the synthesis (at the same times as R. J. Kuhn) of vitamins A and B₂, and for the synthesis of vitamin E; Nobel Laureate in Chemistry (1937) 'for his investigations on carotenoids, flavins, and vitamins A and B₂' [prize shared with W. N. Haworth].

karyo+ or **caryo+** *comb. form* denoting the nucleus of a cell.

karyogram an alternative name for **idiogram**.

karyoid an alternative name for **nucleoid** (def. 1).

karyokinesis 1 an alternative name for **mitosis**. 2 division of the cell nucleus during mitosis. *Compare* **cytokinesis**.

karyon or **caryon** the **nucleus** of a cell.

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karyophilic proteins soluble proteins that accumulate in the nucleus on microinjection into the cytoplasm of amphibian oocytes. They are not bound to chromatin or other nuclear structures. They include **nucleoplamin** and the acidic nuclear proteins N1 to N4 of the *Xenopus* oocyte nucleus.

karyosome a compartment, located on the nucleolus of some plant species, that may support the rRNA genes as they extend from the nucleolar organizer region.

karyotype the visual appearance of the set of chromosomes of a typical somatic eukaryotic cell of a given species, individual, or cell strain. It is expressed in terms of chromosomal sizes, shapes, and number. —**karyotypic adj.**

kasugamine 2,4-diamino-2,3,4,6-tetra-deoxy-D-arabino-hexose; a component of **kasugamycin** and other antibiotics.

kasugamycin an aminoglycoside antibiotic produced by a strain of *Streptomyces kasugaensis* and containing **kasugamine**. It interacts stoichiometrically with the 30S ribosomal subunit, preventing initiation of protein biosynthesis in prokaryotes; it also blocks translation in fungal systems.

kat *symbol* for katal.

katabolism a variant spelling of **catabolism**. —**katabolic adj.**

katacalcine a 21-residue peptide derived from the protein precursor of calcitonin, in which it flanks calcitonin on the C-terminal side. It is potent in lowering plasma calcium levels. It is present in normal plasma in concentrations equimolar with calcitonin, and is released rapidly by calcium infusion.

katal *symbol*: kat; a derived coherent unit for expression of enzyme activities, defined as that **catalytic activity** of an enzyme that will raise the rate of conversion of a specified chemical reaction by one mole per second in a specified assay system. There are several important points regarding usage. (1) This definition supersedes a version expressed in terms of the intensive quantity 'rate of reaction' rather than the extensive quantity 'rate of conversion', as well as earlier versions in which catalytic activity was identified with the overall rate of reaction (including the spontaneous rate) rather than with the increase in rate. (2) The katal is not an SI unit; the SI unit mol s⁻¹ may be used if preferred. (3) The katal is not used to express rates of conversion themselves (for which the unit is mol s⁻¹). (4) The katal is independent of assay conditions (although the measured catalytic activity of a given system depends on assay conditions). (5) The use of **enzyme unit** (*symbol*: U) is discouraged; 1 kat ≈ 6 × 10³ U; 1 U ≈ 16.7 nkat).

katanin a microtubule-stimulated ATPase that severs and disassembles microtubules to tubulin dimers.

Kat.f. an obsolete unit for expressing the specific activity of catalase preparations, originally defined as the rate constant for the catalysed reaction under certain conditions divided by the number of grams of enzyme in the total test volume. The rate constant is calculated from the standard equation for a unimolecular reaction using decadic logarithms and is expressed per minute. Later (apparently unintended) variations of this definition sometimes were used. [From the German *Katalasefähigkeit*, catalase capability.]

katharometer an apparatus used to determine the composition of a gas mixture by measuring its thermal conductivity.

kathode a variant spelling of **cathode**. —**kathodic adj.**

kation a variant spelling of **cation**.

Katz, (Sir) Bernard (1911–), German-born British biophysicist and neurophysiologist distinguished for his studies of the physiology of muscle and nerve, especially regarding the mechanism of release of acetylcholine by neural impulses; Nobel Laureate in Physiology or Medicine (1970) jointly with J. Axelrod and U. S. von Euler 'for their discoveries concerning the humoral transmitters in the nerve terminals and the mechanism for their storage, release, and inactivation'.

Kauffmann-White scheme a classification scheme for the many serotypes of the bacterial genus *Salmonella*, defined in terms of their somatic (surface) antigens and, where appropriate, their capsular and flagellar antigens.

Kautsky effect the characteristic changes with time in the flu-



orescence emission of chlorophyll *a* that occur when dark-adapted cells or isolated chloroplasts are illuminated. The changes may be distinguished into a fast change lasting only a few seconds, and a slower change lasting a few minutes. [After Hans Kautsky (1891–1966).]

kayser a unit of **wave number** expressed as the number of wavelengths of electromagnetic radiation in 1 cm.

Kazal inhibitor an electrophoretically heterogeneous proteinaceous trypsin inhibitor and anticoagulant found in the pancreas and in pancreatic juice of a number of mammalian species. *Compare* **pancreatic trypsin inhibitor**. [After Louis Anthony Kazal (b. 1912), US physiologist.]

kb symbol for kilobase.

K capture *see* **electron capture**.

K cell 1 an immunocytochemically distinguishable, granule-containing cell, found in the duodenum and jejunum, that produces **glucose-dependent insulinotropic peptide**. 2 or **killer cell** a type of large granular mononuclear lymphocyte that can kill target cells sensitized with antibody. The K cells engage the antibody via their **Fc receptors**. They are probably identical with natural killer (NK) cells.

K chromophore any **chromophore** that gives rise to high-intensity absorption bands due to π - π conjugation. *Compare* **R chromophore**.

kDa symbol for kilodalton.

KDEL a tetrapeptide sequence (in the single-letter amino-acid code) at the C terminus of proteins that are to be retained in the endoplasmic reticulum (ER). An equivalent sequence is HDEL, especially in plants. The sequence is recognized by the **KDEL receptor**, and retention is thought to be mediated by continual retrieval from a post-ER compartment, as soluble ER proteins carry carbohydrate modifications assumed to have been acquired in the Golgi complex. *See also* **luminal proteins**.

KDEL receptor a transmembrane protein of the endoplasmic reticulum that recognizes and binds the tetrapeptide **KDEL**, which constitutes the C terminus of proteins destined to be retained in the endoplasmic reticulum. Such receptors are seven-transmembrane-helix proteins. An equivalent receptor in yeast for the HDEL sequence is a product of the gene *ERD2*. Example: database code ER21_HUMAN, 212 amino acids (24.54 kDa).

KDO *abbr.* for ketodeoxyoctanoate.

ke- *abbr.* for keto-.

Keilin-Hartree heart-muscle preparation a preparation of submitochondrial particles from bovine heart muscle that is capable of catalysing the reoxidation by dioxygen of NADH and succinate. [After D. Keilin (1887–1963) and E. F. Hartree (1910–94).]

Kell system a blood-group system whose antigenic determinant depends on the structure of an erythrocyte membrane glycoprotein with molecular mass 93 kDa. This glycoprotein has a long extracellular C-terminal domain with 15 cysteine residues, a short N-terminal cytoplasmic domain, and a zinc binding site of the type found in neutral endopeptidases. Example from human: database code KELL_HUMAN, 732 amino acids (82.82 kDa).

kelvin symbol: K; the SI base unit of thermodynamic temperature, defined as the fraction 1/273.16 of the thermodynamic temperature of the triple point of water. The symbol K is used both for thermodynamic temperature and for thermodynamic temperature interval, and supersedes °K (degree Kelvin). [After William Thomson, Lord Kelvin (1824–1907), British mathematician, physicist, and inventor, who first defined the absolute scale of temperature.]

KEM1 the symbol of a gene of yeast that encodes a nuclease specific for **G-quartets**.

Kendall, Edward Calvin (1886–1972), US physiological chemist noted for his isolation of the thyroid hormone thyroxine (1915) and especially for the isolation of a number of adrenal cortical hormones and the study (with P. S. Hench) of their actions; Nobel Laureate in Physiology or Medicine, (1950).

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jointly with P. S. Hench and T. Reichstein 'for their discoveries relating to the hormones of the adrenal cortex, their structure and biological effects'.

Kendall's compounds a series of naturally occurring steroids of the adrenal cortex and their metabolites; they were provisionally designated by **Kendall**, using letters of the alphabet, before elucidation of their structures. They are: Kendall's compound A, 11-dehydrocorticosterone; Kendall's compound B (Reichstein's substance H), corticosterone; Kendall's compound C (Reichstein's substance C), 5 α -pregnane-3 α ,11 β ,17 α ,21-tetrol-20-one; Kendall's compound D (Reichstein's substance A), 5 α -pregnane-3 β ,11 β ,17 α ,20 β ,21-pentol; Kendall's compound E (Reichstein's substance Fa), cortisone; Kendall's compound F (Reichstein's substance M), cortisol; Kendall's compound G (Reichstein's substance D), 5 α -pregnane-3 β ,17 α ,21-triol-11,20-dione; Kendall's compound H (Reichstein's substance N), 5 α -pregnane-3 β ,21-diol-11,20-dione; Kendall's desoxy compound B (Reichstein's substance Q), cortexone, 11-deoxycorticosterone.

Kendrew, (Sir) John Cowdery (1917–97), British biophysicist and molecular biologist renowned especially for his determination by X-ray analysis of the three-dimensional structure of the myoglobin molecule; Nobel Laureate in Chemistry (1962) the prize being shared with M. F. Perutz 'for their studies of the structures of globular proteins', especially myoglobin and hemoglobin.

kēphalin a variant spelling of **cephalin**.

kerasin or **cerasin** a **cerebroside** in which the fatty acyl moiety is lignoceryl.

keratan sulfate or (formerly) **keratosulfate** any member of a group of glycosaminoglycans with repeat units consisting of (β 1 $\rightarrow$ 3)-linked D-galactopyranosyl-(β 1 $\rightarrow$ 4)-N-acetyl-D-glucosamine 6-sulfate and containing variable amounts of fucose, sialic acid, and mannose units. In keratan sulfate I, isolated from cornea, the carbohydrate is attached to the polypeptide backbone through N-acetylglucosaminyl-asparaginyl linkages, while in keratan sulfate II, isolated from cartilage and bone, the carbohydrate appears to be attached to the polypeptide backbone through N-acetylgalactosamine in O-glycosidic linkage to either serine or threonine residues. The keratan sulfates are often attached to the same polypeptide backbone as the chondroitin sulfates.

keratein the soluble product formed by reductive scission of disulfide cross-links in wool keratin on treatment with sulfhydryl compounds such as thioglycolate.

keratin any member of a class of structural fibrous scleroproteins that occur in vertebrate skin and in various specialized epidermal structures, e.g. feathers, nails, hair, hooves, horns, and quills, as well as in the cytoskeleton. They are classified into **hard keratins**, specific to hair, nails and feathers, and **cytokeratins**, which form part of the cell cytoskeleton. The hard keratins are typically sulfur-rich proteins cross-linked by disulfide bonds, having M_r in the region of 6000–20 000; the cytokeratins are larger (40 000–56 000) low-sulfur proteins. The hard keratins of mammals are largely α helical (' α keratins'), whereas those of birds and reptiles are largely β sheet (' β keratins'). However α keratin can be converted to a β -sheet conformation by reduction and heating (as in hairdressing). Example, keratin of merino wool: database code KRB2_SHEEP, 97 amino acids (10.29 kDa). Cytokeratins are intermediate-filament proteins and are of two types: I (acidic) and II (neutral to basic) (40–55 and 56–70 kDa respectively); these form microfibrillar filaments in epithelial cells. Cytokeratin is composed of complexes formed by the aggregation of at least one of each type: e.g. keratin 1 is generally associated with keratin 10. There is similarity to all other intermediate-filament proteins. Example, keratin 1 (human): database code K2C1_HUMAN, 643 amino acids (65.42 kDa); keratin 10: database code K1CJ_HUMAN, 593 amino acids (59.46 kDa).

keratinization or **keratinisation** 1 the process by which epidermal cells synthesize keratin. 2 the elaboration of structures



keratinocyte

containing keratin, especially in skin. —**keratinize** or **keratinise** *vb.*

keratinocyte or **keratocyte** any epidermal cell that produces keratin.

keratohyalin a material seen in microscopic sections in granular cells of the skin. It is thought to consist mainly of **filaggrin**.

keratosulfate a former name for **keratan sulfate**.

kernicterus an encephalopathy associated with degeneration and yellow pigmentation of basal ganglia and other nerve cells in the spinal cord and brain, caused by the severe unconjugated bilirubinemia occurring in newborn infants and occasionally in older children with hereditary unconjugated hyperbilirubinemia.

kerosene or **kerosine** an alternative name (*esp. US*) for **paraffin** (def. 2).

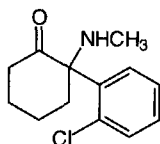
Kerr effect an alternative name for **electric birefringence**. [After John Kerr (1824–1907), Scottish physician.]

kestose 1 or (sometimes) 6-kestose *O*- β -D-fructofuranosyl-(2 $\rightarrow$ 6)-*O*- β -D-fructofuranosyl-(2 $\rightarrow$ 1)- α -D-glucopyranose; a trisaccharide widely distributed in plants, especially in the bulbs and tubers of monocotyledons, and formed artificially when yeast invertase is incubated with a 50% solution of sucrose. The name is derived from the fact that it was discovered at the Tate and Lyle research laboratories, then at Keston, Kent, UK. 2 1-kestose isokestose; inulobiosyl glucose; *O*- β -D-fructofuranosyl-(2 $\rightarrow$ 1)-*O*- β -D-fructofuranosyl-(2 $\rightarrow$ 1)- α -D-glucopyranose; an isomer and congener of 6-kestose. *See also* **neokestose**.

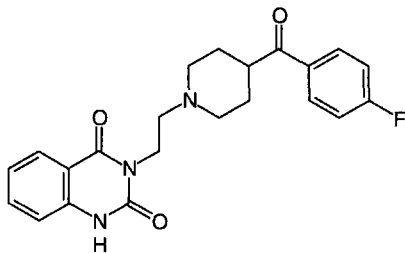
ket+ a variant form of **keto+** (sometimes before a vowel).

ketal *see* **acetal**.

ketamine ($\pm$)-2-(2-chlorophenyl)-2-(methylamino)cyclohexanone; a dissociative anesthetic with cataleptic and analgesic effects, possibly through acting as an *N*-methyl-D-aspartate (excitatory amino acid) receptor antagonist.



ketanserin 3-[2-[4-(4-fluorobenzoyl)-1-piperidinyl]ethyl]-2,4[1*H*,3*H*]-quinazolinone; an antagonist to α_1 **adrenoceptors**, H_1 **histamine receptors**, and 5-HT₂ **5-hydroxytryptamine receptors**. A clinically effective hypotensive agent, it differentiates between 5-HT₁ and 5-HT₂ receptors with high specificity; within adrenoceptors and histamine receptors it is less specific.



ketene 1 ethenone; CH₂=C=O. 2 any substituted **ketene** (def. 1). **ketimine** any **imine** that is an analogue of a ketone; the general structure is R₂C=NR' where R is any organyl group and R' may be any organyl group or H. *Compare* **aldimine**.

keto 1 carbonyl group, =CO, in any ketone. 2 describing a class of chemical compounds of the general formula R-CO-R'

(where R, R' are not H). The term is often used to distinguish such compounds from isomeric compounds of another class, e.g. keto sugar. 3 a nonpreferred term, often still used, to denote the presence of a substituent in an organic compound of a double-bonded oxygen atom that is part of a ketonic structure. The recommended term is **oxo**.

keto+ 1 or (sometimes before a vowel) **ket+** *comb. form* indicating ketonic or derived from a ketone; e.g. ketal, ketoaldose, ketal. 2 or (sometimes before a vowel) **keton+** *comb. form* denoting of or pertaining to a **ketone body** or bodies; e.g. ketoacidosis, ketogenesis, ketonuria. 3 *prefix (in chemical nomenclature)* denoting a ketonic derivative of an indicated organic compound; **oxo** now recommended.

keto- *abbr.*: *ke-*; an optional prefix to the (semi)systematic name of a monosaccharide or monosaccharide derivative containing an uncyclized ketonic carbonyl group, used to stress its acyclic nature. *Compare* **aldehyde-**, **keto**, **keto+**.

keto acid any organic (carboxylic) acid containing a **keto** (def. 3) group; the recommended term is an **oxo acid**.

ketoacidosis a type of **acidosis** caused by enhanced production of **ketone bodies** by the ketogenic pathway. Excessive production of ketone bodies has harmful effects not only in lowering the blood pH but in promoting the excretion of K⁺ and Na⁺ by the kidney, leading to cation depletion. It occurs in conditions where **ketogenesis** occurs faster than the **ketone bodies** produced can be metabolized; this results most often from hormonal imbalance, especially insulin deficiency, when fatty acids are mobilized at an excessive rate from adipose tissue and transported to the liver, where they are converted to ketone bodies. In humans, a plasma level of up to 8 mmol L⁻¹ is considered safe and may be beneficial, but in disease states levels may reach up to 20 mmol L⁻¹, contributing to life-threatening situations. —**ketoacidotic** *adj.*

ketoaldonic acid any monosaccharide formally derived from an aldonic acid by oxidation of a secondary alcoholic hydroxyl group to an oxo group.

ketoaldose any monosaccharide containing both an aldehydic and a ketonic carbonyl group.

ketoconazole a complex derivative of imidazole, clinically useful as an antimycotic. It impairs ergosterol synthesis in fungal cell membranes, thereby increasing their permeability.

ketodeoxyoctanoate *abbr.*: KDO; 2-keto-3-deoxy-D-oxalate; 3-deoxy-D-manno-2-octulosonate; an acid present in lipopolysaccharides of the outer membranes of certain Gram-negative bacteria.

keto-enol tautomerism a type of **tautomerism** in which the keto and enol forms of a compound are in equilibrium, the change corresponding to the migration of a hydron and a shift of bonding electrons. *Compare* **amino-imino tautomerism**.

ketofuranose any **ketose** in the **furanose** form.

ketogenesis the metabolic production of **ketone bodies**. The pathway starts with the formation of acetoacetyl-CoA (I) from two molecules of acetyl-CoA by **acetyl-CoA C-acetyltransferase** (EC 2.3.1.9); I then reacts with a further molecule of acetyl-CoA, in a reaction catalysed by mitochondrial **hydroxymethylglutaryl-CoA synthase**, to form (S)-3-hydroxy-3-methylglutaryl-CoA (II); II is then cleaved to acetoacetate and acetyl-CoA by **hydroxymethylglutaryl-CoA lyase**. The reactions of ketogenesis occur in higher animals, localized entirely in mitochondria; the liver is the main site of synthesis, but kidney, and possibly also small intestine, and adipose tissue may play a part in some situations.

ketogenic forming, or having the quality of being convertible into ketone bodies by metabolic processes.

α -ketoglutarate a nonpreferred (though often used) name for 2-oxoglutarate; 2-oxo-1,5-pentanedioate; a compound that has important roles in carbohydrate and amino-acid metabolism, especially in transamination reactions and as a component of the **tricarboxylic-acid cycle**.

α -ketoglutaric dehydrogenase *see* **oxoglutarate dehydrogenase** (lipoamide).

β -ketothi

NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:



to free enzyme plus the nonproduct C^* , which can further decompose to C . According to kinetic proofreading, the incorrect intermediate, ED^* , diffuses into the solution and decomposes faster than it reacts to give the wrong product, i.e. for $ED^* k_3 \ll k_{-4}$, while for $EC^* k_3 > k_{-4}$. In this way, the better binding of C is used twice, once in the formation of EC and again in the binding of the intermediate EC^* . If the discrimination in the formation of the initial complex is f , a second factor of f may theoretically be attained, giving a total discrimination of f^2 .

kinetics 1 the branch of chemistry concerned with studying rates of chemical reactions, 2 the branch of mechanics concerned with the effects of forces on the motions of material bodies.

kinetin 6-furfurylaminopurine; N^6 -furfuryladenine; an artefactual compound formed by autoclaving DNA. It exhibits **cytokinin** activity in plants and was originally thought to be a naturally occurring substance. *Compare Kinetin.*

Kinetin a proprietary name for **hyaluronidase**. *Compare kinetin.*

kinetochore a specialized assembly of proteins that bind to the **centromere** and also attach to **microtubules** growing out from **centrosomes** during mitosis. They are involved in the separation of chromosomes into the daughter cells. They are composed of **centromere-binding proteins**.

kinetochore microtubule *see* **mitotic spindle**, **prometaphase**.

kinetoplast a DNA-rich organelle located within an expanded region of a mitochondrion lying close to the **basal body** in members of the order **Kinetoplastida**. The kinetoplast is self-replicating and divides before the nucleus in cell division.

Kinetoplastida an order of protozoans whose members are characterized by the presence of a **kinetoplast** in the mitochondrion. It includes the genera *Trypanosoma*, *Leishmania*, and *Crithidia*.

kinetosome an alternative name for **basal body**.

King-Altman method a general method for obtaining steady-state expressions for enzymic mechanisms with n enzyme-containing intermediates $EX_1 \dots EX_n$. First, a simple geometrical pattern describing the mechanism is drawn. Second, all possible patterns, consisting of $(n - 1)$ lines, that interconnect the possible enzyme forms are drawn; closed loops are forbidden. In general there will be $m!/(n - 1)!(m - n + 1)!$ patterns, where m is the number of interconversion steps. Third, n expressions for $(d[EX_i]/dt[E_{tot}]) = 0$ in terms of the rate constants are set out and solved by determinant methods, using the fact that

$$\sum_{i=1}^n ([EX_i]/[E_{tot}]) = 1$$

where $[E_{tot}]$ is the total enzyme concentration. [After E. L. King and C. Altman, who described it in 1956.]

kinin 1 or **kinin-hormone** any member of a varied group of linear polypeptides that constitute a class of so-called tissue or local hormones. Each is locally released from diffuse stores in the body, not having specialized glands of secretion, and is rapidly inactivated at the site of release. Among the most important are **angiotensins**, **bradykinin**, **substance P**, **colostrokinin**, **urinary kinins**, and **secretin**. 2 **kinin-9** an alternative name (sometimes) for **bradykinin**. 3 **kinin-10** an alternative name (sometimes) for **kallidin** (lysylbradykinin). *See* **bradykinin**. 4 an alternative name for **cytokinin**.

kininase any enzyme concerned in the natural breakdown of a **kinin** (def. 1). Included are **kininase I** (or **lysine carboxypeptidase**, EC 3.4.17.3), a carboxypeptidase circulating in plasma that inactivates **bradykinin** and possibly also **kallidin**, and **kininase II** (peptidyl-di-peptidase A, EC 3.4.15.1), also known as **angiotensin-converting enzyme**.

kininogen any inactive precursor of a **kinin** (def. 1).

kininogenin an alternative name for **kallikrein**.

kinked-helix model of duplex DNA a speculative model of the structure of double-stranded DNA in chromatin, proposed in 1975 by F. Crick and A. Klug to account for the high degree of folding of the DNA (to about one seventh of its length). In

this model, based on the B form of duplex DNA, all the base pairs of the double helix remain intact but a sharp turn, or kink, of 95–100° occurs every 20 base pairs along the duplex. At this point one base pair has become unstacked from an adjacent pair through change of the conformation of the deoxyfuranose ring of a nucleotide residue from one of its possible staggered conformations to another. It was suggested that a kink might render several base pairs more easily available for specific interaction with a protein, and that a similar type of folding might occur in double-stranded DNA in other situations, especially when closely associated with protein.

kinomere an alternative term for **centromere**.

Kirkwood-Riseman theory of hydrodynamic properties of complex structures a theory providing an approximate method for computing the friction of a structure made up of identical subunits. A linear or coiled polymer is approximated as a string of beads, each with identical hydrodynamic properties, while an oligomeric protein is modelled as a cluster of identical subunits each with identical hydrodynamic properties.

Kirkwood theory of dielectric polarization of polar liquids an extension of the Onsager theory in which the dielectric constant of the central molecule and its first shell of neighbours is considered to differ from its macroscopic value. [After John Gamble Kirkwood (1907–59).]

Kirkwood theory of ion-dipole interactions a model in which the electrical contribution to the chemical potential of an ion having an arbitrary charge distribution is calculated with the aid of the Debye-Hückel theory and in which the Born relation between the free energy of solvation of a spherical ion and the dielectric constant of the solvent is generalized to include ions of arbitrary charge distribution.

Kirkwood-Westheimer theory of electrostatic effects on acid dissociation constants an extension and amplification of the Bjerrum theory of the influence of substituents on acid dissociation constants, the molecules and ions entering into ionization equilibria being treated as cavities of low dielectric constant. [After John Gamble Kirkwood (1907–59) and Frank Henry Westheimer (1912–), US biochemist.]

kirromycin or **mocimycin** an antibacterial antibiotic isolated from *Streptomyces collinus* and others. It specifically interacts with **elongation factor** EF-Tu (*see* **EF-T**) and prevents the release of EF-Tu from the bacterial ribosome after GTP hydrolysis. In the absence of ribosomes, the antibiotic induces an EF-Tu-dependent hydrolysis of GTP.

kistrin a protein isolated from the venom of the Malayan pit viper, *Agkistrodon rhodostoma*. It contains the **RGD** sequence and binds to a platelet adhesion site. It is a potent platelet aggregation inhibitor and antagonist of a family of glycoproteins that mediate aggregation. *For example, see* **disintegrin**.

KIT a family of genes encoding plasma membrane class III receptor tyrosine kinases. The viral gene, *v-kit*, is the oncogene of the Hardy-Zuckerman 4 strain of acutely transforming feline sarcoma virus. The product of the protooncogene *c-kit* has five immunoglobulin-like loops in the extracellular domain; the cytoplasmic region contains a tyrosine kinase domain with an insert region, and is myristoylated. Interaction with its ligand, steel factor (*see* **stem cell factor**), brings about dimerization, leading to enhanced autophosphorylation and binding of phosphatidylinositol 3-kinase and phospholipase $C\gamma$. Kit and its ligand are highly expressed in small-cell lung cancer and a significant proportion of testicular germ cell tumours. Example from human (precursor): database code KKIT_HUMAN, 976 amino acids (109.74 kDa).

Kjeldahl, Johan Gustav Christoffer Thorsager (1849–1900), Danish brewing chemist; first director of the Carlsberg Laboratory, Copenhagen.

Kjeldahl flask a small pear-shaped glass flask with a long neck, as used in the **Kjeldahl method**.

Kjeldahl method a method for the determination of the total nitrogen in an organic compound or mixture of compounds.



Kleinschmidt technique

The sample, in a **Kjeldahl flask**, is digested in hot, concentrated sulfuric acid in the presence of a catalyst, usually one containing mercury, copper, or selenium; the nitrogen in the sample is thereby converted to ammonium hydrogen sulfate. The ammonium formed may be estimated by neutralization and steam distillation of the resultant ammonia into a known amount of acid, the residual acid being determined by titration, or it may be estimated by **nesslerization** or with an ammonium electrode.

Kleinschmidt technique a technique for preparing monolayers of DNA for electron microscopy in which the DNA molecules are bound to a film of denatured basic protein, e.g. cytochrome *c*, on the surface of an aqueous solution. The DNA molecules are consequently brought from a three-dimensional conformation to a two-dimensional one by adsorption to the protein layer. The DNA-protein complex is then transferred to a solid surface, e.g. an electron microscope grid or collodion film, dried, and contrasted with heavy metals by shadowing or staining. The tendency of **ssDNA** to fold back on itself with the formation of intramolecular base pairs can be overcome by using formamide (which denatures short base-paired regions) or by replacing cytochrome *c* with DNA-unwinding proteins, which have high affinity for ssDNA. The method may also be applied to RNA. [After Albrecht K. Kleinschmidt (1916–), German microbiologist.]

Klenow enzyme or **Klenow fragment** a name sometimes used for the larger (75 kDa) of the two fragments formed by the treatment of *Escherichia coli* DNA polymerase I with subtilisin in the presence of DNA. The Klenow enzyme retains the polymerase activity and the single-strand-specific 3'→5' exonuclease activities but lacks the 5'→3' exonuclease activity of DNA polymerase I. It is useful in sequencing single-stranded DNA. [After Hans Klenow (1923–), Danish biochemist.]

Klotz plot a name sometimes given to a plot of the reciprocal form of an equation describing ligand binding at multiple independent sites on, e.g., a macromolecule. It has the form:

$$1/\bar{v} = (1/nk)(1/[A]) + 1/n,$$

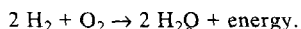
where $\bar{v}$ is the average number of ligand molecules bound per molecule of the macromolecule, n is the number of binding sites per macromolecule, $[A]$ is the molar concentration of free ligand molecules, and k is the intrinsic association constant. If the equation correctly describes the equilibria involved, a plot of $1/\bar{v}$ versus $1/[A]$ gives a straight line of slope $(1/nk)$ and with an intercept on the $1/\bar{v}$ axis of $1/n$ and on the $1/[A]$ axis of $-1/k$. [After Irving Myron Klotz (b. 1916), US biochemist.]

Klug, (Sir) Aaron (1926–), Lithuanian-born South African and subsequently British molecular biologist distinguished for his determination of the structure of transfer RNA and for his contributions to the understanding of the three-dimensional structure of complexes of proteins and nucleic acids, especially tobacco mosaic virus and the nucleosome, through his invention of the technique of image reconstruction; Nobel Laureate in Chemistry (1982) 'for his development of crystallographic electron microscopy and his structural elucidation of biologically important nucleic acid-protein complexes'.

KMEF group *abbr. for* keratin-myosin-epidermin-fibrinogen group; a group of proteins that have in common an α -type X-ray diffraction pattern. Some of these may show a β -type pattern in some conditions. Silk fibroin, which normally occurs with a β structure, is often included in the KMEF group.

knallgas bacteria or **hydrogen bacteria** any of a number of bacterial species that can obtain energy from the **knallgas reaction** i.e. the oxidation of dihydrogen to water by dioxygen. They include some *Alcaligenes* and *Pseudomonas* spp. and *Paracoccus denitrificans*. [From the German, *Knallgas*, an explosive mixture of hydrogen and oxygen.]

knallgas reaction the energy-yielding direct oxidation of dihydrogen by dioxygen occurring in certain bacteria (*see knallgas bacteria*). The reaction is:



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knockout indicating an animal in which function of a gene has been deleted by genetic engineering techniques.

Knoop's hypothesis the hypothesis that fatty acids are metabolically oxidized by the successive removal of two-carbon fragments as acetic acid. *See also beta oxidation*. [After Franz Knoop (1875–1946), German physiologist and biochemist.]

Knop solution an aqueous solution containing the major inorganic constituents required for the growth of cells of higher plants, namely calcium nitrate, potassium nitrate, magnesium sulfate, potassium dihydrogen phosphate, and potassium chloride. [After J. A. L. W. Knop (1817–1901), German chemist.]

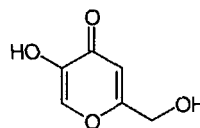
Knox plot (*in high-performance liquid chromatography*) a plot of relative separation efficiency versus flow rate, useful in determining the optimal conditions for a desired separation. [After John H. Knox.]

Koch phenomenon a cell-mediated hypersensitivity reaction in which inflammation of guinea-pig skin, when injected with living or dead *Mycobacterium tuberculosis*, is very much greater in animals actively infected with the organism than in uninfected animals injected with dead organisms. [After (Heinrich Hermann) Robert Koch (1843–1910), German bacteriologist, who isolated the bacillus in 1882.]

Köhler, Georges Jean Franz (1946–95), German immunologist famous for his discovery (with C. Milstein) of the technique of using cell fusion to generate a clonal cell population capable of producing unlimited quantities of a pure antibody; Nobel Laureate in Physiology or Medicine (1984) jointly with N. K. Jerne and C. Milstein 'for theories concerning the specificity in development and control of the immune system and the discovery of the principle for production of monoclonal antibodies'.

kojibiose the disaccharide 2-*O*- α -D-glucopyranosyl-D-glucopyranose.

kojic acid 5-hydroxy-2-(hydroxymethyl)-4H-pyran-4-one; a metabolite from *Aspergillus oryzae* and *A. terreus*. It can be obtained in large amounts and has some industrial application.



Kok effect the sharp difference in yield of photosynthesis above and below the compensation point. Below compensation, where respiration can supply intermediates at a greater rate than the rate of photosynthesis, the yield is about twice as great as above compensation, where CO₂ has presumably to be drawn into the process. [After Bessel Kok (1918–81) Dutch-born US plant physiologist.]

Kornberg 1 Arthur Kornberg (1918–), US biochemist distinguished especially for his discovery and purification of the enzyme DNA polymerase and its use to synthesize segments of DNA molecules and viral DNA; Nobel Laureate (1959) in Physiology or Medicine jointly with S. Ochoa 'for their discovery of the mechanisms in the biological synthesis of ribonucleic acid and deoxyribonucleic [sic] acid'. **2 Hans Leo Kornberg** (1928–) German-born British biochemist notable for his studies of intermediary metabolism in bacteria.

Kornberg enzyme an alternative name for DNA polymerase I. *See DNA polymerase*. [After A. Kornberg.]

Korsakoff's psychosis *see* Wernicke-Korsakoff syndrome.

Koshland model *see* induced-fit theory.

Kossel, Karl Martin Leonhard Albrecht (1853–1927), German physiological chemist distinguished for his researches on nucleins (i.e. nucleoproteins), for his discovery of adenine, cytosine, thymine, and uracil as breakdown products of nucleic

250K protein

acids, for his discoveries of histone and histidine, and for his studies of the composition of proteins and its relation to their biological functions; Nobel Laureate in Physiology or Medicine (1910) 'in recognition of the contributions to our knowledge of cell chemistry made through his work on proteins, including the nucleic substances'.

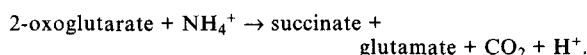
Kr symbol for krypton.

Krabbe's disease an alternative name for **galactosylceramide lipidosis**. [After Knud Haraldsen Krabbe (1885–1961), Danish physician.]

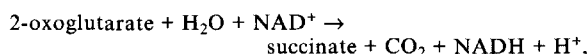
KRB *abbr.* for Krebs Ringer bicarbonate; *see* **Krebs mammalian Ringer**.

Krebs 1 (Sir) **Hans Adolf Krebs** (1900–81), German-born British biochemist renowned for his work on intermediary metabolism and in particular for the discovery of the two cyclic metabolic pathways named after him (*see* **Krebs cycle** def. 1,2) Nobel Laureate in Physiology or Medicine (1953) 'for his discovery of the citric acid cycle' [prize shared with F. A. Lipmann]. **2** **Edwin Gerhard Krebs** (1918–), US biochemist; Nobel Laureate in Physiology or Medicine (1992) jointly with Edmond H. Fischer 'for their discoveries concerning reversible protein phosphorylation as a biological regulatory mechanism'.

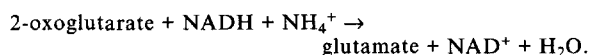
Krebs-Cohen dismutation The anaerobic reaction summarized as:



It has been shown to occur in pigeon heart and breast muscle, rat and guinea-pig renal cortex, and rat-liver mitochondria. It is the sum of two reactions:



and



[After H. A. **Krebs** and Philip Pacy **Cohen** (1908–93), US biochemist.]

Krebs cycle an alternative name (sometimes) for **1 tricarboxylic acid cycle**. **2** **ornithine-urea cycle**.

Krebs-Henseleit cycle an alternative name (sometimes) for the **ornithine-urea cycle**. [After H. A. **Krebs** and Kurt **Henseleit** (1908–73).]

Krebs-Henseleit Ringer bicarbonate *see* **Krebs mammalian Ringer**.

Krebs-Kornberg cycle an alternative name (sometimes) for **glyoxylate cycle**. [After H. A. **Krebs** and H. L. **Kornberg**.]

Krebs mammalian Ringer any of several media devised for the suspension of mammalian tissues. Their composition was essentially based on the available analytical data for the inorganic constituents of blood plasma of a number of mammalian species. In the first version (1932), **Krebs-Henseleit Ringer bicarbonate**, the buffer was bicarbonate-CO₂, while in the second (1933), **Krebs original Ringer phosphate**, it was phosphate. In improved versions (1950), often referred to respectively as **Krebs Ringer bicarbonate** (*abbr.*: **KRB**) and (calcium-free) **Krebs Ringer phosphate** (*abbr.*: **KRP**), a mixture of organic acids and glucose was substituted for part of the NaCl in the first two versions, to resemble as closely as possible the state of the tissues in the intact animal. Also, an additional medium, low in phosphate, bicarbonate, and CO₂, was devised. *See also* **Ringer's solution**.

Krebs' manometer fluid a solution containing 44 g NaBr, 0.3 g Triton X-100, and 0.3 g Evans' Blue in 1000 mL of water; *D* = 1.033. The standard pressure (760 mmHg) in mm of manometer fluid is 10 000. Other detergents may be substituted for the Triton X-100 in equivalent proportions.

K region a region in a carcinogenic polycyclic aromatic hydrocarbon molecule in which there is a carbon-carbon bond having ordinary double-bond rather than aromatic character and that takes part in addition reactions; e.g. the bond between

atoms C-9 and C-10 in phenanthrene. It has been suggested that carcinogenicity in such molecules is determined by the existence in them of an active K region. *Compare* **L region**.

kringle a conserved protein domain of about 85 amino acids that is held in a characteristic three-dimensional conformation by three internal disulfide bridges. It is so-named because its two-dimensional representation resembles that of the kringle, a Scandinavian pastry of characteristic shape. It was first described in prothrombin, but homologous domains, with the characteristic shape and similarly positioned disulfides, are found in plasminogen and (both tissue-type and urokinase-type) plasminogen activators. *See also* **Appendix E**.

Kronig-Kramers transform any of a set of mathematical relations describing the response of a system to a perturbation such as light. These relations allow the calculation of optical rotary dispersion from circular dichroism (ellipticity) and vice versa, and describe the relationship between dispersion and absorption. [After R. de L. **Kronig** and Hendrik Anthony **Kramers** (1894–1952), Dutch physicist, who independently devised them in 1926 and 1927, respectively.]

KRP *abbr.* for Krebs Ringer phosphate; *see* **Krebs mammalian Ringer**.

Krüppel a protein involved in the segmentation of the embryo and in the differentiation of the malpighian tubules in insects (e.g. *Drosophila*). It is a nuclear, chromatin-associated protein, encoded by a **gap gene**, and is a member of a subfamily of C₂H₂-type zinc-finger proteins. Mutant *krüppel* larvae lack eight body segments. Example from *Drosophila melanogaster*: database code **KRUP_DROME**, 466 amino acids (50.74 kDa). [From German *Krüppel*, cripple.]

KS/hst a human oncogene, with no known viral counterpart, whose product is related to **fibroblast growth factor** and is expressed in **Kaposi's sarcoma**. Database code: **HBG4_HUMAN**, 206 amino acids (22.02 kDa).

k(T) symbol for rate constant (alternative to *k*, used when distinction of a rate constant from the Boltzmann constant is necessary).

Kuhn, Richard Johann (1900–67), Austrian-born German chemist and biochemist notable, *inter alia*, for his reintroduction (with H. Lederer) of the technique of adsorption chromatography by demonstrating the resolution of carotene into three isomers, for his synthesis (at the same times as P. Karrer) of vitamins A and B₂, for his isolation of nicotinamide from heart muscle and of riboflavin from milk, and for the synthesis of the coenzyme flavin mononucleotide, riboflavin 5'-phosphate; Nobel Laureate in Chemistry (1938) 'for his work on carotenoids and vitamins'.

Kunitz and Northrop inhibitor *see* **pancreatic trypsin inhibitor**.

Kunitz inhibitor the best known of the three distinguishable **soybean trypsin inhibitors**. The term has been extended to refer to any type of protease inhibitor that contains consensus sequences known as **Kunitz domains** (*see* **Appendix E**). These include some serine protease inhibitors. Such domains have also been identified in some α -amyloid precursor protein isoforms.

Kunitz (tryptic) unit a measure of tryptic activity based on the extent to which trypsin-containing samples can digest casein under standard conditions often used to determine the relative amounts of Kunitz and Bowman-Birk type trypsin inhibitors present in samples, comparison being made between tryptic activity and chymotryptic activity. Crystalline trypsin has an activity of 6 Kunitz units per mg.

Kupffer cell a type of nonmotile, mononuclear phagocytic cell. Such cells line the sinusoids of the liver. [After Karl Wilhelm von **Kupffer** (1829–1903), German anatomist.]

kuru a transmissible **spongiform encephalopathy** of humans, characterized by degeneration of the nervous system and found among the Fore people of Papua New Guinea. It is a **prion** disease, transmitted through funerary practices thought to have involved endocannibalism. The brains of victims of this and **Gerstmann-Sträussler syndrome** contain so-called kuru plaques which have been shown to stain positively with antibodies

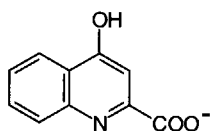


kwashiorkor

against prion protein in patients with Gerstmann-Sträussler syndrome. It has become extremely rare since cannibalism ceased in the late 1950s.

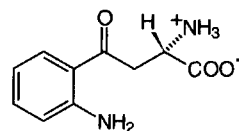
kwashiorkor a form of severe malnutrition of children occurring in tropical or near-tropical regions. It is generally believed to be caused by a dietary deficiency of protein, but not of other energy-producing foodstuffs. The condition commonly develops when, after prolonged breast feeding, the child is weaned onto a diet inadequate in protein. *Compare marasmus.*

kynurenate 4-hydroxyquinaldate; 4-hydroxy-2-quinoline carboxylate; the anion of kynurenic acid and an intermediate in the conversion of **kynurenine** to quinaldate.



kynureninase EC 3.7.1.3; *systematic name*: L-kynurenine hydrolase; a pyridoxal-phosphate-containing enzyme that hydrolyses L-kynurenine to anthranilate and L-alanine. It also acts on 3'-hydroxykynurenine and some other (3-arylcarbonyl)-alanines.

kynurenine 3-anthraniloylalanine; L-kynurenine is a metabolite on the degradative pathway of tryptophan, is found in the urine especially in pyridoxal (vitamin B₆) deficiency, and is an intermediate in the synthesis of nicotinic acid from tryptophan.



kyotorphin the analgesic dipeptide, L-Tyr-L-Arg, isolated from bovine brain; D-kyotorphin is L-Tyr-D-Arg.

L

l symbol for 1 litre (alternative to L). **2** liquid.

l symbol for length (alternative to L).

l- prefix (in chemical nomenclature) denoting levorotatory; now obsolete, use (-); see **optical isomerism**.

L symbol for 1 litre (alternative to l). **2** a residue of the α -amino acid L-leucine (alternative to Leu). **3** an atom of hydrogen that may be one of either protium or deuterium. **4** linking number. **5** light chain (def. 1) (of an immunoglobulin).

L symbol for 1 Avogadro constant (alternative to N_A). **2** latent heat. **3** radiance.

L- a configurational prefix used in designating the absolute configuration of chiral compounds of certain classes; see **D/L convention**.

La symbol for lanthanum.

label **1** or (sometimes informally) **tag** any (radioactive or stable) isotope substituted in a proportion of the molecules of a particular compound so that the translocation and/or the chemical or biochemical transformation of the compound may be followed. See also **tracer**. **2** or (sometimes) **marker** an easily recognizable chemical group (which can be radioactive) attached to a molecule or incorporated within a structure. **3** to attach or apply such a label. The following particular uses of the terms label or labelled have individual entries (see also the list at **labelling**): **affinity label**, **electron-dense label**, **immunoassay**, **mixed labelled**, **multiply labelled**, **nonselectively labelled**, **photoaffinity label**, **radioactive label**, **selectively labelled**, **singly labelled**, **specifically labelled**, **spin label**, **uniformly labelled**.

labelled compound or (US, usually) **labeled compound** a compound bearing a **label** (def. 1, 2). An isotopically labelled compound is a compound in which one or more atoms of a proportion of the molecules have been replaced by a (radioactive or stable) isotope; i.e. it is a mixture of an isotopically unmodified compound with one or more analogous isotopically substituted compound(s). In some cases, the amount of isotopically unmodified compound may be small. This is particularly the case for ^2H compounds, when near 100% replacement is possible. See also **carrier-free**.

labelling or (US, usually) **labeling** **1** the attribute of being labelled. **2** the act or process of chemically introducing a **label** (def. 1, 2) into a compound either by substitution or by synthesis, or of incorporating such a label into a material or structure. The following particular uses of the term labelling have individual entries (see also the list at **label**): **conjugation labelling**, **double labelling**, **enzyme labelling**, **ferritin labelling**, **fluorescein labelling**, **generally labelled**, **hapten-sandwich labelling**, **iodine labelling**.

labile **1** describing a chemical compound that is prone to spontaneous change or decomposition; relatively unstable. **2** describing an atomic or molecular grouping in a chemical compound that unusually readily dissociates from the compound; loosely attached. **3** describing a chemical bond that very easily ruptures or is ruptured. —**lability** *n*.

labile factor an alternative name for factor V; see **blood coagulation**.

labile phosphate the amount or proportion of any phosphate that is readily liberated from a phosphate-containing compound by hydrolysis in 1 M HCl at 100 °C within 7–10 min (hence also sometimes known as **seven-minute phosphate** or **ten-minute phosphate**). These conditions were selected originally for estimating the two terminal phosphate groups of ATP. The following compounds are included in the group (percentage hydrolysis in parentheses): phosphocreatine (100), ATP (67), ADP (50), aldose 1-phosphate (100), pyrophosphate (100).

labilize or **labilise** to render **labile** (esp. def. 3). —**labilization** or **labilisation**, **labilizer** or **labiliser** *n*.

lac any of various resinous substances secreted by certain plants or insects, used in the manufacture of lacquers.

lac symbol for the genetic system of *Escherichia coli* that controls the organism's ability (lac^+) or inability (lac^-) to metabolize lactose. See also **lac operon**, **lac repressor**.

laccase any of a group of copper-protein enzymes (EC 1.10.3.2) of low specificity, that catalyse the oxidation of both *o*- and *p*-quinols by dioxygen to form the corresponding semiquinones and two molecules of water. They are so-called because such an enzyme was first found in the lac (or lacquer) tree (*Rhus verniciflua*); however, they are now known to be widely distributed in plants and fungi.

lacI the structural gene for **lac repressor**, the repressor of the **lac operon**.

lac operon or **lactose operon** a genetic system, or **operon**, of *Escherichia coli* that controls the synthesis of the inducible enzyme β -D-galactosidase (see **galactosidase**). It is probably the most intensively studied group of *E. coli* genes. It consists of a linear sequence (5' end to 3' end of the coding strand) of two control sites, designated *p* and *o*, followed by three structural genes, designated *z*, *y*, and *a*; *p* is the promoter (concerned with RNA-polymerase binding), *o* is the operator (binding site for repressor), *z* is the gene for β -galactosidase, *y* is the gene for the **permease**, and *a* is the gene for thiogalactoside transacetylase (an enzyme of uncertain physiological function). Immediately preceding the *p* site is a regulatory gene designated *i* (concerned with synthesis of **lac repressor**).

lac repressor a protein encoded by the **lacI** gene, that controls the expression of structural genes coding for enzymes involved in lactose metabolism in the *Escherichia coli* genome; in the absence of inducer it binds with high affinity to the operator site of the **lac operon**. It is a homotetramer and specifically recognizes a 23 base-pair DNA sequence in the **lac** operator by means of its four N-terminal domains (headpieces), each of 51 amino-acid residues; each subunit also has one allosteric binding site for an inducer. The repressor is a typical **helix-turn-helix** protein and gives its name to a subfamily. Database code: LACI_ECOLI, 360 amino acids (38.55 kDa).

lact+ a variant form of **lacto+** (before a vowel).

lactalbumin or α -**lactalbumin** a Ca^{2+} -binding protein of the **lactalbumin/lysozyme family**, present in the milk of virtually all mammals. In the case of non-ruminants it is the predominant protein in whey. It contains 123 amino-acid residues, 40 of which are identical with and occupy corresponding positions to those of lysozyme. It is a noncatalytic regulatory protein whose presence is required for **lactose synthase** to transfer a galactose residue from UDPgalactose to D-glucose; hence it is also known as B protein of lactose synthase. It causes the substrate for N-acetyllactosamine synthase to change from N-acetylglucosamine to glucose to give lactose and UDP. Protein B had, therefore, been called a '**specifier**' protein. The genes for lysozyme and α -lactalbumin arose evolutionarily from a common ancestral gene. Example (precursor) from human: database code LCA_HUMAN, 142 amino acids (16.21 kDa); five motifs.

lactalbumin/lysozyme family a protein family whose members are characterized by six conserved motifs. Common members are α -lactalbumin (see **lactalbumin** (def. 2)) and **lysozyme**; 3-D structures are known.

lactam any intramolecular cyclic amide produced by the formal removal of a molecule of water between the amino and the carboxyl groups of an amino acid other than an α -amino acid. A prefix (as in β -lactam) may be used to designate the position of the amino group in the parent compound. A lactam commonly exists in tautomeric equilibrium with its corresponding **lactim**.

β -lactam antibiotic any member of a class of natural or semi-synthetic antibiotics whose characteristic feature is a strained,